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The Homotopy Theory of Operator Algebras

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Abstract

In this thesis, we undertake a systematic study of the homotopy theory of operator algebras, focusing on the category of equivariant C^* -algebras and Banach algebras. The first part of the thesis mainly focuses on the classical aspects of homotopy theory in these categories, setting the stage for the later development of abstract homotopy theory. We also address several open problems in the homotopy theory of C^* -algebras. Firstly, we introduce a new class of C^* -algebras, the Schochet C^* -algebras, in relation to a conjecture of Schochet about the Calkin projection on an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space. While this conjecture already has a positive solution, we manage to provide a more precise obstruction by showing, explicitly, the existence of a non-Schochet C^* -algebra, which also gives a simpler proof of the conjecture. The example we construct is related to an open problem posed by Blackadar and we prove a theorem which connects this problem to other open problems in the literature. Secondly, we address a question of Schochet & Kaminker on the homotopy invariance of the Brown-Douglas-Fillmore Ext groups and resolve this in the negative. Thirdly, we partially address a question of Conway's with regards to the Schochet property for C^* -algebras of certain spaces. Finally, we relate the existing notion of semiprojectivity to the Schochet property & use our preceding work to construct an obstruction to a conjecture of Blackadar's on approximately semiprojective C^* -algebras.

In the second part of the thesis, we move on to abstract homotopy theory and introduce structures which are meant to model homotopy theory on the equivariant category of C^* -algebras and Banach algebras. We develop some of the theory associated to these categories, proving some new results which are analogues of known results for model categories. This theory makes a conceptual appearance in our treatment of axiomatic (co)homology on C^* -algebras, where we present a refinement of Cuntz's proof of the Bott Periodicity Theorem. Finally, we also highlight an important result about categories of fibrant objects, the Spanier-Whitehead construction, which says that there exists a stable homotopy category associated to every category of fibrant objects and that it has a triangulated structure. This result was used without formal proof in the literature and we prove a partial version of it by using tools from higher category theory.

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1 Introduction

In this introduction, we will provide information relevant to this thesis in the following order:

1. Our overarching goals.
2. Prerequisites and reading recommendations.
3. Conventions & general structure.
4. Acknowledgements.

We strongly recommend reading these sections as they contain information relevant for pleasant reading of this thesis.

1.1 Background, Overarching Goals & Structure

The intersection between operator algebras and homotopy theory is not a new one; it has existed since the days when the Brown-Douglas-Fillmore $\text{Ext}(-)$ groups were invented to study the classification theory of essentially normal operators and when Atiyah's topological K-theory was extended to the realm of C^* -algebras & Banach algebras. Since then, there has been a wave of interest in the use of methods from algebraic topology to study operator algebraic structures. The advent of bivariant homology theories on categories of operator algebras and the crucial role they play in the formulation of important conjectures has become a large part of modern research in the theory of operator algebras.

On the other side of the bridge, algebraic topology has also undergone rapid change. Indeed, the notion of “homotopy”, once thought to be something useful only for studying spaces, has now gained a life of its own and is the main notion studied in *homotopy theory*. Over the years, homotopy theory has found itself becoming a branch of mathematics which is distinct from topology. In a rather famous letter entitled *The Future of Homotopy Theory*, Clark Barwick actually makes a much stronger claim about homotopy theory in relation to topology, among other important topics.

Some of us call ourselves *algebraic topologists*, but this has the unhelpful effect of making the subject appear to be an area of topology, which I think is profoundly inaccurate. It so happens that one way (and historically the first way) to model homotopical thinking is to employ a very particular class of topological spaces. Today, the praxis of homotopy theory interacts with topology no more often than it does with arithmetic geometry and category theory, and the interactions with areas like representation theory are growing rapidly. *Homotopy theory is not a branch of topology.* [10, p. 1]

Indeed, it is certainly true that the notion of “homotopy” is pervasive across different parts of mathematics, even when there are no spaces in sight, and that it plays a crucial role in constructing invariants which are useful, for example, in solving classification problems. In that sense, it would not be correct to say that it is a purely topological notion. Nevertheless, homotopy theory *did* arise from algebraic topology, so one might expect to see many interactions between homotopy theory and operator algebras. Indeed, there have been *some* developments over the years but it has only recently become the case that we are deliberately using homotopy-theoretic methods for the purpose of understanding (and potentially solving) conjectures in operator algebra theory. So, this requires some rationalization.

One possible reason for the lack of such interactions until recent years has to do with the structures that are studied in homotopy theory. The classical structures of interest in homotopy theory, such as the stable homotopy category, did not have obvious analogues in the setting of C^* -algebras. Moreover, the methods stable homotopy theory, which used a lot of homological algebra and category theory, were only barely being introduced into operator algebra theory by the work of (Joachim) Cuntz ([28]), (Nigel) Higson ([43]), (Jonathan) Rosenberg ([84]) and others¹, though it took a number of years before their work was pushed forward *using ideas from homotopy theory*. Of course, one should also be aware that the current status of homotopy theory also took a lot of time to properly develop. For instance, it may surprise the reader to know that model categories were not viewed favorably for a long time by many homotopy theorists. Indeed, Andrew Blumberg, another famous homotopy theorist, mentions as much in a set of lecture notes on equivariant homotopy theory.

Elmendorf's original proof of his theorem was in the 1980s and did not use model categories, even though Quillen had already introduced them at the time. Until the mid-1990s (30 years after Quillen introduced model categories in [76]), many homotopy theorists avoided them, thinking of them as formal gobbledygook. However, around the time [34] introduced a symmetric monoidal category of spectra, people began realizing they were unavoidable. [31, p. 9]

So, it is not so surprising that model categories have only “recently” become structures of interest to operator algebraists. In fact, some operator algebraists might find it somewhat surprising to learn that the possible use of model categories in operator algebra theory was already considered as early as 1984 in the work of (Claude) Schochet, which we quote below.

We have not yet formally defined “admissible category of C^* -algebras”. Roughly, a category of C^* -algebras is admissible if it is closed under the various homotopy constructions required to develop elementary homotopy theory. Here are some admissible categories:

1. all C^* -algebras,
2. separable C^* -algebras,
3. separable nuclear C^* -algebras,

¹The references we provide are some examples of older work by these mathematicians which have a strong homological/categorical flavor.

In each case, we allow all C^* -maps as morphisms. This is a provisional definition. It should be replaced by the closed model categories of Quillen. However, there is as yet no good notion of loop space or fibration. Thus, we proceed in an ad hoc manner... [85, p. 402]

It suffices to say that there was much potential for early interactions between homotopy theory and operator algebras. Of course, we must emphasize that recent years have brought many more interactions. In fact, we should mention that newer developments in homotopy theory have brought the so-called $(\infty, 1)$ -categories to the frontier. These are, in a sense, supposed to “replace” model categories as the “correct” setting in which one should do homotopy theory. Nevertheless, we emphasize that model categories are often used in tandem with $(\infty, 1)$ -categories and, in fact, some results about the latter actually rely on the existence of certain model structures on concrete categories of interest.

Of course, none of this also accounts for various other structures that were of interest in homotopy theory and still continue to be studied today. For instance, triangulated categories, which arose from algebraic geometry and classical stable homotopy theory, could also be viewed as “precursors” to $(\infty, 1)$ -categories and, in fact, are the main structure used in the study of bivariant homology theories on C^* -algebras. Indeed, it was realized very early on, due to the work of (Ralf) Meyer and (Ryszard) Nest in [63], that Kasparov’s KK-category actually has the structure of a triangulated category and this structure has been concretely helpful in recent K-theory computations, as well as in presenting a homological picture of the Baum-Connes Conjecture. The close relationship between triangulated categories and (stable) $(\infty, 1)$ -categories has, through the work of (Ulrich) Bunke, (Alexander) Engel and (Markus) Land in [20], [21], [19] and [22], resulted in certain bivariant homology theories attaining an $(\infty, 1)$ -categorical enhancement. This seems to have potential for applications towards, say, K-theory computations of crossed products and the UCT problem.

At this point, we are not so interested in giving a detailed account of the history² of the interactions between homotopy theory & operator algebras, nor are we going to outline all (possible) applications of homotopy theory to operator algebras. Our **main goal**, in fact, is to *show* that homotopy-theoretic reasoning provides an additional, useful perspective through which we can solve problems in operator algebra theory and through which we can understand existing constructions/results. We also aim to show that the homotopy theory of operator algebras actually provides motivation for the development of some abstract homotopy theory which, in turn, gives us a better understanding of existing homotopical structures on categories of operator algebras. While this thesis is not a complete, systematic development of “noncommutative algebraic topology” or “noncommutative homotopy theory”, we hope that the results we discuss serve as motivation for further work in this direction. Let us, now, describe the structure of the thesis.

Section 2 In this section, we discuss some concrete homotopical constructions in the two main categories of equivariant operator algebras of interest in this thesis. One of the main reasons we care about developing basic homotopy theory is due to our interest in studying (co)homology theories; most of the material in Section 2.1 will be used in our survey of axiomatic (co)homology on C^* -algebras in Section 4. In Section 2.2, we turn towards some conjectures/questions in the literature on (homotopy) lifting problems and address them to some degree.

Section 3 In this section, we turn towards the study of abstract homotopy theory. We begin by giving a survey of model category theory in Section 3.1, working our way up to its definition and surveying some standard examples. This culminates in some basic exposition about a result which essentially tells us that the category of C^* -algebras does not admit a model structure with the usual weak equivalences and the Schochet fibrations. In Section 3.2, we build up the theory of categories of fibrant objects & survey some known results about them before we present new results based on constructions inspired by the homotopy theory of equivariant operator algebras developed in Section 2. We finish the section off by discussing (but not proving) a variation of the Spanier-Whitehead Construction for categories of fibrant objects.

Section 4 In this section, we address some remaining questions which did not fit into the exposition of Sections 2 and 3. Section 4.1 focuses on studying the notion of an admissible subcategory, which we define, of the category of C^* -algebras and showing that this is the correct notion by applying it to obtain a more refined proof of the Bott Periodicity Theorem as well as other results in the axiomatic (co)homology theory of C^* -algebras. We also establish a slightly weaker version of the Spanier-Whitehead Construction in Section 4.2 for categories of fibrant objects using the methods of higher category theory, thereby partially fixing a gap in the literature.

Section 5 In this section, we discuss a number of natural questions that arose in the course of completing this thesis. We offer plenty of motivation and context behind the problems proposed, as well as strategies for solving them whenever this is relevant/available. We focus much more on problems which encourage further interaction between homotopy theory and operator algebras.

Section A This section contains miscellaneous proofs that did not fit naturally into the narrative of the main thesis but should be included in one form or another. Sometimes, we only provide proof sketches when certain results have technical proofs that do not get applied later. We will discuss the specific way this section should be used in Section 1.2. Nevertheless, this is an important part of the thesis which makes it complete.

We say a few words about conventions in Section 1.2. We discuss the prerequisites for the thesis and some reading recommendations in Section 1.3. Finally, we discuss some aspects of foundations in Section 1.4.

²We will do this throughout the thesis at appropriate points

1.2 Conventions

We introduce notation, acronyms and specific categories that may be used throughout the thesis.

General Mathematical Notation

$\mathbb{K}$ denotes either $\mathbb{R}$ or $\mathbb{C}$.

$\mathbb{F}$ denotes any field.

$\mathbb{N}$ denotes the set of natural numbers (excluding 0)

$\mathbb{N}_0$ is the set of natural numbers including 0

$\mathbb{R}_0^+$ denotes the set of all non-negative real numbers.

$\mathbb{R}^+$ denotes the set of all positive real numbers.

If A, B are mathematical objects, then $A \stackrel{\text{def}}{=} B$ means that A is defined to be equal to B .

If A, B are statements, then $A \stackrel{\text{def}}{\Leftrightarrow} B$ means that A is defined to be equivalent to B .

Common Acronyms

WLOG - Without loss of generality.

TFAE - The following are equivalent.

TFSH - The following statements hold.

nbhd - neighborhood.

Some Categories

Set is the category of sets (with functions as morphisms).

Grp is the category of groups (with group homomorphisms as morphisms).

AbGrp is the full subcategory of Grp consisting of abelian groups.

Top is the category of topological spaces (with continuous maps as morphisms).

CHaus is the full subcategory of Top consisting of compact Hausdorff spaces.

Grp(Top) is the category of topological groups (with continuous group homomorphisms as morphisms).

Ab(Top) is the full subcategory of Grp(Top) consisting of topological abelian groups.

Grp(LCH) is the full subcategory of Grp(Top) consisting of locally compact Hausdorff groups.

Grp(CHaus) is the full subcategory of Grp(Top) consisting of compact Hausdorff groups.

Ab(LCH) is the full subcategory of Grp(Top) consisting of locally compact Hausdorff groups.

Ab(CHaus) is the full subcategory of Grp(Top) consisting of compact Hausdorff abelian groups.

Ban $_{\mathbb{C}}$ is the category of $\mathbb{C}$ -Banach spaces (with continuous linear maps as morphisms).

$C^*\text{Alg}$ is the category of C^* -algebras (with $\star$ -homomorphisms as morphisms).

${}^1C^*\text{Alg}$ is the subcategory of $C^*\text{Alg}$ consisting of unital C^* -algebras & unital $\star$ -homomorphisms.

${}_{\omega}C^*\text{Alg}$ is the category of separable C^* -algebras.

BanAlg is the category of $\mathbb{C}$ -Banach algebras (with continuous algebra homomorphisms as morphisms).

${}^1\text{BanAlg}$ is the subcategory of BanAlg consisting of unital Banach algebras and unital morphisms.

Categorical Notation

If $\mathcal{C}$ is a category and x is an object in $\mathcal{C}$, then we denote that by writing $x \in \mathcal{C}$.

If $\mathcal{C}$ is a category, then $\text{Mor}(\mathcal{C})$ refers to the set^a of all morphisms of $\mathcal{C}$.

If $\mathcal{C}$ is a category and $x, y \in \mathcal{C}$, then $\mathcal{C}(x, y)$ is the set of all morphisms from x to y .

If $\mathcal{C}$ is a category and $x \in \mathcal{C}$, then $\text{Aut}(x)$ is the set of all isomorphisms with (co)domain x .

If $\mathcal{C}$ is a pointed category, then the zero object will typically be denoted by 0 unless otherwise stated.

^aThis will make sense through the notion of a Grothendieck universe, as discussed in Section 1.4.

We also mention that there are several places where we have labeled certain definitions/results with the name of the author. This has only been done just for the sake of clarity about our contributions. In other places where we reprove results or simply extend what is in the literature, we do state that very clearly. Let us also explain Section A a little more. Throughout the thesis, we will have boxes of the following form which appear in the main text but also in proofs:

| **Claim 1.1.** THIS IS A TEST BOX.

These are supposed to encode claims whose proofs we do want to include but doing so in the main thesis breaks the flow of the exposition. So, instead, we simply include their statements in the main thesis so that the reader will certainly come across them but we place the proofs in Section A. For electronic usage, the box is hyperlinked, so clicking on “Claim [Number]” will lead you to proof of the claim and similarly, clicking on “Proof [Number]” will lead you back to the claim. For the reader who prefers to use a printed copy, we will state the page number where the proof can be found.

1.3 Prerequisites & Reading Recommendations

Due to the fact that some parts of this thesis involve new results, it will not always be possible to write in a self-contained way. For full appreciation of everything written here, we will need basic background in functional analysis, operator algebras and category theory. Here are some reading recommendations for these topics:

Functional Analysis: One can certainly use [73] because it is extremely well-written.

Operator Algebras: [66] is a more modern reference but [48] was a more pleasant read, even with its clearly analytic tilt. It also contains a nice introduction to the necessary concepts from functional analysis.

Category Theory: [79] is accessible but lacks examples from functional analysis and these might be very interesting for readers who would like an early glimpse into the relevance of category theory towards analytic topics. For that, it is a good idea to look into the rather accessible resource [50]; this also doubles up as an accessible treatment of functional analysis.

We do use homotopy theory in Sections 3 and 4 but we introduce most of the structures we use, with the sole exception of $(\infty, 1)$ -category theory. Nevertheless, we do try to make our exposition intuitive and readable.

1.4 A Word About Foundations

This thesis does make extensive use of category theory and, to a much lesser extent, some $(\infty, 1)$ -category theory. Therefore, it is necessary to say a few words about the foundations we will use. This will effectively play no major role for us, so we mainly just include this as a formality. Nevertheless, there may be instances where use phrases like “consider a set of morphisms in a category” and since we are not using the word “set” or “collection” in just the sense of ZFC set theory, it is best to clarify what exactly we mean. Indeed, we need slightly more than the axioms of ZFC set theory to formalize category theory. This is typically done with the notion of a *Grothendieck universe*.

Definition 1.1

A Grothendieck universe is a set U such that:

1. If $S \in U$ and $x \in S$, then $x \in U$.
2. If $S \in U$, then $\mathcal{P}(S) \in U$.
3. $\mathbb{N} \in U$.
4. If $I \in U$ and $f : I \rightarrow U$ is a map, then $\bigcup_{i \in I} f(i) \in U$.

If U is a Grothendieck universe, then its elements are known as U -small sets.

One may, now, state the additional axiom of interest.

Axiom 1.2. The Existence of Grothendieck Universes

Let x be a set. Then, there exists a Grothendieck universe that contains x .

The typical way in which this axiom is used is that one will typically fix finitely many universes $U_1, \dots, U_n$ such that U_i is contained in U_{i+1} for each i and simply keep track of constructions which may result in one needing to go to a “higher universe”. Such constructions include localizations of categories (this applies both for usual categories and $(\infty, 1)$ -categories). Again, we do not wish to make a big deal out of this because it will not matter for our purposes. Indeed, we will not even explicitly use this material later in the thesis because it simply hampers readability.

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2 Homotopical Constructions of Equivariant Operator Algebras

Summary

In this section, we discuss some concrete homotopical constructions in the two main categories of equivariant operator algebras of interest in this thesis. One of the main reasons we care about developing basic homotopy theory is due to our interest in studying (co)homology theories; most of the material in Section 2.1 will be used in our survey of axiomatic (co)homology on C^* -algebras in Section 4. In Section 2.2, we turn towards some conjectures/questions in the literature on (homotopy) lifting problems and address them to some degree.

2.1 Homotopies & Fibrations of Equivariant C^* -Algebras & Banach Algebras

Since we want to build up the language of homotopy theory for equivariant C^* -algebras and Banach algebras, it is reasonable to start with the definition of an equivariant C^* -algebra/Banach algebra. To facilitate a clean presentation of the definitions/results, we will often mainly state the results for G - C^* -algebras and in remarks afterwards, we will say something about how those definitions/results can be adapted to the Banach algebra situation. **All algebras we work with are over $\mathbb{C}$.** With that said, let us define equivariant C^* -algebras.

Definition 2.1

Let $G \in \text{Grp}(\text{Top})$. A G - C^* -algebra is a pair (A, α) consisting of:

1. A C^* -algebra A .
2. A continuous group action $\alpha : G \times A \rightarrow A$ such that for each $g \in G$, the map:

$$\alpha(g, \cdot) : A \rightarrow A,$$

is a $\star$ -automorphism. We refer to this as a continuous action by G on A via automorphisms.

If $(A, \alpha), (B, \beta)$ are G - C^* -algebras, then a morphism:

$$\phi : (A, \alpha) \rightarrow (B, \beta),$$

is a $\star$ -homomorphism $\phi : A \rightarrow B$ such that the following diagram:

$$\begin{array}{ccc} G \times A & \xrightarrow{\alpha} & A \\ \text{Id}_G \times \phi \downarrow & & \downarrow \phi \\ G \times B & \xrightarrow{\beta} & B \end{array}$$

commutes. ϕ is, then, said to be a G -equivariant $\star$ -homomorphism.

We have a remark about this definition.

Remark 2.2. Definition 2.1 also makes sense when A is a $\mathbb{C}$ -Banach algebra and when the action is by automorphisms of A in the category BanAlg . The morphisms are required to satisfy the same condition as given. For the pair (A, α) where $A \in \text{BanAlg}$ and α being an action with the required properties, we will call this a G -Banach algebra.

Let us form a category out of this data.

Example 2.3. Let $G \in \text{Grp}(\text{Top})$. We form a structure G - $C^*\text{Alg}$ as follows:

1. The objects are G - C^* -algebras.
2. The morphisms are G -equivariant $\star$ -homomorphisms, with the usual composition.

We have the following claim.

Claim. This structure forms a category.

The identity map on any G - C^* -algebra is an equivariant $\star$ -homomorphism. Moreover, equivariance is preserved under composition and since $\star$ -homomorphisms compose to form $\star$ -homomorphisms, it follows that the composition of two equivariant $\star$ -homomorphisms is an equivariant $\star$ -homomorphism.

This structure forms a category; it is the category of G - C^* -algebras. We can also define the equivariant categories:

$$G\text{-BanAlg}, G\text{-BanAlg}_{\leq 1},$$

of G -Banach algebras, with the main difference between them being the morphisms.

At this point, we will present a number of standard constructions that will be crucial for a systematic treatment of homotopy theory on these categories of equivariant operator algebras. It is also a chance to carry out some explicit verifications of facts which are used in the literature but which are not given proofs.

Construction 2.4. Let $G \in \text{Grp}(\text{LCH})$ and let $A \in G\text{-}C^*\text{Alg}$. Define:

$$\text{Cyl}_G(A) \stackrel{\text{def}}{=} \mathcal{C}([0, 1], A),$$

$$\text{Cone}_G(A) \stackrel{\text{def}}{=} \{\phi \in \text{Cyl}_G(A) : \phi(0) = 0\},$$

$$\text{Sus}_G(A) \stackrel{\text{def}}{=} \{\phi \in \text{Cyl}_G(A) : \phi(1) = \phi(0) = 0\}.$$

These constructions also make sense when $A \in G\text{-BanAlg}$. In all of these cases, the various operations of interest are induced point-wise on elements of each of these sets. Moreover, we define:

$$\forall g \in G : \forall f \in \text{Cyl}_G(A) : \forall t \in [0, 1] : (g \cdot f)(t) \stackrel{\text{def}}{=} gf(t).$$

Let us check that this goes into $\text{Cyl}_G(A)$.

Claim 2.1. If $g \in G$ and $f \in \text{Cyl}_G(A)$, then $gf \in \text{Cyl}_G(A)$.

This action is also naturally induced on Cone_G and Sus_G . We name these constructs as follows:

1. $\text{Cyl}_G(A)$ is the (equivariant) cylinder of A .
2. $\text{Cone}_G(A)$ is the (equivariant) cone of A .
3. $\text{Sus}_G(A)$ is the (equivariant) suspension of A .

We have the following claim.

Claim 2.2. Let $A \in G\text{-C}^*\text{Alg}$. Then:

$$\text{Cyl}_G(A), \text{Cone}_G(A), \text{Sus}_G(A),$$

are $G\text{-C}^*\text{-algebras}$. The same proof goes through when $A \in G\text{-BanAlg}$.

This is proved on Page 100. Moreover, we define for each $t \in [0, 1]$ a map:

$$p_t^A : \text{Cyl}_G(A) \rightarrow A, \quad \phi \mapsto p_t^A(\phi) \stackrel{\text{def}}{=} \phi(t).$$

This is known as the t -th canonical inclusion^a or the evaluation map at^b t .

Claim 2.3. For each $t \in [0, 1]$ and $A \in G\text{-C}^*\text{Alg}$, the map p_t^A is an equivariant $\star$ -homomorphism. When $A \in G\text{-BanAlg}$, then this is an equivariant continuous algebra homomorphism.

This is proved on Page 101. The evaluation map is useful specifically in providing an alternative characterization of two equivariant $\star$ -homomorphisms being homotopic; see Proposition 2.18.

^aWe'll assume that the cylinder of A is always equipped with this family of maps.

^bThis terminology makes sense when $A \in G\text{-BanAlg}$ without any change.

In the next few examples, we will construct functors associated with the cylinder, cone & suspension.

Example 2.5. Let $G \in \text{Grp}(\text{LCH})$. Define a map of categories:

$$\text{Cyl}_G : G\text{-C}^*\text{Alg} \rightarrow G\text{-C}^*\text{Alg},$$

as follows:

1. If $A \in G\text{-C}^*\text{Alg}$, then $\text{Cyl}_G(A)$ is just the cylinder of A as defined above.
2. If $A, B \in G\text{-C}^*\text{Alg}$ and $f : A \rightarrow B$ is an equivariant $\star$ -homomorphism, then we define:

$$\text{Cyl}_G(f) : \text{Cyl}_G(A) \rightarrow \text{Cyl}_G(B), \quad \phi \mapsto \text{Cyl}_G(f)(\phi) \stackrel{\text{def}}{=} f \circ \phi.$$

It's easily checked that this is well-defined.

Let us show that this is a functor:

1. Let $A \in G\text{-C}^*\text{Alg}$. Then:

$$\begin{aligned} \forall \phi \in \text{Cyl}_G(A) : \text{Cyl}_G(\text{Id}_A)(\phi) &= \text{Id}_A \circ \phi \\ &= \phi \\ &= \text{Id}_{\text{Cyl}_G(A)}(\phi), \end{aligned}$$

so $\text{Cyl}_G(\text{Id}_A) = \text{Id}_{\text{Cyl}_G(A)}$.

2. Let $A, B, C \in G\text{-C}^*\text{Alg}$, let $f \in G\text{-C}^*\text{Alg}(A, B)$ and $g \in G\text{-C}^*\text{Alg}(B, C)$. Then:

$$\begin{aligned} \forall \phi \in \text{Cyl}_G(A) : \text{Cyl}_G(g \circ f)(\phi) &= (g \circ f) \circ \phi \\ &= g \circ (f \circ \phi) \\ &= g \circ \text{Cyl}_G(f)(\phi) \\ &= \text{Cyl}_G(g)(\text{Cyl}_G(f)(\phi)) \\ &= (\text{Cyl}_G(g) \circ \text{Cyl}_G(f))(\phi), \end{aligned}$$

from which we conclude that $\text{Cyl}_G(g \circ f) = \text{Cyl}_G(g) \circ \text{Cyl}_G(f)$.

This proves that this is a functor; we will refer to it as the cylinder functor.

Example 2.6. Let $G \in \text{Grp}(\text{LCH})$. Define a map of categories:

$$\text{Cone}_G : G\text{-C}^*\text{Alg} \rightarrow G\text{-C}^*\text{Alg},$$

as follows:

1. If $A \in G\text{-C}^*\text{Alg}$, then $\text{Cone}_G(A)$ is the cone of A as defined above.
2. If $A, B \in G\text{-C}^*\text{Alg}$ and $f \in G\text{-C}^*\text{Alg}(A, B)$, then we define:

$$\text{Cone}_G(f) \stackrel{\text{def}}{=} \text{Cyl}_G(f)|_{\text{Cone}_G(A)}.$$

Now, we need to check that $\text{Cone}_G(f)$ actually maps into $\text{Cone}_G(B)$; let $\phi \in \text{Cone}_G(A)$. Then:

$$\begin{aligned} \text{Cone}_G(f)(\phi)(0) &= (f \circ \phi)(0) \\ &= f(\phi(0)) \\ &= f(0) \\ &= 0, \end{aligned}$$

as was desired.

Since this is just a restriction of the cylinder functor, it is also a functor; we refer to it as the cone (functor).

Example 2.7. Let $G \in \text{Grp}(\text{LCH})$. Define a map of categories:

$$\text{Sus}_G : G\text{-C}^*\text{Alg} \rightarrow G\text{-C}^*\text{Alg},$$

as follows:

1. If $A \in G\text{-C}^*\text{Alg}$, then $\text{Sus}_G(A)$ is the suspension of A as defined above.
2. If $A, B \in G\text{-C}^*\text{Alg}$ and $f : A \rightarrow B$ is an equivariant $\star$ -homomorphism, then we define:

$$\text{Sus}_G(f) \stackrel{\text{def}}{=} \text{Cyl}_G(f)|_{\text{Sus}_G(A)}.$$

We need to show that $\text{Sus}_G(f)$ maps suspensions into suspensions and that is similar to [Example 2.6](#).

Since this is just a restriction of the cylinder functor, it is also a functor; we refer to it as the suspension (functor).

With the functors above, we can show that the time evaluation maps in [Construction 2.4](#) assemble into a natural transformation. In fact, there are a few technical results which are useful for us to establish right now.

Claim 2.4. Let $G \in \text{Grp}(\text{LCH})$ and for each $t \in [0, 1]$, define a map of functors:

$$\alpha_t : \text{Cyl}_G \Rightarrow \text{Id}, \quad \alpha_t^A \stackrel{\text{def}}{=} p_t^A,$$

on $G\text{-C}^*\text{Alg} / G\text{-BanAlg}$. This map of functors is a natural transformation. This is proved on [Page 101](#).

It is also a notable result that the cylinder and suspension “commute” naturally.

Claim 2.5. Let $G \in \text{Grp}(\text{LCH})$. There exists a natural isomorphism^a $\alpha : \text{Sus}_G \circ \text{Cyl}_G \Rightarrow \text{Cyl}_G \circ \text{Sus}_G$.

^aThis is proved on [Page 101](#).

Claim 2.6. Let $G \in \text{Grp}(\text{LCH})$ and let $A \in G\text{-C}^*\text{Alg}$ or $A \in G\text{-BanAlg}$. Then:

$$\forall t \in [0, 1] : p_t^{\text{Sus}_G(A)} \circ \alpha_A = \text{Sus}_G(p_t^A),$$

where α is the natural isomorphism of [Claim 2.5](#). This is proved on [Page 102](#).

In the next construction, we will place the cone and suspension into exact sequences.

Construction 2.8. Let $G \in \text{Grp}(\text{LCH})$ and let $A \in G\text{-C}^*\text{Alg}$. In the notation of [Construction 2.4](#), we have:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Cone}_G(A) & \xrightarrow{\iota} & \text{Cyl}_G(A) & \xrightarrow{p_1^A} & A \longrightarrow 0 \\ \\ 0 & \longrightarrow & \text{Sus}_G(A) & \xrightarrow{\iota} & \text{Cone}_G(A) & \xrightarrow{p_0^A} & A \longrightarrow 0 \end{array}$$

where the maps ι are two different inclusion maps which simply include an element of the cone/suspension into the cylinder. These sequences are defined even when A is a G -Banach algebra. We have the following notable result.

Claim 2.7. The sequences above are exact, even when $A \in G\text{-BanAlg}$.

This is proved on [Page 102](#). These facts are useful to have on standby.

As we mentioned in the introduction, our interest in the homotopy theory of operator algebras partly exists because it is useful for understanding various (co)homology theories on admissible subcategories of $G\text{-C}^*\text{Alg}$ and $G\text{-BanAlg}$. Regarding the former, this is a legitimate “problem” that we describe formally as a question.

Question 2.9. Is there an abstract framework which describes the homotopy theory of the following categories:

1. C^*Alg ,
2. $\omega\text{C}^*\text{Alg}$, the full subcategory of separable C^* -algebras,
3. $\omega\text{NucC}^*\text{Alg}$, the full subcategory of separable nuclear C^* -algebras,

all at once?

We have a remark about this question.

Remark 2.10. This question was implicitly posed as a remark in [\[85, p.408\]](#). In the same paper, it is suggested that one should use the closed model categories of Quillen to define the notion of an admissible subcategory and even provides some explicit conditions that one could impose on a full subcategory to make it admissible in [\[85, Remark 2.12, p.420-421\]](#). Nevertheless, the matter is essentially left unresolved. In [Section 4.1](#), we will propose a working definition of an admissible subcategory of C^*Alg which addresses [Question 2.9](#) in the non-equivariant case.

In spite of the fact that we do not yet have the definition of an admissible subcategory ([Definition 4.5](#)), we will still use it for the purposes of explaining the notions that will be needed to motivate our work. This is not detrimental to our exposition since we will only use it as motivation, for now. In that case, let us introduce some notation.

Notation 2.11. Let $\mathcal{A}$ be an abelian category. We use the notation:

$$F : \text{C}^*\text{Alg} \rightarrow \mathcal{A},$$

to denote a functor which is defined on some admissible subcategory of C^*Alg . We will usually not need to have separate notation for such a category, because we can always use $\text{dom}(F)$ to specify it.

With that said, let us introduce some standard properties of functors from an admissible category of C^* -algebras.

Terminology 2.12. Let $\mathcal{A}$ be an abelian category & let $F : C^*Alg \rightarrow \mathcal{A}$ be a functor.

1. F is exact iff for any exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in the domain, the following sequence:

$$0 \longrightarrow F(A) \xrightarrow{F(\phi)} F(B) \xrightarrow{F(\psi)} F(C) \longrightarrow 0$$

is exact in $\mathcal{A}$.

2. F is split-exact iff for any split-exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightleftharpoons[\alpha]{\psi} C \longrightarrow 0$$

in the domain, the following sequence:

$$0 \longrightarrow F(A) \xrightarrow{F(\phi)} F(B) \xrightleftharpoons[F(\alpha)]{F(\psi)} F(C) \longrightarrow 0$$

is split exact in $\mathcal{A}$.

F is half-exact iff for any exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in the domain, the following sequence:

$$F(A) \xrightarrow{F(\phi)} F(B) \xrightarrow{F(\psi)} F(C)$$

is exact in $\mathcal{A}$.

We have a remark about Terminology 2.12.

Remark 2.13. The given terminology makes sense when $\mathcal{A}$ is replaced with any category for which we have a reasonable notion of *exact sequence*; there are many such categories which are not abelian (or even pre-additive) for which this is the case. It also makes sense when C^*Alg is replaced by^a $BanAlg$. Finally, the given terminology also makes sense in the equivariant case, without any modification other than the morphisms we consider.

^aWe will, however, not discuss this situation explicitly in this thesis.

An important (and independently interesting) property of the suspension functor is that it is exact. This is sometimes used in technical arguments, so we will include a proof of this fact. To do this, we need some notation.

Notation 2.14. Let $X \in LCH$ and let $A \in Ban_C$ with $a \in A$ and $f \in \mathcal{C}_0(X)$. Define:

$$fa : X \rightarrow A, \quad x \mapsto (fa)(x) \stackrel{\text{def}}{=} f(x)a.$$

These are known as the simple tensors in $\mathcal{C}_0(X, A)$; it is clear that they belong to this algebra.

The following technical lemma tells us that the span above generates $\mathcal{C}_0(X, A)$ with the given assumptions on X and A .

Lemma 2.15. [83, Lemma 10.1.1, p.175-176] Let $X \in LCH$ and let $A \in Ban_C$. Then:

$$\text{Span}\{fa : f \in \mathcal{C}_0(X) \wedge a \in A\},$$

is dense in $\mathcal{C}_0(X, A)$.

The proof of the lemma in the reference is for the case where A is a C^* -algebra but the result holds in the stated generality by the same proof. The difficulty in showing exactness is proving the surjectivity of the last map in the image sequence. The lemma above allows us to obtain an approximate solution which is upgraded to a full solution.

Proposition 2.16

Let $G \in \text{Grp}(LCH)$. Then, Sus_G is an exact functor.

Proof. Consider the following exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in $G\text{-}C^*Alg$. Then, we obtain the following sequence:

$$0 \longrightarrow \text{Sus}_G(A) \xrightarrow{\text{Sus}_G(\phi)} \text{Sus}_G(B) \xrightarrow{\text{Sus}_G(\psi)} \text{Sus}_G(C) \longrightarrow 0$$

whose exactness needs to be checked. We'll do this in three separate steps.

1. Let $f \in \text{Sus}_G(A)$ so that $\text{Sus}_G(\phi)(f) = 0$. This means that $\phi \circ f = 0$ so $\text{ran}(f) \subseteq \ker(\phi) = 0$. That is, $f = 0$.
2. We need to show that $\text{Sus}_G(\psi)$ is surjective. Note that the following set:

$$\text{Span}\{fc : f \in \mathcal{C}_0(0, 1) \wedge c \in C\},$$

approximates every element in $\text{Sus}_G(C)$ so it remains to show that the elements in the dense subspace above belong to $\text{ran}(\text{Sus}_G(\psi))$. Indeed, let $c \in C$. Since ψ is surjective, there is a $b \in B$ such that:

$$\psi(b) = c.$$

So, we see that if $f \in \mathcal{C}_0(0, 1)$, then:

$$\begin{aligned} \text{Sus}_G(\psi)(fb) &= \psi \circ fb \\ &= f\psi(b) \\ &= fc, \end{aligned}$$

which proves that $\text{Sus}_G(\psi)$ is surjective.

3. Let's show that the second sequence is exact at $\text{Sus}_G(B)$. By functoriality, we have that:

$$\begin{aligned} 0 &= \text{Sus}_G(0) \\ &= \text{Sus}_G(\psi \circ \phi) \\ &= \text{Sus}_G(\psi) \circ \text{Sus}_G(\phi), \end{aligned}$$

which implies that $\text{ran}(\text{Sus}_G(\phi)) \subseteq \ker(\text{Sus}_G(\psi))$.

This completes the proof. ■

One property that functors of interest should have is *homotopy invariance*. For this to be defined, we need the notion of homotopy and luckily, the two categories of interest have enough structure to accommodate such a definition.

Terminology 2.17. Let $A, B \in G\text{-}C^*\text{Alg}$ and let $f, g \in G\text{-}C^*\text{Alg}(A, B)$.

1. A homotopy from f to g is a family $(\phi_t : A \rightarrow B)_{t \in [0,1]}$ in $G\text{-}C^*\text{Alg}(A, B)$ such that:

$$H_a : [0, 1] \rightarrow B, t \mapsto H_a(t) \stackrel{\text{def}}{=} \phi_t(a),$$

is continuous for each $a \in A$. Moreover, $\phi_0 = f$ and $\phi_1 = g$.

2. f is homotopic to g iff there exists a homotopy from f to g . We denote this by writing $f \simeq_h g$.

3. A and B are homotopy equivalent iff there exist $f \in G\text{-}C^*\text{Alg}(A, B)$ and $g \in G\text{-}C^*\text{Alg}(B, A)$ so that:

$$g \circ f \simeq_h \text{Id}_A,$$

$$f \circ g \simeq_h \text{Id}_B.$$

In this case, we refer to g as the homotopy inverse of f and f as the homotopy inverse of g .

Finally, A and B are homotopically equivalent iff there is an equivariant $\star$ -homomorphism:

$$f : A \rightarrow B,$$

which is also a homotopy equivalence. The same definition makes sense in $G\text{-BanAlg}$.

There is an alternative, useful characterization of homotopy that we present in the next result.

Proposition 2.18

Let $A, B \in G\text{-}C^*\text{Alg}$ and let $f, g \in G\text{-}C^*\text{Alg}(A, B)$. TFAE:

1. f and g are homotopic.

2. There is an equivariant $\star$ -homomorphism $h : A \rightarrow \text{Cyl}_G(B)$ such that $p_0^B \circ h = f$ and $p_1^B \circ h = g$.

The morphism h in (2) is also called a homotopy from f to g . The result also holds when $A, B \in G\text{-BanAlg}$.

Proof. Let's prove this for equivariant C^* -algebras first.

(1) $\Rightarrow$ (2): By definition, for each $t \in [0, 1]$, we have a $\star$ -homomorphism $\phi_t : A \rightarrow B$ such that:

$$H_a : [0, 1] \rightarrow B, t \mapsto H_a(t) \stackrel{\text{def}}{=} \phi_t(a),$$

is continuous for each $a \in A$. Moreover, we have that $\phi_0 = f$ and $\phi_1 = g$. Now, we define:

$$h : A \rightarrow \text{Cyl}_G(B), a \mapsto h(a) \stackrel{\text{def}}{=} H_a.$$

Let's show that h is an equivariant $\star$ -homomorphism.

| **Claim 2.8.** h is an equivariant $\star$ -homomorphism.

This is proved on Page 10. Next, we will show that $p_0^B \circ h = f$. This is a calculation:

$$\begin{aligned} \forall a \in A : (p_0^B \circ h)(a) &= p_0^B(h(a)) \\ &= p_0^B(H_a) \\ &= H_a(0) \\ &= \phi_0 \\ &= f. \end{aligned}$$

A similar calculation with p_1^B yields the implication.

(2) $\Rightarrow$ (1): Let $t \in [0, 1]$. Define:

$$\phi_t : A \rightarrow B, a \mapsto \phi_t(a) \stackrel{\text{def}}{=} h(a)(t).$$

Let's check that $\phi_0 = f$. We have the following computation:

$$\begin{aligned}\forall a \in A : \phi_0(a) &= h(a)(0) \\ &= (p_0^B \circ h)(a) \\ &= f(a),\end{aligned}$$

as was desired. A similar calculation gives us that $\phi_1 = g$. Finally, for each $a \in A$, the following map:

$$H_a : [0, 1] \rightarrow B, \quad t \mapsto H_a(t) \stackrel{\text{def}}{=} \phi_t(a),$$

is continuous because h maps into the cylinder of B .

For equivariant Banach algebras, the only technical point lies in $(1) \Rightarrow (2)$. In that case, we would've had to explicitly prove that h is continuous since that would not follow from its algebraic structure. For that, we note that $(\phi_t)_{t \in [0,1]}$ is a family of continuous linear maps which is pointwise bounded since each H_a is a continuous map on a compact space. By the Uniform Boundedness Principle, we see that there exists a $c > 0$ such that:

$$\forall t \in [0, 1] : \|\phi_t\| \leq c.$$

So, this means that:

$$\begin{aligned}\forall a \in A : \|H_a\| &= \sup_{t \in [0,1]} \|H_a(t)\| \\ &= \sup_{t \in [0,1]} \|\phi_t(a)\| \\ &\leq c\|a\|,\end{aligned}$$

which is exactly the statement that h is bounded. ■

It is slightly better to have a proof of the fact that homotopy is an equivalence relation using the second characterization of homotopy as presented in Proposition 2.18. We will also show that isomorphisms are homotopy equivalences.

Claim 2.9. Let $G \in \text{Grp}(\text{LCH})$. Then, $\simeq_h$ is an equivalence relation in ${}^a G\text{-C}^*\text{Alg} / G\text{-BanAlg}$.

^aWe prove this on Page 102.

Claim 2.10. If $G \in \text{Grp}(\text{LCH})$, then isomorphisms in $G\text{-C}^*\text{Alg} / G\text{-BanAlg}$ are homotopy equivalences^a.

^aWe prove this on Page 103.

We can also produce a neat and simple example of a homotopy equivalence.

Example 2.19. Let $A \in G\text{-C}^*\text{Alg}$ and define a map:

$$s : A \rightarrow \text{Cyl}_G(A), \quad a \mapsto s(a),$$

where $s(a) : [0, 1] \rightarrow A$ is the constant function with value a . We have the following claim.

Claim 2.11. s is a homotopy equivalence and p_0^A is a homotopy inverse. We prove this on Page 103.

The result also holds when $A \in G\text{-BanAlg}$, simply because every map we define remains continuous.

Equipped with the notion of homotopy, we can also define what it means for a functor to be homotopy-invariant. We will be concerned with the basic study of homotopy-invariant functors later in Section 4.1.

Terminology 2.20. Let $\mathcal{C}$ be a full subcategory of $G\text{-C}^*\text{Alg}$, let $\mathcal{D}$ be a category and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$. F is homotopy-invariant iff for two parallel morphisms $\phi, \psi \in \mathcal{C}$, we have that:

$$\phi \sim_h \psi \Rightarrow F(\phi) = F(\psi).$$

This also works for $G\text{-BanAlg}$.

This is a reasonable place to introduce the definition of contractibility of a C^* -algebra.

Terminology 2.21. Let $A \in G\text{-C}^*\text{Alg} / G\text{-BanAlg}$. A is contractible iff Id_A is homotopic to 0.

In the next claim, we will establish the contractibility of the cone over any equivariant operator algebra.

Claim 2.12. Let $A \in G\text{-C}^*\text{Alg} / G\text{-BanAlg}$. Then, $\text{Cone}_G(A)$ is contractible. We prove this on Page 103.

One should note that contractibility of Banach algebras/ C^* -algebras is not the same as contractibility of spaces.

Example 2.22. Let $A \in {}^1\text{BanAlg} / {}^1\text{C}^*\text{Alg}$.

Claim. A is not contractible.

If it was contractible, then $K_0(A) \simeq 0$ since K_0 is homotopy-invariant. In particular:

$$[1_A]_{K_0} = [0]_{K_0},$$

which implies that $1_A = 0$ and this is impossible.

In particular, $\mathbb{C}$ is **not** contractible as a Banach algebra.

The main result about homotopies between morphisms of equivariant operator algebras is given by the next proposition. There are two conceptual reasons why this result will be significant for us:

1. From this result, we can deduce all formal properties of homotopy in the two categories of interest.
2. This result allows us to define homotopy categories.

We will see, later in the thesis, that the following result also has explicit influence on how we equip categories of fibrant objects with sufficient structure to ensure that their internal notion of homotopy is compatible with compositions.

Proposition 2.23

Let $A, B, C \in G\text{-C}^*\text{Alg}$, let $f_1, f_2 \in G\text{-C}^*\text{Alg}(A, B)$ and let $g_1, g_2 \in G\text{-C}^*\text{Alg}(B, C)$ so that:

$$f_1 \simeq_h f_2,$$

$$g_1 \simeq_h g_2.$$

Then, $g_1 \circ f_1 \simeq_h g_2 \circ f_2$. This result also holds in $G\text{-BanAlg}$.

Proof. We will proceed in three steps.

1. Show that the result holds when $f_1 = f_2$.
2. Show that the result holds when $g_1 = g_2$.
3. Deduce the general case from the two specific situations above.

With this plan, we proceed with the proof.

1. Let $f_1 = f_2 = f$. Then, there is a homotopy $h : B \rightarrow \text{Cyl}_G(C)$ such that:

$$g_1 = p_0^C \circ h,$$

$$g_2 = p_1^C \circ h.$$

Now, we define:

$$H \stackrel{\text{def}}{=} h \circ f.$$

This is an equivariant $\star$ -homomorphism $A \rightarrow \text{Cyl}_G(C)$. It is clear that this is a

$$\begin{aligned} \forall i \in \{0, 1\} : p_i^C \circ H &= p_i^C \circ (h \circ f) \\ &= (p_i^C \circ h) \circ f \\ &= g_{i+1} \circ f, \end{aligned}$$

which means that H is a homotopy from $g_1 \circ f$ to $g_2 \circ f$.

2. Let $g_1 = g_2 = g$. Then, there is a homotopy $h : A \rightarrow \text{Cyl}_G(B)$ such that:

$$f_1 = p_0^B \circ h,$$

$$f_2 = p_1^B \circ h.$$

Now, we define:

$$H \stackrel{\text{def}}{=} \text{Cyl}_G(g) \circ h.$$

This is an equivariant $\star$ -homomorphism. Let's show that this is a homotopy from $g \circ f_1$ to $g \circ f_2$:

$$\begin{aligned} p_0^C \circ H &= p_0^C \circ (\text{Cyl}_G(g) \circ h) \\ &= (p_0^C \circ \text{Cyl}_G(g)) \circ h \\ &= (g \circ p_0^B) \circ h \\ &= g \circ (p_0^B \circ h) \\ &= g \circ f_1, \end{aligned}$$

where we used the naturality of the time evaluation maps as proved in **Claim 2.3**. Showing that:

$$p_1^C \circ H = g \circ f_2,$$

is a similar computation.

3. We have that:

$$\begin{aligned} g_1 \circ f_1 &\simeq_h g_2 \circ f_1 \\ &\simeq_h g_2 \circ f_2, \end{aligned}$$

as was desired.

This completes the proof. ■

As mentioned above, we can establish three other formal properties of homotopy on categories of operator algebras.

Claim 2.13. Let $G \in \text{Grp}(\text{LCH})$ and let $\mathcal{C}$ be either $G\text{-C}^*\text{Alg}$ or $G\text{-BanAlg}$. TFSH:

1. If $f, g_1, g_2 \in \text{Mor}(\mathcal{C})$ with g_1, g_2 being homotopy inverses of f , then $g_1 \simeq_h g_2$.
2. Let $f, g_1, g_2 \in \text{Mor}(\mathcal{C})$ where g_1 is a left homotopy inverse of f and g_2 is a right homotopy inverse of f . Then, we have that $g_1 \simeq_h g_2$. In particular, f is a homotopy equivalence.
3. Let $f, g \in \text{Mor}(\mathcal{C})$ such that any two of $f, g, g \circ f$ are homotopy equivalences, then so is the third.

We prove this on Page 104.

In light of the facts above, we can now introduce homotopy categories.

Construction 2.24. We define:

$$\mathbf{Ho}(G\text{-}\mathbf{C}^*\mathbf{Alg}) \stackrel{\text{def}}{=} G\text{-}\mathbf{C}^*\mathbf{Alg} / \sim,$$

where $\sim$ is the equivalence relation above on morphism classes. This is known as the homotopy category of $G\text{-}\mathbf{C}^*\text{Alg}$. The same construction can be adapted for $G\text{-}\mathbf{BanAlg}$ without change.

In classical homotopy theory, it is useful to introduce the notion of (co)fibration alongside the different notions of “weak equivalence” available (typically, homotopy equivalence and weak homotopy equivalence). The reasons for introducing these objects is variable but for our purposes, they will be tools which will assist us in “controlling” the homotopy categories associated to the categories above. In this case, we will focus primarily on a notion of *fibration* which seems to have been first introduced by (Claude) Schochet in [85]. Actually, he referred to these as “cofibrations” due to the fact that they actually correspond, in the commutative case, to cofibrations in topology.

Definition 2.25

Let $G \in \text{Grp}(\text{LCH})$, let $A, B \in G\text{-C}^*\text{Alg}$ and let $p \in G\text{-C}^*\text{Alg}(A, B)$. p is a Schochet fibration iff for any $D \in G\text{-C}^*\text{Alg}$, for any equivariant homotopy $(h_t : D \rightarrow B)_{t \in [0,1]}$ and for any $f \in G\text{-C}^*\text{Alg}(D, A)$ such that the diagram:

A commutative triangle diagram with vertices A , B , and D . A horizontal arrow from D to B is labeled h_t . A diagonal arrow from D to A is labeled f . A vertical arrow from A to B is labeled p .

commutes when $t = 0$, then there exists an equivariant homotopy $(H_t : D \rightarrow A)_{t \in [0,1]}$ such that the following diagram:

A commutative diagram with three nodes: D at the bottom left, A at the top right, and B at the bottom right. A dashed arrow labeled H_t points from D to A . A solid arrow labeled f points from D to A . A solid arrow labeled h_t points from D to B . A solid arrow labeled p points from A to B .

commutes for each $t \in [0, 1]$ and $H_0 = f$. This definition also makes sense in $G\text{-BanAlg}$.

Let us mention an important example of a Schochet fibration.

Example 2.26. Let $G \in \text{Grp}(\text{LCH})$ and let $A \in G\text{-C}^*\text{Alg}$. Here's the main claim.

Claim. For each $t \in [0, 1]$, the canonical inclusion:

$$p_t : \text{Cyl}_G(A) \rightarrow A,$$

as defined in [Construction 2.4](#), is a Schochet fibration. This result also holds when $A \in G\text{-BanAlg}$.

This is proved in the non-equivariant case in [85, Lemma 1.3, p.407]. To apply that to the equivariant case, one needs to check that the calculated lift is equivariant and it is, because the morphisms we start with are equivariant.

Our eventual goal is to use the notion above to show that $G\text{-}C^*\text{Alg}$ and $G\text{-BanAlg}$ are categories of fibrant objects; see Proposition 4.1. For this to happen, we need to establish some key permanence properties of Schochet fibrations. These were actually “proved” in the non-equivariant case in [85], though full proofs were not always supplied so we will fill in those gaps. One novelty in our approach is that we will consistently use a characterization of Schochet fibrations which was mentioned in [85, Definition 1.2, p.406] for $\star$ -homomorphisms between C^* -algebras but was not explicitly established.

Claim 2.14. Let $G \in \text{Grp}(\text{LCH})$, $A, B \in G\text{-C}^*\text{Alg}$ and let $p \in G\text{-C}^*\text{Alg}(A, B)$. TFAE:

1. p is a Schochet fibration.
2. If $D \in G\text{-}\mathbf{C}^*\mathbf{Alg}$, $f : D \rightarrow A$ is a $\star$ -homomorphism and $h : D \rightarrow \mathrm{Cyl}_G(B)$ is a homotopy of the composite $p \circ f$, then there exists a homotopy $H : D \rightarrow \mathrm{Cyl}_G(A)$ of f such that the following diagram:

$$\begin{array}{ccc}
 & & A \\
 & \nearrow f & \uparrow p_0 \\
 D & \xrightarrow{H} & \text{Cyl}_G(A) \\
 & \searrow h & \downarrow \text{Cyl}_G(p) \\
 & & \text{Cyl}_G(B)
 \end{array}$$

commutes.

The same result holds in G -BanAlg. The proof is given on Page 105.

We have two remarks at this stage.

Remark 2.27. In statement (2) of Proposition 2.18, we characterized homotopies between morphisms of algebras completely via morphisms and compositions. Statement (2) of **Claim 2.14** does the same thing for Schochet fibrations.

Remark 2.28. There are modern reasons as to why it would not make sense to refer to Schochet fibrations as “cofibrations”. One reason for this is the duality between spaces and operator algebras. Indeed, cofibrations over spaces should correspond to fibrations & fibrations over spaces should correspond to cofibrations. Moreover, cofibrations are typically thought of as morphisms which are closed under pushout-type constructions, even in model category theory. As we will see through Proposition 2.29, this is not the case for Schochet fibrations.

With this said, let us deal with our required permanence properties.

Proposition 2.29

Let $G \in \text{Grp}(\text{LCH})$. TFSH:

1. Schochet fibrations in $G\text{-C}^*\text{Alg}$ are closed under composition.
2. Let $A \in G\text{-C}^*\text{Alg}$ and let $f \in G\text{-C}^*\text{Alg}(A, 0)$. Then, f is a Schochet fibration.
3. Let $A, B \in G\text{-C}^*\text{Alg}$ and let $f \in G\text{-C}^*\text{Alg}(A, B)$ be an isomorphism. f is a Schochet fibration.
4. The cylinder functor preserves pullbacks.
5. Schochet fibrations are preserved by base extensions.

These results also hold in $G\text{-BanAlg}$.

Let us recall the meaning of some property being preserved by base extensions.

Terminology 2.30. Let $\mathcal{C}$ be a category and let P be a property of morphisms in $\mathcal{C}$. We say that P is preserved by base extensions iff for any pullback diagram in $\mathcal{C}$ of the following form:

$$\begin{array}{ccc} p & \longrightarrow & x \\ \downarrow & & \downarrow \\ z & \longrightarrow & y \end{array}$$

if $x \rightarrow y$ has property P , then so does $p \rightarrow z$.

Now, we can prove our result.

Proof of Proposition 2.29. We will prove each statement one-by-one, mainly using **Claim 2.14**.

1. Let $A, B, C \in G\text{-C}^*\text{Alg}$, let $p \in G\text{-C}^*\text{Alg}(A, B)$ and $q \in G\text{-C}^*\text{Alg}(B, C)$ be Schochet fibrations. This part is just a two-step lifting and we approach by starting with the following diagram which we will explain afterwards:

$$\begin{array}{ccccc} & & A & & \\ & \nearrow f & \uparrow p_0^A & & \\ & & \text{Cyl}_G(A) & & \\ & \nearrow H & \downarrow \text{Cyl}_G(p) & & \\ D & \xrightarrow{\tilde{H}} & \text{Cyl}_G(B) & & \\ & \searrow h & \downarrow \text{Cyl}_G(q) & & \\ & & \text{Cyl}_G(C) & & \end{array}$$

Let us ignore the dotted arrows for the moment. We note that h is a homotopy of $(q \circ p) \circ f$ and that this is the same diagram as in statement (2) of **Claim 2.14**, except that we have used the functoriality of Cyl_G to separate p and q . Now, we want to fill in the diagram by the dotted arrows and let us explain why this is possible. So, h is, in particular, a homotopy of $q \circ (p \circ f)$. Since q is a Schochet fibration, we can find a homotopy lift $\tilde{H}$ to $\text{Cyl}_G(B)$ and this is a homotopy of $p \circ f$. Since p is a Schochet fibration, we can find a homotopy lift H to $\text{Cyl}_G(A)$ and this is a homotopy of f . But this was the desired lift for the original lifting problem.

2. Let $D \in G\text{-C}^*\text{Alg}$ and suppose we have a commutative diagram:

$$\begin{array}{ccccc} & & A & & \\ & \nearrow p & \uparrow p_0^A & & \\ & & \text{Cyl}_G(A) & & \\ & \nearrow h & \downarrow \text{Cyl}_G(f) & & \\ D & \xrightarrow{\quad} & \text{Cyl}_G(0) & & \end{array}$$

where h is a homotopy of $f \circ p$. We define:

$$s : D \rightarrow \text{Cyl}_G(D), \quad d \mapsto s(d),$$

where $s(d)$ is the constant function from $[0, 1]$ into D with value d . Now, we define:

$$H \stackrel{\text{def}}{=} \text{Cyl}_G(p) \circ s.$$

Let us check that this works.

| **Claim 2.15.** H is a homotopy of p and it is the required homotopy lift. This is proved on Page 105.

This proves the result.

3. Consider the following homotopy lifting problem:

$$\begin{array}{ccc} & & A \\ & \nearrow g & \uparrow p_0^A \\ D & & \text{Cyl}_G(A) \\ & \searrow h & \downarrow \text{Cyl}_G(f) \\ & & \text{Cyl}_G(B) \end{array}$$

where h is a homotopy of $f \circ g$. Since f is an isomorphism, we simply define:

$$H \stackrel{\text{def}}{=} \text{Cyl}_G(f^{-1}) \circ h.$$

Let us check that this works.

| **Claim 2.16.** H is a homotopy of g and is the desired homotopy lift. This is proved on Page 105.

This proves the result.

4. Consider the following diagrams:

$$\begin{array}{ccc} P & \xrightarrow{\pi_1} & A \\ \pi_2 \downarrow & & \downarrow f \\ B & \xrightarrow{g} & C \end{array} \quad \begin{array}{ccc} \text{Cyl}_G(P) & \xrightarrow{\text{Cyl}_G(\pi_1)} & \text{Cyl}_G(A) \\ \text{Cyl}_G(\pi_2) \downarrow & & \downarrow \text{Cyl}_G(f) \\ \text{Cyl}_G(B) & \xrightarrow{\text{Cyl}_G(g)} & \text{Cyl}_G(C) \end{array}$$

where the left diagram is a pullback in $G\text{-C}^*\text{Alg}$ or $G\text{-BanAlg}$. We want to show that the diagram on the right is a pullback diagram. Let Q denote the actual pullback of the diagram on the right and note that the universal property of the pullback gives us a unique morphism $i : \text{Cyl}_G(P) \rightarrow Q$ such that the diagram:

$$\begin{array}{ccccc} & & \text{Cyl}_G(P) & \xrightarrow{\text{Cyl}_G(\pi_1)} & \text{Cyl}_G(A) \\ & & \downarrow i & & \downarrow \text{Cyl}_G(f) \\ & & Q & \xrightarrow{p_1} & \text{Cyl}_G(A) \\ & & \downarrow p_2 & & \downarrow \text{Cyl}_G(f) \\ & & \text{Cyl}_G(B) & \xrightarrow{\text{Cyl}_G(g)} & \text{Cyl}_G(C) \end{array}$$

(Note: In the original image, there is a curved arrow from $\text{Cyl}_G(P)$ to $\text{Cyl}_G(B)$ labeled $\text{Cyl}_G(\pi_2)$ and a curved arrow from $\text{Cyl}_G(P)$ to $\text{Cyl}_G(A)$ labeled $\text{Cyl}_G(\pi_1)$. The dashed arrow i points from $\text{Cyl}_G(P)$ to Q .)

commutes. Since we have an explicit form for P and for Q , we do have an explicit form for i . Namely:

$$\forall \phi \in \text{Cyl}_G(P) : i(\phi) = (\pi_1 \circ \phi, \pi_2 \circ \phi).$$

Now, we wish to define a map $j : Q \rightarrow \text{Cyl}_G(P)$, where we note that:

$$Q = \{(\phi, \psi) \in \text{Cyl}_G(A) \times \text{Cyl}_G(B) : f \circ \phi = g \circ \psi\}.$$

For $(\phi, \psi) \in Q$, we define:

$$j(\phi, \psi) : [0, 1] \rightarrow A \times B, \quad t \mapsto j(\phi, \psi)(t) \stackrel{\text{def}}{=} (\phi(t), \psi(t)).$$

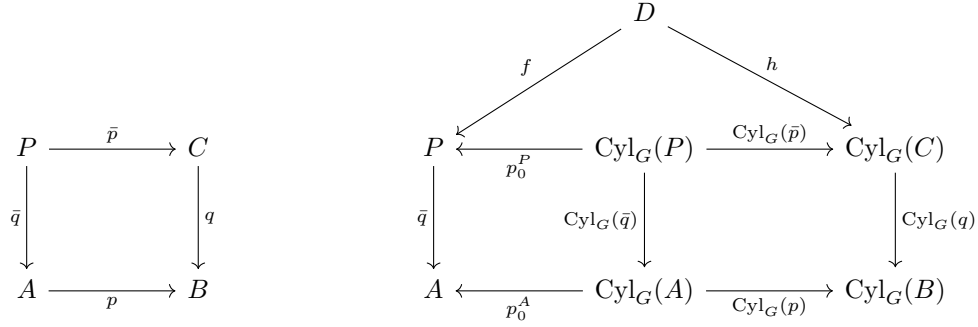
Let us check that this is valued in P .

| **Claim 2.17.** $j(\phi, \psi)$ is valued in P . This is proved on Page 106.

$j(\phi, \psi)$ is clearly continuous since its components are continuous, so j is a well-defined function into $\text{Cyl}_G(P)$. It is clear that j is a homomorphism. Finally, we check that i and j are inverse to each other.

So, i and j are inverse to each other. Equivariance follows from the fact that all morphisms involved are equivariant. In the case of G -BanAlg, the continuity of j follows from the continuity of its components.

5. Consider the following diagrams in G -C*Alg or G -BanAlg:



The left diagram is a pullback which we start with, where p is a Schochet fibration. To obtain the diagram on the right, we will first apply the functor Cyl_G to the left diagram. Then, we will extend the resulting diagram from the top by copying and pasting the diagram in (2) of **Claim 2.14**. Now, the proof proceeds as follows:

- Compute $\text{Cyl}_G(q) \circ h$ as a homotopy of the composite $p \circ (\bar{q} \circ f)$.
- Since p is a Schochet fibration, this induces a morphism from D to $\text{Cyl}_G(A)$ which is equal to $\text{Cyl}_G(q) \circ h$.
- The above creates a pullback cone for the pullback diagram of cylinders and since the pullback of this diagram is $\text{Cyl}_G(P)$, this induces a morphism from $D \rightarrow \text{Cyl}_G(P)$ which is the morphism we wanted.

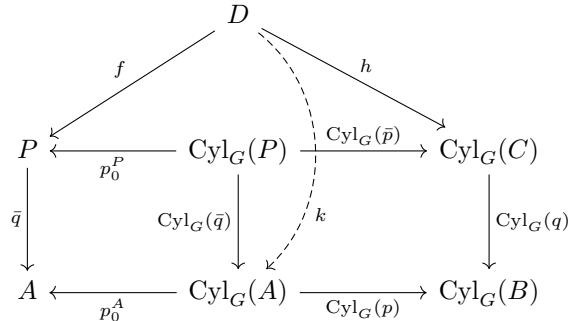
With this outline, let us proceed with filling in details.

- We have the following computation:

$$\begin{aligned}
 p_0^B \circ \text{Cyl}_G(q) \circ h &= q \circ p_0^C \circ h \\
 &= q \circ (\bar{p} \circ f) \\
 &= (q \circ \bar{p}) \circ f \\
 &= (p \circ \bar{q}) \circ f \\
 &= p \circ (\bar{q} \circ f),
 \end{aligned}$$

as was desired. So, $\text{Cyl}_G(q) \circ h$ is a homotopy of $p \circ (\bar{q} \circ f)$.

- As we mentioned above, p being a Schochet fibration gives us a diagram:



with $k : D \rightarrow \text{Cyl}_G(A)$ being a homotopy of $\bar{q} \circ f$. Moreover, we have that:

$$k \circ \text{Cyl}_G(p) = \text{Cyl}_G(q) \circ h.$$

by statement (2) of our characterization of Schochet fibrations.

- There is a pullback cone formed from $D \rightarrow \text{Cyl}_G(A) \rightarrow \text{Cyl}_G(B)$ and $D \rightarrow \text{Cyl}_G(C) \rightarrow \text{Cyl}_G(B)$. By the universal property of the pullback, there is a unique morphism from $D \rightarrow \text{Cyl}_G(P)$ such that all sub-diagrams formed commute. So, this morphism is a homotopy for f and $\bar{p}$ is a Schochet fibration.

This completes the proof. ■

Here is one more notable result which allows us to conclude that a $\star$ -homomorphism has a split from knowing that it is a Schochet fibration. We do not believe that this result is actually useful in practice, as we will eventually see that checking that a given morphism is a Schochet fibration can be a rather difficult problem. In fact, we are barely able to do it in an elementary way for even the Calkin projection on an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space; we will discuss this extensively in Section 2.2 and also when we discuss open problems in Section 5.1.

Proposition 2.31. A. Vats, 2026

Let $G \in \text{Grp}(\text{LCH})$, let $\mathcal{C}$ be either G -C*Alg or G -BanAlg, let $A, B \in \mathcal{C}$ and let $f \in \mathcal{C}(A, B)$ be a Schochet fibration such that there is a $g \in \mathcal{C}(B, A)$ which is a right homotopy inverse of f . Then, f is split epi in G -C*Alg and any left homotopy inverse of f is homotopic to its section.

Proof. Let us start with the following lifting problem:

$$\begin{array}{ccc}
 & & A \\
 & \nearrow g & \uparrow p_0^A \\
 B & \dashrightarrow H & \text{Cyl}_G(A) \\
 & \searrow h & \downarrow \text{Cyl}_G(f) \\
 & & \text{Cyl}_G(B)
 \end{array}$$

where h is a homotopy from $f \circ g$ to Id_B . Since f is a Schochet fibration, we have a homotopy lift H and this is a homotopy of g . Now, we have the following computation:

$$\begin{aligned}
 f \circ (p_1^A \circ H) &= (f \circ p_1^A) \circ H \\
 &= (p_1^B \circ \text{Cyl}_G(f)) \circ H \\
 &= p_1^B \circ (\text{Cyl}_G(f) \circ H) \\
 &= p_1^B \circ h \\
 &= \text{Id}_B
 \end{aligned}$$

so f is split surjective. Now, let p be a left homotopy inverse of f and let g be the section of f . Then:

$$\begin{aligned}
 p &= p \circ \text{Id}_B \\
 &\simeq_h p \circ (f \circ g) \\
 &= (p \circ f) \circ g \\
 &\simeq_h g,
 \end{aligned}$$

as was desired. ■

This result is interesting and we have a remark about it that we will provide momentarily. Before that, we wish to introduce another class of morphisms which can be used as the “category of fibrations” over the category of C^* -algebras.

Terminology 2.32. Let $G \in \text{Grp}(\text{LCH})$, let $A, B \in G\text{-}C^*\text{Alg}$ and let $\phi \in G\text{-}C^*\text{Alg}(A, B)$. ϕ is a semisplit surjection iff it is a surjection which has an equivariant contractive & completely positive (ccp) linear section.

Here is a corollary of the proposition above.

Corollary 2.33. A. Vats, 2026

Let $G \in \text{Grp}(\text{LCH})$, let $A, B \in G\text{-}C^*\text{Alg}$ and let $f \in G\text{-}C^*\text{Alg}(A, B)$ be a Schochet fibration which admits a right homotopy inverse. Then, f is a semisplit surjection.

Proof. Indeed, the proposition above implies that f has a right inverse in $G\text{-}C^*\text{Alg}$ so we’re done. ■

Checking that some morphism is a Schochet fibration is difficult and one would like to construct homotopy equivalences without having to construct such fibrations. So, we introduce good deformation retracts to facilitate this process.

Definition 2.34

Let $G \in \text{Grp}(\text{LCH})$, let $\mathcal{C}$ be $G\text{-}C^*\text{Alg}$ or $G\text{-}Ban\text{Alg}$, let $A, B \in \mathcal{C}$ and let $f \in \mathcal{C}(A, B)$ be surjective. f is a good deformation retract iff there exists a homotopy $F : A \rightarrow \text{Cyl}_G(A)$ such that:

1. $p_0^A \circ F = \text{Id}_A$.
2. $\forall t \in [0, 1] : p_t^B \circ \text{Cyl}_G(f) \circ F = f$.
3. $(p_1^A \circ F)(\ker(f)) = 0$.

We will abbreviate good deformation retracts as GDRs.

The next result tells us that surjective GDRs give us homotopy equivalences. It is important to note that the next result **does not** necessarily hold for equivariant Banach algebras; see [Remark 2.36](#) for elaboration.

Proposition 2.35

Let $G \in \text{Grp}(\text{LCH})$. A surjective GDR in $G\text{-}C^*\text{Alg}$ is a homotopy equivalence.

Proof. Let $A, B \in G\text{-}C^*\text{Alg}$ and let $f \in G\text{-}C^*\text{Alg}(A, B)$ be a surjective GDR. There is a homotopy:

$$F : A \rightarrow \text{Cyl}_G(A),$$

such that the conditions of [Definition 2.34](#) hold. Let us factorize $p_1^A \circ F$ so that it is the composition $r \circ f$. To do this, we need to define r on B so let $b \in B$. Since f is surjective, we find $a \in A$ so $f(a) = b$. We define:

$$r(f(a)) \stackrel{\text{def}}{=} (p_1^A \circ F)(a).$$

Indeed, this is well-defined because if $a' \in A$ so that $b = f(a')$, then $a - a' \in \ker(f)$ and we know that $p_1^A \circ F$ vanishes on $\ker(f)$. The fact that r is a $\star$ -homomorphism follows by direct calculation with its definition and, in particular, the fact that f, p_1^A and F are $\star$ -homomorphisms. The rest of the proof involves establishing the next claim.

| **Claim.** $r \circ f \simeq_h \text{Id}_A$ and $f \circ r = \text{Id}_B$.

The first statement holds because the map F satisfies the following relations:

$$\begin{aligned} p_0^A \circ F &= \text{Id}_A, \\ p_1^A \circ F &= r \circ f. \end{aligned}$$

For the second relation, let $b \in B$. Then, we find $a \in A$ so that $b = f(a)$. This leads us to a computation:

$$\begin{aligned} (f \circ r)(b) &= (f \circ r \circ f)(a) \\ &= (f \circ p_1^A \circ F)(a) \\ &= (p_1^B \circ \text{Cyl}_G(f) \circ F)(a) \\ &= f(a) \\ &= b, \end{aligned}$$

as was desired. ■

We have a remark about this result.

Remark 2.36. The proposition above is interesting for two reasons:

1. So far, most of the definitions and results we've presented work in the category of equivariant Banach algebras. It is not clear, however, that the next result holds in that category; the problem is that we construct a candidate homotopy inverse in the proof and there is no reason why this needs to be continuous.
2. In [85, p.412, Proposition 1.13a], it is claimed that if we have a Schochet fibration, then that morphism is a homotopy equivalence iff it is a GDR. We were not able to establish this directly ourselves and we do not think that it is true. However, we were not able to construct an explicit counterexample.

We will somewhat return to the issue in (2) within Remark 2.37.

The non-equivariant version of the next result is in [85, Proposition 1.13b, p.412]. However, a proof is not provided and a reference is given to the proof of an analogous result on spaces. That reference proves a more general result which does not seem to have an analogue in our setting. For this reason, we provide a self-contained full proof with details.

| **Claim 2.19.** GDRs are preserved by base extensions.

At this point, we come to another result which we wanted to show but could not find a suitable argument to do it.

Remark 2.37. Let us state this as a claim.

| **Claim.** Consider the following pullback diagram in $G\text{-}\mathbf{C}^*\mathbf{Alg}$ or $G\text{-}\mathbf{BanAlg}$:

$$\begin{array}{ccc} P & \xrightarrow{\bar{p}} & C \\ \bar{q} \downarrow & & \downarrow q \\ A & \xrightarrow{p} & B \end{array}$$

| If p is a homotopy equivalence and a Schochet fibration, then $\bar{p}$ is a homotopy equivalence and a Schochet fibration.

For the non-equivariant case and in the context of $\mathbf{C}^*\mathbf{Alg}$ specifically, this result has been proved in [92, Theorem 2.19, p.10]. The proof uses a topological characterization of homotopy equivalences and Schochet fibrations:

1. the homotopy equivalences are characterized in terms of weak homotopy equivalences of mapping spaces and;
2. the Schochet fibrations are characterized in terms of Serre fibrations between the mapping spaces.

These characterizations exist because $\mathbf{C}^*\mathbf{Alg}$ is enriched in the category of compactly-generated weakly Hausdorff spaces; we believe that these results extend to $\mathbf{BanAlg}$. However, the proof of the claim above in these cases uses the fact that the category of compactly-generated weakly Hausdorff spaces equipped with the weak homotopy equivalences and Serre fibrations must be a *category of fibrant objects*. On the other hand, the presence of an action complicates matters for us; at the moment, it is unclear to us if the characterizations above extend to $G\text{-}\mathbf{C}^*\mathbf{Alg}$ or $G\text{-}\mathbf{BanAlg}$. If it were the case that a Schochet fibration were a homotopy equivalence iff it was a GDR, then we would be able to provide a very easy proof of the claim. Since this is meant to emphasize a gap in the literature which does not seem to have an obvious resolution, let us present this alternative argument now.

Alternative Argument for Claim. The proof proceeds in just three steps:

1. Since p is a Schochet fibration, we see that $\bar{p}$ is a Schochet fibration.
2. To show that $\bar{p}$ is a homotopy equivalence, it's enough to show that it's a GDR.
3. Since p is a Schochet fibration and a homotopy equivalence, it also a GDR.
4. Since GDRs are preserved by base extensions, we see that $\bar{p}$ is a GDR.

This completes the proof.

This alternative is simple but it does not immediately work in $G\text{-}\mathbf{BanAlg}$ since Proposition 2.35 may not hold. Now, we attempted to concoct a noncommutative argument to prove the claim above but there is an obstruction towards the end which we could not overcome. It effectively has to do with keeping a homotopy within a pullback.

Proof Attempt. We will prove this in the following concrete model for the pullback:

$$P = \{(a, c) \in A \times C : p(a) = q(c)\}.$$

Since p is a Schochet fibration, we see that $\bar{p}$ is a Schochet fibration by statement (5) of Proposition 2.29. Let $r : B \rightarrow A$ be a homotopy inverse of p ; by Proposition 2.31, r can be chosen so that it is a section of p . Define a map:

$$i_C : C \rightarrow P, \quad c \mapsto i_C(c) \stackrel{\text{def}}{=} ((r \circ q)(c), c).$$

Since r is a section of p , we see that i_C does, indeed, map into P and we also note that it is an equivariant $\star$ -homomorphism, since all maps involved in defining it are equivariant $\star$ -homomorphisms.

Subclaim. i_C is a homotopy inverse of $\bar{p}$.

Indeed, we have the following computation:

$$\begin{aligned} \forall c \in C : (\bar{p} \circ i_C)(c) &= \bar{p}((r \circ q)(c), c) \\ &= c \\ &= \text{Id}_C(c), \end{aligned}$$

which shows that i_C is a right inverse of $\bar{p}$. Moreover:

$$\begin{aligned} \forall (a, c) \in P : (i_C \circ \bar{p})(a, c) &= i_C(\bar{p}(a, c)) \\ &= i_C(c) \\ &= ((r \circ q)(c), c) \\ &= (r(q(c)), c) \\ &= (r(p(a)), c) \\ &= ((r \circ p) \times \text{Id}_C)|_P(a, c). \end{aligned}$$

So, it is sufficient for us to show that:

$$((r \circ p) \times \text{Id}_C)|_P \simeq_h \text{Id}_P.$$

Let $h : A \rightarrow \text{Cyl}_G(A)$ be a homotopy from Id_A to $r \circ p$. We could define:

$$H : P \rightarrow A \times B, \quad (a, c) \mapsto H(a, c) \stackrel{\text{def}}{=} (h_t(a), c),$$

but this does not lie in P for each $t \in [0, 1]$. We had the idea to define a two-parameter family:

$$(\Phi_{(s,t)} : P \rightarrow A \times C)_{(s,t) \in [0,1]^2},$$

where $\Phi_{(0,t)} = H_t$ for each $t \in [0, 1]$ and $\Phi_{(1,t)}$ lies in P for each $t \in [0, 1]$. We define:

$$\forall (a, c) \in P : \forall (s, t) \in [0, 1]^2 : \Phi_{(s,t)}(a, c) \stackrel{\text{def}}{=} \begin{cases} H_t(a, c) & \text{if } s = 0, \\ (a, c) & \text{if } t = 0, \\ H_{1-s}((r \circ p)(a), c) & \text{if } t = 1, \\ H_{(1-s)t}(a, c) & \text{otherwise} \end{cases}.$$

This was inspired by [96, Theorem 5.9, p.25]. However, $(\Phi_{(1,t)})_{t \in [0,1]}$ does not seem to be our desired homotopy. Indeed, for $t < 1$, this is just $H_0 = \text{Id}_P$. On the other hand, at $t = 1$, this is just $((r \circ p) \times \text{Id}_C)|_P$. The point, then, is that we cannot have point-norm continuity of this family of maps and the failure occurs at $t = 1$.

In spite of the obstruction laid out above, we firmly believe that the claim does hold. We believe this to be the case because the two categories in question do satisfy all other axioms of a category of fibrant objects & their structure is not so far from $C^*\text{Alg}$ that we expect this claim to fail.

At this stage, we have successfully developed the basics of equivariant homotopy theory on categories of operator algebras. Let us turn our attention towards a concrete conjecture which seems to have some unexpected applications.

2.2 Lifting Problems, Schochet C^* -Algebras & Semiprojectivity

Let us start by setting some notation.

Notation 2.38. Unless otherwise stated, H is an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space. Moreover:

$$\mathcal{Q}(H) \stackrel{\text{def}}{=} \mathcal{B}(H)/\mathcal{K}(H).$$

Finally, $\pi : \mathcal{B}(H) \rightarrow \mathcal{Q}(H)$ denotes the Calkin projection unless otherwise stated.

Our first goal is to provide a solution to the following conjecture due to Schochet.

Conjecture 2.39 (C. Schochet, 1984, [85, p.407]). The Calkin projection $\pi : \mathcal{B}(H) \rightarrow \mathcal{Q}(H)$ is not a Schochet fibration.

We will provide further motivation for interest in this conjecture momentarily. Here is an important remark.

Remark 2.40. Strictly speaking, **Conjecture 2.39** has actually been resolved (positively) in [37, Example 6.3, p. 153]. In that paper, Loring and Exel introduce the notion of a spherical filtration of the K-theory of a unital C^* -algebra and go on to prove that if the surjective $\star$ -homomorphism in a given exact sequence is a Schochet fibration, then the associated K-theory boundary map respects the given spherical filtration (see Theorem 5.6 in the same paper, p.151-152). In the case of the Calkin extension, they show explicitly that the boundary map does not respect the given spherical filtration, so the Calkin projection cannot be a Schochet fibration. Nevertheless, a rather strong downside to their result is that an explicit C^* -algebra for which the fibration property of the Calkin projection fails is not provided. This leads us into the novelty of our approach, of which there are three main points worth noting:

1. **Conjecture 2.39** is actually intimately related to an older question of Schochet's and (Jerome) Kaminker,
2. we will give an explicit example of a C^* -algebra for which the fibration property fails and;
3. our approach serves as a possible template for constructing counterexamples for homotopy-lifting in general.

Beyond these facts, we will also use our methods, which revolve around basic Fredholm theory, to resolve **Conjecture 2.39** in the case where H is non-separable. We will see that this has some natural implications for extensions of this problem which one can consider; we will discuss these in Section 5.1.

We will start by providing an extended discussion of the context behind the conjecture and its relationship with older problems in operator theory. Let us record down a useful piece of terminology that will fix language.

Terminology 2.41. Let P be a property that is satisfied by elements of a C^* -algebra and let $T \in \mathcal{B}(H)$. Then, T is essentially P iff $\pi(T)$ is P in $\mathcal{Q}(H)$. Moreover, the essential spectrum of T is defined by $\sigma_{\text{ess}}(T) \stackrel{\text{def}}{=} \sigma_{\mathcal{Q}(H)}(\pi(T))$.

We will mainly be concerned with essentially normal operators³. It is a stunning result of Brown-Douglas-Fillmore theory that there is a classification of such operators. To state this theorem formally, we will need a construction.

Construction 2.42. Let H be a $\mathbb{C}$ -Hilbert space and let $T \in \mathcal{B}(H)$ be Fredholm. We define:

$$\phi_T : \mathbb{C} \setminus \sigma_{\text{ess}}(T) \rightarrow \mathbb{Z}, \lambda \mapsto \phi_T(\lambda) \stackrel{\text{def}}{=} \text{Index}(T - \lambda \text{Id}_H).$$

This is known as the index function of T . This makes sense, because $\lambda \notin \sigma_{\text{ess}}(T)$ so $\pi(T) - \lambda 1_{\mathcal{Q}(H)}$ is invertible. Hence, $T - \lambda \text{Id}_H$ is a Fredholm operator and the index above is well-defined.

Now, we can state our result.

Theorem 2.43. The Brown-Douglas-Fillmore Theorem, [38, Corollary 11.2, p.118], [39]

Let H be an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space and let $T \in \mathcal{B}(H)$ be essentially normal. TFAE:

1. $T \in N_{\mathcal{B}(H)} + \mathcal{K}(H)$.
2. ϕ_T is trivial.

The key tool used to prove the theorem above (and related theorems classifying such operators) are the so-called Brown-Douglas-Fillmore (BDF) Ext groups. We will, now, provide some intuition for how these groups arise naturally; our exposition is heavily inspired by the introduction of [17]. If $T \in \mathcal{B}(H)$ is essentially normal, then T (along with $\mathcal{K}(H)$ and Id_H) generates a C^* -subalgebra A of $\mathcal{B}(H)$. With this data, we can construct a morphism $A \rightarrow C^*(\pi(T), \text{Id}_{\mathcal{Q}(H)})$, whose kernel is $\mathcal{K}(H)$. On the other hand, the C^* -algebra on the right-hand side of the arrow is a commutative unital C^* -algebra. By Gelfand Duality and some more concrete results about functional calculus, we see that:

$$C^*(\pi(T), \text{Id}_{\mathcal{Q}(H)}) \simeq \mathcal{C}(\sigma_{\text{ess}}(T)).$$

So, this data fits into an exact sequence:

$$0 \rightarrow \mathcal{K}(H) \rightarrow A \rightarrow \mathcal{C}(\sigma_{\text{ess}}(T)) \rightarrow 0.$$

In other words, essentially normal operators naturally give rise to extensions of the form given above and now, it becomes reasonable to replace $\sigma_{\text{ess}}(T)$ by some compact metrizable space X and study extensions of the following form:

$$0 \rightarrow \mathcal{K}(H) \rightarrow A \rightarrow \mathcal{C}(X).$$

This leads us to the following definition.

³For an extended discussion of essentially normal operators & their history, as well as recent developments, see [30].

Definition 2.44

Let X be a compact, metrizable space.

1. An extension of $\mathcal{K}(H)$ by $\mathcal{C}(X)$ is a pair:

$$(A, \phi),$$

where A is a C^* -subalgebra of $\mathcal{B}(H)$ containing $\mathcal{K}(H)$ and $\text{Id}_H \in \phi \in C^*\text{Alg}(A, \mathcal{C}(X))$ with $\ker(\phi) = \mathcal{K}(H)$.

2. If $(A_1, \phi_1), (A_2, \phi_2)$ are extensions of $\mathcal{K}(H)$ by $\mathcal{C}(X)$, then these extensions are strongly equivalent^a iff there exists a $\star$ -isomorphism $\psi : A_2 \rightarrow A_1$ such that $\phi_2 = \phi_1 \circ \psi$.

The set of equivalence classes of extensions of $\mathcal{K}(H)$ by $\mathcal{C}(X)$ is denoted by $\text{Ext}(X)$.

^aThis can be checked directly to be an equivalence relation.

We will not spend any time discussing the proofs of various results in BDF theory since that would take up too much space. The main result that needs to be known is that $\text{Ext}(X)$ can be made into a group and that Ext is functorial.

Proposition 2.45. [17, Corollary 1.18, Theorem 1.23, p. 272-273, p. 275]

We have a functor $\text{Ext}(-)$ from the category of compact metrizable spaces to AbGrp .

In fact, this functor happens to be homotopy-invariant. The proof of homotopy-invariance as discussed in [17, Section 2, p.275-286] is somewhat tricky and an alternative, simpler proof was desired. To that end, there was another functor constructed by Schochet & Kaminker which homotopy-invariance is “obvious”.

Construction 2.46. Let X be a compact metrizable space. We define:

$$\text{inj}^h(\mathcal{C}(X), \mathcal{Q}(H)),$$

to be the set of all homotopy classes of injective $\star$ -homomorphisms $\mathcal{C}(X) \rightarrow \mathcal{Q}(H)$. Using the methods of [49], this can be shown to be an abelian group. This is also a functor^a, denoted by inj^h .

^aFrom the category of compact metrizable spaces to AbGrp .

These functors coincide on a very special class of spaces. To define them, we will need some terminology.

Terminology 2.47. Let $A, B \in C^*\text{Alg}$ and let $f, g \in C^*\text{Alg}(A, B)$. f and g are $\star$ -homotopic iff there is a continuous map^a $\phi : A \times [0, 1] \rightarrow B$ such that $\phi(\cdot, t)$ is a $\star$ -homomorphism for each $t \in [0, 1]$, with $\phi(\cdot, 0) = f$ and $\phi(\cdot, 1) = g$.

^aUnsurprisingly, we call this a $\star$ -homotopy.

This notion was introduced in [26] and [27]. Let us show that it is not different from the usual notion of homotopy. While this has not been shown explicitly in the literature, the computation used to establish it is standard.

Claim 2.20. Let $A, B \in C^*\text{Alg}$ and let $f, g \in C^*\text{Alg}(A, B)$. TFAE:

1. f and g are homotopic.
2. f and g are $\star$ -homotopic.

We also get explicit ways of defining homotopies from $\star$ -homotopies and vice versa. This is proved on Page 107.

Now, we can introduce the C^* -Covering Homotopy Property.

Definition 2.48. J. Conway, [26]

Let $A \in C^*\text{Alg}$. A has the C^* -Covering Homotopy Property (or satisfies the C^* -CHP or is C^* -CHP) iff for any $\star$ -homotopy $\phi : A \times [0, 1] \rightarrow \mathcal{Q}(H)$ and any $f \in C^*\text{Alg}(A, \mathcal{B}(H))$ such that:

$$\pi \circ f = \phi(\cdot, 0),$$

there is a $\star$ -homotopy $\Phi : A \times [0, 1] \rightarrow \mathcal{B}(H)$ such that $\pi \circ \Phi = \phi$ and $\Phi(\cdot, 0) = f$.

The following result explains the relationship between the BDF Ext groups and the functor in Construction 2.46.

Proposition 2.49. C. Schochet, J. Kaminker, [49, p.139]

If X is a compact metrizable space which is C^* -CHP, then:

$$\text{Ext}(X) \simeq \text{inj}^h(\mathcal{C}(X), \mathcal{Q}(H)),$$

and the isomorphism is natural on the full subcategory of compact metrizable C^* -CHP spaces.

In light of the fact that inj^h is homotopy-invariant, Schochet and Kaminker asked the following question.

| **Question 2.50** ([49, p.139]). Is every compact metrizable space C^* -CHP?

In the aforementioned papers [26] and [27], (John) Conway actually showed that there are a number of interesting examples of C^* -algebras which satisfy the C^* -CHP. In particular, he showed that AF algebras are C^* -CHP, $\mathcal{C}(S^1)$ is C^* -CHP and $\mathcal{C}([0, 1])$ is C^* -CHP. He also posed the following question in less formal form.

Question 2.51 (J. Conway, [26, p.136]). Are the following spaces:

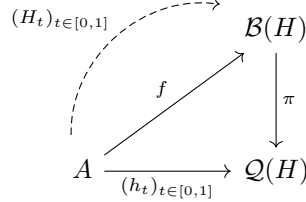
1. $S^1 \vee S^1$,
2. $S^1 \vee [1, 2]$,
3. $[-1, 0] \cup \{2^{-n} : n \in \mathbb{N}_0\}$,

C^* -CHP?

It seems difficult to construct many examples of C^* -algebras which are C^* -CHP. It is likely that the spaces in Question 2.51 were chosen since they were just out of reach using the methods of the aforementioned papers. This question will be addressed later, when we have introduced semiprojectivity into our exposition. Now, we have to relate the discussion above to Conjecture 2.39. Let us introduce the first main class of C^* -algebras whose properties will concern us.

Definition 2.52. A. Vats, 2026

Let $A \in C^* \text{Alg}$. A is Schochet iff for any infinite-dimensional, separable $\mathbb{C}$ -Hilbert space H and for any solid diagram:



which commutes for $t = 0$, there exists a homotopy $(H_t)_{t \in [0,1]}$ such that:

$$H_0 = f,$$

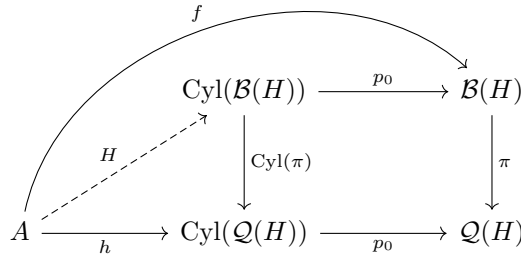
$$\forall t \in [0, 1] : \pi \circ H_t = h_t.$$

Let $X \in \text{LCH}$ and $G \in \text{Grp}(\text{LCH})$. X is Schochet iff $\mathcal{C}_0(X)$ is Schochet. G is Schochet iff $C_{\text{red}}^*(G)$ is Schochet.

Let us characterize the Schochet property for C^* -algebras in a way that will be useful.

Claim 2.21. Let $A \in C^* \text{Alg}$. TFAE:

1. A is Schochet.
2. The following lifting problem (indicated by the solid diagram):



admits a solution (indicated by the dotted arrows).

This is proved on Page 107.

In the next result, we will clarify the relationship between Schochet C^* -algebras and C^* -CHP C^* -algebras.

Claim 2.22. Let $A \in C^* \text{Alg}$. TFAE:

1. A is Schochet.
2. A is C^* -CHP.

Having given all of this background and having posed some interesting questions, it is time that we started working towards answering them and also developing this theory a bit more systematically. Let us, before doing anything else, mention how Conjecture 2.39 can be understood in the context of the material introduced previously.

Proposition 2.53. A. Vats, 2026

The Calkin projection is not a GDR.

Proof. Let us suppose that it's a GDR. By Proposition 2.35, the Calkin projection must be a homotopy equivalence so it follows that $\mathcal{B}(H)$ and $\mathcal{Q}(H)$ have isomorphic K_1 -groups. However, this is clearly not the case as:

$$K_1(\mathcal{B}(H)) \simeq 0,$$

$$\text{and } K_1(\mathcal{Q}(H)) \simeq \mathbb{Z}.$$

■

We have two remarks about this proof.

Remark 2.54. In fact, we can use a similar argument to show that the canonical projection cannot even have a right homotopy inverse. So, there does not seem to be any hope in utilizing Proposition 2.31 to obtain a contradiction.

Remark 2.55. The argument used above can be generalized to show that other morphisms of C^* -algebras are not GDRs. Indeed, we can do this for morphisms where the domain and codomain do not have isomorphic K-groups.

Let us give our positive solution to [Conjecture 2.39](#).

Theorem 2.56. A. Vats, 2026

The following universal C^* -algebra:

$$A \stackrel{\text{def}}{=} C^*(x | xx^* = x^*x \wedge \|x\| \leq 1).$$

is not Schochet. That is, [Conjecture 2.39](#) holds.

Proof. Let $H = \ell^2(\mathbb{N})$, let S be the unilateral shift in $\mathcal{B}(H)$ and let s be the image of S in $\mathcal{Q}(H)$ under the Calkin projection. Then, the C^* -algebra generated by S in $\mathcal{B}(H)$ is the Toeplitz C^* -algebra $\mathcal{T}$ and it contains $\mathcal{K}(H)$. Moreover, it is a known fact that $\mathcal{T}/\mathcal{K}(H) \simeq \mathcal{C}(S^1)$. Now, we will need two subclaims.

Subclaim. There is no normal element $a \in \mathcal{B}(H)$ so that $\pi(a) = s$.

This simple fact is stated without proof in [16, p.1]; we will establish it at the end of this proof, after we have dealt with the main result. Now, the second subclaim is much more standard so we will point to a reference.

Subclaim. Let $E = \{x_i : i \in I\}$ be a set of generators, let R be relations between these generators such that the universal C^* -algebra $C^*(E|R)$ exists and let $B \in C^*\text{Alg}$ contain a set $E' = \{y_i : i \in I\}$. If the elements of E' satisfy the relations R , there is a unique $\star$ -homomorphism $\phi : C^*(E|R) \rightarrow B$ sending x_i to y_i for each $i \in I$.

For the proof and further systematic discussion about universal C^* -algebras, see [52, Proposition 6.7, p.71]. By subclaim (1), s does not have a unitary preimage in $\mathcal{B}(H)$ via the Calkin projection. Now, we define a map

$$\phi(x) : [0, 1] \rightarrow \mathcal{Q}(H), \quad t \mapsto \phi(x)(t) \stackrel{\text{def}}{=} ts.$$

This is continuous since:

$$\begin{aligned} \forall t, r \in [0, 1] : \|\phi(x)(t) - \phi(x)(r)\| &= \|ts - rs\| \\ &= |t - r| \cdot \|s\| \\ &\leq |t - r|. \end{aligned}$$

So, $\phi(x) \in \text{Cyl}(\mathcal{Q}(H))$. Moreover, $\phi(x)$ actually maps into $\text{Cyl}(\mathcal{T}/\mathcal{K}(H))$ and this is commutative so it follows that $\phi(x)$ is normal in $\mathcal{Q}(H)$. Now, we also have the following estimate:

$$\begin{aligned} \|\phi(x)\| &= \sup_{t \in [0, 1]} \|\phi(x)(t)\| \\ &= \sup_{t \in [0, 1]} \|ts\| \\ &\leq 1. \end{aligned}$$

So, $\phi(x)$ is a normal contraction in $\text{Cyl}(\mathcal{Q}(H))$. By the universal property of universal C^* -algebras (subclaim (2)), this induces a $\star$ -homomorphism $\phi : A \rightarrow \text{Cyl}(\mathcal{Q}(H))$. By construction, this is a homotopy from ϕ_0 to ϕ_1 . Now, the key point is that ϕ_0 lifts to $\mathcal{B}(H)$ while ϕ_1 does not lift to $\mathcal{B}(H)$ due to subclaim (1). Indeed, if ϕ_1 did have a lift $\psi : A \rightarrow \mathcal{B}(H)$, then $\pi(\psi(x)) = s$ with $\psi(x)$ being a normal contraction because ψ is contractive as a $\star$ -homomorphism. So, we have constructed a homotopy from ϕ_0 to ϕ_1 with ϕ_0 being liftable and ϕ_1 not being liftable, so we're done.

Now, let us deal with the proof of subclaim (1). A very classical and direct index theory calculation reveals that:

$$\text{Index}(S) = -1.$$

If $T \in \mathcal{B}(H)$ is normal, then we know that:

$$\text{coker}(T) \simeq \ker(T^*).$$

So, we see that:

$$\begin{aligned} \text{Index}(T) &= \dim(\ker(T)) - \dim(\text{coker}(T)) \\ &= \dim(\ker(T)) - \dim(\ker(T^*)) \\ &= 0, \end{aligned}$$

where the last statement holds, again, because normality is equivalent to the assertion that:

$$\forall x \in H : \|T(x)\| = \|T^*(x)\|.$$

On the other hand, let $T \in \mathcal{B}(H)$ so that:

$$\pi(T) = s = \pi(S),$$

which implies that $T - S \in \mathcal{K}(H)$. But this means that their Fredholm indices coincide but this is impossible. ■

As a consequence, we also resolve [Question 2.50](#).

Corollary 2.57. A. Vats, 2026

There exists a compact metrizable space which is not C^* -CHP. So, [Question 2.50](#) has a negative answer.

Proof. Let us take the compact unit disk $\mathbb{D}$. Then:

$$\mathcal{C}(\mathbb{D}) \simeq C^*(x, 1 | xx^* = x^*x \wedge \|x\| \leq 1).$$

This C^* -algebra is actually the unitization of the one used in [Theorem 2.56](#). In that case, we simply construct a homotopy from the $\star$ -homomorphism which takes x to 0 and 1 to $\text{Id}_{\mathcal{Q}(H)}$ to the $\star$ -homomorphism which takes x to s (as defined in the proof of the aforementioned theorem) and 1 to $\text{Id}_{\mathcal{Q}(H)}$. The first $\star$ -homomorphism is clearly liftable while the latter is not liftable, but both are homotopic. So, the given homotopy cannot lift. ■

Let us describe the inspiration behind the construction above.

Remark 2.58. Our argument in [Theorem 2.56](#) was inspired by the proof of (Bruce) Blackadar that the C^* -algebra constructed in the theorem is not l -closed. We will introduce this notion in [Definition 2.64](#) and explain why it made sense to look at different examples of l -closed C^* -algebras; see [\[13, Example 6.5\]](#).

As we promised in [Remark 2.40](#), let us show that the Calkin projection is not a fibration for **any** infinite-dimensional $\mathbb{C}$ -Hilbert space. The proof requires just a little more input from Fredholm theory.

Proposition 2.59. A. Vats, 2026

Let H be any infinite-dimensional $\mathbb{C}$ -Hilbert space. Then, the Calkin projection over H is not a Schochet fibration.

Proof. We will adapt the argument of [Theorem 2.56](#) to this situation. In particular, define:

$$A \stackrel{\text{def}}{=} C^*(x, 1 | x^*x = xx^* \wedge \|x\| \leq 1).$$

We want to show that the Calkin projection fails the fibration property for this C^* -algebra. We have already shown this in the separable case, so assume that H is non-separable. We will construct a $T \in \mathcal{B}(H)$ such that:

1. T is Fredholm;
2. $\text{Index}(T) \neq 0$ and;
3. $\pi(T)$ is a normal contraction.

If we can do this, then exactly the same argument as in the separable case will work. We need a general claim.

Claim. Let $X, X', Y, Y' \in \text{Ban}_{\mathbb{C}}$, let $T \in \text{Ban}_{\mathbb{C}}(X, Y)$ and let $S \in \text{Ban}_{\mathbb{C}}(X', Y')$.

1. The linear map $T \oplus S$ is bounded, with $\|T \oplus S\| = \max\{\|T\|, \|S\|\}$.
2. If T and S are Fredholm operators, then $T \oplus S$ is a Fredholm operator.
3. If (2) holds, then $\text{Index}(T \oplus S) = \text{Index}(T) + \text{Index}(S)$.

We refer to $T \oplus S$ as the direct sum of T and S .

This can be found in [\[66, Exercise 19, p.33\]](#); the proof involves relating $\ker(T \oplus S)$ to $\ker(T)$ and $\ker(S)$, while also relating $\text{ran}(T \oplus S)$ to $\text{ran}(T)$ and $\text{ran}(S)$. Now, let $M \subseteq H$ be a separable, closed subspace and write $H = M \oplus N$, where this is a topological direct sum; this is possible because closed subspaces of $\mathbb{C}$ -Hilbert spaces are complemented. Next, let S be the unilateral shift on M with respect to some orthonormal basis of M and define:

$$T \stackrel{\text{def}}{=} S \oplus \text{Id}_N.$$

Since S and Id_N are Fredholm operators, the claim implies that T is a Fredholm operator. Moreover:

$$\text{Index}(T) = \text{Index}(S) + \text{Index}(\text{Id}_N) = -1.$$

Let us also note that:

$$\|T\| = \|S \oplus \text{Id}_N\| = \max\{\|S\|, \|\text{Id}_N\|\} = 1.$$

Next, we have the following claim.

Claim. $\pi(T)$ is a normal contraction.

The fact that this is a contraction is easy:

$$\|\pi(T)\| \leq \|T\| = 1.$$

As for normality, we have the following computation:

$$\begin{aligned} TT^* - T^*T &= (S \oplus \text{Id}_N)(S^* \oplus \text{Id}_N) - (S^* \oplus \text{Id}_N)(S \oplus \text{Id}_N) \\ &= SS^* \oplus \text{Id}_N - \text{Id}_M \oplus \text{Id}_N \\ &= (SS^* - \text{Id}_M) \oplus 0, \end{aligned}$$

where $SS^* - \text{Id}_M$ is a compact operator. So, the above is a finite-rank operator and, therefore, compact. Hence, its image in the Calkin algebra vanishes and this implies that $\pi(T)$ is normal. So, the claim holds.

So, we have constructed our operator with the desired properties and we are done. ■

We have a remark about the result above.

Remark 2.60. We have written out the argument above in excess detail just to make sure that each step is convincing. It will also lead rather nicely into our discussion of a more general version of this problem in [Section 5.1](#).

It is interesting to note that we do not even know if $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is l -closed. To explain what this means, we need a bit of preparation. Let us introduce two pieces of convenient terminology and a construction.

Terminology 2.61. Let $A \in C^*Alg$ and let $I \subseteq A$ be a closed ideal. Then, the pair:

$$(A, I),$$

is known as a marked C^* -algebra.

Terminology 2.62. Let $A \in C^*Alg$, let (B, I) be a marked C^* -algebra, let $\pi_I : B \rightarrow B/I$ be the canonical projection and let $f \in C^*Alg(A, B/I)$. f is liftable iff there exists a $\star$ -homomorphism $\bar{f}$ such that the following diagram:

$$\begin{array}{ccc} & & B \\ & \nearrow \bar{f} & \downarrow \pi_I \\ A & \xrightarrow{f} & B/I \end{array}$$

commutes.

Here is our construction.

Construction 2.63. Let $A, B \in C^*Alg$. For each $a \in A$, we define a semi-metric:

$$p_a : \text{Ban}_{\mathbb{C}}(A, B) \times \text{Ban}_{\mathbb{C}}(A, B) \rightarrow [0, \infty), \quad (f, g) \mapsto p_a(f, g) \stackrel{\text{def}}{=} \|f(a) - g(a)\|.$$

This gives us a family of seminorms on $\text{Ban}_{\mathbb{C}}(A, B)$ and this generates a locally convex topology on $\text{Ban}_{\mathbb{C}}(A, B)$; we will refer to this as the point-norm topology. Now, we take the subspace topology on $C^*Alg(A, B)$ and this set equipped with this topology will be denoted by $\text{Hom}(A, B)$. If A and B are unital, then we denote the (topological) subspace of unital $\star$ -homomorphisms by $\text{Hom}_1(A, B)$. Finally, if $I \subseteq B$ is a closed ideal, then we define:

$$\text{Hom}(A, B, I) \stackrel{\text{def}}{=} \pi_I \circ \text{Hom}(A, B),$$

as the topological subspace of liftable morphisms in $\text{Hom}(A, B/I)$.

Now, we can define l -openness and closedness.

Definition 2.64

Let $A \in C^*Alg$ and let (B, I) be a marked C^* -algebra.

1. A is (B, I) - l -open iff $\text{Hom}(A, B, I)$ is open in $\text{Hom}(A, B/I)$.
2. A is (B, I) - l -closed iff $\text{Hom}(A, B, I)$ is closed in $\text{Hom}(A, B/I)$.

Finally, A is l -open (closed) iff it is (B, I) - l -open (closed) for every marked C^* -algebra^a (B, I) .

^aIf we are working with the marked C^* -algebra $(\mathcal{B}(H), \mathcal{K}(H))$, then we will say that A is either H - l -open or H - l -closed.

Let us state our desired question more formally.

Question 2.65 (B. Blackadar, 2016, [13, Example 6.5, p.8-9]). Is $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ l -closed^a?

^a $\mathbb{D}$ is the compact unit disk in $\mathbb{C}$ centered at the origin.

We were not able to answer this question fully but we have managed to “explain” why one may expect it to be a difficult problem. Before we get to that, let us review what is known in the literature, starting with some terminology.

Terminology 2.66. Let (A, I) be a marked C^* -algebra and let $b \in A/I$ satisfy some property P which is satisfied by elements of C^* -algebras. b is P -liftable iff there exists an $a \in A$ such that $\pi(a) = b$ and a satisfies P .

We will also need some notation.

Notation 2.67. Let $A \in C^*Alg$. Then, the set of normal elements in A is denoted by N_A .

The following result characterizes when Question 2.65 has a positive answer.

Proposition 2.68. B. Blackadar (2016), H. Thiel (2020), [13, Example 6.5, p.8-9], [90]

TFAE:

1. $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is l -closed.
2. If (A, I) is a marked C^* -algebra and let $(x_n)_{n \in \mathbb{N}}$ be a sequence of normal elements in A/I which converges to some $x \in A/I$. If x_n is normal-liftable for each $n \in \mathbb{N}$, then x is also normal-liftable.
3. If (A, I) is a marked C^* -algebra, then $\pi(N_A)$ is closed in B/I , where $\pi : A \rightarrow A/I$ is the canonical projection.

Now, we state our result.

Theorem 2.69. A. Vats, 2026

Let (A, I) be a marked C^* -algebra. TFAE:

1. The sum $N_A + I$ is closed in A .
2. Any convergent sequence in $N_{A/I}$ whose terms are normal-liftable must have a normal-liftable limit.

The proof of this result requires the C^* -Bartle-Graves Theorem. The Banach-space version of this result is an old theorem proved in [64, p.364], though the version that we will provide now is a much newer result due to (Tatiana) Shulman which we hope to use later as a way of simplifying some proofs involving the Schochet property.

Theorem 2.70. The C^* -Bartle-Graves Theorem, T. Shulman, [86, Theorem 5, p.7-8]

Let $A, B \in C^*Alg$ and let $\pi \in C^*Alg(A, B)$ be surjective. Then, there exists a $s \in \mathcal{C}(B, A)$ so that:

1. s is contractive and;
2. s is unital (if A and B are unital, with π also being unital) and;
3. $\forall b \in B : s(b^*) = s(b)^*$ and;
4. $\forall b \in B^+ : s(b) \in A^+$.

Moreover, if $a_0 \in A$ and $b_0 \in B$ with $\pi(a_0) = b_0$, then s can be chosen so that:

$$s(b_0) = a_0,$$

and so that it still satisfies the above properties. We refer to s as the C^* -section of π .

This leads us to the proof of our result.

Proof of Theorem 2.69.

(1) $\Rightarrow$ (2): We will use the C^* -Bartle-Graves Theorem as stated in Theorem 2.70; since π is a surjective $\star$ -homomorphism, it has a C^* -section $s : A/I \rightarrow A$. Now, let $(x_n + I)_{n \in \mathbb{N}}$ be a convergent sequence in A/I with each x_n being normal in A and let $x + I$ be the limit. We define:

$$\begin{aligned} \forall n \in \mathbb{N} : y_n &\stackrel{\text{def}}{=} s(x_n + I), \\ y &\stackrel{\text{def}}{=} s(x + I). \end{aligned}$$

Since s is continuous, we see that $y_n \rightarrow y$. Now, observe that:

$$\begin{aligned} \forall n \in \mathbb{N} : y_n + I &= \pi(y_n) \\ &= \pi(s(x_n + I)) \\ &= x_n + I, \end{aligned}$$

which means that $y_n - x_n \in I$ for each $n \in \mathbb{N}$. In other words, each $y_n \in N_A + I$; we use the normality of each x_n over here. Since $(y_n)_{n \in \mathbb{N}}$ is convergent and $N_A + I$ is closed, we see that $y \in N_A + I$. So, we write:

$$y = a + k,$$

for some normal a and $k \in I$. But now, we see that:

$$\begin{aligned} x + I &= (\pi \circ s)(x + I) \\ &= \pi(s(x + I)) \\ &= \pi(y) \\ &= \pi(a + k) \\ &= \pi(a) + \pi(k) \\ &= \pi(a), \end{aligned}$$

which proves the result.

(2) $\Rightarrow$ (1): Let $(a_n + k_n)_{n \in \mathbb{N}}$ be a sequence in $N_A + I$ which converges to some $x \in A$, where each a_n is normal and each $k_n \in I$. Then, we have that $(\pi(a_n + k_n))_{n \in \mathbb{N}}$ is a sequence in A/I which converges to $\pi(x)$. Moreover, this sequence in A/I consists of terms which are normal-liftable, so its limit is normal-liftable. That is:

$$\exists a \in N_A : \pi(a) = \pi(x).$$

This implies that $x - a \in I$ so $x \in N_A + I$. ■

The following is an immediate corollary.

Corollary 2.71. A. Vats, 2026

Let (A, I) be a marked C^* -algebra. TFAE:

1. $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is (A, I) - l -closed.
2. $N_A + I$, as defined in Theorem 2.69, is closed in A .

So, $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is l -closed iff for every marked C^* -algebra (A, I) , we have that $N_A + I$ is closed in A .

Proof. This follows by combining Proposition 2.68 and Theorem 2.69. ■

This leads us to introduce the following piece of terminology.

Terminology 2.72. Let $A \in C^*Alg$. A is normally closed iff for every closed ideal $I \subseteq A$, we have that:

$$N_A + I,$$

is closed in A .

In fact, it turns out that there are many examples of normally closed C^* -algebras.

Example 2.73. Let $A \in \text{AbC}^*\text{Alg}$. Then, A is clearly normally closed, since the normal elements of A are all elements of A . So, if $I \subseteq A$ is a closed ideal, then $N_A + I = A + I$ and this is closed.

Example 2.74. Let $A \in \text{C}^*\text{Alg}$ be simple. Then, A is clearly normally closed. Indeed, let $I \subseteq A$ be a closed ideal and note that $I = 0$ or $I = A$. If I is trivial, then we see that $N_A + I = N_A$ and this is closed. If $I = A$, then $N_A + A$ is also closed; the limit x of any convergent sequence in $N_A + A$ can be written just as $0 + x$.

Example 2.75. Let H be an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space. Then, $\mathcal{B}(H)$ is normally closed. Of course, the main point is to simply prove that $N_{\mathcal{B}(H)} + \mathcal{K}(H)$ is closed in $\mathcal{B}(H)$. Indeed, that is true but the classical proof uses Brown-Douglas-Fillmore theory; see [38, Corollary 11.4, p.119].

Example 2.76. Let H be an infinite-dimensional, nonseparable $\mathbb{C}$ -Hilbert space and let $I \subseteq \mathcal{B}(H)$ be a closed ideal which is not the compact operators in $\mathcal{B}(H)$. The results of [97] tell us that every normal element in $\mathcal{B}(H)/I$ is normal-liftable through the canonical projection. So, $N_{\mathcal{B}(H)} + I$ is closed in $\mathcal{B}(H)$. The problem for the compact operators is still open and the proof in the aforementioned paper does not seem to generalize to other C^* -algebras.

Normally closed C^* -algebras actually arise naturally in the context of lifting problems. In [90], Thiel introduces the notion of a universally liftable normal element in a C^* -algebra and presents some reasonable questions about this notion. Let us refine his definition just a bit and examine it more closely.

Definition 2.77

Let $\mathcal{C}$ be a full subcategory of C^*Alg , let $A \in \text{C}^*\text{Alg}$ and let $a \in A$ be normal.

1. a is universally liftable for $\mathcal{C}$ iff for every $B \in \mathcal{C}$ and every surjective $\star$ -homomorphism:

$$\pi : B \rightarrow A,$$

there exists a normal $b \in B$ such that $\pi(b) = a$.

2. a is universally liftable iff it is universally liftable for C^*Alg .

The set of normal elements in A which are universally liftable for $\mathcal{C}$ is denoted by $^a \text{UnivLift}_{\mathcal{C}}(A)$.

^aWe omit the $\mathcal{C}$ when $\mathcal{C}$ is just C^*Alg .

There are two main questions of interest regarding universally liftable elements.

Question 2.78 (H. Thiel, 2020, [90]). Let $A \in \text{C}^*\text{Alg}$. Is $\text{UnivLift}(A)$ closed?

Question 2.79 (H. Thiel, 2020, [90]). Let $A \in \text{C}^*\text{Alg}$. Is $\text{UnivLift}(A)$ open?

The next result indicates progress on Question 2.78.

Proposition 2.80. A. Vats, 2026

Let $A \in \text{C}^*\text{Alg}$. The set of universally liftable elements for normally closed C^* -algebras is closed.

Proof. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of universally liftable normal elements^a in A which converges to some normal $a \in A$. Let $B \in \text{C}^*\text{Alg}$ be normally closed and let $\pi : B \rightarrow A$ be a surjective $\star$ -homomorphism. For each $n \in \mathbb{N}$, there is a normal $b_n \in B$ so that $\pi(b_n) = a_n$. Now, we identify A with $B/\ker(\pi)$. Since B is normally closed, we can simply use Theorem 2.69 to conclude that a is normal-liftable so we're done. ■

^aFor normally closed C^* -algebras.

Due to the availability of stronger theorems and more palatable descriptions of certain C^* -algebras, it is actually possible for us to establish that the examples highlighted by Conway in [26] and [27] are actually Schochet C^* -algebras. Indeed, his methods were rather technical and we are in a position where we can give some simplified proofs. To illustrate that it can be completely non-trivial to verify the Schochet property, let us do it in the case of $\mathbb{C}$.

Example 2.81. We want to show that $\mathbb{C}$ is a Schochet C^* -algebra. We will rely on the following subclaim.

Subclaim. Let $A, B \in {}^1\text{C}^*\text{Alg}$, let $\pi \in \text{C}^*\text{Alg}(A, B)$ be surjective and let $(p_t)_{t \in [0,1]}$ be a path of projections in B such that p_0 lifts to a projection $P_0 \in A$ through π . The entire path lifts to a path $(P_t)_{t \in [0,1]}$ of projections in A .

The proof of this is somewhat technical, so we defer it to a reference [35, Exercise 8.1.28, p. 311], which contains essentially the whole proof. It is useful for us to simply sketch out the main points of the proof:

1. We will start by noticing that π is unital, since it is surjective so $\pi(1_A)$ is an identity for B .
2. Paths of projections in a unital C^* can be given by unitary conjugation. To be precise, there exists a path $(u_t)_{t \in [0,1]}$ of unitaries in B such that $p_t = u_t p_1 u_t^*$. This follows essentially from the fact that two projections which are sufficiently close must be unitarily equivalent.
3. So, the problem, now, reduces to lifting paths of unitaries. This can be done essentially by writing $u_t = \exp(ix_t)$ for some path $(x_t)_{t \in [0,1]}$ of self-adjoint elements (using the logarithm) and reducing the problem to lifting this path of self-adjoint elements to a path of self-adjoint elements in A , which is easy.

With this in mind, we start with the following lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ \mathbb{C} & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

Now, $(h_t(1))_{t \in [0,1]}$ is a path of projections in $\mathcal{Q}(H)$ so the subclaim tells us that there is a path $(S_t)_{t \in [0,1]}$ of projections in $\mathcal{B}(H)$ which is a lift for the given path in $\mathcal{Q}(H)$, with $S_0 = f(1)$. Now, we define the following map for each $t \in [0, 1]$:

$$H_t : \mathbb{C} \rightarrow \mathcal{B}(H), \quad z \mapsto H_t(z) \stackrel{\text{def}}{=} zS_t.$$

We observe that this is a $\star$ -homomorphism^a. In fact, the family $(H_t)_{t \in [0,1]}$ forms a homotopy because $(S_t)_{t \in [0,1]}$ is a path and scalar multiplication is continuous. Now, let us check that this works:

$$\begin{aligned} \forall z \in \mathbb{C} : H_0(z) &= zS_0 \\ &= zf(1) \\ &= f(z), \end{aligned}$$

which means that $H_0 = f$. Moreover:

$$\begin{aligned} \forall t \in [0, 1] : \forall z \in \mathbb{C} : (\pi \circ H_t)(z) &= \pi(H_t(z)) \\ &= \pi(zS_t) \\ &= z\pi(S_t) \\ &= zh_t(1) \\ &= h_t(z), \end{aligned}$$

which shows that $\pi \circ H_t = h_t$ so we are done.

So, even with a C^* -algebra as simple as $\mathbb{C}$, it does take some effort to prove the Schochet property directly.

^aLinearity is clear simply by the properties of scalar multiplication of operators. The fact that this preserves products follows from the fact that S_t is idempotent. Finally, the fact that this preserves the involution follows from the self-adjointness of S_t .

The next example generalizes the preceding one and it was one of those shown by Conway which we will not reprove here, as there is a simpler argument which already exists. So, we will point to that reference.

Example 2.82. Let A be an AF algebra.

| **Claim.** A is Schochet.

While this was proved by Conway in [27], the proof seems to not be so amenable to generalization. There is another proof in [69, Theorem 2.2.9] and it seems more reasonable to attempt to generalize this proof. The main obstruction to this is Lemma 2.2.7 in the same reference. We could not find the cited paper which proves that lemma, so we point to the proof of [74, Lemma 1.3, p.958-959] instead. Now, we also have the following claim.

| **Claim.** All totally disconnected compact Hausdorff spaces are Schochet spaces.

Indeed, continuous functions over such spaces are unital commutative AF algebras so we're done.

In fact, all commutative von Neumann algebras are Schochet C^* -algebras.

Example 2.83. Let $A \subseteq \mathcal{B}(H)$ be a commutative von Neumann algebra.

| **Claim.** A is a Schochet C^* -algebra.

Indeed, the Gelfand spectrum of any commutative von Neumann algebra is extremally disconnected and Hausdorff, so it is totally disconnected. Hence, it is a Schochet space and this implies the desired claim.

This argument works for any commutative C^* -algebra with extremally disconnected spectrum.

In the next example, we will show that $[0, 1]$ is a Schochet space.

Example 2.84. Let $A = \mathcal{C}([0, 1])$.

| **Claim.** $\mathcal{C}([0, 1])$ is a Schochet C^* -algebra. That is, $[0, 1]$ is a Schochet space.

Observe that we can write $\mathcal{C}([0, 1]) = C^*(u, 1 | 0 \leq u \leq 1)$. Next, we consider the following lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ \mathcal{C}([0, 1]) & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

Observe that $(h_t(u))_{t \in [0,1]}$ is a path of positive contractions in $\mathcal{Q}(H)$ with an initial lift given by $f(u)$ and so, the problem reduces to lifting paths of positive contractions. We will accomplish this by using a continuous section of π . By the C^* -Bartle-Graves Theorem, there is a unital C^* -section $s : \mathcal{Q}(H) \rightarrow \mathcal{B}(H)$ such that:

$$s(h_0(u)) = f(u).$$

So, now, we consider the path $(s(h_t(u)))_{t \in [0,1]}$ and let us show that this is a path of positive contractions in $\mathcal{B}(H)$. Indeed, the positivity of s implies that this is a path of positive elements. So, it remains to show that:

$$\forall t \in [0, 1] : s(h_t(u)) \leq 1.$$

But this is equivalent to the claim that^a:

$$\forall t \in [0, 1] : \|s(h_t(u))\| \leq 1.$$

This holds because s is a contraction and each h_t is a contraction. So, we define:

$$H(u) : [0, 1] \rightarrow \mathcal{B}(H) , \quad t \mapsto H(u)(t) \stackrel{\text{def}}{=} s(h_t(u)).$$

This is a continuous map because it is a composition of continuous maps. Moreover, our work above shows that it is a positive contraction. On the other hand, we have a path $(h_t(1))_{t \in [0,1]}$ of projections with initial lift $f(1)$ & using the subclaim of Example 2.81, we can lift this to a path $(S_t)_{t \in [0,1]}$ of projections in $\mathcal{B}(H)$ with:

$$S_0 = f(1),$$

$$\forall t \in [0, 1] : \pi(S_t) = h_t(1).$$

This allows us to define a $\star$ -homomorphism:

$$H : \mathcal{C}([0, 1]) \rightarrow \text{Cyl}(\mathcal{B}(H)),$$

by the universal property of $\mathcal{C}([0, 1])$. This sends u to $H(u)$ as defined above and sends 1 to the map $t \mapsto S_t$. Let us check that this is the desired homotopy lift. We will do this in the following claim.

Claim 2.23. H is the desired homotopy lift. This is proved on Page 108.

This completes the proof.

Again, the proof here is only simplified by the fact that we are using a much stronger form of the Bartle-Graves Theorem which tells us so much more about the continuous section of our surjective $\star$ -homomorphism.

^aThis is a general fact. Let $A \in {}^1C^*\text{Alg}$ and let $x \in A^+$. Let's prove that $x \leq 1_A$ iff $\|x\| \leq 1$. Indeed, let B be the unital commutative C^* -algebra generated by x and 1_A . By the Gelfand-Naimark Theorem, this is $\star$ -isomorphic to $\mathcal{C}(X)$ for some compact Hausdorff space X . Now, observe that x corresponds to some map f in $\mathcal{C}(X)$ and we note that this result is clearly true for f . So, it is also true for x .

Since $[0, 1]$ is an example of a Schochet space, we would like to conclude that $(0, 1]$ is also Schochet. Since $[0, 1]$ is just the one-point compactification of $(0, 1]$, we will need to demonstrate some compatibility between the Schochet property and one-point compactifications. At the level of C^* -algebras, this amounts to showing some compatibility between the Schochet property and unitizations. To start discussing this compatibility, we will need to introduce unital homotopies into the picture. This also leads us to introducing a unital version of the Schochet property.

Terminology 2.85. Let $A, B \in {}^1C^*\text{Alg}$.

1. If $(\phi_t : A \rightarrow B)_{t \in [0,1]}$ is a homotopy, we say that it is unital iff ϕ_t is unital for each $t \in [0, 1]$.
2. A is 1-Schochet iff for the following lifting problem^a:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \overset{(H_t)_{t \in [0,1]}}{\curvearrowright} & \uparrow \pi \\ A & \xrightarrow{f} & \mathcal{Q}(H) \\ & \xrightarrow{(h_t)_{t \in [0,1]}} & \end{array}$$

where $(h_t)_{t \in [0,1]}$ is unital and f is unital, we always have a homotopy lift satisfying the usual conditions.

Note that the notion of being 1-Schochet only applies to unital C^* -algebras.

^aFor convenience, we will refer to this as a unital lifting problem.

We have a remark about this terminology.

Remark 2.86. The lift above does not have to be unital but it's not far away from it. Indeed:

$$\forall t \in [0, 1] : H_t(1_A) + \mathcal{K}(H) = \pi(H_t(1_A)) = h_t(1_A) = \text{Id}_H + \mathcal{K}(H),$$

which implies that $H_t(1_A) - \text{Id}_H$ is compact for each $t \in [0, 1]$.

Even though this is very easy to check, we still need to show that the unitization of a homotopy is a homotopy.

Claim 2.24. Let $A, B \in {}^1C^*\text{Alg}$ and let $(\phi_t : A \rightarrow B)_{t \in [0,1]}$ be a homotopy. Then, the unitization:

$$(\phi_t^\sim : A^\sim \rightarrow B^\sim)_{t \in [0,1]},$$

is a homotopy.

Now, we should also discuss some permanence properties since this will give us access to some examples.

TFSH:

1. If $A, B \in \mathbf{C^*Alg}$ are Schochet, then $A \star B$ is Schochet.
2. If $A, B \in \mathbf{C^*Alg}$ with A being Schochet and a split epi $\phi \in \mathbf{C^*Alg}(A, B)$, then B is Schochet.
3. If $A, B \in \mathbf{C^*Alg}$ with A and B being $\star$ -isomorphic and A being Schochet, then B is Schochet.
4. If $A, B \in \mathbf{C^*Alg}$ with A being Schochet and a Schochet fibration $f \in \mathbf{C^*Alg}(A, B)$ so that:

$$\exists g \in \mathbf{C^*Alg}(B, A) : f \circ g \simeq_h \text{Id}_B,$$
then B is Schochet.
5. If $A, B \in \mathbf{C^*Alg}$ with $A \oplus B$ being Schochet, then A and B are Schochet.
6. If $A \in \mathbf{C^*Alg}$ and $M_{n \times n}(A)$ is Schochet for some $n \in \mathbb{N}$, then A is Schochet.
7. If $A \in \mathbf{C^*Alg}$ and $\text{Cyl}(A)$ is Schochet, then A is Schochet.
8. If $A, B \in {}^1\mathbf{C^*Alg}$ with A being 1-Schochet and a split epi $\phi \in {}^1\mathbf{C^*Alg}(A, B)$, then B is 1-Schochet.
9. If $A, B \in {}^1\mathbf{C^*Alg}$ with A and B being $\star$ -isomorphic and A being 1-Schochet, then B is 1-Schochet.

Proof. We'll prove each statement one-by-one. In each case, we will start with the following diagram:

$$\begin{array}{ccc} & \mathcal{B}(H) & \\ & \downarrow \pi & \\ D & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array} \quad \begin{array}{c} \nearrow f \\ \end{array} \quad (1)$$

where $(h_t)_{t \in [0,1]}$ is a homotopy such that the diagram above commutes when $t = 0$.

1. Let $D = A \star B$ and let ι_A, ι_B be the respective inclusion maps. In the case of A , we have a diagram:

$$\begin{array}{ccc} & \mathcal{B}(H) & \\ & \downarrow \pi & \\ A & \xrightarrow{(h_t \circ \iota_A)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array} \quad \begin{array}{c} \nearrow f \circ \iota_A \\ \end{array}$$

which commutes at $t = 0$ because the one in (1) commutes when $t = 0$; a similar diagram exists in the case of B . Since A is Schochet, there is a homotopy lift $(H_t^A)_{t \in [0,1]}$ such that $H_0^A = f \circ \iota_A$ and we have that:

$$\forall t \in [0,1] : \pi \circ H_t^A = h_t \circ \iota_A.$$

Similarly, we have a homotopy lift $(H_t^B)_{t \in [0,1]}$ for the corresponding diagram for B . By the universal property of the free product, we obtain a unique homotopy $(H_t : A \star B \rightarrow \mathcal{B}(H))_{t \in [0,1]}$ such that each H_t restricts to H_t^A and H_t^B . Now, we need to check that this works. Indeed, we need to check that $H_0 = f$. For this, we note that:

$$H_0 \circ \iota_A = H_0^A = f \circ \iota_A,$$

$$H_0 \circ \iota_B = H_0^B = f \circ \iota_B.$$

By the uniqueness part of the universal property for the free product, we see that $H_0 = f$. Let's check that:

$$\forall t \in [0,1] : \pi \circ H_t = h_t.$$

So, let $t \in [0,1]$. Then:

$$(\pi \circ H_t) \circ \iota_A = \pi \circ (H_t \circ \iota_A) = \pi \circ H_t^A = h_t \circ \iota_A,$$

$$(\pi \circ H_t) \circ \iota_B = \pi \circ (H_t \circ \iota_B) = \pi \circ H_t^B = h_t \circ \iota_B,$$

which implies, by uniqueness again, that our desired equality holds.

2. Let $D = B$ and consider the following diagram:

$$\begin{array}{ccc} & \mathcal{B}(H) & \\ & \downarrow \pi & \\ A & \xrightarrow{(h_t \circ \phi)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array} \quad \begin{array}{c} \nearrow f \circ \phi \\ \end{array}$$

Since A is Schochet, we obtain a homotopy lift $(F_t : A \rightarrow \mathcal{B}(H))_{t \in [0,1]}$ such that:

$$F_0 = f \circ \phi,$$

$$\forall t \in [0,1] : \pi \circ F_t = h_t \circ \phi.$$

Now, let ψ be a split for ϕ . Then, we define:

$$\forall t \in [0,1] : H_t \stackrel{\text{def}}{=} F_t \circ \psi.$$

In that case, we see that:

$$H_0 = F_0 \circ \psi = (f \circ \phi) \circ \psi = f,$$

and we also have that:

$$\forall t \in [0, 1] : \pi \circ H_t = \pi \circ F_t \circ \psi = h_t \circ \phi \circ \psi = h_t,$$

which proves the result.

3. This follows from the previous statement, since $\star$ -isomorphisms are split epis.
4. We see that f is split epi by Proposition 2.31 so we're done.
5. Let us consider the projection $\pi_A : A \oplus B \rightarrow A$ and define:

$$\iota_A : A \rightarrow A \oplus B, \quad a \mapsto \iota_A(a) \stackrel{\text{def}}{=} (a, 0).$$

Then, we see that:

$$\begin{aligned} \forall a \in A : (\pi_A \circ \iota_A)(a) &= \pi_A(a, 0) \\ &= a, \end{aligned}$$

which means that π_A is a split-epi since ι_A is a $\star$ -homomorphism. So, A is Schochet and similarly, so is B .

6. We consider the following homotopy lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ A & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

where $h_0 = \pi \circ f$. Now, we apply the functor $M_{n \times n}(-)$ to this diagram so we obtain:

$$\begin{array}{ccc} & & M_{n \times n}(\mathcal{B}(H)) \\ & \nearrow M_{n \times n}(f) & \downarrow M_{n \times n}(\pi) \\ M_{n \times n}(A) & \xrightarrow{(M_{n \times n}(h_t))_{t \in [0,1]}} & M_{n \times n}(\mathcal{Q}(H)) \end{array}$$

Next, we will make some identifications as follows:

- (a) We will identify $M_{n \times n}(\mathcal{B}(H))$ with $\mathcal{B}(H^{\oplus n})$, where we note that $H^{\oplus n}$ is still an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space so we are still in the same set-up.
- (b) We will identify $M_{n \times n}(\mathcal{Q}(H))$ with $\mathcal{Q}(H^{\oplus n})$.
- (c) We will identify $M_{n \times n}(\pi)$ with the Calkin projection on $\mathcal{B}(H^{\oplus n})$.
- (d) If we view the homotopy $(h_t)_{t \in [0,1]}$ as a map from A into $\text{Cyl}(\mathcal{Q}(H))$, then we can view $(M_{n \times n}(h_t))_{t \in [0,1]}$ as a homotopy from $M_{n \times n}(A)$ into $\text{Cyl}(M_{n \times n}(\mathcal{Q}(H)))$ and due to our identification above, this is just a homotopy from $M_{n \times n}(A)$ into $\text{Cyl}(\mathcal{Q}(H^{\oplus n}))$.

With these identifications, we see that we just have a lifting problem for $M_{n \times n}(A)$ and since this is Schochet, we find a homotopy lift $(F_t : M_{n \times n}(A) \rightarrow \mathcal{B}(H^{\oplus n}))_{t \in [0,1]}$. Now, we precompose each F_t with the left-upper corner inclusion of A into $M_{n \times n}(A)$ and postcompose it with the canonical isomorphism between $M_{n \times n}(\mathcal{B}(H))$ and $\mathcal{B}(H^{\oplus n})$. This is the homotopy lift for the lifting problem associated to A .

7. Let's consider the $\star$ -homomorphism $p_0^A : \text{Cyl}(A) \rightarrow A$ and define:

$$s : A \rightarrow \text{Cyl}(A), \quad a \mapsto s(a),$$

where $s(a)$ is the constant function on $[0, 1]$ with value a . In Example 2.19, we actually showed that s is a section for p_0^A . But this implies that A is Schochet.

Finally, statements 8 and 9 follow in the same way as statements (2) and (3). ■

The next corollary is due to Conway but we prove it using the proposition.

Corollary 2.88

Let $X, Y \in \text{LCH}$ be homeomorphic, with X being Schochet. Then, Y is Schochet.

Proof. Since X and Y are homeomorphic, we see that $\mathcal{C}_0(X)$ and $\mathcal{C}_0(Y)$ are $\star$ -isomorphic and $\mathcal{C}_0(X)$ is Schochet. Hence, it follows that $\mathcal{C}_0(Y)$ is Schochet and this means that Y is Schochet. ■

The next one gives us more examples of Schochet spaces.

Corollary 2.89. A. Vats, 2026

Let $X \in \text{CHaus}$. TFAE:

1. X is Schochet.
2. Every clopen subset Y of X is Schochet.

Proof. (2) $\Rightarrow$ (1) is clear, since X is clopen in itself. So, it remains to prove (1) $\Rightarrow$ (2). We define:

$$\phi : \mathcal{C}(X) \rightarrow \mathcal{C}(Y), \quad f \mapsto \phi(f) \stackrel{\text{def}}{=} f|_Y.$$

Note that this is a $\star$ -homomorphism. Now, we define:

$$\psi : \mathcal{C}(Y) \rightarrow \mathcal{C}(X), \quad f \mapsto \psi(f),$$

where we simply take a function and extend it to X through continuation by 0; this extension will be continuous^a and the map above is a $\star$ -homomorphism. Now, we observe that ψ is a split for ϕ and since X is Schochet, it follows from Proposition 2.87 that Y is Schochet. $\blacksquare$

^aWe illustrate the argument for points in $X \setminus Y$ and note that the argument for continuity at points in Y will rely on it being an open set. Let $p \in X \setminus Y$ and let $(p_\delta)_{\delta \in D}$ be a net that converges to p . Since Y is closed, we see that $X \setminus Y$ is open so the net eventually lies in $X \setminus Y$. But this means that $f(p_\delta) = 0$ for each $\delta \in D$ and this implies that $f(p) = 0$.

In order to relate the Schochet property to the 1-Schochet property, we will need the following theorem.

Theorem 2.90. A. Vats, 2026

Let $A, B \in {}^1\mathcal{C}^*\text{Alg}$ be 1-Schochet. Then, $A \oplus B$ is 1-Schochet.

To prove this, we will need two technical lemmas about paths of projections.

Lemma 2.91. J. Conway, [26, Lemma 5, p.138–139]

Let $(Q_t)_{t \in [0,1]}$ be a path of projections in $\mathcal{B}(H)$ and let $(p_t)_{t \in [0,1]}$ be a path of projections in $\mathcal{Q}(H)$ satisfying the following conditions:

- $\forall t \in [0, 1] : \pi(Q_t) \geq p_t$ and;
- there is a projection $P_0 \in \mathcal{B}(H)$ such that $P_0 \leq Q_0$ and $\pi(P_0) = p_0$.

Then, there is a path $(P_t)_{t \in [0,1]}$ of projections in $\mathcal{B}(H)$ so that:

1. $\forall t \in [0, 1] : \pi(P_t) = p_t$,
2. $\forall t \in [0, 1] : P_t \leq Q_t$.

In other words, paths of projections which are order-bounded by a liftable path of projections are liftable.

Actually, it turns out that Lemma 2.91 is an essential step in Conway's proof that totally disconnected compact Hausdorff spaces are C^* -CHP. So, we could replace the use of the lemma by that result instead but we will just use the lemma as we have stated it. The next result lets us construct paths of partial isometries from a given path of projections.

Lemma 2.92. J. Conway, [26, Lemma 7, 140–141]

Let $(p_t)_{t \in [0,1]}$ be a path of projections in $\mathcal{B}(H)$, then there exist paths of partial isometries $(u_t)_{t \in [0,1]}, (v_t)_{t \in [0,1]}$ such that the following relations are satisfied:

$$\begin{aligned} u_0 &= p_0 \\ v_0 &= 1 - p_0 \\ \forall t \in [0, 1] : u_t^* u_t &= p_0 \\ \forall t \in [0, 1] : v_t^* v_t &= 1 - p_0 \\ \forall t \in [0, 1] : u_t u_t^* &= p_t \\ \forall t \in [0, 1] : v_t v_t^* &= 1 - p_t, \end{aligned}$$

where 1 is used in place of Id_H above.

Our proof of Theorem 2.90 is modelled after the proof of the analogous fact in [26, Theorem 8, p.141]. There is another proof of a similar result due to (Dolapo) Oyetunbi in [69, Theorem 2.2.10, p.40] but we will not proceed by that approach.

Proof of Theorem 2.90. Let us consider the following unital lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ A \oplus B & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

Let $i_A : A \rightarrow A \oplus B$ and $i_B : B \rightarrow A \oplus B$ the canonical inclusions; these are not unital $\star$ -homomorphisms. We will view the homotopy as a morphism into $\text{Cyl}(\mathcal{Q}(H))$. Since $(1_A, 0)$ and $(0, 1_B)$ are orthogonal projections in $A \oplus B$, we can define the following orthogonal projections in $\text{Cyl}(\mathcal{Q}(H))$:

$$\begin{aligned} Q_1 &\stackrel{\text{def}}{=} h(1_A, 0), \\ Q_2 &\stackrel{\text{def}}{=} h(0, 1_B), \end{aligned}$$

such that $Q_1 + Q_2 = 1$, since $(1_A, 0) + (0, 1_B) = (1_A, 1_B)$. These are paths of projections which are order-bounded by a liftable (constant) path. Moreover, both of these paths lift at $t = 0$ in a way which satisfies the conditions of Lemma 2.91. So, we can find a path $(P_t^1)_{t \in [0,1]}$ of projections in $\mathcal{B}(H)$ such that:

1. $\forall t \in [0, 1] : \pi(P_t^1) = Q_1(t)$,

2. $\forall t \in [0, 1] : P_t^1 \leq 1$,
3. $P_0^1 = f(1_A, 0)$.

If we define:

$$\forall t \in [0, 1] : P_t^2 \stackrel{\text{def}}{=} 1 - P_t^1,$$

then we see that $(P_t^2)_{t \in [0, 1]}$ is also a path of projections such that the following facts hold:

1. $\forall t \in [0, 1] : \pi(P_t^2) = \pi(1) - \pi(P_t^1) = 1 - Q_1(t) = Q_2(t)$.
2. $\forall t \in [0, 1] : P_t^2 \leq 1$,
3. $P_0^2 = 1 - f(1_A, 0) = f(1_A, 1_B) - f(1_A, 0) = f(0, 1_B)$.

For $i \in \{1, 2\}$, we may now define the following objects:

$$P^i \stackrel{\text{def}}{=} P_0^i$$

$$H_i \stackrel{\text{def}}{=} P^i(H).$$

In particular, we can identify $\mathcal{B}(H_i)$ with $P^i \mathcal{B}(H) P^i$ and we can identify $\mathcal{B}(H_i)/\mathcal{K}(H_i)$ with $\pi(P^i) \mathcal{Q}(H) \pi(P^i)$; we will denote the latter by $\mathcal{Q}(H_i)$. Now, we have two paths of projections in $\mathcal{B}(H)$ so we can use [Lemma 2.92](#) to find two paths of partial isometries $(U_i(t))_{t \in [0, 1]}$ for $i \in \{1, 2\}$ such that:

1. $U_i(0) = P^i$,
2. $\forall t \in [0, 1] : U_i(t)^* U_i(t) = P^i$,
3. $\forall t \in [0, 1] : U_i(t) U_i(t)^* = P_t^i$.

Next, let $t \in [0, 1]$ be fixed. We will define four maps:

$$\Phi_t^A : A \rightarrow \mathcal{Q}(H_1), \quad a \mapsto \Phi_t^A(a) \stackrel{\text{def}}{=} \pi(U_1(t))^* h_t(a, 0) \pi(U_1(t)),$$

$$\Phi_t^B : B \rightarrow \mathcal{Q}(H_2), \quad b \mapsto \Phi_t^B(b) \stackrel{\text{def}}{=} \pi(U_2(t))^* h_t(0, b) \pi(U_2(t)),$$

$$\psi^A : A \rightarrow \mathcal{B}(H_1), \quad a \mapsto \psi^A(a) \stackrel{\text{def}}{=} P^1 f(a, 0) P^1,$$

$$\psi^B : B \rightarrow \mathcal{B}(H_2), \quad b \mapsto \psi^B(b) \stackrel{\text{def}}{=} P^2 f(0, b) P^2.$$

Next, we have the following claim.

Claim 2.25. The maps above are $\star$ -homomorphisms and the families:

$$(\Phi_t^A)_{t \in [0, 1]}, \quad (\Phi_t^B)_{t \in [0, 1]},$$

are homotopies. This is proved on Page 108.

Now, we want to fit these into lifting problems whose solutions can be glued together to form a solution for the original lifting problem. In particular, we have the following candidate diagrams under consideration:

$$\begin{array}{ccc} & \mathcal{B}(H_1) & \\ \psi^A \nearrow & \downarrow \pi & \\ A & \xrightarrow{(\Phi_t^A)_{t \in [0, 1]}} & \mathcal{Q}(H_1) \end{array} \quad \begin{array}{ccc} & \mathcal{B}(H_2) & \\ \psi^B \nearrow & \downarrow \pi & \\ B & \xrightarrow{(\Phi_t^B)_{t \in [0, 1]}} & \mathcal{Q}(H_2) \end{array}$$

Next, we will verify the following claim.

| **Claim 2.26.** The two diagrams given commute when $t = 0$. This is proved on Page 109.

So, this shows that we have two lifting problems for A and B respectively. Since A and B are Schochet, these two lifting problems have solutions $(\Psi_t^A)_{t \in [0, 1]}$ and $(\Psi_t^B)_{t \in [0, 1]}$. For each $t \in [0, 1]$, we now define:

$$\Psi_t : A \oplus B \rightarrow \mathcal{B}(H), \quad (a, b) \mapsto \Psi_t(a, b) \stackrel{\text{def}}{=} U_1(t) \Psi_t^A(a) U_1(t)^* + U_2(t) \Psi_t^B(b) U_2(t)^*.$$

A similar computation as in [Claim 2.25](#) shows that Ψ_t is a $\star$ -homomorphism. In fact, it is a homotopy, as seen by a similar argument in the aforementioned claim. Finally, we have our main claim.

| **Claim 2.27.** $(\Psi_t)_{t \in [0, 1]}$ is a solution to the original lifting problem. This is proved on Page 109.

This claim finishes the proof of this implication. ■

We can use this theorem to deduce the relationship between the 1-Schochet property & Schochet property.

Corollary 2.93. A. Vats, 2026

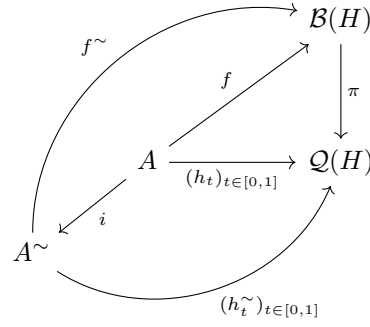
Let $A \in {}^1\mathbf{C}^*\mathbf{Alg}$. TFAE:

1. A is Schochet.
2. A is 1-Schochet.

Proof.

(1) $\Rightarrow$ (2): This follows clearly from the definition of being Schochet.

(2) $\Rightarrow$ (1): We consider the following diagram:



where $i : A \rightarrow A^\sim$ is the canonical inclusion, the triangle is a lifting problem for A and the outer arrows are the universal extensions of f and each h_t to the unitization through i . Since A is unital, we see that:

$$A^\sim \equiv_{1\text{-C}^*\text{Alg}} A \oplus \mathbb{C},$$

with the right-hand side being 1-Schochet so $A^\sim$ is 1-Schochet^a. In that case, we can find a homotopy lift $(F_t : A^\sim \rightarrow \mathcal{B}(H))_{t \in [0,1]}$ which solves the unital lifting problem. Now, we define:

$$\forall t \in [0, 1] : H_t \stackrel{\text{def}}{=} F_t \circ i.$$

Let us check that this works:

$$\begin{aligned} H_0 &= F_0 \circ i = f^\sim \circ i = f, \\ \forall t \in [0, 1] : \pi \circ H_t &= \pi \circ (F_t \circ i) = h_t^\sim \circ i = h_t, \end{aligned}$$

as was desired. ■

^aThis is where we use [Theorem 2.90](#).

As an immediate corollary, we obtain an even stronger result.

Corollary 2.94

Let $A, B \in {}^1\text{C}^*\text{Alg}$ be Schochet. Then, $A \oplus B$ is Schochet.

Proof. Since A, B are unital and Schochet, it is the case that they are 1-Schochet. Hence, $A \oplus B$ is 1-Schochet and since it is unital, it follows that it is Schochet. Over here, we simply apply Corollary 2.93 multiple times. ■

The corollary above is rather pivotal in proving the next result.

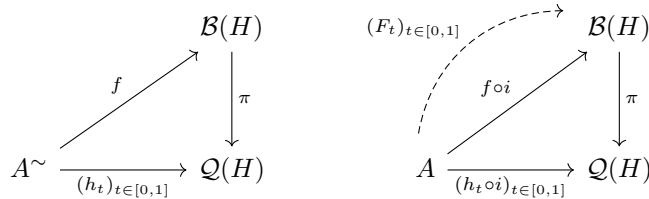
Corollary 2.95. A. Vats, 2026

Let $A \in \text{C}^*\text{Alg}$. TFAE:

1. A is Schochet.
2. $A^\sim$ is Schochet.

Proof.

(1) $\Rightarrow$ (2): Since $A^\sim$ is unital, we can just show that it is 1-Schochet. Consider the following diagrams:



where the diagram on the left is a unital lifting problem and the one on the right is obtained by composing the relevant morphisms with the canonical inclusion $i : A \rightarrow A^\sim$. In particular, the one on the right is a lifting problem for A . Since A is Schochet, we have a homotopy lift $(F_t)_{t \in [0,1]}$. Now, we define:

$$\forall t \in [0, 1] : H_t \stackrel{\text{def}}{=} F_t^\sim.$$

By [Claim 2.24](#), this is a homotopy so let's check that it works. If $t = 0$, then:

$$H_0 = F_0^\sim = (f \circ i)^\sim = f,$$

because f is already unital and defined on $A^\sim$. Next, let $t \in [0, 1]$ and consider the following computation:

$$\begin{aligned} \forall a + \lambda 1_{A^\sim} \in A^\sim : (\pi \circ H_t)(a + \lambda 1_{A^\sim}) &= \pi(H_t(a)) + \lambda \pi(H_t(1_{A^\sim})) \\ &= h_t(a) + \lambda \pi(\text{Id}_H) \\ &= h_t(a) + \lambda (\text{Id}_H + \mathcal{K}(H)) \\ &= h_t(a) + \lambda h_t(1_{A^\sim}) \\ &= h_t(a + \lambda 1_{A^\sim}), \end{aligned}$$

which proves that we have the required lift for the original problem.

(2) $\Rightarrow$ (1): Consider the canonical projection:

$$p : A^\sim \rightarrow A, (a, \lambda) \mapsto p(a, \lambda) \stackrel{\text{def}}{=} a.$$

One observes that:

$$\forall a \in A : (p \circ i)(a) = p(a, 0) = a,$$

which proves that p is a split epi. Since $A^\sim$ is Schochet, we see that A is Schochet. ■

Here is a corollary of the result above.

Corollary 2.96. A. Vats, 2026

Let $X \in \text{LCH}$. TFAE:

1. X is Schochet.
2. X^+ , its one-point compactification, is Schochet.

Proof. We know that:

$$\mathcal{C}_0(X)^\sim \simeq \mathcal{C}(X^+).$$

Now, X is Schochet iff $\mathcal{C}_0(X)$ is Schochet iff $\mathcal{C}_0(X)^\sim$ is Schochet iff $\mathcal{C}(X^+)$ is Schochet iff X^+ is Schochet. ■

With this corollary, we can show that the half-open unit interval is a Schochet space.

Example 2.97. We have the following main claim.

| **Claim.** $(0, 1]$ is a Schochet space.

We note that $(0, 1]$ is Schochet iff $(0, 1]^+ \simeq [0, 1]$ is Schochet. But the latter statement is true.

At this stage, we have developed most of the tools needed to show that S^1 is a Schochet space; this is one of the major examples of Conway and we give a slightly simplified proof. The point is that $\mathcal{C}(S^1)$ is a unital C^* -algebra so it is sufficient to show that it is 1-Schochet. This simplification will allow us to use lifting results for paths of unitaries.

Example 2.98. We have the following claim.

| **Claim.** S^1 is a Schochet space.

Observe that we can write:

$$\mathcal{C}(S^1) = C^*(u, 1 | uu^* = u^*u = 1).$$

It is sufficient for us to show that $\mathcal{C}(S^1)$ is 1-Schochet. With the universal description of $\mathcal{C}(S^1)$, we see that the proof reduces to lifting paths of unitaries. In fact, this already been “used” in [Example 2.81](#).

Subclaim. Let $A, B \in {}^1C^*\text{Alg}$ and let $\pi \in C^*\text{Alg}(A, B)$ be surjective, with $(u_t)_{t \in [0, 1]}$ being a path of unitaries in B so that there exists a unitary $U_0 \in A$ such that $\pi(U_0) = u_0$. Then, there exists a path $(U_t)_{t \in [0, 1]}$ of unitaries in A such that $\pi(U_t) = u_t$ for each $t \in [0, 1]$. This is proved in [\[44, Proposition 4.3.14a, p.95-96\]](#).

^aThe statement can be phrased a bit more succinctly by saying that $\pi|_{\text{Unitary}(A)}$ has the path-lifting property.

We consider the following unital lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ \mathcal{C}(S^1) & \xrightarrow{(h_t)_{t \in [0, 1]}} & \mathcal{Q}(H) \end{array}$$

Observe that $\mathcal{C}(S^1)$ is the universal C^* -algebra generated by a unitary; let u denote a generator. Since each h_t is unital, we see that $(h_t(u))_{t \in [0, 1]}$ is a path of Calkin unitaries. Using the lemma, we obtain a unitary lift $(v_t)_{t \in [0, 1]}$ through the Calkin projection, with $v_0 = f(u)$. Now, we just define for each $t \in [0, 1]$:

$$H_t(u) \stackrel{\text{def}}{=} v_t,$$

and this defines, by universal property, a $\star$ -homomorphism $H_t : \mathcal{C}(S^1) \rightarrow \text{Cyl}(\mathcal{B}(H))$ with the desired property.

Ideally, we want to extend [Example 2.97](#) to $(0, 1)$ as well and we can do it.

| **Example 2.99.** $(0, 1)$ is a Schochet space because $(0, 1)^+ \simeq S^1$ is a Schochet space.

We have discussed the Schochet property relative to spaces. One can ask what the basic situation looks like for groups and we have some results in that direction. In the next example, we see that all locally finite groups are Schochet.

| **Example 2.100.** Let $G \in \text{Grp}$ be locally finite (that is, every finitely-generated subgroup of G is finite). Then, we note that $C_{\text{red}}^*(G)$ is an AF algebra and this is a Schochet C^* -algebra. So, we see that G is a Schochet group.

Which can also characterize Schochet groups among the LCH abelian groups.

Proposition 2.101. A. Vats, 2026

Let $G \in \text{Ab}(\text{LCH})$. TFSH:

1. TFAE:
 - (a) G is a Schochet group.
 - (b) $\widehat{G}$, the Pontryagin dual of G , is a Schochet space.
- In particular, $\widehat{G}$ is a Schochet group iff G is a Schochet space.
2. If G is compact, then G is Schochet.

Proof. We will prove each statement one-by-one.

1. Since G is abelian, we have the following chain of isomorphisms in $C^*\text{Alg}$:

$$C_{\text{red}}^*(G) \simeq C_{\text{max}}^*(G) \simeq C_0(\widehat{G}),$$

where the first one holds because G is amenable and the second one holds as a general result of harmonic analysis. Now, G is Schochet iff $C_{\text{red}}^*(G)$ is Schochet iff $C_0(\widehat{G})$ is Schochet iff $\widehat{G}$ is Schochet. The last statement follows via Pontryagin duality, as well as the fact that the Schochet property for groups is preserved by isomorphisms.

2. This follows from statement (1), since $\widehat{G}$ is discrete^a so it is Schochet. ■

^aThey are, in particular, locally compact Hausdorff totally disconnected spaces.

With this corollary, we can provide more interesting examples.

Example 2.102. We have that $\mathbb{T}$ is a Schochet group since it is a compact Hausdorff abelian group.

Example 2.103. Let $G = \mathbb{Z}$, the additive group of the integers. Then, we know that:

$$\widehat{G} \simeq S^1,$$

and the latter is a Schochet space. So, we see that G is a Schochet group.

Example 2.104. Let $G = \mathbb{R}$, the additive group of $\mathbb{R}$. Then, we know that:

$$\widehat{G} \simeq \mathbb{R},$$

and the latter is a Schochet space because $(0, 1)$ is a Schochet space. So, G is a Schochet group.

It would, of course, be very desirable for us to have more examples (or non-examples) of Schochet groups. We will discuss a possible way forward in Section 5.1. For now, we want to turn towards partially resolving [Question 2.51](#). Actually, it turns out that the theory we have developed so far is not sufficient in tackling this problem. For that purpose, we will need to introduce the notion of *semiprojectivity*. This notion has its origins in the shape theory of C^* -algebras initiated by (Bruce) Blackadar in [\[12\]](#) (see [Remark 2.109](#), however). We have a more practical reason to be interested in this notion. To explain this, we need some terminology which generalizes what we have done with the Schochet property.

Terminology 2.105. A pair (A, π) consisting of:

1. an $A \in C^*\text{Alg}$ and;
2. a surjective $\pi \in C^*\text{Alg}(E, B)$;

satisfies the homotopy lifting property iff the following lifting problem^a:

$$\begin{array}{ccc} & & E \\ & \nearrow f & \downarrow \pi \\ A & \xrightarrow{(h_t)_{t \in [0,1]}} & B \end{array}$$

(The diagram shows a dashed curved arrow from A to E labeled $(H_t)_{t \in [0,1]}$ above it.)

always admits a solution in $C^*\text{Alg}$. We also say that (A, π) is HLP in this case.

^aThis is defined in exactly the same sense in which lifting problems are defined for the Schochet property.

Our main reason for being interested in semiprojectivity is the following theorem.

Theorem 2.106. Blackadar's Homotopy Lifting Theorem, [\[13, Theorem 5.1, p.297-298\]](#)

Let $A \in C^*\text{Alg}$ be semiprojective. Then, every pair:

$$(A, \pi : E \rightarrow B),$$

where π is a surjective $\star$ -homomorphism, is HLP.

Before we explain (approximate) semiprojectivity, we have a remark regarding our solution to [Conjecture 2.39](#).

Remark 2.107. So, in principle, [Theorem 2.106](#) provides us a way to show that certain C^* -algebras are not semiprojective; we just have to show that they fail the HLP. For instance, this provides a proof that:

$$A = C^*(x | xx^* = x^*x \wedge \|x\| \leq),$$

is not semiprojective (this can be shown directly). We note that showing the failure of the HLP is difficult.

Let us explain semiprojectivity and also introduce approximate semiprojectivity simultaneously. These notions are mainly interesting for separable C^* -algebras, so this is the realm in which we'll consider them.

Definition 2.108

Let $A \in {}_{\omega}C^*Alg$.

1. A is semiprojective iff for any $B \in C^*Alg$, any increasing sequence $(J_n)_{n \in \mathbb{N}}$ of closed ideals in B with J being the closed ideal generated by the sequence and any $\star$ -homomorphism $\phi : A \rightarrow B/J$, we have that this can be lifted to a $\star$ -homomorphism $\psi : A \rightarrow B/J_n$ for sufficiently large n .
 2. A is approximately semiprojective iff it is isomorphic to a sequential inductive limit of semiprojective C^* -algebras.
- In the literature, an approximately semiprojective C^* -algebra is often also said to have a strong shape system.

In the next remark, we will provide some context for these notions.

Remark 2.109. In [12], it is claimed that a paper, “Homotopy Continuity & Shape Theory for C^* -Algebras”, by (Edward) Effros and Kaminker was responsible for developing a noncommutative version of shape theory. This theory was originally introduced in the topological setting to study certain spaces by focusing on homotopy types of the spaces that are used to construct them; for more details on these matters and history, see [40]. We could not locate the aforementioned paper but it is referenced in a paper by Blackadar ([12]) which seems to be the first one which introduced the notion of semiprojectivity which is in use within modern^a research.

^aSee Definition 2.10, p.256 of the aforementioned paper.

With this background, we can finally (partially) address Question 2.51 through the following theorem.

Theorem 2.110. A. Vats, 2026

The spaces in statements (1) and (2) of Question 2.51 are Schochet spaces.

The proof relies on a difficult permanence property of semiprojectivity that we will not prove.

Proposition 2.111. D. Enders, 2014, [36, Corollary 3.4, p.8]

Suppose we have the following pullback square:

$$\begin{array}{ccc} A \oplus_F B & \longrightarrow & B \\ \downarrow & & \downarrow \\ A & \longrightarrow & F \end{array}$$

in C^*Alg , where F is finite-dimensional. If A and B are semiprojective, then $A \oplus_F B$ is semiprojective.

Here is the proof of our theorem.

Proof of Theorem 2.110. We will deal with each of these spaces individually.

1. Observe that:

$$\mathcal{C}(S^1 \vee S^1) \simeq \mathcal{C}(S^1) \oplus_{\mathbb{C}} \mathcal{C}(S^1).$$

It is known that $\mathcal{C}(S^1)$ is semiprojective. So, the above is a pullback of two semiprojective C^* -algebras over a finite-dimensional C^* -algebra so Proposition 2.111 tells us that it must be semiprojective and, hence, Schochet.

2. Observe that:

$$\mathcal{C}(S^1 \vee [0, 1]) \simeq \mathcal{C}(S^1) \oplus_{\mathbb{C}} \mathcal{C}([0, 1]).$$

By the same argument as in the proof of statement (1), we can conclude that this pullback is Schochet. ■

Let us mention a recent result which improves our intrinsic understanding of the homotopy lifting property. In fact, it was partly this result which prompted us to look into l -closed C^* -algebras when it came to our resolution of Conjecture 2.39.

Theorem 2.112. D. Oyetunbi, A. Tikuisis, [68, Theorem 3.8, p.7]

Let $A \in C^*Alg$. TFAE:

1. A is l -open.
2. The system of maps:

$$\text{Hom}(A, B) \rightarrow \text{Hom}(A, B/J),$$

for all marked C^* -algebras (B, J) , is uniformly open^a.

3. If (B, I) is a marked C^* -algebra, then $(A, \pi : B \rightarrow B/I)$ is HLP and $\text{Hom}(A, B)$ is locally path-connected.

^aWe will not define this term as it will not be needed; one can find it in the reference.

In the last result of this section, we will collect some facts that are derived from our work and that of others.

Let $A \in C^*Alg$. TFSH:

1. (A, Id_A) is HLP.
2. A is Schochet iff (A, π) is HLP, with π being the Calkin projection.
3. A , as defined in [Theorem 2.56](#), is approximately semiprojective.
4. The Schochet property is not closed under sequential inductive limits.
5. The HLP is not closed under sequential inductive limits.
6. If A is l -open, then it is Schochet.
7. There exists a Schochet C^* -algebra which is not l -open.
8. There exists a Schochet C^* -algebra which is not semiprojective.
9. The full subcategories of ${}_{\omega}C^*Alg$:
 - (a) the category of Schochet C^* -algebras and;
 - (b) the category of approximately semiprojective C^* -algebras;
 do not coincide.
10. If (b) is a subcategory of (a) , then [Conjecture 2.114](#) is false.
11. Consider the full subcategories of ${}_{\omega}C^*Alg$:
 - (a) the category of approximately Schochet C^* -algebras and;
 - (b) the category of approximately semiprojective C^* -algebras.
 Then, (b) is a subcategory of (a) .

Proof. We will prove each statement one-by-one.

1. This is clear; we just need to take the lift to be the original homotopy.
2. This is the definition of the Schochet property.
3. This was a difficult result but it has been recently shown to be the case; see [\[54, p. 723\]](#).
4. Since semiprojective C^* -algebras are Schochet, the C^* -algebra of [Theorem 2.56](#) is exactly an example of a sequential inductive limit of Schochet C^* -algebras which is not Schochet.
5. Since there is a non-Schochet C^* -algebra, this C^* -algebra also cannot be HLP and one can construct such an example which is approximately semiprojective, so it is an inductive limit of HLP C^* -algebras^a.
6. This follows by [Theorem 2.112](#).
7. In [\[68, Example 3.9, p.7\]](#), a unital AF algebra is given which is not l -open.
8. We simply need a Schochet C^* -algebra which is not l -open and we are done^b.
9. Since there exists a non-Schochet C^* -algebra which is approximately semiprojective, we see that the category of Schochet C^* -algebras is not closed under sequential inductive limits. On the other hand, the category of approximately semiprojective C^* -algebras is closed under sequential inductive limits; this was shown in [\[89, Theorem 4.9, p.13\]](#). So, it follows that these two categories cannot coincide.
10. Indeed, if every approximately semiprojective C^* -algebra is Schochet, then we can just take a separable non-Schochet C^* -algebra and this cannot be approximately semiprojective so we're done.
11. This follows from the fact that semiprojective C^* -algebras are Schochet. ■

^aThis follows by Blackadar's Homotopy Lifting Theorem.

^bThis is because semiprojectivity implies l -openness.

Before we end this section, let us briefly explain why Proposition 2.113 is significant. The answer has to do with a long-standing conjecture on semiprojectivity due to Blackadar. We have all the language necessary to state this conjecture.

| **Conjecture 2.114** (B. Blackadar, 1985, [\[12, 4.4, p.266\]](#)). Every separable C^* -algebra is approximately semiprojective.

Through our work with Schochet C^* -algebras, we have shown that the approximate Schochet property serves as a possible obstruction to [Conjecture 2.114](#). That is, if we can prove that there exists an approximately Schochet separable C^* -algebra which is not approximately semiprojective, then this would resolve the conjecture in the negative. In Section 5.1, we will discuss our reasons for believing that this could be a useful obstruction and this will involve relating it to other obstructions in the literature. This brings us to the end of our exposition on the concrete aspects of the homotopy theory on categories of operator algebras. We will, now, begin to explore abstract homotopy theory, so the tone of the thesis will change. Nevertheless, we are guided by much of the work done here and will refer back to it at key points.

3 Introducing Categories of Fibrant Objects

Summary

In this section, we turn towards the study of abstract homotopy theory. We begin by giving a survey of model category theory in Section 3.1, working our way up to its definition and surveying some standard examples. This culminates in some basic exposition about a result which essentially tells us that the category of C^* -algebras does not admit a model structure with the usual weak equivalences and the Schochet fibrations. In Section 3.2, we build up the theory of categories of fibrant objects & survey some known results about them before we present new results based on constructions inspired by the homotopy theory of equivariant operator algebras developed in Section 2. We finish the section off by discussing (but not proving) a variation of the Spanier-Whitehead Construction for categories of fibrant objects.

3.1 A Survey of Model Category Theory

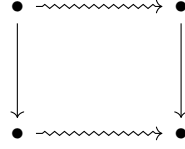
Let us start by recalling the definition of the localization of a category with respect to some set of morphisms. To state this definition, we need a bit of preparation. To be precise, we need a construction that will be used later.

Construction 3.1. Let $\mathcal{C}$ be a category. We define a structure:

$$\text{Arr}(\mathcal{C}),$$

as follows:

1. An object is an arrow^a in $\mathcal{C}$.
2. A morphism in this structure is a commutative diagram:



where the arrow on the left is the domain of the morphism, the arrow on the right is the codomain of the morphism and the two arrows on top & below form the data of the morphism. We define composition by the concatenation of commutative diagrams in the usual way through composition of morphisms in $\mathcal{C}$.

This data is easily shown to form a category; we refer to this as the arrow category of $\mathcal{C}$. In particular, if f, g are morphisms in $\mathcal{C}$ and $\alpha : f \Rightarrow g$ is a morphism in $\text{Arr}(\mathcal{C})$, then α is known as a lifting problem for f and g .

^aMore formally, it is a triple (x, y, f) with $x, y \in \mathcal{C}$ and $f \in \mathcal{C}(x, y)$ being an arrow.

We also need some terminology.

Terminology 3.2. Let $\mathcal{C}, \mathcal{D}$ be categories, let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$ and let W be a full subcategory of $\text{Arr}(\mathcal{C})$. We say that F inverts W iff for any $f \in W$, we have that $F(f)$ is an isomorphism. We define:

$$\text{Fun}^W(\mathcal{C}, \mathcal{D}),$$

to be the full subcategory of $\text{Fun}(\mathcal{C}, \mathcal{D})$ consisting of functors which invert W .

The universal property of the localization is easy to describe.

Definition 3.3

Let $\mathcal{C}$ be a category and let W be a full subcategory of $\text{Arr}(\mathcal{C})$. The Gabriel-Zisman localization/1-categorical localization of $\mathcal{C}$ with respect to W is a pair $(\mathcal{C}[W^{-1}], \gamma : \mathcal{C} \rightarrow \mathcal{C}[W^{-1}])$ consisting of:

1. a category $\mathcal{C}[W^{-1}]$,
2. a functor $\gamma : \mathcal{C} \rightarrow \mathcal{C}[W^{-1}]$ which inverts W .

Moreover, for any category $\mathcal{D}$, the following precomposition functor:

$$\text{Fun}(\mathcal{C}[W^{-1}], \mathcal{D}) \rightarrow \text{Fun}^W(\mathcal{C}, \mathcal{D}),$$

is an equivalence of categories.

We have two remarks about this notion.

Remark 3.4. The universal property above is just about all that we will ever need, though we will certainly have to reference the calculus-of-fractions approach to understanding localizations. For a conceptual exposition of localizations, see [53, Chapter 3.3, p.26-35]. For the proof of existence, see the sketch presented in [51, Chapter 1.1, p.3-5].

Remark 3.5. We will typically refer to the Gabriel-Zisman localization as just the localization, at least until Section 4.2. In that subsection, we will introduce localizations of ∞ -categories & since one can take the ∞ -categorical localization of a category with respect to some full subcategory of morphisms, we distinguish between the two localizations by referring to the one in Definition 3.3 by the 1-categorical localization.

Let us introduce some further notation which fits in better with the perspective we will take.

Notation 3.6. Let $\mathcal{C}$ be a category and let W be a full subcategory of $\text{Arr}(\mathcal{C})$. We define:

$$\mathbf{Ho}(\mathcal{C}) \stackrel{\text{def}}{=} \mathcal{C}[W^{-1}],$$

as the homotopy category of $\mathcal{C}$, equipped also with the localization functor $\gamma : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{C})$. If $x, y \in \mathbf{Ho}(\mathcal{C})$, then:

$$[x, y]_{\mathcal{C}} \stackrel{\text{def}}{=} \mathbf{Ho}(\mathcal{C})(x, y),$$

are known as the homotopy classes of morphisms from x to y . Sometimes, we omit the $\mathcal{C}$ on the left-hand side.

Here is a remark regarding this piece of notation.

Remark 3.7. We stress that the construction $\mathbf{Ho}(-)$ as used above applies to the pair $(\mathcal{C}, W)$ and not just to $\mathcal{C}$. This has, however, been suppressed in the notation. For the most part, this will not be a problem but we will use the usual notation for the localization whenever we do need to emphasize the category W .

Before we introduce any of the homotopical structures we will study in this section, it is useful to have an overview of their place in history and to also have some insight into the general philosophy of homotopy theory.

Remark 3.8. It is difficult to give a full, historical account of the development of homotopy theory, so we will start with the creation of model categories. These were introduced by (Daniel) Quillen in 1967 in his book on homotopical algebra; see [76]. At that time, it was recognized that the general goal of homotopy theory was to study categories $\mathcal{C}$ which were equipped with some collection W of morphisms of $\mathcal{C}$ which were known, abstractly, as the “weak equivalences”. Of course, it is easy to cook up examples for which we have this kind of situation:

1. Top, with the homotopy equivalences.
2. Top, with the weak homotopy equivalences.
3. Chain complexes over abelian categories, with the quasi-isomorphisms.
4. Simplicial sets, with homotopy equivalences of simplicial sets.
5. $G\text{-}\mathcal{C}^*\text{Alg}$ or $G\text{-}\text{BanAlg}$, with the equivariant homotopy equivalences.

The last example is, of course, more recent than the other 3 but it was being considered (in the non-equivariant set-up) as far back as the 1970s, with the development of the Brown-Douglas-Fillmore Ext groups. It was realized that the goal of homotopy theory was to have some measure of control over the localization $\mathcal{C}[W^{-1}]$. As we have discussed above, this is a category which is equipped with a functor $i : \mathcal{C} \rightarrow \mathcal{C}[W^{-1}]$ such that the objects in both categories are the same and i takes morphisms in W to isomorphisms. Now, the main point is that the structure of a model category allows us to have some measure of control over the localization and this is afforded to us by various structure theorems about the mapping spaces. This seems to be somewhat close to the speculated reason for why Quillen introduced these structures; see [46]. Even though Quillen’s work on model categories seems to not have directly influenced his work on higher algebraic K-theory ([75]), it is clear that later work done on the homotopical foundations of algebraic K-theory, such as in [47], was influenced by the development of model categories. Moreover, modern approaches to algebraic K-theory via the theory of ∞ -categories certainly do rely on model structures on categories of (bi)simplicial sets, particularly for explicit descriptions of the underlying K-theory space.

As we will see later in Section 3.1, the definition of a model category is rather complex. This is unfortunate since much of the difficulty lies in verifying that some choice of category has a model structure. So, even though model categories do provide a fairly satisfactory setting in which we can “do” homotopy theory, they do not capture all examples sufficiently well. Indeed, this was first recognized in the study of the homotopy theory of sheaves of simplicial sets and sheaves of spectra by (Ken) Brown in [15]. In the same paper, he proposed a significant weakening of a model category and studied the so-called categories of fibrant objects, proving that their structure is sufficient in controlling the localization. On the other hand, developments in algebraic K-theory due to (Friedhelm) Waldhausen in [93] indicated that categories with cofibrations and weak equivalences are the right input for algebraic K-theory. In fact, it is true that there were many intermediate structures between a relative category (a category equipped with some collection W of morphisms) and a model category which were proposed to deal with various problems in stable homotopy theory; we will see one example of this in Section 3.2. It is not possible for us to give a systematic breakdown of their individual influences on modern homotopy theory. What is clear, however, is that the model categories of Quillen did have a substantial influence on later work done by homotopy theorists.

We are aware that we have not spoken about the influence of model categories on operator algebras; we will speak about this towards the end of this subsection, after we have given the definition of a model category. For now, we should proceed towards our survey of this topic and we will mainly be following [24, Chapter 2.1-2.2, p.28-41]. Our starting point is actually an abstract description of lifting problems in categories. As it turns out, this is not exactly the most natural approach towards the study of model categories. However, it is the fastest way to get to the point and highlight the main features of the theory which will come up again later in our study of categories of fibrant objects. In the next definition, we will state what it means for a lifting problem, within some arbitrary category, to have a solution.

Definition 3.9

Let $\mathcal{C}$ be a category, let $x, y, x', y' \in \mathcal{C}$ and let $f : x \rightarrow y, g : x' \rightarrow y'$ be morphisms in $\mathcal{C}$. f has the left-lifting property (LLP) with respect to g (or g has the right-lifting property (RLP) with respect to f) iff any lifting problem:

$$\begin{array}{ccc} x & \xrightarrow{\quad} & x' \\ f \downarrow & & \downarrow g \\ y & \xrightarrow{\quad} & y' \end{array}$$

has a solution $\phi : y \rightarrow x'$. That is, the following diagram:

$$\begin{array}{ccc} x & \xrightarrow{\quad} & x' \\ f \downarrow & \nearrow \phi & \downarrow g \\ y & \xrightarrow{\quad} & y' \end{array}$$

commutes. We denote this by writing $f \square g$.

We have a remark about this definition.

Remark 3.10. The solution ϕ must work for all possible choices of morphisms from x to x' and y to y' . So, even though lifting problems do cover a wide range of phenomena, the requirements of a solution are rather stringent.

We introduce two pieces of convenient terminology associated to the definition.

Terminology 3.11. Let $\mathcal{C}$ be a category and let f, g be morphisms such that:

$$f \square g.$$

Then, we say that f left-lifts g or g right-lifts f .

Terminology 3.12. Let $\mathcal{C}$ be a category and let $S, T \subseteq \text{Mor}(\mathcal{C})$. We say that S has the LLP with respect to T (or T has the RLP with respect to S) iff for every $f \in S$ and $g \in T$, we have that $f \square g$. We denote this by writing $S \square T$. Also, if $T = \{g\}$, then we'll often just write $S \square g$ instead of $S \square \{g\}$; the same is true when S is a singleton.

Here are some basic examples of lifting problems.

Example 3.13. Let $A, B \in \text{Set}$ and let $f \in \text{Set}(A, B)$.

Claim. TFAE:

1. f is surjective.
2. $(\emptyset \rightarrow *) \square f$.

We did not label the map from $\emptyset$ to $*$ because there is only one such map.

Before we prove this equivalence, here's the diagram lurking in the background.

$$\begin{array}{ccc} \emptyset & \longrightarrow & A \\ \downarrow & & \downarrow f \\ * & \longrightarrow & B \end{array}$$

Now, we prove each implication separately.

- (1) $\Rightarrow$ (2): Take the map from $*$ to B and let b be in its range. Then, there is an $a \in A$ such that $f(a) = b$. Now, define a map from $*$ to A whose value is a ; this is a solution to the lifting problem.
- (2) $\Rightarrow$ (1): Let $b \in B$. Then, we pick a map from $*$ to B with value b and we select the unique map from $\emptyset$ to A . This, of course, gives us a lifting problem which has a solution $\phi : * \rightarrow A$. The solution is literally the condition that $f(a) = b$, where the solution has value $a \in A$.

This completes the proof.

Example 3.14. Let $R \in \text{Rng}^1$ and let $M \in {}_R\text{Mod}$. We recall that M is projective iff:

$${}_R\text{Mod}(M, -),$$

is an exact functor.

Claim. TFAE:

1. M is projective.
2. $(0 \rightarrow M) \square \{\text{epimorphisms in } \mathcal{A}\}$.

This is essentially a standard characterization of projective modules.

Example 3.15. Let $\mathcal{C}, \mathcal{D}$ be categories and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$.

Claim. TFAE:

1. F is surjective on objects.
2. F has the RLP with respect to $\emptyset \rightarrow [0]$.

We will prove each implication separately.

- (1) $\Rightarrow$ (2): Consider the following commutative diagram:

$$\begin{array}{ccc} \emptyset & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow F \\ [0] & \xrightarrow{d} & \mathcal{D} \end{array}$$

where d represents the constant functor which selects an object $d \in \mathcal{D}$. Since F is surjective on objects, there is a $c \in \mathcal{C}$ such that $F(c) = d$. Now, we define a lift $[0] \rightarrow \mathcal{C}$ which simply selects the object c in $\mathcal{C}$.

- (2) $\Rightarrow$ (1): Let $d \in \mathcal{D}$. Then, the commutative diagram above has a lift; this lift just selects an object $c \in \mathcal{C}$. Since the lift causes the diagram to commute, we see that $F(c) = d$.

Our earlier purpose in defining the arrow category was to specify retracts of morphisms in categories.

Terminology 3.16. Let $\mathcal{C}$ be a category and let $x, y \in \mathcal{C}$. x is a retract of y iff there is a commutative diagram:

$$\begin{array}{ccccc} & & \text{Id}_x & & \\ & \nearrow & & \searrow & \\ x & \xrightarrow{f} & y & \xrightarrow{g} & x \end{array}$$

Also, if $f, g \in \text{Mor}(\mathcal{C})$, then f is a retract of g iff f is a retract of g in $\text{Arr}(\mathcal{C})$. Finally, if $S \subseteq \text{Mor}(\mathcal{C})$, then S is closed/stable under retracts iff every morphism in $\mathcal{C}$ which is a retract of some morphism in S must also lie in S .

In fact, an object being a retract of another object does not change under dualization.

Claim 3.1. Let $\mathcal{C}$ be a category and let $x, y \in \mathcal{C}$ with x being a retract of y . Then, x is a retract of y in $\mathcal{C}^{\text{op}}$.

The following result is standard and easy to prove.

Proposition 3.17. The Retract Lemma

Let $\mathcal{C}$ be a category, let $f : x \rightarrow y$ be a morphism and suppose we have a factorization:

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ & \searrow i \quad \nearrow p & \\ & z & \end{array}$$

Then, TFSH:

1. If $i \boxdot f$, then f is a retract of p .
2. If $f \boxdot p$, then f is a retract of i .

Proof. We'll prove each statement one-by-one.

1. We need it to be the case that f is a retract of p in $\text{Arr}(p)$. We would like to construct a commutative diagram:

$$\begin{array}{ccccc} x & \xrightarrow{i} & z & \xrightarrow{\quad} & x \\ \downarrow f & & \downarrow p & & \downarrow f \\ y & \xrightarrow{\text{Id}_y} & y & \xrightarrow{\text{Id}_y} & y \end{array}$$

and to do this, we only need to specify the morphism $z \rightarrow x$. The above diagram is supposed to represent the composition of two morphisms $f \Rightarrow p \Rightarrow f$. We start by re-working the commutative triangle in the statement:

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ & \searrow i \quad \nearrow p & \\ & z & \end{array} \Longrightarrow \begin{array}{ccc} x & & \\ \downarrow i & \searrow f & \\ z & \xrightarrow{p} & y \end{array} \Longrightarrow \begin{array}{ccc} x & \xrightarrow{\text{Id}_x} & x \\ \downarrow i & & \downarrow f \\ z & \xrightarrow{p} & y \end{array}$$

Since $i \boxdot f$, it follows that there is a solution $\phi : z \rightarrow x$. But this just means that $\phi \circ i = \text{Id}_x$.

2. If $f \boxdot p$ in $\mathcal{C}$, then $p \boxdot f$ in $\mathcal{C}^{\text{op}}$. So, f is a retract of i in $\mathcal{C}^{\text{op}}$ but that means that f is a retract of i in $\mathcal{C}$. ■

With retracts out of the way, we can specify weak factorization systems.

Terminology 3.18. Let $\mathcal{C}$ be a category and let $S, T \subseteq \text{Mor}(\mathcal{C})$ so that:

1. S and T are closed under retracts.
2. $S \boxtimes T$.
3. If $f \in \text{Mor}(\mathcal{C})$, then it has a factorization $f = p \circ i$ with $p \in T$ and $i \in S$.

The pair (S, T) is known as a weak factorization system (WFS).

With this, we can finally define model structures; this is different from the one given by Quillen in [76].

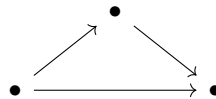
Definition 3.19

Let $\mathcal{C}$ be a category. A model structure on $\mathcal{C}$ consists of a triple:

$$(\text{Cof}, W, \text{Fib}),$$

where W, Fib are full subcategories of $\text{Arr}(\mathcal{C})$ such that:

1. W has the 2-out-of-3 property. Namely, if we have a commutative triangle:



with 2-out-of-3 morphisms in this triangle being in W , then the third is also in W .

2. The pairs $(\text{Cof}, W \cap \text{Fib})$ and $(W \cap \text{Cof}, \text{Fib})$ are WFSs.

A model category is the datum of a finitely bicomplete category $\mathcal{C}$ equipped with a fixed model structure.

We have some terminology that goes along well with the definition of a model category.

Terminology 3.20. Let $\mathcal{C}$ be a category with some model structure $(\text{Cof}, W, \text{Fib})$. Then:

1. Fib is the set of fibrations.
2. Cof is the set of cofibrations.
3. W is the set of weak equivalences.
4. $W \cap \text{Fib}$ is the set of all trivial fibrations.
5. $W \cap \text{Cof}$ is the set of all trivial cofibrations.

Moreover, (Fib, Cof) is known as the phantom structure on $\mathcal{C}$.

We have a remark about examples and our work so far.

Remark 3.21. In general, it is very difficult to provide examples of model categories because the verification that the appropriate datum on a category actually provides a model structure can be extremely difficult. There are three classes of examples that are most heavily prevalent in the literature, partly due to their classical nature but also because they get used in modern work in homotopy theory. These classes are listed as follows:

1. Topological spaces.
2. Simplicial sets.
3. Chain complexes.

Actually, [76] also gives other interesting examples of model categories, though complete verifications of such categories actually having a model structure are not provided. With regards to the three examples above, there are now many resources which give a systematic overview of various interesting model structures related to the objects listed above. Therefore, we will not repeat any of that over here, especially since we are barely going to need any of it^a. Instead, we refer the reader to the very detailed webpages [4], [5] and [3], which also contain more detailed references.

^aIn fact, we are strictly **not** going to need any of it for the explicit work that we do in this thesis. On the other hand, we do use the theory of ∞ -categories later in this thesis and while we use it as a “machine”, this machine does rely on several model structures on categories of simplicial sets which we do not discuss here, though we do mention some work done by (Denis-Charles) Cisinski on the construction of such model structures in Section 5.2. This is related to some of our work in the next subsection.

One may ask where operator algebras stand with respect to these developments. As we described in Remark 2.10, it was already noticed by Schochet that one might reasonably want to bring the homotopy theory of C^* -algebras into the framework of Quillen’s model categories. The point was to find a model structure on $C^*\text{Alg}$ such that the W would be the homotopy equivalences and Fib would consist of the Schochet fibrations. In fact, no such model structure exists.

Theorem 3.22

There is no model structure on $C^*\text{Alg}$ such that:

1. the weak equivalences are homotopy equivalences and;
2. the fibrations are Schochet fibrations.

Let us discuss this theorem and literature related to it in the next remark.

Remark 3.23. A variant of this theorem was proved by Andersen and Grodal in the unpublished preprint [1]; they showed that $C^*\text{Alg}$ can be equipped with the structure of a Baues fibration category and demonstrated the analogous impossibility result. The theorem above, however, was shown by (Otgonbayar) Uuye in [92, Theorem A.1, p.15-16]; he proves it by showing that the loop functor on the homotopy category of commutative C^* -algebras does not admit a left adjoint. Nevertheless, recent work by (Michael) Joachim and (Anupam) Datta suggests that it is possible to place a model structure on the category of locally multiplicative convex C^* -algebras; see [29]. Similar work of this type has been done in the past involving projective systems of C^* -algebras; see [8].

The benefit of Joachim and Datta’s work is that it addresses the homotopical nature of the so-called Cuntz q -construction; this is used in the algebraic construction of Kasparov’s KK-theory functor in [28]. Introducing this construction here would be a distraction but we will note that their (Joachim & Datta) work tells us that the Cuntz q should be viewed, up to stabilization, as a cofibrant replacement functor; see Corollary 8.10 on page 37 of their paper. This seems to address a question of Meyer as stated in [61, p.19], so it is clear that a homotopical perspective elucidates known constructions and give us indications of the role they should play in the theory.

As mentioned in Remark 3.21, model structures are not commonly found among the operator-algebraic categories of interest. So, we have to be content with studying homotopical structures that are weaker but encapsulate the two major examples we have discussed thoroughly in Section 2. It was first recognized by Uuye in [92] that the appropriate structures are the *categories of fibrant objects*. We have mentioned these before and in Section 3.2, we will present a rather extensive discussion of their basic theory & our contributions to it. We will use this to resolve Question 2.9 in Section 4.

3.2 Properties of Categories of Fibrant Objects

Ultimately, we wish to study categories $\mathcal{C}$ which are equipped with two full subcategories:

$$W, \text{Fib},$$

of $\text{Arr}(\mathcal{C})$; the first will be a category of *weak equivalences* and the second will be a category of *fibrations*. The literature which deals with these categories systematically is somewhat sparse and we aim to extend some results about these categories while taking inspiration from our work on the homotopy theory of operator algebras as done in Section 2. With that general aim in mind, let us introduce some terminology that will only be relevant later in this subsection.

Terminology 3.24. Let $\mathcal{C}$ be a category and let W be a full subcategory of $\text{Arr}(\mathcal{C})$.

1. W satisfies the 2-out-of-3 property iff for any diagram of two composable arrows:

$$x \xrightarrow{f} y \xrightarrow{g} z$$

where any two of f, g and $g \circ f$ lies in W , then the third lies in W .

2. W satisfies the 2-out-of-6 property iff for a diagram of three composable arrows:

$$w \xrightarrow{f} x \xrightarrow{g} y \xrightarrow{h} z$$

where $g \circ f$ and $h \circ g$ lies in W , we have that f, g, h and $h \circ g \circ f$ lie in W .

If $\gamma : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{C})$ is the localization of $\mathcal{C}$ with respect to W , then we say that W is saturated in $\mathcal{C}$ iff for any morphism $f \in \text{Mor}(\mathcal{C})$ with $\gamma(f)$ being an isomorphism, we have that $f \in W$.

Here are some basic (but useful facts) which are strewn across the literature.

Claim 3.2. Let $\mathcal{C}$ be a category and let W be a full subcategory of $\text{Arr}(\mathcal{C})$.

1. The isomorphisms of $\mathcal{C}$, viewed as a full subcategory of $\text{Arr}(\mathcal{C})$, satisfy the 2-out-of-6 property.
2. If W satisfies the 2-out-of-6 property and contains the identity morphisms, then it is 2-out-of-3.
3. If W satisfies the 2-out-of-6 property and contains the identity morphisms, then it contains all isomorphisms.

This is proved on Page 109.

Claim 3.3. Let $\mathcal{C}, \mathcal{D}$ be categories, let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$ and let W be a full subcategory of $\text{Arr}(\mathcal{C})$ consisting of all morphisms whose images under F become isomorphisms in $\mathcal{D}$. Then, W satisfies the 2-out-of-6 property^a.

^aThis is proved on Page 110.

Claim 3.4. Let $\mathcal{C}$ be a category and let W be a saturated full subcategory of $\text{Arr}(\mathcal{C})$. Then^a, W is 2-out-of-6.

^aThis is proved on Page 110.

We will work our way up to the definition of a category of fibrant objects, starting with the notion of a fibration category.

Definition 3.25

A fibration category is a pair:

$$(\mathcal{C}, \text{Fib}_{\mathcal{C}}),$$

of a pointed category $\mathcal{C}$ with a full subcategory $\text{Fib}_{\mathcal{C}}$ of $\text{Arr}(\mathcal{C})$ consisting of *fibrations* such that:

1. The isomorphisms in $\mathcal{C}$ are in $\text{Fib}_{\mathcal{C}}$.
2. The composition of any two fibrations is a fibration.
3. If $x \in \mathcal{C}$, then the unique morphism $x \rightarrow 0$ lies in $\text{Fib}_{\mathcal{C}}$.
4. Pullbacks by fibrations exist and fibrations are preserved by base extensions (recall Terminology 2.30).

A cofibration category is a category with cofibrations as defined on [93, p.320], being exactly dual to fibration categories.

Here are two simple results.

Claim 3.5. Let $\mathcal{C}$ be a category.

1. If $\mathcal{C}$ is a fibration category, then $\mathcal{C}^{\text{op}}$ is a cofibration category when paired with $\text{Fib}_{\mathcal{C}}^{\text{op}}$.
2. If $\mathcal{C}$ is a cofibration category, then $\mathcal{C}^{\text{op}}$ is a fibration category when paired with $\text{Cof}_{\mathcal{C}}^{\text{op}}$.

This is proved on Page 110.

Claim 3.6. Let $\mathcal{C}$ be a fibration category. Then, $\mathcal{C}$ admits all finite products^a.

^aThis is proved on Page 110.

Of course, one can also define structure-preserving functors, even though we will not use this notion.

Terminology 3.26. Let $\mathcal{C}, \mathcal{D}$ be fibration categories and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$. Then, F is exact iff it preserves the zero object, if it preserves fibrations and if it preserves base extensions along a fibration.

Here are two nice constructions of fibration categories.

Construction 3.27. Let $\mathcal{C}$ be a fibration category and let I be a small category. We define:

$$\text{Fib}_{\text{Fun}(I, \mathcal{C})},$$

as follows: if $F, G \in \text{Fun}(I, \mathcal{C})$ and $\alpha \in \text{Nat}(F, G)$, then α is a fibration iff its components are fibrations in $\mathcal{C}$.

| **Claim.** $\text{Fun}(I, \mathcal{C})$ becomes a fibration category with the data of $\text{Fib}_{\text{Fun}(I, \mathcal{C})}$.

We will show each axiom of a fibration category one-by-one.

1. If α is an isomorphism in the category $\text{Fun}(I, \mathcal{C})$, then its components are isomorphisms in $\mathcal{C}$ and these are certainly fibrations in $\mathcal{C}$. So, it follows that α is a fibration.
2. Let α and β be natural transformations so that $\beta \circ \alpha$ is well-defined and so that each of them is a fibration in $\text{Fun}(I, \mathcal{C})$. Now, let $x \in \mathcal{C}$ and note that $(\beta \circ \alpha)_x = \beta_x \circ \alpha_x$. Since β_x and α_x are fibrations, their composition is a fibration and this means that $\beta \circ \alpha$ is a fibration on its components so it is a fibration in $\text{Fun}(I, \mathcal{C})$.
3. Let $F \in \text{Fun}(I, \mathcal{C})$ and let $\alpha : F \Rightarrow 0$, where the zero functor in $\text{Fun}(I, \mathcal{C})$ is a zero object which takes every object in I to the zero object in $\mathcal{C}$. Now, the components of α are the unique morphisms from each $F(i)$ to the zero object of $\mathcal{C}$ so these are fibrations in $\mathcal{C}$. But this means that α is a fibration in $\mathcal{C}$.
4. Limits in $\text{Fun}(I, \mathcal{C})$ are computed point-wise, so fibrations are preserved by base extensions.

This completes the proof.

Construction 3.28. Let $\mathcal{C}$ be a pointed model category. We define a full subcategory:

$$\mathcal{C}_f,$$

consisting of the fibrant objects^a of $\mathcal{C}$.

| **Claim.** If we restrict the fibrations of $\mathcal{C}$ to fibrations between objects of $\mathcal{C}_f$, then $\mathcal{C}_f$ is a fibration category.

With the definition of a model category that we have given in [Definition 3.19](#), it seems non-trivial to verify that restricting to the fibrant objects as we have described above gives us a fibration category. Nevertheless, one can show that our definition is equivalent to the original one due to Quillen ([\[76, Definition 1.1.1, p. 14-15\]](#)) and that definition makes this result obvious. This equivalence is discussed to some degree in [\[24, Remark 2.2.3, p.33-34\]](#) and since we will not need this result beyond its philosophical significance, we choose to say nothing more about it.

^aAn object $x \in \mathcal{C}$ is fibrant iff the unique morphism $x \rightarrow 0$ is a fibration.

Finally, we have two examples coming from operator algebra theory.

Example 3.29. We have the following claim.

| **Claim.** TFSH:

1. $G\text{-C}^*\text{Alg}$ is a fibration category when equipped with the Schochet fibrations.
2. $G\text{-BanAlg}$ is a fibration category when equipped with the Schochet fibrations.

We'll verify each of the axioms as listed in [Definition 3.25](#) one-by-one.

1. Clearly, $G\text{-C}^*\text{Alg}$ and $G\text{-BanAlg}$ have zero objects; it's just $\{0\}$.
2. In both cases, axiom (1) is taken care of by statement (3) of [Proposition 2.29](#).
3. In both cases, axiom (2) is taken care of by statement (1) of [Proposition 2.29](#).
4. In both cases, axiom (3) is taken care of by statement (2) of [Proposition 2.29](#).
5. In both cases, axiom (4) is taken care of by statement (5) of [Proposition 2.29](#).

This proves the result.

For the next example, we will need the notion of a semisplit surjection as introduced in [Terminology 2.32](#)

Example 3.30. Let $G \in \text{Grp}(\text{LCH})$ and define:

$$\text{Fib} \stackrel{\text{def}}{=} \{\text{Semisplit surjections in } G\text{-C}^*\text{Alg}\}.$$

Here's the main claim.

| **Claim.** The pair $(G\text{-C}^*\text{Alg}, \text{Fib})$ forms a fibration category.

We'll verify each of the axioms as listed in [Definition 3.25](#) one-by-one.

1. Clearly, $G\text{-C}^*\text{Alg}$ and $G\text{-BanAlg}$ have zero objects; it's just $\{0\}$.
2. In both cases, axiom (1) is taken care of by statement (3) of [Proposition 2.29](#).
3. In both cases, axiom (2) is taken care of by statement (1) of [Proposition 2.29](#).
4. In both cases, axiom (3) is taken care of by statement (2) of [Proposition 2.29](#).
5. This is the most difficult one to verify and a complete verification has been done in the proof of [\[21, Proposition 2.10, p.26-27\]](#). That result is stated for discrete, countable groups but the proof works for LCH groups.

This proves the result.

Once we have a fibration category, we also would want to add an additional bit of data about *weak equivalences*; this is a second subcategory of the arrow category that we include in our data.

Definition 3.31

A weak fibration category is a triple:

$$(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}}),$$

such that:

1. $(\mathcal{C}, \text{Fib}_{\mathcal{C}})$ is a fibration category.
2. $W_{\mathcal{C}}$ is a full subcategory of $\text{Arr}(\mathcal{C})$; this is known as the subcategory of weak equivalences.
3. All isomorphisms in $\mathcal{C}$ are contained in $W_{\mathcal{C}}$.
4. Suppose we have a commutative diagram in $\mathcal{C}$:

$$\begin{array}{ccccc} b & \xrightarrow{\quad\quad} & a & \xleftarrow{\quad\quad} & c \\ \downarrow & & \downarrow & & \downarrow \\ b' & \xrightarrow{\quad\quad} & a' & \xleftarrow{\quad\quad} & c' \end{array}$$

where the red arrows are fibrations and the vertical arrows are weak equivalences. Then, the induced map:

$$b \times_a c \rightarrow b' \times_{a'} c',$$

is in $W_{\mathcal{C}}$. This requirement is known as the Gluing Lemma.

A weak cofibration category is a category with cofibrations and weak equivalences as defined in [93, p.326].

Since our notation for the two subcategories of a weak fibration category involve the notation for the category itself, we will find it reasonable to mostly omit explicit mention of the subcategories when referring to a weak fibration category. Here's some further terminology that will be repeated later for categories of fibrant objects.

Terminology 3.32. Let $\mathcal{C}$ be a category equipped with two full subcategories:

$$\text{Fib}_{\mathcal{C}}, W_{\mathcal{C}},$$

of $\text{Arr}(\mathcal{C})$. Then, the morphisms in the subcategory $\text{Fib}_{\mathcal{C}} \cap W_{\mathcal{C}}$ are known as trivial fibrations.

Before we give examples, let us develop some theory associated to this definition since it is rather interesting. In particular, the Gluing Lemma is ideally something we'd like to establish, rather than postulate. We have the following result, which is an amalgamation of facts that are scattered throughout the literature but are useful to have in a single place.

Proposition 3.33

Let $\mathcal{C}$ be a category and suppose that there are two full subcategories Fib, W of $\text{Arr}(\mathcal{C})$ so that:

- All isomorphisms of $\mathcal{C}$ are in W .
- W satisfies the 2-out-of-3 property.
- If $f \in \text{Mor}(\mathcal{C})$, then $f = p \circ i$, where $p \in \text{Fib}$ and $i \in W$.
- Fibrations are preserved by base extensions.

Then, the following statements hold:

1. If $f \in \text{Mor}(\mathcal{C})$ is a section of a morphism in W , then $f \in W$.
2. If $f \in \text{Mor}(\mathcal{C})$ with factorization:

$$f = p \circ i,$$

then $f \in W$ iff $p \in \text{Fib} \cap W$.

3. If $\mathcal{C}$ is closed under pullbacks, then TFAE:
 - (a) The Gluing Lemma holds.
 - (b) A pullback of a morphism in W along a morphism in Fib is in W .

Proof. We'll prove each statement one-by-one.

1. Let $g \in W$ so that $g \circ f = \text{Id}$. Since Id is an isomorphism, it is in W . On the other hand, $g \in W$ so the 2-out-of-3 property implies that $f \in W$.
2. We'll prove each implication separately.
 - $\Rightarrow$: If $f \in W$, then the 2-out-of-3 property implies that $p \in W$.
 - $\Leftarrow$: If $p \in W$, then the 2-out-of-3 property means that $f \in W$.

This completes the proof.

3. This is the dual of the result proved in [60]. ■

Here are some examples.

Example 3.34. Let $\mathcal{C}$ be a fibration category & let $W_{\mathcal{C}}$ be the full subcategory of $\text{Arr}(\mathcal{C})$ of all isomorphisms in $\mathcal{C}$.

Claim 3.7. The triple $(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}})$ is a weak fibration category.

This is proved on Page 110; it is known as the minimal weak fibration structure on a fibration category.

Example 3.35. Let $\mathcal{C}$ be a fibration category & let $W_{\mathcal{C}}$ just be $\text{Arr}(\mathcal{C})$.

Claim. The triple $(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}})$ is a weak fibration category.

This is completely clear; it is known as the maximal weak fibration structure on a fibration category.

Our interest is mainly in categories of fibrant objects; we will relate these to weak fibration categories later. For now, here is some terminology which will lead us towards categories of fibrant objects.

Notation 3.36. Let $\mathcal{C}$ be a category and suppose we have the following commutative diagram:

$$\begin{array}{ccc} x & \xrightarrow{\beta} & y \\ \alpha \downarrow & & \downarrow f \\ x' & \xrightarrow{g} & y' \end{array}$$

If the fibre product of the diagram above exists, then the commutative diagram above is a pullback cone. So, there exists a unique morphism $(\alpha, \beta) : x \rightarrow x' \times_{y'} y$ such that the resulting diagram formed is commutative. In particular, let $f : x \rightarrow y$ be a morphism in $\mathcal{C}$. Then, the diagonal induced by f is defined by:

$$\Delta_f \stackrel{\text{def}}{=} (\text{Id}_x, \text{Id}_x).$$

Diagram-wise, the one to imagine is the following:

$$\begin{array}{ccc} x & \xrightarrow{\text{Id}_x} & x \\ \text{Id}_x \downarrow & & \downarrow f \\ x & \xrightarrow{f} & y \end{array}$$

commutative diagram.

Terminology 3.37. Let $\mathcal{C}$ be a category with finite products, final object e and two full subcategories:

$$\text{Fib}, W \subseteq \text{Arr}(\mathcal{C}).$$

If $x \in \mathcal{C}$, then a path space of x is an object, usually called a path object, x^I together with morphisms:

$$x \xrightarrow{s} x^I \xrightarrow{(d_0, d_1)} x \times x$$

such that $s \in W$, $(d_0, d_1) \in \text{Fib}$ and the composite is the diagonal morphism^a.

^aThe notation (s, x^I, d_0, d_1) will be used for a given path space of x .

Here's some notation involving path spaces.

Notation 3.38. Let $\mathcal{C}$ be a category as in Terminology 3.37 and let $x \in \mathcal{C}$. From now on, if (x^I, s, d_0, d_1) is a path space for x , then we will simply denote this by x^I unless it is absolutely necessary to specify the other morphisms. Moreover, if there is one path space being considered, then the notation as a quadruple above will be valid. So, we may say “Let x^I be the path space for x ” and later in a given proof, we might use d_0, d_1 whenever necessary.

With this preparation at hand, we can define categories of fibrant objects.

Definition 3.39

Let $\mathcal{C}$ be a pointed category and two full subcategories of $\text{Arr}(\mathcal{C})$:

$$\text{Fib}, W,$$

called *fibrations* and *weak equivalences* respectively. Consider the following axioms:

1. $(\mathcal{C}, \text{Fib})$ is a fibration category.
2. Any isomorphism in $\mathcal{C}$ is in W .
3. W satisfies the 2-out-of-3 property.
4. If $x \in \mathcal{C}$, there exists a path space (s, x^I, d_0, d_1) .
5. Morphisms in $\text{Fib} \cap W$ are preserved by base extensions.

The triple $(\mathcal{C}, \text{Fib}, W)$ is called a Brown category iff it satisfies axioms (1), (2), (3), (4). If such a category also satisfies axiom (5), then we refer to it as a category of fibrant objects.

We have a few remarks about this definition.

Remark 3.40. The main reason we isolate axiom (5) above is because it is usually the most difficult axiom to verify, with the path space axiom being comparable. Moreover, as we will see in Proposition 4.1, for $G \in \text{Grp}(\text{LCH})$, we have only been able to prove that $G\text{-C}^*\text{Alg}$ and $G\text{-BanAlg}$ form Brown categories. As we specified towards the end of Remark 2.37, we do expect this result to be true even for these equivariant categories.

Remark 3.41. In the original definition of Brown, the category $\mathcal{C}$ is not required to be pointed but simply has a final object. Also, the way we've stated the definition above is much less explicit than the one given by Brown. We have this assumption because it is a reasonable simplification given the main examples of interest to us.

In the next remark, we will discuss some of the history behind the notions we have introduced so far, though we will refrain from speaking of the connections to operator algebras since that has been discussed in [Remark 3.23](#).

Remark 3.42. Categories of fibrant objects were first introduced by (Kenneth) Brown in [15] for the purpose of studying the homotopy theory of sheaves of simplicial sets and sheaves of spectra. It turns out that their homotopy theory is not satisfactorily described by the usual axioms of a model category and for the purpose of developing a cohomology theory for the aforementioned categories, Brown introduced categories of fibrant objects and proved results about them which are analogous to known statements about model categories. We also mention that weak fibration categories were essentially introduced in [93] by Waldhausen. Part of the underlying goal, at least of Chapter 1 of the paper, was to introduce an extension of Quillen's definition of higher algebraic K-theory as described in [75]. In particular, Waldhausen demonstrated that the $S_\bullet$ -construction, as attributed to (Graeme) Segal, on categories with cofibrations and weak equivalences actually “recovers” the Q -construction on exact categories. We emphasize that Waldhausen's approach to K-theory has been very influential on modern developments of algebraic K-theory, particularly those which involve its universal characterization. For more information in this direction, see [9] and [14].

Here is an immediate construction.

Construction 3.43. Let $\mathcal{C}$ be a pointed model category and consider the fibration category $\mathcal{C}_f$ of [Construction 3.28](#).

| **Claim.** With the weak equivalences of $\mathcal{C}$ restricted to the objects of $\mathcal{C}_f$, then $\mathcal{C}_f$ is a category of fibrant objects.

Again, our definition of a model category does not help us in verifying this claim. Nevertheless, the equivalent definition of a model category due to Quillen makes this claim obvious. In particular, this gives us a number of useful examples of categories of fibrant objects. Note that pointedness is not actually necessary; see [Remark 3.41](#).

Of course, one would hope that we can construct categories of fibrant objects by looking at special subcategories. This leads us to the notion of *fibrant subcategory* which we introduce as a definition. This seems to not be in the literature.

Definition 3.44. A. Vats, 2026

Let $\mathcal{C}$ be a Brown category and let $\mathcal{D}$ be a full subcategory of $\mathcal{C}$ so that:

1. $\mathcal{D}$ retains the zero object of $\mathcal{C}$.
2. $W_{\mathcal{D}}$ is a full subcategory of $W_{\mathcal{C}}$ with objects being morphisms in $\mathcal{D}$.
3. $\text{Fib}_{\mathcal{D}}$ is a full subcategory of $\text{Fib}_{\mathcal{C}}$ with objects being morphisms in $\mathcal{D}$.
4. $\mathcal{D}$ is closed under pullbacks along morphisms in $\text{Fib}_{\mathcal{D}}$ and the inclusion $\mathcal{D} \rightarrow \mathcal{C}$ preserves pullbacks.
5. Fibrations in $\mathcal{D}$ are preserved by base extensions.
6. If $x \in \mathcal{D}$, then there is a path space of x which lies in $\mathcal{D}$.

In that case, $(\mathcal{D}, W_{\mathcal{D}}, \text{Fib}_{\mathcal{D}})$ is known as a fibrant subcategory.

In fact, fibrant subcategories certainly do inherit appropriate structure.

Proposition 3.45. A. Vats, 2026

Let $\mathcal{C}$ be a Brown category.

1. Any fibrant subcategory of $\mathcal{C}$ is a Brown category.
2. If $\mathcal{C}$ is a category of fibrant objects, then any fibrant subcategory of $\mathcal{C}$ is a category of fibrant objects^a.

^aOf course, equipped with the specified structure in [Definition 3.44](#).

Proof. Let $\mathcal{D}$ be a fibrant subcategory of $\mathcal{C}$. In both cases above, we will need the following claim.

| **Claim.** $\mathcal{D}$ is a fibration category.

We note that $\mathcal{D}$ is a pointed category since it is a full subcategory of $\mathcal{C}$ which retains the zero object of $\mathcal{C}$. So, it remains for us to verify the axioms listed in [Definition 3.25](#).

1. Let f be any isomorphism in $\mathcal{D}$. Then, $f \in \text{Fib}_{\mathcal{C}}$ so $f \in \text{Fib}_{\mathcal{D}}$.
2. If $f, g \in \text{Fib}_{\mathcal{D}}$ such that $g \circ f$ exists, then $g \circ f \in \text{Fib}_{\mathcal{C}}$ so it lies in $\text{Fib}_{\mathcal{D}}$.
3. If $x \in \mathcal{D}$, then $x \rightarrow 0$ is a fibration in $\mathcal{C}$ and since $0 \in \mathcal{D}$, this is a fibration in $\mathcal{D}$.
4. Consider the following pullback diagram in $\mathcal{D}$:

$$\begin{array}{ccc} P & \longrightarrow & B \\ \bar{p} \downarrow & & \downarrow p \\ A & \longrightarrow & C \end{array}$$

where $p \in \text{Fib}_{\mathcal{D}}$. We know that this exists in $\mathcal{D}$ and we also note that this is also a pullback diagram in $\mathcal{C}$ (since the inclusion functor preserves pullbacks), so we see that $\bar{p} \in \text{Fib}_{\mathcal{C}}$. But this means that $\bar{p} \in \text{Fib}_{\mathcal{D}}$.

This proves the claim.

Now, assume that $\mathcal{C}$ is a Brown category. Let us verify each of the axioms listed in [Definition 3.39](#).

1. $(\mathcal{D}, \text{Fib}_{\mathcal{D}})$, as shown by the claim above, form a fibration category.
2. Let f be an isomorphism in $\mathcal{D}$. Then, it is an isomorphism in $\mathcal{C}$ and $f \in W_{\mathcal{C}}$, so $f \in W_{\mathcal{D}}$.
3. Let $f, g \in \text{Mor}(\mathcal{D})$ so that $g \circ f$ is defined and suppose that 2-of-3 of these are in $W_{\mathcal{D}}$. So, 2-of-3 of these are in $W_{\mathcal{C}}$, so we conclude that the third must be in $W_{\mathcal{C}}$. Hence, the third must also be in $W_{\mathcal{D}}$.
4. This is clear.

This proves statement (1).

Now, assume that $\mathcal{C}$ is a category of fibrant objects. Then, we have to verify axiom (5) of [Definition 3.39](#), so let $f \in \text{Fib}_{\mathcal{D}} \cap W_{\mathcal{D}}$ and suppose that we take a pullback along f . Then, $f \in \text{Fib}_{\mathcal{C}} \cap W_{\mathcal{D}}$ and this pullback is also one in $\mathcal{C}$ (since the inclusion functor preserves pullbacks), so the morphism opposite to it is a trivial fibration in $\mathcal{C}$. But that morphism is one between objects in $\mathcal{D}$, so it is a trivial fibration in $\mathcal{D}$. This proves statement (2). ■

Whenever we have a functor between categories of fibrant objects and this functor preserves the weak equivalences of both categories, we can always induce a functor between the respective homotopy categories.

Construction 3.46. Let $\mathcal{C}, \mathcal{D}$ be categories of fibrant objects and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor such that:

$$F(W_{\mathcal{C}}) \subseteq W_{\mathcal{D}}.$$

Then, we note that $\gamma_{\mathcal{D}} \circ F : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{D})$ is a functor which inverts morphisms in $W_{\mathcal{C}}$. So, the universal property of the localization yields that there exists a unique functor $\mathbf{Ho}(\mathcal{C}) : \mathbf{Ho}(\mathcal{C}) \rightarrow \mathbf{Ho}(\mathcal{D})$ such that the following diagram:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} & \mathcal{D} \\ \gamma_{\mathcal{C}} \downarrow & & \downarrow \gamma_{\mathcal{D}} \\ \mathbf{Ho}(\mathcal{C}) & \xrightarrow{\mathbf{Ho}(F)} & \mathbf{Ho}(\mathcal{D}) \end{array}$$

commutes. This is known as the homotopy functor of F between homotopy categories. In fact, we can prove more.

Claim. For the construction above to go through, it is sufficient to assume that $F(W_{\mathcal{C}} \cap \text{Fib}_{\mathcal{C}}) \subseteq W_{\mathcal{D}}$.

This follows immediately from [Corollary 3.53](#) and the fact that $W_{\mathcal{D}}$ satisfies the 2-out-of-3 property.

One may notice that categories of fibrant objects satisfy assumptions (1), (2) and (4) of [Proposition 3.33](#). Our first goal will be to show that it satisfies (3) as well. Indeed, this result ends up being a standard tool of the theory.

Proposition 3.47. Brown's Factorization Lemma

Let $\mathcal{C}$ be a category of fibrant objects and let $f \in \text{Mor}(\mathcal{C})$. Then:

$$\exists p \in \text{Fib} : i \in W : f = p \circ i.$$

Such a factorization of f will be called a Brown factorization of f .

The proof of this result is not so difficult but it does require a bit of preparation. The first bit of preparation involves the following lemma, where we say a bit more about the evaluation maps in a path space.

Lemma 3.48 Let $\mathcal{C}$ be a category of fibrant objects, let $x \in \mathcal{C}$ and select a path space:

$$(s, x^I, d_0, d_1).$$

Then, $d_0, d_1 : x^I \rightarrow x$ are trivial fibrations.

This lemma was originally found in [\[76, Lemma 2, p. 6-7\]](#); that result is about cofibrant objects which have cylinders in a model category. However, the structure of a category of fibrant objects is rich enough to yield the result.

Proof of Lemma 3.48. Let 0 be the zero object. Since $\mathcal{C}$ is a fibration category, we see that the unique morphism $x \rightarrow 0$ is a fibration. Now, we can fit the unique morphism $x \rightarrow 0$ in the following pullback diagram with $x \times_0 x$:

$$\begin{array}{ccc} x \times_0 x & \xrightarrow{\pi_2} & x \\ \pi_1 \downarrow & & \downarrow \\ x & \xrightarrow{\quad} & 0 \end{array}$$

and we note that since fibrations are preserved by base extensions, it follows that π_1 and π_2 are fibrations.

Claim. d_0 is a fibration.

This follows from the fact that (d_0, d_1) is a fibration so composing it with π_1 produces a fibration. However, this composition is just d_0 so the claim follows, since fibrations compose to form fibrations.

Claim. d_0 is a weak equivalence.

The identity map is a trivial fibration, so diagonal maps must also be trivial fibrations. Now, $(d_0, d_1) \circ s$ is a weak equivalence because it is a diagonal map and since s is a weak equivalence, we conclude by Axiom 3 of Definition 3.39 that (d_0, d_1) is a weak equivalence. But this means that d_0 is a weak equivalence^a and we're done. ■

^aThe analogous proof for d_1 will go through as well.

A reasonably big component to the proof of the factorization lemma is the so-called mapping cylinder construction.

Construction 3.49. Let $\mathcal{C}$ be a category of fibrant objects, let $f \in \mathcal{C}(x, y)$ and let:

$$(s, y^I, d_0^y, d_1^y),$$

be a path space for y . We consider the following pullback^a:

$$\begin{array}{ccc} \text{cyl}(f) & \xrightarrow{\bar{f}} & y^I \\ \bar{d}_f \downarrow & & \downarrow d_0^y \\ x & \xrightarrow{f} & y \end{array}$$

$\text{cyl}(f)$ is called the mapping cylinder of f relative to the given path space.

^aThe pullback exists because d_0^y is a fibration.

We have a further remark about this construction, since we will need it later.

Remark 3.50. For some specific categories of fibrant objects, there might very well be a canonical choice of path space for any given object and in those cases, one computes the mapping cylinder with respect to that path space.

Here's another lemma where we record some basic properties of the construction.

Lemma 3.51 In the notation of Construction 3.49, TFSH:

1. $\bar{d}_f$ is a trivial fibration.
2. If f is a fibration, then $d_i^y \circ \bar{f}$ is a fibration for $i \in \{0, 1\}$.
3. If f is arbitrary, then $d_1^y \circ \bar{f}$ is a fibration.

Proof. We will prove each statement one-by-one.

1. This follows from the fact that d_0^y is a trivial fibration and that trivial fibrations are preserved by base extensions.
2. If f is a fibration, then $\bar{f}$ is a fibration. Since each d_i^y is a fibration, the result holds.
3. We will rewrite $d_1^y \circ \bar{f}$ as a composition of two fibrations. That is, we write it as the composition:

$$\text{cyl}(f) \xrightarrow{g} x \times y \longrightarrow y$$

where the second morphism is a projection. The projection fits into a pullback diagram with the unique morphisms $x \rightarrow 0$ and $y \rightarrow 0$ as the legs. These are fibrations, so it follows that $x \times y \rightarrow y$ is a fibration^a. To deal with the first morphism, let us construct it by introducing a diagram that will be discussed throughout the proof. We will first draw the diagram and then explain its role in the proof:

$$\begin{array}{ccccc} \text{cyl}(f) & \longrightarrow & x \times y & \xrightarrow{\pi_x} & x \\ \downarrow & & \downarrow f \times \text{Id}_y & & \downarrow f \\ y^I & \xrightarrow{(d_0^y, d_1^y)} & y \times y & \xrightarrow{\pi_1} & y \end{array}$$

We will describe this diagram in a few steps:

- (a) We first take the pullback of (d_0^y, d_1^y) against $f \times \text{Id}_y$; this produces an object p in the upper left corner which we will identify with $\text{cyl}(f)$ later in the proof.
- (b) The right square is a part of the diagram used to construct $f \times \text{Id}_y$ and it is a pullback square.
- (c) Since both squares are pullbacks, the pasting law says that the outer rectangle is a pullback diagram. However, the outer rectangle is the pullback diagram for the mapping cylinder of f . In other words, there is an isomorphism from $\text{cyl}(f)$ to p . So, we will just replace p by $\text{cyl}(f)$ and we obtain the diagram above.

Let g be the morphism from $\text{cyl}(f)$ to $x \times y$. This is a fibration because the left square is a pullback and (d_0^y, d_1^y) is a fibration. We need to show that $\pi_y \circ g$ is $d_1^y \circ \bar{f}$. The commutativity of the resulting diagram gives us:

$$(f \times \text{Id}_y) \circ g = (d_0^y, d_1^y) \circ \bar{f}.$$

Next, we note that the construction of $f \times \text{Id}_y$ entails that:

$$\pi_2 \circ (f \times \text{Id}_y) = \text{Id}_y \circ \pi_y = \pi_y,$$

where π_2 and π_y are projection maps. This means that we have the following computation:

$$\begin{aligned}\pi_y \circ g &= \pi_2 \circ (f \times \text{Id}_y) \circ g \\ &= \pi_2 \circ (d_0^y, d_1^y) \circ \bar{f} \\ &= d_1^y \circ \bar{f},\end{aligned}$$

as was desired. ■

^aWe have used an analogous argument in [Lemma 3.48](#).

We are ready to prove Brown's Factorization Lemma.

Proof of Proposition 3.47. The proof is conceptually simple. We will divide it into two parts: we'll first construct the decomposition of f and then show that it has the required properties. With that said, let us proceed.

1. Let $f : x \rightarrow y$ be a morphism and choose a path space:

$$(s, y^I, d_0^y, d_1^y).$$

Let us consider the mapping cylinder $\text{cyl}(f)$ relative to this path space. Consider the following diagram:

$$\begin{array}{ccccc} x & & & & \\ & \searrow^{s \circ f} & & & \\ & & \text{cyl}(f) & \xrightarrow{\bar{f}} & y^I \\ & \searrow^{\text{Id}_x} & \downarrow \bar{d}_f & & \downarrow d_0^y \\ & & x & \xrightarrow{f} & y \end{array}$$

where the square is a pullback. We will define morphisms $i : x \rightarrow \text{cyl}(f)$ and $p : \text{cyl}(f) \rightarrow y$ so that:

$$f = p \circ i.$$

For p , this is easy to do:

$$p \stackrel{\text{def}}{=} d_1^y \circ \bar{f}.$$

Now, we want to show that the outer diagram above commutes:

$$\begin{aligned}d_0^y \circ (s \circ f) &= (d_0^y \circ s) \circ f \\ &= \text{Id}_y \circ f \\ &= f \\ &= f \circ \text{Id}_x.\end{aligned}$$

So, the whole diagram commutes and by the universal property of the pullback, there is a unique morphism $i : x \rightarrow \text{cyl}(f)$ so that the resulting diagram commutes. Finally, we check that this factorization works:

$$\begin{aligned}p \circ i &= (d_1^y \circ \bar{f}) \circ i \\ &= d_1^y \circ (\bar{f} \circ i) \\ &= d_1^y \circ (s \circ f) \\ &= (d_1^y \circ s) \circ f \\ &= \text{Id}_y \circ f \\ &= f,\end{aligned}$$

as was desired.

2. By statement (3) of [Lemma 3.51](#), $p \in \text{Fib}_{\mathcal{C}}$ so we should show that $i \in W$. We know that $\text{Id}_x = \bar{d}_f \circ i$. Since $\bar{d}_f$ is a trivial fibration, i is a section of a trivial fibration so statement (1) of [Proposition 3.33](#) tells us that $i \in W$.

This completes the proof. ■

The next result is mentioned in some places in the literature but it is never proved, so we fill in the gap.

Proposition 3.52

Let $\mathcal{C}$ be a pointed category and suppose that there are two full subcategories:

$$\text{Fib}_{\mathcal{C}}, W_{\mathcal{C}},$$

satisfying axioms (1) – (4) in [Definition 3.39](#). TFAE:

1. This structure satisfies axiom (5).
2. If $f \in \text{Mor}(\mathcal{C})$, then there is a $p \in \text{Fib}_{\mathcal{C}}$ and $i \in W$ such that $f = p \circ i$.

In fact, (2) $\Rightarrow$ (1) holds even when $\mathcal{C}$, with $\text{Fib}_{\mathcal{C}}$, is just a fibration category.

Proof. (1) $\Rightarrow$ (2) is simply Brown's Factorization Lemma. For (2) $\Rightarrow$ (1), let $x \in \mathcal{C}$ and consider the diagonal map:

$$\Delta_x : x \rightarrow x \times x,$$

where this is well-defined because $\mathcal{C}$ is closed under finite products. Now, we factorize this as follows:

$$\Delta_x = p_x \circ i_x,$$

where $p_x \in \text{Fib}_{\mathcal{C}}$ and $i_x \in W$. This leads us to the following claim which will finish the proof.

Claim. The quadruple $(i_x, \text{dom}(p_x), \pi_1 \circ p_x, \pi_2 \circ p_x)$ is a path space for x , where $\pi_1, \pi_2 : x \times x \rightarrow x$ are projections.

Indeed, we note that $s \stackrel{\text{def}}{=} i_x$ is a weak equivalence by construction. If we define:

$$d_0 \stackrel{\text{def}}{=} \pi_1 \circ p_x$$

$$d_1 \stackrel{\text{def}}{=} \pi_2 \circ p_x,$$

then $(d_0, d_1) = p_x$ and this is a fibration by construction so we're done. The last statement is clear. $\blacksquare$

Let us discuss some corollaries of the Brown Factorization Lemma. The first two results below appear in [15, Lemma 1, Lemma 2, p.422-423] with one-line indications towards proofs, so we fill in this gap.

Claim 3.8. Let $\mathcal{C}$ be a category of fibrant objects. Then, any commutative diagram:

$$\begin{array}{ccc} x & \xrightarrow{\quad} & y \\ i \downarrow & & \downarrow p \\ x' & \xrightarrow{\quad} & y' \end{array}$$

with i being a weak equivalence and p being a fibration, can be embedded in a commutative diagram:

$$\begin{array}{ccccc} x & \xrightarrow{\quad} & z & \xrightarrow{\quad} & y \\ & \searrow i & \downarrow t & & \downarrow p \\ & & x' & \xrightarrow{\quad} & y' \end{array}$$

where t is a trivial fibration. This is proved on Page 111.

Claim 3.9. Let $\mathcal{C}$ be a category of fibrant objects, let $x, y \in \mathcal{C}$, let $u : x \rightarrow y$ be a morphism and suppose that we have two path spaces $(x^I, s, d_0, d_1), (y^I, s^y, d_0^y, d_1^y)$. Then, there is a path space (x^J, s^x, d_0^x, d_1^x) , a trivial fibration $t : x^J \rightarrow x^I$ and a morphism $\bar{u} : x^J \rightarrow y^I$ such that the following diagram:

$$\begin{array}{ccccc} & & & & y \\ & & & & \downarrow s^y \\ x & \xrightarrow{s^x} & x^J & \xrightarrow{\bar{u}} & y^I \\ \downarrow s & \swarrow t & \downarrow (d_0^x, d_1^x) & & \downarrow (d_0^y, d_1^y) \\ x^I & \xrightarrow{(d_0, d_1)} & x \times x & \xrightarrow{u \times u} & y \times y \end{array}$$

commutes. This is proved on Page 111.

The next result is a neat corollary of the factorization lemma.

Corollary 3.53. Ken Brown's Lemma

Let $\mathcal{C}$ be a category of fibrant objects, let $\mathcal{D}$ be a category with a set:

$$W \subseteq \text{Mor}(\mathcal{D}),$$

satisfying the 2-out-of-3 property and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$ so that $F(W_{\mathcal{C}} \cap \text{Fib}_{\mathcal{C}}) \subseteq W$. Then, $F(W_{\mathcal{C}}) \subseteq W$.

Proof. Let $f \in \text{Mor}(\mathcal{C})$ be a weak equivalence. We write:

$$f = p \circ i,$$

where p is a trivial fibration and i is a weak equivalence. So, $F(p) \in W$ and it would be sufficient to show that $F(i) \in W$. To do this, we note that i is right-inverse to a trivial fibration q so we must have that $q \circ i = \text{Id}$. Since Id is an isomorphism, it is a trivial fibration so we see that $F(q) \circ F(i) = F(\text{Id})$ with $F(q)$ and $F(\text{Id})$ being in W . It follows that $F(i) \in W$. But this means that $F(f) \in W$ and we're done. $\blacksquare$

The next result seems to have first been proved in the C^* -algebra context in [85, Proposition 1.13, p.412-413]. However, the proof actually generalizes so that the analogue of that result holds in any category of fibrant objects.

Corollary 3.54

Let $\mathcal{C}$ be a category of fibrant objects and suppose we have a pullback diagram in $\mathcal{C}$:

$$\begin{array}{ccc} u & \xrightarrow{g_1} & x \\ g_2 \downarrow & & \downarrow f_1 \\ y & \xrightarrow{f_2} & z \end{array}$$

where f_1 is a fibration and f_2 is a weak equivalence. Then, g_1 is a weak equivalence.

Proof. Let us proceed in two steps.

1. We'll prove this result assuming that f_2 is also a fibration.
2. We'll extend the result to f_2 just being a weak equivalence.

Again, we deal with these two steps separately:

1. If we assume that f_2 is also a fibration, then we have a pullback of a trivial fibration and this means that g_1 is a trivial fibration. In particular, g_1 is a weak equivalence.
2. We factorize f_2 as $j \circ r$, with j being a fibration and r being a weak equivalence. Now, we have a diagram:

$$\begin{array}{ccccc} u & \xrightarrow{g_1} & x & & \\ & \searrow \bar{r} & \nearrow \bar{j} & & \\ & & m' & & \\ g_2 \downarrow & & \downarrow & & \downarrow f_1 \\ y & \xrightarrow{r} & m & \xrightarrow{j} & z \end{array}$$

Let us explain what's going on in the diagram:

- (a) m' is the pullback $x \times_z m$.
- (b) The maps g_1 and $r \circ g_2$ give rise to a commutative square and $\bar{r}$ is the morphism formed from the commutative square via the universal property of the pullback.

Having explained the diagram, we can now present it more cleanly:

$$\begin{array}{ccccc} u & \xrightarrow{\bar{r}} & m' & \xrightarrow{\bar{j}} & x \\ g_2 \downarrow & & \downarrow & & \downarrow f_1 \\ y & \xrightarrow{r} & m & \xrightarrow{j} & z \end{array}$$

Since f_2 is a weak equivalence and since r is a weak equivalence, we use the 2-out-of-3 property to conclude that j is a weak equivalence. So, j is a trivial fibration and we use (1) above to conclude that $\bar{j}$ is a weak equivalence. But this means that $\bar{j} \circ \bar{r} = g_1$ is a weak equivalence.

This completes the proof. ■

As a consequence, we have the following corollary.

Corollary 3.55. A. Vats, 2026

Let $\mathcal{C}$ be a category of fibrant objects. Then, it is a weak fibration category.

Proof. Since a pullback of a morphism in $W_{\mathcal{C}}$ along a morphism in Fib is in W , we see that $W_{\mathcal{C}}$ satisfies the Gluing Lemma. It follows that our triple $(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}})$ is a weak fibration subcategory. ■

The next result generalizes what has been proved for $C^* \text{Alg}$ in [85, Proposition 1.8, p.410].

Proposition 3.56. A. Vats, 2026

In the notation of Construction 3.49, TFSH:

1. f is a weak equivalence iff $\bar{f}$ is a weak equivalence.
2. A Brown factorization of f produces a Brown factorization of $\bar{f}$.
3. If there is a morphism $\phi : x \rightarrow y^I$ such that $f = d_0^y \circ \phi$, then $\bar{d}_f$ is a split epi.
4. The morphism ϕ in statement (3) exists, so $\bar{d}_f$ is always split epi.

Proof. We will prove each statement one-by-one.

1. We'll prove each implication separately.

$\Rightarrow$: This is an application of Corollary 3.54.

$\Leftarrow$: We consider the pullback diagram defining the mapping cylinder and note that $d_0^y \circ \bar{f}$ is a weak equivalence since each factor is a weak equivalence. Hence, we see that $f \circ \bar{d}_f$ is a weak equivalence and since $\bar{d}_f$ is a

weak equivalence, it follows by the 2-out-of-3 property that f is a weak equivalence.

This proves the result.

- Let us factorize f as $j \circ r$, where $j : m \rightarrow y$ is a fibration and $r : x \rightarrow m$ is a weak equivalence. As in the proof of Corollary 3.54, we can put this in a large commutative diagram:

$$\begin{array}{ccccc} \text{cyl}(f) & \xrightarrow{\bar{r}} & m' & \xrightarrow{\bar{j}} & y^I \\ \downarrow \bar{d}_f & & \downarrow & & \downarrow d_0^y \\ x & \xrightarrow{r} & m & \xrightarrow{j} & y \end{array}$$

The key observation is that the right square above is a pullback, so the entire rectangle is a pullback iff the left square is a pullback^a. Since the entire rectangle is a pullback, we see that the left square is a pullback. Now:

$$\bar{f} = \bar{j} \circ \bar{r}.$$

Since j is a fibration, we see that $\bar{j}$ is a fibration. So, the morphism in the middle of the diagram is a fibration because d_0^y is a fibration. Since r is a weak equivalence, we can use Corollary 3.54 to conclude that $\bar{r}$ is a weak equivalence. This produces a Brown factorization for $\bar{f}$ so we're done.

- The point is to write the given information into the diagram of Construction 3.49. To be precise, we have:

$$\begin{array}{ccccc} & & \phi & & \\ & \searrow & & \searrow & \\ x & & & & y^I \\ & \swarrow r & & \swarrow \bar{f} & \\ & \text{cyl}(f) & \xrightarrow{\bar{f}} & y^I & \\ & \downarrow \bar{d}_f & & \downarrow d_0^y & \\ & x & \xrightarrow{f} & y & \\ & \swarrow \text{Id}_x & & \swarrow & \end{array}$$

So, the given condition on ϕ ensures that the outer diagram above commutes. By the universal property of the pullback, we obtain a morphism $r : x \rightarrow \text{cyl}(f)$ such that the entire diagram above commutes. We see that:

$$\bar{d}_f \circ r = \text{Id}_x,$$

as was desired.

- Let $s : y \rightarrow y^I$ be the morphism into the path object which comes as part of the data of the path space. Then:

$$d_0^y \circ s = \text{Id}_y,$$

so we see that:

$$d_0^y \circ s \circ f = \text{Id}_y \circ f = f.$$

So, we simply define $\phi \stackrel{\text{def}}{=} s \circ f$ and this works, so $\bar{d}_f$ is split epi. ■

^aWe are using the pasting law for pullbacks here; see [6].

At this point, we would like to bring actual “homotopies” into the picture. As motivation, recall that in Section 2, we established that homotopies and Schochet fibrations in the realm of equivariant operator algebras can be characterized entirely in terms of morphisms, compositions and certain functors & natural transformations. For homotopies, we showed, in Proposition 2.18, having “cylinders” and “time evaluation maps” is good enough. We will, now, employ this philosophy in studying morphisms within categories of fibrant objects. Here is some reasonable terminology.

Terminology 3.57. Let $\mathcal{C}$ be a category of fibrant objects and let $f, g : x \rightarrow y$ be morphisms.

- f is right-homotopic to g iff there exists a path space (y^I, s, d_0, d_1) and a morphism $h : x \rightarrow y^I$ such that:

$$f = d_0 \circ h,$$

$$g = d_1 \circ h,$$

We denote this by writing $f \simeq_{h,r} g$ and we say that h is a right-homotopy from f to g .

- f is homotopic to g iff there exists a weak equivalence $t : z \rightarrow x$ such that:

$$f \circ t \simeq_{h,r} g \circ t.$$

We denote this by writing $f \simeq_h g$ and we say that f is homotopic to g .

Clearly, if two morphisms are right-homotopic, then they are homotopic.

The next result was essentially mentioned as a comment in [15, p.422]. Nevertheless, we furnish it with a detailed proof.

Claim 3.10. Let $\mathcal{C}$ be a category of fibrant objects, let $\gamma : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{C})$ be the localization and let $f, g \in \mathcal{C}(x, y)$ so:

$$f \simeq_{h,r} g,$$

in $\mathcal{C}$. Then, $\gamma(f) = \gamma(g)$. In particular, this is true when f and g are homotopic.

The following somewhat technical result is also essential.

Proposition 3.58. K. Brown, 1973, [15, Proposition 1, p.423-424]

Let $\mathcal{C}$ be a category of fibrant objects and let $f, g \in \mathcal{C}(x, y)$ so that:

$$f \simeq_{h,r} g.$$

TFSH:

1. If $u \in \mathcal{C}(z, x)$, then $f \circ u \simeq_{h,r} g \circ u$.
2. If $u \in \mathcal{C}(y, z)$, then there is a trivial fibration $t : x' \rightarrow x$ so that $u \circ f \circ t \simeq_{h,r} u \circ g \circ t$.

As a consequence, both $\simeq_{h,r}$ and $\simeq_h$ are equivalence relations^a on $\text{Mor}(\mathcal{C})$. Moreover, the relation $\simeq_h$ is compatible with compositions in such a way that the quotient category $\pi(\mathcal{C}) \stackrel{\text{def}}{=} \mathcal{C} / \simeq_h$ is well-defined. This category comes equipped with a canonical functor $\pi : \mathcal{C} \rightarrow \pi(\mathcal{C})$, which we will refer to as the projection.

^aFormally, what we mean is that this forms an equivalence relation on each hom-class.

With the result above in hand, we can set some notation which also doubles up as terminology.

Notation 3.59. Let $\mathcal{C}$ be a category of fibrant objects and let $f \in \text{Mor}(\mathcal{C})$. f is a homotopy equivalence in $\mathcal{C}$ iff $\pi(f)$ is an isomorphism. The full subcategory of homotopy equivalences in $\text{Arr}(\mathcal{C})$ is denoted by $W_{\mathcal{C},h}$ or W_h .

Here is a notable result.

Claim 3.11. Let $\mathcal{C}$ be a category of fibrant objects. Then, $W_{\mathcal{C},h}$ is 2-out-of-6. This is proved on Page 112.

We have a remark about this simple result.

Remark 3.60. So, even though we do not necessarily assume that the weak equivalences in a category of fibrant objects is 2-out-of-6, it is the case that it produces two subcategories of morphisms which are 2-out-of-6.

One of the most important results about categories of fibrant objects which was proved by Brown has to do with a way to control the size of the homotopy category of a category of fibrant objects. This goes back to the reason why Section 1.4 exists in this thesis. What makes this notable is that such a result is true for the homotopy category of a model category and categories of fibrant objects can be quite far from being model categories.

Theorem 3.61. K. Brown, [15, Theorem 1, p.425]

Let $\mathcal{C}$ be a category of fibrant objects. TFSH:

1. $\pi(W) \subseteq \pi(\mathcal{C})$ admits a calculus of right fractions.
2. $\forall x, y \in \mathcal{C} : [x, y]_{\mathcal{C}} \simeq \text{colim}_{x' \xrightarrow{\sim} x} \pi(\mathcal{C})(x', y)$.

Moreover, if $\gamma : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{C})$ is the localization functor, then TFSH:

1. If $x, y \in \mathbf{Ho}(\mathcal{C})$, then any morphism in $[x, y]_{\mathcal{C}}$ can be written as a right fraction:

$$x \xleftarrow{\gamma(t)} x' \xrightarrow{\gamma(f)} y$$

where $t \in W$ is a weak equivalence.

2. If $x, y \in \mathcal{C}$ and $f_0, f_1 \in \mathcal{C}(x, y)$, then $\gamma(f_0) = \gamma(f_1)$ iff f_0 and f_1 are homotopic.

Even though several terms are used in this theorem which we have not defined, this should not be viewed as a problem because we are not going to use this theorem. With that said, let us return to the fact that Proposition 3.58 is rather unappealing. Indeed, it does not allow for compatibility of right-homotopy with compositions to the highest degree. Let us fix this by taking some inspiration from our work with homotopy theory on equivariant operator algebras.

Definition 3.62. A. Vats, 2026

Let $\mathcal{C}$ be a category of fibrant objects and suppose that we have further data of:

1. a functor $T : \mathcal{C} \rightarrow \mathcal{C}$ such that for each $x \in \mathcal{C}$, $T(x)$ gives us a path space (the canonical path space) and;
2. two natural transformations $\alpha_0, \alpha_1 : T \Rightarrow \text{Id}_{\mathcal{C}}$ such that for each $x \in \mathcal{C}$, the morphisms:

$$\alpha_0^x, \alpha_1^x : T(x) \rightarrow x,$$

are the evaluation morphisms in the path space of $T(x)$,

such that for any $x, y \in \mathcal{C}$ and $f, g \in \mathcal{C}(x, y)$, the following are equivalent:

1. $f \simeq_{h,r} g$.
2. There exists a morphism $h : x \rightarrow T(y)$ such that $\alpha_0^y \circ h = f$ and $\alpha_1^y \circ h = g$.

Then, the quadruple $(\mathcal{C}, T, \alpha_0, \alpha_1)$ forms a h-category of fibrant objects^a.

^aAll constructions involving path spaces will, then, occur with respect to the canonical path space.

In such a category, we see that right-homotopy is fully compatible with compositions.

Proposition 3.63. A. Vats, 2026

Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects, let $x, y, z \in \mathcal{C}$, let $f_1, f_2 \in \mathcal{C}(x, y)$ and let $g_1, g_2 \in \mathcal{C}(y, z)$ so that:

$$f_1 \simeq_{h,r} f_2,$$

$$g_1 \simeq_{h,r} g_2.$$

Then, $g_1 \circ f_1 \simeq_{h,r} g_2 \circ f_2$.

Proof. The key is to proceed exactly as in the proof of Proposition 2.23. The first step of the argument is simply statement (1) of Proposition 3.58, while the last step is purely formal and proceeds as written. So, what we need to do is to clarify step (2). So, let $g_1 = g_2 = g$ and let $h : x \rightarrow T(y)$ be a homotopy from f_1 to f_2 . We define:

$$H \stackrel{\text{def}}{=} T(g) \circ h.$$

This is a morphism from x to $T(z)$. Now, let us show that this is a homotopy from $g \circ f_1$ to $g \circ f_2$ by a computation:

$$\begin{aligned} \alpha_i^z \circ H &= \alpha_i^z \circ T(g) \circ h \\ &= g \circ \alpha_i^y \circ h \\ &= g \circ f_{i+1}, \end{aligned}$$

for $i \in \{0, 1\}$, which completes the proof. ■

We have a remark about this result and Definition 3.62.

Remark 3.64. Looking at the proof above, we see that it is the functoriality of the path space and the naturality of the evaluation morphisms which allowed us to conclude that right-homotopies are compatible with post-composition.

It is the case that a h -category of fibrant objects admits an internal notion of homotopy which agrees with the notion of homotopy defined in concrete categories. It is also the case that such a category will admit an internal notion of fibration. To define this, we will utilize statement (2) of Claim 2.14 as inspiration.

Terminology 3.65. Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects and let $p \in \mathcal{C}(x, y)$. p is an internal fibration (int-fibration) iff for any $z \in \mathcal{C}$, any $f \in \mathcal{C}(z, x)$ and for any $h \in \mathcal{C}(z, T(y))$ which is a homotopy of $p \circ f$:

$$\alpha_0^y \circ h = p \circ f,$$

there exists a $H \in \mathcal{C}(z, T(x))$ such that the following diagram:

$$\begin{array}{ccc} & & x \\ & \nearrow f & \uparrow \alpha_0^x \\ z & \cdots \cdots H \cdots \cdots & T(x) \\ & \searrow h & \downarrow T(p) \\ & & T(y) \end{array}$$

commutes.

With a h -category of fibrant objects, we can also characterize fibrations in a different way. The discussion here was inspired by [85, Proposition 1.10, p.410-411]. We will, first of all, require a simple construction.

Construction 3.66. Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects, let $f \in \mathcal{C}(x, y)$ and consider the diagram:

$$\begin{array}{ccccc} & & & \alpha_0^x & \\ & & & \curvearrowright & \\ T(x) & & & & x \\ & \searrow v & & \bar{d}_f & \downarrow f \\ & \text{cyl}(f) & \xrightarrow{\quad} & & y \\ & \downarrow \bar{f} & & & \\ T(y) & \xrightarrow{\quad} & & \alpha_0^y & \end{array}$$

So, the key point is that we have the pullback square for the mapping cylinder and the outer diagram commutes due to the naturality of α_0 . By the universal property of the pullback, we induce a unique morphism $v : T(x) \rightarrow \text{cyl}(f)$ such that the entire diagram commutes. We refer to this as the cyl-mapcyl morphism.

We should collect some properties of internal fibrations. Actually, these properties will show that a suitable h -category of fibrant objects actually admits **another** fibration category structure.

Proposition 3.67. A. Vats, 2026

Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects so that T preserves pullbacks & zero objects. TFSH:

1. The int-fibrations are closed under composition.
2. If $x \in \mathcal{C}$, then the unique map $x \rightarrow 0$ is an int-fibration.
3. Let $f \in \text{Mor}(\mathcal{C})$ be an isomorphism. Then, it is an int-fibration.
4. Int-fibrations are preserved by base extensions.

In particular, $\mathcal{C}$ is a fibration category when equipped with its int-fibrations.

Proof. Due to the work we have already put into making the proof of Proposition 2.29 as formal as possible, so the proof of this result is simply the proof of the aforementioned proposition with the cylinder replaced with T and with the time-evaluation maps replaced with the components of α_0 and α_1 . The additional requirement on T to preserve pullbacks ensures that the proof of statement (4) in the aforementioned result also carries over here. ■

Let us characterize internal fibrations in a h -category of fibrant objects among the morphisms.

Proposition 3.68. C. Schochet, [85, Proposition 1.10, p.410-412], A. Vats, 2026

In the notation of Construction 3.66 & assuming that T preserves pullbacks, TFAE:

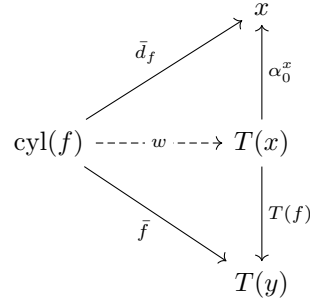
1. f is an int-fibration.
2. The cyl-mapcyl morphism v is a split epi.

Proof. We will use the notation of Construction 3.66 throughout the proof.

(1) $\Rightarrow$ (2): Observe that:

$$\alpha_0^y \circ \bar{f} = f \circ \bar{d}_f.$$

So, this means that $\bar{f}$ is a homotopy of $\bar{d} \circ f$. Now, we consider the following homotopy lifting problem:



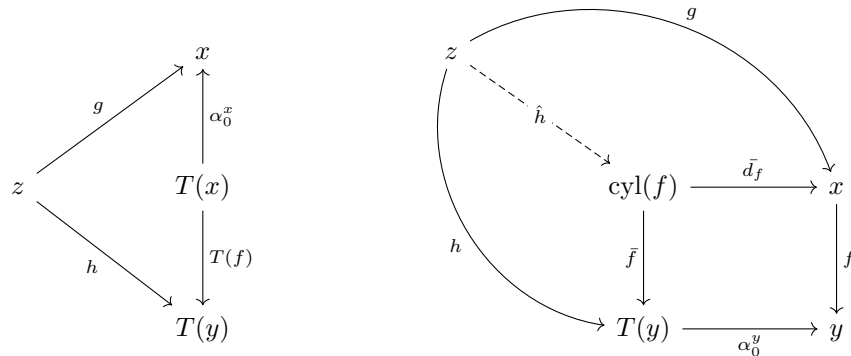
Since f is an internal fibration, we see there exists a homotopy lift $w : \text{cyl}(f) \rightarrow T(x)$. Let us check that this is, in fact, a right inverse to the map v . Observe that we have the following equations:

$$\bar{d}_f = \alpha_0^x \circ w = \bar{d}_f \circ (v \circ w),$$

$$\bar{f} = T(f) \circ w = \bar{f} \circ (v \circ w).$$

Now, we apply the universal property of the pullback to conclude that $v \circ w = \text{Id}_{\text{cyl}(f)}$.

(2) $\Rightarrow$ (1): Let $w : \text{cyl}(f) \rightarrow T(x)$ be a split for w and consider the following diagrams following lifting problem:



The diagram on the left is a lifting problem, where h is a homotopy of $f \circ g$. The diagram on the right is a pullback diagram for the mapping cylinder of f and the outer diagram commutes because h is a homotopy of $f \circ g$. By the universal property of the pullback, we have a morphism $\hat{h} : z \rightarrow \text{cyl}(f)$. We define:

$$k \stackrel{\text{def}}{=} w \circ \hat{h}.$$

This is a morphism from z to $T(x)$, so it is a candidate homotopy lift. Let's show that it works.

| **Claim 3.12.** k is the desired lift.

This is proved on Page 112. So, f is an internal fibration. ■

Let us turn towards a more general result about categories of fibrant objects which deserves some attention. Introducing this result ([Theorem 3.79](#)) requires some preparation and in fact, we will not discuss all of the terminology used within the theorem within this subsection, seeing as it is too much to cover. We will mainly use the remainder of this subsection to work towards the statement of that theorem and provide some context around it. Let us start with some terminology.

Terminology 3.69. Let $\mathcal{C}$ be a pointed category and let $f \in \mathcal{C}(x, y)$. Then, we define:

$$\text{Fibre}(f) \stackrel{\text{def}}{=} \text{Eq}(f, 0),$$

the fibre of f , if this exists. The cofibre of f , denoted by $\text{Cofibre}(f)$, is its fibre in $\mathcal{C}^{\text{op}}$.

Now, we use this to construct the loop of an object in a category of fibrant objects.

Construction 3.70. Let $\mathcal{C}$ be a category of fibrant objects and let $x \in \mathcal{C}$. The loop-object of x is the fibre:

$$\Omega(x) \stackrel{\text{def}}{=} \text{Fibre}((d_0, d_1)),$$

where (s, x^I, d_0, d_1) is a path-space of x in $\mathcal{C}$.

In fact, this construction is not functorial on the underlying category of fibrant objects in general. The next result, however, tells us that the loop can be defined as a functor on the *homotopy category* of a category of fibrant objects.

Proposition 3.71. K. Brown, [15, Theorem 3, Proposition 3, p.430-432]

Let $\mathcal{C}$ be category of fibrant objects. Then, Ω defines a functor:

$$\Omega : \mathbf{Ho}(\mathcal{C}) \rightarrow \mathbf{Ho}(\mathcal{C}),$$

called the loop-object functor. This functor satisfies the following properties:

1. If $x \in \mathcal{C}$, then $\Omega(x)$ is a group object in $\mathbf{Ho}(\mathcal{C})$ and $\Omega^2(x)$ is an abelian group object in $\mathbf{Ho}(\mathcal{C})$.
2. If $p : x \rightarrow y$ is a fibration with fibre F , then there is a natural right action:

$$F \times \Omega(y) \rightarrow F,$$

in $\mathbf{Ho}(\mathcal{C})$. In particular, we have a natural morphism $\Omega(y) \rightarrow F$ in $\mathbf{Ho}(\mathcal{C})$.

Let us introduce (abelian) group objects in a category. The idea is that the properties of (abelian) groups can be represented by the commutativity of some diagrams and this can be mimicked in a fairly general way within a category. Our treatment of these notions is based on the excellent exposition in [33, Part I, Section 2, p.3-4].

Terminology 3.72. Let $\mathcal{C}$ be a pointed category^a with finite products. A group object in $\mathcal{C}$ is a quadruple:

$$(G, m, e, i),$$

consisting of $G \in \mathcal{C}$ and three morphisms:

$$m : G \times G \rightarrow G,$$

$$e : 1 \rightarrow G,$$

$$i : G \rightarrow G,$$

such that the following diagrams:

$$\begin{array}{ccc} G \times (G \times G) & \xrightarrow{\cong} & (G \times G) \times G \xrightarrow{m \times \text{Id}_G} G \times G \\ \text{Id}_G \times m \downarrow & & \downarrow m \\ G \times G & \xrightarrow{m} & G \end{array}$$

$$\begin{array}{ccccc} 1 \times G & \xrightarrow{e \times \text{Id}_G} & G \times G & \xleftarrow{\text{Id}_G \times e} & G \times 1 \\ & \searrow \cong & \downarrow m & \swarrow \cong & \\ & & G & & \end{array}$$

$$\begin{array}{ccc} G & \xrightarrow{(1, i)} & G \times G \\ \downarrow & & \downarrow m \\ 1 & \xrightarrow{e} & G \end{array}$$

commute. This group object is an abelian group object iff the following diagram:

$$\begin{array}{ccc} G \times G & \xrightarrow{T} & G \times G \\ & \searrow m & \downarrow m \\ & & G \end{array}$$

commutes, where T is the interchange map^b.

^aThe zero object is denoted by 1.

^bThis is constructed by the universal property of the product on the canonical projections.

Of course, the (abelian) group objects in $\mathbf{Set}$ are just (abelian) groups, because the diagrams above precisely encode the defining axioms of (abelian) groups. Now, we should also explain the notion of an action of a group object on another object in a category. Again, group actions of usual groups on sets are encoded by maps and equations, the latter of which are encoded by diagrams. So, we simply adapt those diagrams to a category.

Terminology 3.73. Let $\mathcal{C}$ be a category with finite products and a final object 1 , let G be a group object in $\mathcal{C}$ and let $x \in \mathcal{C}$. A (right) action of G on x is a map $\nu : x \times G \rightarrow x$ such that the following diagram:

$$\begin{array}{ccccccc} (x \times G) \times G & \xrightarrow{\cong} & x \times (G \times G) & \xrightarrow{\text{Id}_x \times m} & x \times G & \xleftarrow{\text{Id}_x \times e} & x \times 1 \\ \downarrow \nu \times \text{Id}_G & & & & \downarrow \nu & \nearrow \cong & \\ x \times G & \xrightarrow{\nu} & x & & & & \end{array}$$

commutes.

Now, there is a classical result of model category theory which says that if $\mathcal{M}$ is a model category and $x, y \in \mathcal{M}$, then $[y, \Omega(x)]$ has the structure of a group and $[y, \Omega^2(x)]$ has the structure of an abelian group. In that setting, this result is difficult to prove. For instance, it is essentially shown within [45, Corollary 6.1.6, p.151] and makes use of the full structure available on model categories. However, it seems that a significant simplification of that proof is available and it just uses Proposition 3.71 and a formal characterization of (abelian) group objects. Let us sketch the latter first.

Proposition 3.74

Let $\mathcal{C}$ be a locally small category with a final object and have finite products & let $x \in \mathcal{C}$. TFAE:

1. x has a group structure in $\mathcal{C}$.
2. The functor $\mathcal{C}(-, x)$ has a group structure in $\mathbf{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Set})$.

If either (1) or (2) holds, then x is an abelian group object iff $\mathcal{C}(-, x)$ is an abelian group object in $\mathbf{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Set})$.

Proof Sketch. This is explained in [59, Proposition 1, p.75-76]. However, the basic idea is that we can translate between morphisms in $\mathcal{C}$ and morphisms between hom-sets by using the Yoneda Lemma. For example, let $m : x \times x \rightarrow x$ be the multiplication morphism if x is a group object and let $y \in \mathcal{C}$. Then, we have the following composition:

$$\mathcal{C}(y, x) \times \mathcal{C}(y, x) \rightarrow \mathcal{C}(y, x \times x) \rightarrow \mathcal{C}(y, x),$$

where the first map is a natural isomorphism because representable functors preserve products while the second map is obtained by applying $\mathcal{C}(y, -)$ to the morphism m . So, the above forms a “multiplication” map for the hom-set $\mathcal{C}(y, x)$ and one checks that this multiplication actually satisfies the relevant properties. Going backwards, suppose that $\mathcal{C}(y, x)$ is a group for each $y \in \mathcal{C}$. So, this has a multiplication map denoted as follows:

$$\bar{m} : \mathcal{C}(y, x) \times \mathcal{C}(y, x) \rightarrow \mathcal{C}(y, x).$$

We precompose this with the natural isomorphism $p : \mathcal{C}(y, x) \times \mathcal{C}(y, x) \simeq \mathcal{C}(y, x \times x)$ and we get:

$$\bar{m} \circ p : \mathcal{C}(y, x \times x) \rightarrow \mathcal{C}(y, x).$$

Due to the naturality of $\bar{m}$ and the naturality of p , we see that the above is a component of a natural transformation $\mathcal{C}(-, x \times x) \Rightarrow \mathcal{C}(-, x)$. Now, we apply the Yoneda Lemma and note that this natural transformation corresponds precisely to a morphism in $\mathcal{C}(x \times x, x)$ and this is our multiplication map. ■

Now, we can state and prove our result.

Theorem 3.75

Let $\mathcal{C}$ be a category of fibrant objects and let $x, y \in \mathcal{C}$.

1. $[y, \Omega(x)]$ is a group.
2. $[y, \Omega^2(x)]$ is an abelian group.

Proof. Observe that $\mathbf{Ho}(\mathcal{C})$ is a locally small category with a final object and admits finite products. Since $\Omega(x)$ is a group object in $\mathbf{Ho}(\mathcal{C})$, we see that the functor $\mathbf{Ho}(\mathcal{C})(-, \Omega(x))$ is a group object in $\mathbf{Fun}(\mathbf{Ho}(\mathcal{C})^{\text{op}}, \mathbf{Set})$. So, $\mathbf{Ho}(\mathcal{C})(y, \Omega(x)) = [y, \Omega(x)]$ is a group, which proves statement (1). For statement (2), the argument is precisely analogous, except that we use the fact that $\Omega^2(x)$ is an abelian group object in $\mathbf{Ho}(\mathcal{C})$. ■

As is the case with homotopy theory on spaces, there is a long-exact sequence obtained from a fibration.

Proposition 3.76. Barratt-Puppe Fibration Sequence, [76, Proposition 1.3.3-1.3.4, p.42-46]

Let $\mathcal{C}$ be a category of fibrant objects and let $p : x \rightarrow y$ be a fibration with fibre F . There is an exact sequence:

$$\dots \rightarrow [z, \Omega^2(y)]_{\mathcal{C}} \rightarrow [z, \Omega(F)]_{\mathcal{C}} \rightarrow [z, \Omega(x)]_{\mathcal{C}} \rightarrow [z, \Omega(y)]_{\mathcal{C}} \rightarrow [z, F]_{\mathcal{C}} \rightarrow [z, x]_{\mathcal{C}} \rightarrow [z, y]_{\mathcal{C}},$$

for any $z \in \mathcal{C}$.

We have a remark about this proposition.

Remark 3.77. We need to explain what exactness means in the context above, because the first three terms of the sequence are simply pointed sets. Let us define exactness here. Consider the following sequence of pointed sets:

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

The sequence is exact at Y iff:

$$\text{ran}(f) = \ker(g) \stackrel{\text{def}}{=} g^{\leftarrow}(\{z\}),$$

where $z \in Z$ is the basepoint. In the case of groups and abelian groups, these come equipped with a natural basepoint (the identity) and one recovers the natural notion of exactness in those cases by using what we have defined above. Of course, we should also mention that for the latter maps which appear in the fibration sequence of Proposition 3.76, it is desirable to have them be group homomorphisms and, in fact, it turns out that they are group homomorphisms.

With the results above, we can construct the stable homotopy category of a category of fibrant objects. We note that this is the main category of interest in [Theorem 3.79](#) and we take its construction from [\[92, Definition 1.23, p.5-6\]](#).

Construction 3.78. Let $\mathcal{C}$ be a category of fibrant objects. We define the stable homotopy category of $\mathcal{C}$:

$$\mathbf{SHo}(\mathcal{C}) \stackrel{\text{def}}{=} \mathbf{Ho}(\mathcal{C})[\Omega^{-1}],$$

obtained from $\mathbf{Ho}(\mathcal{C})$ by inverting the endofunctor Ω . To be precise:

1. The objects are pairs:

$$(x, n),$$

where $x \in \mathbf{Ho}(\mathcal{C})$ and $n \in \mathbb{Z}$.

2. If $(x, n), (y, m) \in \mathbf{SHo}(\mathcal{C})$, then we define:

$$\mathbf{SHo}(\mathcal{C})((x, n), (y, m)) \stackrel{\text{def}}{=} \text{colim}_{k \rightarrow \infty} [\Omega^{n+k}(x), \Omega^{m+k}(y)]_{\mathcal{C}},$$

where the colimit on the right-hand side is taken in the category of sets.

For $n \in \mathbb{Z}$, we have a natural functor:

$$\Omega^n : \mathbf{Ho}(\mathcal{C}) \rightarrow \mathbf{SHo}(\mathcal{C}),$$

which takes an object $x \in \mathbf{Ho}(\mathcal{C})$ and maps it to the pair (x, n) .

This leads us into the major theorem that we wanted to discuss.

Theorem 3.79. The Spanier-Whitehead Construction

Let $\mathcal{C}$ be a category of fibrant objects. Then, the stable homotopy category:

$$\mathbf{SHo}(\mathcal{C}),$$

is a triangulated category with shift:

$$\Sigma = \Omega^{-1} : \mathbf{SHo}(\mathcal{C}) \rightarrow \mathbf{SHo}(\mathcal{C}),$$

given by $(x, n) \mapsto (x, n - 1)$ and the exact triangles given by triangles isomorphic to triangles of the form:

$$(\Omega(y), n) \rightarrow (F, n) \rightarrow (x, n) \rightarrow (y, n),$$

where $n \in \mathbb{Z}$ and $x \rightarrow y$ a fibration, $F \rightarrow x$ a fibre inclusion & $\Omega(y) \rightarrow F$ is the morphism in Proposition 3.71.

This “result” is interesting for a few different reasons and we will expand on these reasons in the next few remarks. The first of these focuses on the history of triangulated categories; this type of category appears in the statement above and we will introduce them in Section 4. Nevertheless, it is good to get a bird’s eye view of the role that they have played in mathematics and, in particular, operator algebra theory.

Remark 3.80. We will introduce triangulated categories in Section 4.2 but it is good to give a basic overview of their history & the specific ways in which they come up naturally in the study of operator algebraic structures. Of course, it is impossible (and also inappropriate) for us to give a full historical account of these topics when such accounts do exist in the literature. For a detailed dive into the classical origins of triangulated categories, see [\[95\]](#). The important point is that this theory was independently developed in the 1960s by (Dieter) Puppe (with a focus towards understanding the classical stable homotopy category) and (Alexander) Grothendieck & (Jean-Louis) Verdier (with a focus towards understanding derived categories of abelian categories). Indeed, these were the initial, classical examples of triangulated categories that have been studied extensively.

Since then, this theory has found applications in many other parts of mathematics such as operator algebras, modular representation theory, symplectic geometry and many more. To our knowledge, the PhD thesis of Andreas Thom, [\[91\]](#), used triangulated categories for the first time in relation to operator algebras. Thom uses these to study bivariant homology theories^a on ${}_{\omega}\mathbf{C}^*\mathbf{Alg}$ and shows that stable homotopy for separable C^* -algebras is a triangulated homology theory. Since then, it has become known that Kasparov’s KK-theory and Higson’s E-theory both carry a natural triangulated structure (essentially through [\[63\]](#)) & this has played a role in recent computations of K-theory for certain Hecke C^* -algebras; see [\[77\]](#). This does not summarize all of the different ways in which this theory has interacted with operator algebras but it should give the reader a nice set of reasons to engage with this framework.

^aKasparov’s KK-Theory is an example and Connes-Higson’s E-Theory is another major example.

In the next remark, we will discuss some concrete (operator-algebraic) applications of the Spanier-Whitehead construction.

Remark 3.81. The Spanier-Whitehead Construction is a classical construction of stable homotopy theory. In operator algebra theory, we are aware of two direct applications of this construction. We will describe them below:

1. In the usual (equivariant) Kasparov KK-category, it turns out that the suspension functor (the shift) is not an automorphism but an auto-equivalence. So, the Spanier-Whitehead construction is applied to this category to produce a slightly modified category where the suspension is actually an automorphism; this is discussed in [63, p.7-8]. One might think that this is a technical point but it is rather necessary for certain K-theory computations. For instance, the K-theory computations of certain right-angled Hecke C^* -algebras in [77, Proposition 3.1, p.7-8] uses this triangulated structure to great effect to show that a certain map is a KK-equivalence.
2. The construction of KK-theory for Banach algebras in [70]. The Spanier-Whitehead construction is used as an intermediate step in a localization-centric construction of Banach KK-theory. However, we wish to emphasize that the approach used there is technically quite different from the one here. That approach uses the Spanier-Whitehead construction as concocted in [32, Appendix A, p.125-148], which is centered around working with *admissible categories of algebras*^a. Indeed, these are shown to have their own well-behaved homotopy theory and it is not completely clear that this homotopy theory is captured by a model structure or even the structure of a category of fibrant objects. Nevertheless, the category produced in Construction A.4.1 (p.134-135) is the same as the one in Construction 3.78, at least on the level of categories.

Of course, this also does not include the role played by this construction in [92]. In that case, the result is used to provide as a basis for an abstract treatment of homology theories defined on the stable homotopy category & it is eventually used to show, in Theorem 1.35 (p.7) of the aforementioned paper, that a certain stable homotopy category of a localization is triangulated equivalent to the Verdier localization. The main point is that this result is used to show that the triangulated categories associated to various bivariant homology theories like KK-theory or E-theory can be recovered from these more abstract results.

^aThis is **not** the notion of admissible subcategory we will introduce later.

Nevertheless, there seems to be a problem with Theorem 3.79 as it is currently stated and we will spend the rest of this subsection essentially discussing some of these issues. Let us outline the main problems in the next remark.

Remark 3.82. This theorem was stated in the form above in [92, Theorem 1.24, p.6] and as we stated in Remark 3.81, it seems to be one of the main results of the paper. Nevertheless, it is not furnished with an explicit proof. The author seems to simply recourse to citing three different resources, with only one seeming to be remotely useful in recovering a full proof of this result. This paper that we refer to is (Alex) Heller’s paper entitled “Stable Homotopy Categories” and it seems to be one of the earliest papers which attempted to generalize the Spanier-Whitehead construction to a wider range of categories. It seems to be the case that Uuye obtained the idea that Theorem 3.79 might be true specifically from [41, Section 9, p.42-43], where something similar does hold for the categories which Heller considers. Nevertheless, there are a few important issues that one must keep in mind:

1. It is not clear that categories of fibrant objects are examples of categories under consideration by Heller. In fact, we will show that the (dualized versions of the) categories of Heller are actually categories of fibrant objects but this may not be true the other way around.
2. The result which Heller shows, more explicitly, is that his construction of the stable homotopy category is a *pre-triangulated category* (see Definition 4.39). Indeed, Heller explicitly omits the Octahedral Axiom (see Definition 4.47) from his definition of a triangulated structure on a category. In fact, this is typically the most difficult axiom to verify among the axioms defining a triangulated category, so this is a non-trivial omission.
3. The existence of a *canonical* triangulated structure on the stable homotopy category can be shown and we will certainly do it later. Nevertheless, the second part of Theorem 3.79 is somewhat perplexing, because it is not clear how one can show that all exact triangles are just given by fiber sequences of fibrations. This might be reasonable if we had Brown factorization but we do not have it in the stable homotopy category.

There is evidence which suggests that the first part of Theorem 3.79 should be true. A variant of that result is true for pointed model categories, as is shown by Ivo Dell’Ambrogio in [33]. His strategy is described as follows:

1. He defines the Spanier-Whitehead construction for a pair:

$$(\mathcal{C}, \Sigma),$$

of a category $\mathcal{C}$ and an endofunctor $\Sigma : \mathcal{C} \rightarrow \mathcal{C}$. He shows that if this pair satisfies some list of axioms, then the Spanier-Whitehead construction yields a triangulated structure.

2. He shows, then, that the homotopy category of a pointed model category equipped with its suspension satisfies the list of axioms above. This is actually the more difficult step, as he uses the explicit fact that the homotopy category of a pointed model category is equipped with a closed (right) action of the homotopy category of pointed simplicial sets with its standard model structure (see [5]). Indeed, we were not able to prove this for categories of fibrant objects and there is no indication that this should be true^a.

One might hope to deduce the theorem from the one for pointed model categories but the natural way to do this would be to find an “enveloping pointed model category” of our category of fibrant objects. However, this requires that the objects be different, because there are results (like Theorem 3.22) which suggest that such an enveloping category should not exist without further conditions. So, it seems that the classical strategies do not work.

^aDell’Ambrogio mentions on [33, p.36] that “any attempt to avoid [the closedness of the action above] failed miserably”. So, this seems to be an essential technical difficulty that must be overcome if one proceeds by that approach.

Since Uuye seems to take inspiration from Heller’s work, it seems reasonable to explicitly address where this could potentially go wrong. Some of the open questions we propose in Section 5.2 could be relevant in resolving the problems we outline. To prove his theorem involving the stable homotopy category, Heller proceeds via the notion of a *h-c-category*; this stands for a “homotopy-cofibration category” and it is meant to axiomatize the properties of homotopy. In our case, we will dualize those axioms and we will give a brief argument (in Remark 3.85) for why the axioms of a category of fibrant objects do not satisfy the axioms of Heller. Before we do this, we need the following result.

Proposition 3.83

Let $\mathcal{C}$ be a cofibration category^a. Then, $\mathcal{C}$ is a category with a system of cofibrations in the sense of Heller^b.

^aRecall Definition 3.25.

^bSee [41, p.34].

Proof. This follows literally by dualizing the axioms of a fibration category and comparing them to the axioms C0 and C1 as laid out in Heller’s paper cited above. In particular, one needs the fact that cofibrations are preserved by pushouts in a cofibration category but this follows because the pullback-analogue holds in a fibration category so dualizing that axiom will give precisely the pushout-analogue which is exactly what we require. ■

With this, we can appropriately dualize the axioms of Heller.

Definition 3.84

Let $\mathcal{C}$ be a fibration category equipped with an equivalence relation $\simeq$ on all morphism classes. We define:

$$\mathcal{C}_{\square} \stackrel{\text{def}}{=} \mathcal{C} / \simeq,$$

and set aside some terminology:

1. A morphism in $\mathcal{C}$ is a homotopy-equivalence iff its image in $\mathcal{C}_{\square}$ is an isomorphism.
2. If $x \in \mathcal{C}$, then x is contractible iff the morphism $x \rightarrow 0$ is a homotopy equivalence.
3. A morphism in $\mathcal{C}$ is a hom-fibration^a iff it is a fibration and a homotopy equivalence.

Now, suppose that $\simeq$ satisfied the following axioms:

1. If $x, y, z \in \mathcal{C}$ and $f, f' \in \mathcal{C}(z, x)$ and $g, g' \in \mathcal{C}(z, y)$ so that:

$$f \simeq f',$$

$$g \simeq g',$$

then the natural maps $(f, g) \simeq (f', g')$, where these morphisms are defined from^b $z \rightarrow x \times y$.

2. If we have a morphism (f', f, f'') in $\text{Fib}_{\mathcal{C}}$ and there is a morphism g'' in $\mathcal{C}$ such that $f'' \simeq g''$, then there exist morphisms g', g in $\mathcal{C}$ such that $g' \simeq f'$ and $g \simeq f$, with the triple (g', g, g'') being a morphism in $\text{Fib}_{\mathcal{C}}$.
3. Hom-fibrations are preserved by base extensions.
4. For every morphism f in $\mathcal{C}$, there is a factorization:

$$f = p \circ i,$$

where p is a fibration and i is a homotopy equivalence.

This is known as a h-f-category^c. The set of all homotopy equivalences is denoted by W_h .

^aThe reason for this name is that Heller uses the term “homotopy cofibration” for something else in his paper, so we avoid using the obvious analogue of that piece of terminology just to prevent confusion. Also, we do not use “trivial fibration” for obvious reasons.

^bThis makes sense because fibration categories admit finite products.

^cThat is, a “homotopy-fibration-category”.

In the next remark, we will explain why Heller’s formalism seems to not be sufficient for proving our theorem.

Remark 3.85. The explanation ultimately boils down to the fact that not all categories of fibrant objects are h-f categories. Indeed, the right way to approach the above would be to use the internal notion of homotopy which is available in a category of fibrant objects and to show that we have a h-f category with respect to these homotopy equivalences & fibrations. However, it is not clear that we should expect the homotopy equivalences to behave well enough with respect to the fibrations so that all of the axioms above are satisfied. Indeed, it is not clear that the factorization axiom would be satisfied, for example. Even if we switch to the weak equivalences, it is not clear that the path space object would behave well with respect to products; this would be needed to verify Axiom (1) of Definition 3.84. Finally, Heller’s paper does not even show that the Spanier-Whitehead construction applied to h-f categories gives triangulated categories; he shows that it gives a pre-triangulated category and establishing the octahedral axiom does not seem to be an easy task. It seems to not be the case that proceeding via h-f categories will give us the desired result, even if it might be desirable for us to obtain our construction in that way.

On the other hand, we can easily show that h-f categories are categories of fibrant objects. So, one may reasonably wonder when it is the case that a category of fibrant objects, with its internal notion of homotopy equivalence, becomes a h-f-category. It seems that one might hope that this happens when the weak equivalences are exactly the homotopy equivalences but we do not know if there are conditions where this can happen. We discuss this further in Section 5.2.

Proposition 3.86. A. Vats, 2026

Let $\mathcal{C}$ be a h-f category. Then, $\mathcal{C}$ is a category of fibrant objects where:

1. the weak equivalences are homotopy equivalences and;
2. the fibrations are just the given one on $\mathcal{C}$ as a h-f category.

In fact, only axioms (3) and (4) are needed to ensure we have the structure of a category of fibrant objects. Moreover, the full subcategory W_h actually satisfies the 2-out-of-6 property.

Proof. Let's start by noting that the homotopy equivalences do form a full subcategory because functors map identity morphisms to identity morphisms and isomorphisms to isomorphisms. In particular, the canonical functor from $\mathcal{C}$ to $\mathcal{C}_\square$ does this so we do have full subcategories of $\text{Arr}(\mathcal{C})$ in our hands. Now, we will show that the axioms listed in [Definition 3.39](#) are satisfied. We will do this by verifying them one-by-one.

1. By definition, a h-f category is a fibration category.
2. Again, isomorphisms are sent to isomorphisms by functors so an isomorphism is a homotopy equivalence.
3. In fact, the 2-out-of-3 property will follow if we show the 2-out-of-6 property; this is just statement (2) of [Claim 3.2](#). On the other hand, the 2-out-of-6 property follows from [Claim 3.4](#).
4. This follows because hom-fibrations are preserved by base extensions.
5. This follows from [Proposition 3.52](#).

This proves the result. ■

Our lengthy discussion here should convince the reader that there is a gap in the literature in relation to [Theorem 3.79](#). Given that this construction is used in several places of interest in operator algebra theory, it seems reasonable to attempt to fix it. It will also become part of our list of results which are true for pointed model categories and also hold for categories of fibrant objects. We intend to resolve this issue, among other issues from [Section 2](#), in the next section.

4 Admissibility of Subcategories & The Spanier-Whitehead Construction

Summary

In this section, we address some remaining questions which did not fit into the exposition of Sections 2 and 3. Section 4.1 focuses on studying the notion of an admissible subcategory, which we define, of the category of C^* -algebras and showing that this is the correct notion by applying it to obtain a more refined proof of the Bott Periodicity Theorem as well as other results in the axomatic (co)homology theory of C^* -algebras. We also establish a slightly weaker version of the Spanier-Whitehead Construction in Section 4.2 for categories of fibrant objects using the methods of higher category theory, thereby partially fixing a gap in the literature.

4.1 Admissible Subcategories & Applications Towards Axiomatic (Co)Homology

We will start with the following proposition which is known in the non-equivariant case for C^* -algebras.

Proposition 4.1. A. Vats (2026)

Let $G \in \text{Grp}(\text{LCH})$ and let $\mathcal{C}$ be either $G\text{-}C^*\text{Alg}$ or $G\text{-}Ban\text{Alg}$ and define:

$$W \stackrel{\text{def}}{=} \{\text{homotopy equivalences in } \mathcal{C}\},$$

$$\text{Fib} \stackrel{\text{def}}{=} \{\text{Schochet fibrations in } \mathcal{C}\}.$$

Then, $(\mathcal{C}, W, \text{Fib})$ forms a Brown category.

Proof. We just need to verify the axioms one-by-one, in the order presented within Definition 3.39.

1. $\mathcal{C}$ has a final object; it's just $\{0\}$.
2. $(\mathcal{C}, \text{Fib})$ is a fibration category, as established in Example 3.29.
3. Every isomorphism in $\mathcal{C}$ is in W by Claim 2.10.
4. W satisfies the 2-out-of-3 property by statement (3) of Claim 2.13.
5. If $A \in \mathcal{C}$, then we can easily specify a quadruple:

$$(s, \text{Cyl}_G(A), p_0^A, p_1^A),$$

where $s : A \rightarrow \text{Cyl}_G(A)$ is the inclusion as constant functions & we could try to show that this is a path space for A . However, this seems to be difficult and, in fact, Uuye does not seem to give a complete argument for this as well; see Remark 4.2. As a consequence, we will use (2) $\Rightarrow$ (1) of Proposition 3.52 to deduce the existence of a path space for any $A \in \mathcal{C}$. That is, let us construct a factorization of any morphism in $\mathcal{C}$ and we do this by taking inspiration from the proof of the Brown Factorization Lemma. If $A, B \in \mathcal{C}$ and $f \in \mathcal{C}(A, B)$, then its mapping cylinder (with respect to Cyl_G) is still defined as in Construction 3.49:

$$\begin{array}{ccc} \text{cyl}(f) & \xrightarrow{\bar{f}} & \text{Cyl}_G(B) \\ \bar{d}_f \downarrow & & \downarrow p_0^B \\ A & \xrightarrow{f} & B \end{array}$$

Now, we define:

$$p \stackrel{\text{def}}{=} p_1^B \circ \bar{f}.$$

This will be our Schochet fibration; we will show this later. To construct the homotopy equivalence $i : A \rightarrow \text{cyl}(f)$ such that $p \circ i = f$, we will use the universal property of the pullback. To be precise, we define a map:

$$s : B \rightarrow \text{Cyl}_G(B), \quad b \mapsto s(b),$$

where $s(b)$ is the constant function on $[0, 1]$ with value b . This fits into the following diagram:

$$\begin{array}{ccccc} & & & s \circ f & \\ & & & \searrow & \\ A & & & & \text{Cyl}_G(B) \\ & \searrow i & & \bar{f} & \\ & \text{cyl}(f) & \xrightarrow{\bar{f}} & & \\ & \bar{d}_f \downarrow & & \downarrow p_0^B & \\ & A & \xrightarrow{f} & B & \\ & \text{Id}_A \nearrow & & & \end{array}$$

where the solid part of the diagram commutes; this follows from the definition of s . The universal property of the pullback induces a unique morphism $i : A \rightarrow \text{cyl}(f)$ such that the entire diagram commutes. So, we have:

$$\begin{aligned} p \circ i &= (p_1^B \circ \bar{f}) \circ i \\ &= p_1^B \circ (\bar{f} \circ i) \\ &= p_1^B \circ (s \circ f) \\ &= (p_1^B \circ s) \circ f \\ &= \text{Id}_B \circ f \\ &= f, \end{aligned}$$

where the fact that $p_1^B \circ s$ can be deduced quickly from the explicit definitions of p_1^B and s . So, it remains for us to show that p is a Schochet fibration and i is a homotopy equivalence.

- (a) Let us show that i is a homotopy equivalence. Observe that:

$$\bar{d}_f \circ i = \text{Id}_A,$$

since the diagram above commutes. To find a right homotopy inverse, let us make $\text{cyl}(f)$ more explicit:

$$\text{cyl}(f) = \{(\phi, a) \in \text{Cyl}_G(B) \times A : \phi(0) = f(a)\}.$$

Using this, we can deduce the exact formula for i from the commuting diagram as follows:

$$\forall a \in A : i(a) = (s(f(a)), a).$$

Let us consider $i \circ \bar{d}_f$ as follows:

$$\begin{aligned} \forall (\phi, a) \in \text{cyl}(f) : (i \circ \bar{d}_f)(\phi, a) &= i(\bar{d}_f(\phi, a)) \\ &= i(a) \\ &= (s(f(a)), a) \\ &= (s(\phi(0)), a). \end{aligned}$$

This motivates us to define a homotopy $h : \text{cyl}(f) \rightarrow \text{Cyl}_G(\text{cyl}(f))$ as we will do now, so let $(\phi, a) \in \text{cyl}(f)$. For each $s \in [0, 1]$, we can define the following function:

$$\phi_s : [0, 1] \rightarrow A, \quad t \mapsto \phi_s(t) \stackrel{\text{def}}{=} \phi(st).$$

Since ϕ is continuous, we see that ϕ_s is continuous. Moreover:

$$\phi_s(0) = \phi(0) = f(a),$$

which means that $(\phi_s, a) \in \text{cyl}(f)$ for each $s \in [0, 1]$. Now, we consider the map:

$$h(\phi, a) : [0, 1] \rightarrow \text{cyl}(f), \quad s \mapsto h(\phi, a)(s) \stackrel{\text{def}}{=} (\phi_s, a).$$

We have to argue that this is continuous. If it is continuous, then we can form a homotopy:

$$h : \text{cyl}(f) \rightarrow \text{Cyl}_G(\text{cyl}(f)),$$

which will be a $\star$ -homomorphism by direct computation. More importantly, it does what we need it to do because if $s = 0$, then this gives us $i \circ \bar{d}_f$ and when $s = 1$, this gives us $\text{Id}_{\text{cyl}(f)}$. So, it remains to argue that $h(\phi, a)$ is continuous. This follows from the following estimate:

$$\begin{aligned} \forall s, s' \in [0, 1] : \|h(\phi, a)(s) - h(\phi, a)(s')\| &= \|(\phi_s, a) - (\phi_{s'}, a)\| \\ &= \|(\phi_s - \phi_{s'}, 0)\| \\ &= \sup_{t \in [0, 1]} \|\phi(st) - \phi(s't)\|. \end{aligned}$$

Next, we observe that $|st - s't| = |s - s'| \cdot |t| \leq |s - s'|$. As a consequence, if we have s and s' close enough, then st is close enough to $s't$ and this implies, by the continuity of ϕ , that $h(\phi, a)$ is continuous. We want to explicitly note that all morphisms formed above are equivariant because they are formed from equivariant morphisms. So, this proves that the map i is a homotopy equivalence.

- (b) Let us prove that p is a Schochet fibration. Let us consider the following lifting problem:

$$\begin{array}{ccc} & & \text{cyl}(f) \\ & \nearrow k & \uparrow p_0^{\text{cyl}(f)} \\ D & & \text{Cyl}_G(\text{cyl}(f)) \\ & \searrow h & \downarrow \text{Cyl}_G(p) \\ & & \text{Cyl}_G(B) \end{array}$$

where h is a homotopy of $p \circ k$. Since $\text{cyl}(f) \subseteq \text{Cyl}_G(B) \times A$, we can project k down to each component and write it as a pair $k = (k', k'')$. Let us record that this pair satisfies the following property:

$$\forall d \in D : k'(d)(0) = f(k''(d)).$$

So, we will define a homotopy lift H into $\text{Cyl}_G(\text{Cyl}_G(B) \times A)$ and show that this is contained in $\text{Cyl}_G(\text{cyl}(f))$. Since Cyl_G preserves pullbacks, we will conveniently identify $\text{Cyl}_G(\text{Cyl}_G(B) \times A)$ with $\text{Cyl}_G(\text{Cyl}_G(B)) \times \text{Cyl}_G(A)$. For any function into this algebra, we simply recover the situation in $\text{Cyl}_G(B) \times A$ by evaluating that function additionally at a pair $(s, t) \in [0, 1] \times [0, 1]$; this will be illustrated below. We define:

$$\forall d \in D : \forall (s, t) \in [0, 1] \times [0, 1] : h'(d)(s, t) \stackrel{\text{def}}{=} \begin{cases} k'(d)\left(\frac{2s}{2-t}\right) & \text{if } 2s \leq 2-t, \\ h(d)(2s+t-2) & \text{if } 2s > 2-t \end{cases}.$$

If we fix d and s , then this defines a map $[0, 1] \rightarrow B$ and this is continuous everywhere except for possibly at $t = 2 - 2s$. At that point, the right-hand limit exists and it is equal to $k'(d)(1)$, while the left-hand limit is equal to $h(d)(0)$. But we see that $h(d)(0) = p(k(d))$ and we note that this is just $k'(d)(1)$. Since the two one-sided limits coincide, we see that the map $[0, 1] \rightarrow B$ is continuous. For similar reasons, we have continuity when we fix d so that $h'(d) \in \text{Cyl}_G(\text{Cyl}_G(B))$ and since k' & h are equivariant $\star$ -homomorphisms,

we see that h' is an equivariant $\star$ -homomorphism. Next, we define a morphism from A to $\text{Cyl}_G(A)$:

$$h'' \stackrel{\text{def}}{=} s^A \circ k'',$$

where we now use the morphism $s^A : A \rightarrow \text{Cyl}_G(A)$ which takes an element of A to the constant function on $[0, 1]$ with the value a . We observe that h'' is an equivariant $\star$ -homomorphism. We define:

$$H \stackrel{\text{def}}{=} (h', h''),$$

and this is an equivariant $\star$ -homomorphism from D to $\text{Cyl}_G(\text{Cyl}_G(B)) \times \text{Cyl}_G(A)$. Again, we specify values in $\text{Cyl}_G(B) \times A$ by evaluating H as $H(-)(-, -)$, where it is the second variable in time which controls the homotopy. Now, we need to check that this lies in the mapping cylinder of f and that this is a homotopy of k . Let $d \in D$ and $s, t \in [0, 1]$. Then, we want to show that:

$$H(d)(s, t) = (h'(d)(s, t), h''(d)(t)) \in \text{cyl}(f).$$

Let $t = 0$. In that case, we see that:

$$H(d)(s, 0) = (h'(d)(s, 0), h''(d)(0)) = (k'(d)(s), k''(d)) = k(d)$$

So, H is a homotopy of k . Next, we let s, t be arbitrary. Then, we have the following computation:

$$\begin{aligned} f(h''(d)(t)) &= f(k''(d)) \\ &= k'(d)(0) \\ &= h'(d)(0, t), \end{aligned}$$

which proves that we are in the mapping cylinder throughout the homotopy. Finally, we will show that:

$$\text{Cyl}_G(p) \circ H = h.$$

So, let $d \in D$ and observe that:

$$\begin{aligned} \forall t \in [0, 1] : h(d)(t) &= h(d)(2 \cdot 1 + t - 2) \\ &= h'(d)(1, t). \end{aligned}$$

On the other hand, we have that:

$$\begin{aligned} \forall t \in [0, 1] : (\text{Cyl}_G(p) \circ H)(d)(t) &= (p \circ H(d))(t) \\ &= (p_1^B \circ \bar{d}_f \circ H(d))(t) \\ &= h'(d)(1, -)(t) \\ &= h'(d)(1, t), \end{aligned}$$

from which we conclude that $\text{Cyl}_G(p) \circ H = h$.

This completes our construction of the Brown factorization of f . Now, we are in a position to apply Proposition 3.52 to conclude that each object in $\mathcal{C}$ admits a path space.

This proves the claim. ■

^aNote that when $t = 2 - 2s$, $s \neq 0$ since that would imply that $t = 2$ and this cannot happen. So, $\frac{2s}{2-t} = 1$.

We have two remarks about this proof.

Remark 4.2. The result above was deduced in the non-equivariant set-up by Uuye in [92, Theorem 2.19, p.10]. However, we want to point out that his proof of the existence of path spaces (in particular, all of the necessary verifications) seems to be incomplete. This is actually related to a significant disadvantage of his approach which we will discuss in Remark 4.3. As the reader can see, our solution to this problem was to simply establish Brown factorizations for $\star$ -homomorphisms in the same way that was done in [85, Proposition 1.8, p.410] for the non-equivariant set-up, supplemented with many more details, and then utilize Proposition 3.52. This proof seems to have a good chance of generalizing to other categories of equivariant topological algebras.

Remark 4.3. There are two reasons why the approach used by Uuye does not generalize to the equivariant case:

1. The proof relies on purely topological characterizations of homotopy equivalences and Schochet fibrations in terms of weak homotopy equivalences and Serre fibrations between spaces. For the latter, a necessary component of the proof seems to be to use the fact that the maximal tensor product of C^* -algebras preserves pullbacks. This does not seem to be true for the projective tensor product of Banach algebras and there seems to be no easy fix for this defect in the setting of Banach algebras. One might consider using the injective tensor product of Banach algebras but it is not clear that the resulting tensor product is actually an algebra.
2. The proof does not account for equivariance. Indeed, it is not clear that these characterizations hold when one needs to account for equivariance. In fact, to work with the model structures available for categories of equivariant spaces, it is necessary that even stronger characterizations hold^a. It is not clear that this is the case.

Nevertheless, these (potential) hurdles with Uuye's proof seem to suggest that more work needs to be done to relate equivariant classical homotopy theory and equivariant homotopy theory on C^* -algebras^b.

^aSee [2].

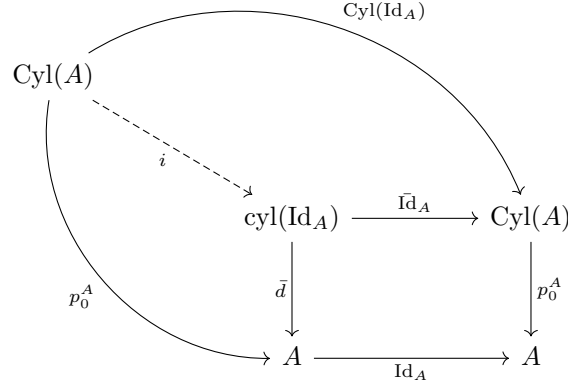
^bWe will discuss some questions in this direction in Section 5.2.

Given the role that the mapping cylinder plays in the construction of the factorization above, it is useful to have a computation of it in a very concrete example, so we will discuss that now. Indeed, the mapping cylinder of the diagonal morphism on a C^* -algebra can be used to explain why it has that name in the first place.

Example 4.4. Let $A \in C^*Alg$, for simplicity^a. We have the following claim.

| **Claim.** $Cyl(A)$ is isomorphic to $cyl(Id_A)$.

Let us consider the following diagram:



So, the main square is the pullback diagram for the cylinder, of course. The outer diagram is actually the naturality square for the time evaluation natural transformation $p_0 : Cyl \Rightarrow Id$, so it commutes. Therefore, the universal property of the pullback gives a natural morphism from $Cyl(A) \rightarrow cyl(Id_A)$. We do have an explicit description of $cyl(Id_A)$:

$$cyl(Id_A) = \{(a, \phi) \in A \times Cyl(A) : a = \phi(0)\}.$$

In particular, this allows us to have an explicit form for the map i :

$$i : Cyl(A) \rightarrow cyl(Id_A), \phi \mapsto i(\phi) \stackrel{\text{def}}{=} (\phi(0), \phi).$$

This is clearly actually valued in $cyl(Id_A)$ and it is a $\star$ -homomorphism^b. Now, we define:

$$j : cyl(Id_A) \rightarrow Cyl(A), (a, \phi) \mapsto j(a, \phi) \stackrel{\text{def}}{=} \phi,$$

where $\pi_1 : A \times A \rightarrow A$ is the projection onto the first factor. This is a $\star$ -homomorphism, as can be checked with a direct computation; as above, one will need that π_1 is a $\star$ -homomorphism. One can check that these are inverses.

| **Claim 4.1.** i and j are inverses.

This is proved on Page 112. So, this proves that $Cyl(A)$ and $cyl(Id_A)$ are $\star$ -isomorphic.

^aWe can also do this with equivariant C^* -algebras or Banach algebras but this keeps notation simple.

^bChecking this amounts to a direct computation with the formula.

At this stage, we are ready to define what it means for a given full subcategory of C^*Alg to be admissible.

Definition 4.5. A. Vats, 2026

Let $\mathcal{C}$ be a full subcategory of C^*Alg . We say that $\mathcal{C}$ is admissible iff:

1. $\mathcal{C}$ is closed under the cylinder.
2. $\mathcal{C}$ is closed under tensoring by $M_{n \times n}(\mathbb{C})$ for each $n \in \mathbb{N}$.
3. $\mathcal{C}$ is closed under sequential inductive limits.
4. $\mathcal{C}$ is closed under ideals, quotients and extensions.

Moreover, we require that $\mathcal{C}$ also contains 0 and $\mathbb{C}$.

In fact, we can show that any admissible subcategory of C^*Alg will be closed under a selection of constructions. Before we prove this, let us introduce another construction which will be relevant for later work in this subsection.

Construction 4.6. Let $A, B \in C^*Alg$ and let $f \in C^*Alg(A, B)$. We define:

$$cone(f),$$

to be a C^* -algebra that fits into the following pullback:

$$\begin{array}{ccc} cone(f) & \longrightarrow & Cone(B) \\ \downarrow & & \downarrow p_0^B \\ A & \xrightarrow{f} & B \end{array}$$

We refer to this as the mapping cone of f .

One of the main reasons why admissible categories are crucial for us is because we can deduce that they contain many interesting C^* -algebras. In particular, they should be closed under “homotopical constructions” and should contain sufficiently many C^* -algebras such that the classical results of axiomatic (co)homology as deduced by Schochet on C^*Alg actually apply to those subcategories as well. The next proposition partially handles this for us.

Proposition 4.7. A. Vats, 2026

Let $\mathcal{A}$ be an admissible subcategory of C^*Alg .

1. $\mathcal{A}$ is closed under pullbacks.
2. $\mathcal{A}$ is closed under the cone and suspensions.
3. $\mathcal{A}$ is closed under the mapping cone and mapping cylinder.
4. $\mathcal{A}$ is closed under the minimal unitization.
5. If $A, B \in C^*Alg$ with $A \in \mathcal{A}$ and if $f \in C^*Alg(A, B)$, then $\text{ran}(f), \ker(f) \in \mathcal{A}$.
6. $\mathcal{C}(S^1) \in \mathcal{A}$.
7. $\mathcal{K} \in \mathcal{A}$.
8. $\mathcal{T} \in \mathcal{A}$; this is the Toeplitz C^* -algebra.

Moreover, $\mathcal{A}$ is a fibrant subcategory of C^*Alg .

Proof. Let's prove each statement one-by-one.

1. This is [72, Proposition 3.4, p.259]. It is only implied by Axiom 4 of Definition 4.5.
2. Let $A \in \mathcal{A}$. Then, we have the following pullback diagrams:

$$\begin{array}{ccc} \text{Cone}(A) & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ \text{Cyl}(A) & \xrightarrow{p_1^A} & A \end{array} \quad \begin{array}{ccc} \text{Sus}(A) & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ \text{Cyl}(A) & \xrightarrow{(p_0^A, p_1^A)} & A \oplus A \end{array}$$

Since these pullback diagrams live in $\mathcal{A}$, it follows that $\text{Cone}(A)$ and $\text{Sus}(A) \in \mathcal{A}$.

3. These are defined by pullback diagrams whose vertices are in $\mathcal{A}$, so they must lie in $\mathcal{A}$.
4. If $A \in \mathcal{A}$, then we have the following extension:

$$0 \longrightarrow A \xrightarrow{i} A^\sim \xrightarrow{p} \mathbb{C} \longrightarrow 0$$

and since $\mathbb{C} \in \mathcal{A}$, it follows that $A^\sim \in \mathcal{A}$.

5. This follows from the fact that $\ker(f)$ and $\text{ran}(f)$ are an ideal and quotient of A , respectively.
6. We consider the following commutative diagram:

$$\begin{array}{ccc} \mathcal{C}(S^1) & \longrightarrow & \mathcal{C}([0, 1]) \\ \downarrow & & \downarrow \\ \mathbb{C} & \xrightarrow{\Delta_{\mathbb{C}}} & \mathbb{C} \oplus \mathbb{C} \end{array}$$

where $\Delta_{\mathbb{C}}$ is the diagonal map and the vertical map takes a function $f \in \mathcal{C}([0, 1])$ & maps it to $(f(0), f(1))$. Since $\mathbb{C} \in \mathcal{A}$ and since $\mathcal{A}$ is closed under cylinders, we know that $\mathcal{C}([0, 1]) = \text{Cyl}(\mathbb{C}) \in \mathcal{A}$. By closure under pullbacks, we see that $\mathbb{C} \oplus \mathbb{C} \in \mathcal{A}$. This implies, by closure under pullbacks, that $\mathcal{C}(S^1) \in \mathcal{A}$.

7. $\mathcal{K}$ can be written as an inductive limit of full matrix algebras and these belong to $\mathcal{A}$, so $\mathcal{K}$ belongs to $\mathcal{A}$.
8. The Toeplitz C^* -algebra fits into the following exact sequence:

$$0 \rightarrow \mathcal{K} \rightarrow \mathcal{T} \rightarrow \mathcal{C}(S^1) \rightarrow 0,$$

and since $\mathcal{K}, \mathcal{C}(S^1)$ lie in $\mathcal{A}$, we see that $\mathcal{T} \in \mathcal{A}$.

Finally, the last statement holds because $\mathcal{A}$ is closed under pullbacks. ■

The next result is a neat observation which is clear.

Claim 4.2. Let $\mathcal{C}, \mathcal{D}$ be admissible subcategories of C^*Alg . Then, $\mathcal{C} \cap \mathcal{D}$ is also admissible.

Of course, our definition also captures the usual examples.

Example 4.8. C^*Alg is certainly an admissible subcategory of C^*Alg .

Example 4.9. ${}_{\omega}C^*Alg$ is an admissible subcategory. We'll have to verify each of the required conditions one-by-one.

1. Clearly, 0 and $\mathbb{C}$ lie in ${}_{\omega}C^*Alg$.
2. ${}_{\omega}C^*Alg$ is closed under the cylinder because the tensor product of two separable C^* -algebras is separable^a.
3. ${}_{\omega}C^*Alg$ is closed under tensoring by $M_{n \times n}(\mathbb{C})$ for the same reasons as in point^b (2).
4. Let $A \in C^*Alg$ be the inductive limit of a sequence of separable C^* -algebras. Then, A is the closure of the directed countable union of a sequence of separable C^* -subalgebras.
5. ${}_{\omega}C^*Alg$ is clearly closed under ideals, quotients and extensions.

This proves the result.

^aOver here, we don't have to fuss over which tensor product is used since $\mathcal{C}[0, 1]$ is commutative and, therefore, nuclear.

^bAgain, no need to fuss over the tensor product used because $M_{n \times n}(\mathbb{C})$ is nuclear.

Example 4.10. NucC^*Alg is an admissible subcategory^a of C^*Alg . Let us verify each property one-by-one.

1. 0 and $\mathbb{C}$ are nuclear C^* -algebras.
2. Let $A \in \text{NucC}^*\text{Alg}$. Then, its tensor product with any other nuclear C^* -algebra will be nuclear; see [18, Proposition 10.1.7, p.302]. So, we see that $\text{Cyl}(A) \in \text{NucC}^*\text{Alg}$.
3. Since $M_{2 \times 2}(A) \simeq M_{n \times n}(\mathbb{C}) \otimes A$ for any $A \in \text{C}^*\text{Alg}$, the result follows by the statement of the previous point.
4. Let's show that NucC^*Alg is closed under ideals, quotients and extensions. This is the most difficult bit because it is a consequence of deep theorems. For closure under ideals, see [18, Exercise 2.3.4, p.36]. For closure under extensions, see [18, Exercise 3.8.1, p.105]. For closure under quotients, see [18, Corollary 9.4.4, p. 297].

This proves the result.

Using **Claim 4.2** and **Example 4.9**, give us that the category of separable nuclear C^* -algebras is also admissible.

^aThis is the full subcategory of C^*Alg consisting of nuclear C^* -algebras.

In particular, we see that there are three other examples of categories of fibrant objects added to the list of such examples in the literature. At this stage, we are ready to move on to a more explicit set of applications of the ideas introduced above. In particular, we want to demonstrate that it is possible to do axiomatic (co)homology on fibrant subcategories of C^*Alg . This provides an enhancement of some of the basic results found in [85].

Since that paper is quite lengthy, our treatment will not be complete. Instead, we wish to simply convince the reader that this should be possible by showing that a number of basic results follow without trouble & that even the Bott Periodicity Theorem follows without any trouble. We will mainly focus on homology; since we are dealing with functors valued in abelian categories, these will simply encode the same basic results for cohomology theories by duality. Before we describe these C^* -homology theories, let us recall Notation 2.11 and also introduce another piece of notation.

Notation 4.11. Unless stated otherwise, $\mathcal{A}$ will be an abelian category.

Abelian categories will be the codomains of the functors we consider in this subsection. This is mainly a formality so that the results of this subsection also apply to cohomology theories on C^*Alg by a duality argument. By the Freyd-Mitchell Embedding Theorem, we will always pretend that our functors are valued (at least, locally) in some module category. Now, we shall give the definition of an equivariant C^* -homology theory.

Definition 4.12

Let $(h_n : \text{C}^*\text{Alg} \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a sequence of functors^a. It is a C^* -homology theory iff:

1. Each h_n is a homotopy-invariant functor.
2. For any exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in $\text{dom}(h_n)$, there exists a sequence $(\partial : h_n(C) \rightarrow h_{n-1}(A))_{n \in \mathbb{Z}}$ of morphisms and a long-exact sequence:

$$\dots \longrightarrow h_n(A) \xrightarrow{h_n(\phi)} h_n(B) \xrightarrow{h_n(\psi)} h_n(C) \xrightarrow{\partial} h_{n-1}(A) \longrightarrow \dots$$

such that ∂ is natural with respect to morphisms of exact sequences.

The maps ∂ are known as boundary homomorphisms.

^aThe domain of each is the same, so it does not vary with n ! Moreover, we also allow for one-sided C^* -homology theories.

We have a remark about this definition.

Remark 4.13. Let's be more precise about what it means for ∂ to be natural with respect to morphisms of exact sequences. Indeed, suppose we have the following commutative diagram in C^*Alg with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' \longrightarrow 0 \end{array}$$

then, the resulting squares:

$$\begin{array}{ccc} h_n(C) & \longrightarrow & h_{n-1}(A) \\ \downarrow & & \downarrow \\ h_n(C') & \longrightarrow & h_{n-1}(A') \end{array}$$

commute for every $n \in \mathbb{Z}$.

Sometimes, one assumes that a homology theory is, at least, finitely-additive. Let us introduce this notion somewhat generally. One could generalize this to countable additivity or arbitrary additivity but we will not need that here, so we will not work with that level of generality. We will need the following basic construction.

Construction 4.14. Let $(A_i)_{i \in I}$ be a finite family in an admissible subcategory of $C^* \text{Alg}$. Then, we know that:

$$A = \bigoplus_{i \in I} A_i \in C^* \text{Alg},$$

and that it lies in the given admissible subcategory. Now, we have inclusions:

$$\iota_j : A_j \rightarrow A.$$

If $(h_n : C^* \text{Alg} \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ is a sequence of functors^a, then we get maps:

$$h_n(\iota_j) : h_n(A_j) \rightarrow h_n(A).$$

Now, the universal property of the coproduct in $\mathcal{A}$ gives a morphism:

$$\bigoplus_{j \in I} h_n(\iota_j) : \bigoplus_{j \in I} h_n(A_j) \rightarrow h_n(A),$$

for each $n \in \mathbb{Z}$.

^aThe domains are all the same.

This construction leads us to the following bit of terminology.

Terminology 4.15. Let $(h_n : C^* \text{Alg} \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a sequence of functors (the domains are all assumed to be the same) and let $(A_i)_{i \in I}$ be a finite family in the common domain of the given sequence of functors. The given sequence is finitely-additive iff the canonical morphism in [Construction 4.14](#) is an isomorphism for each $n \in \mathbb{Z}$.

Here are some basic properties that we can already derive.

Proposition 4.16

Let $(h_n : C^* \text{Alg} \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a C^* -homology theory.

1. The boundary homomorphisms obtained from a split exact sequence are zero.
2. The given sequence is finitely-additive.
3. Each h_n preserves finite coproducts.
4. $h_n(0) = 0$.
5. If $n \in \mathbb{Z}$, $A \in C^* \text{Alg}$ and $h_n(\mathbb{C}) \simeq 0$, then $h_n(\iota)$ is an isomorphism^a.

^a $\iota : A \rightarrow A^\sim$ is the canonical inclusion.

Proof. We'll prove each statement one-by-one.

1. Consider a split-exact sequence in $C^* \text{Alg}$.

$$0 \longrightarrow A \xrightarrow{\phi} B \begin{array}{c} \xrightarrow{\psi} \\ \xleftarrow{\alpha} \end{array} C \longrightarrow 0$$

This implies that $\psi \circ \alpha = \text{Id}_C$. Now, let $n \in \mathbb{Z}$ be fixed. By functoriality, we conclude that $h_n(\psi)$ is surjective. But this means that its range is just $h_n(C)$. On the other hand, the split-exact sequence above gives us a long-exact sequence with boundary homomorphisms $(\partial_n : h_n(C) \rightarrow h_{n-1}(A))_{n \in \mathbb{Z}}$. Since ∂_n connects to $h_n(\psi)$ and the long-exact sequence is exact at $h_n(C)$, it follows that $\ker(\partial_n) = h_n(C)$. That is, $\partial_n = 0$.

2. Consider a split-exact sequence in $C^* \text{Alg}$.

$$0 \longrightarrow A \xrightarrow{\phi} B \begin{array}{c} \xrightarrow{\psi} \\ \xleftarrow{\alpha} \end{array} C \longrightarrow 0$$

This induces a long-exact sequence:

$$\dots \longrightarrow h_n(A) \xrightarrow{h_n(\phi)} h_n(B) \xrightarrow{h_n(\psi)} h_n(C) \xrightarrow{\partial} h_{n-1}(A) \longrightarrow \dots$$

By (1), we know that $\partial = 0$ and we know that $h_n(\psi)$ is a split epimorphism by functoriality. So, the homology theory takes the split-exact sequence above to a split-exact one.

3. Let $A, B \in C^* \text{Alg}$ and let us be given the following maps:

$$\begin{aligned} \pi_A : A \oplus B &\rightarrow A, (a, b) \mapsto \pi_A(a, b) \stackrel{\text{def}}{=} a, \\ \pi_B : A \oplus B &\rightarrow B, (a, b) \mapsto \pi_B(a, b) \stackrel{\text{def}}{=} b, \\ \iota_A : A &\rightarrow A \oplus B, a \mapsto \iota_A(a) \stackrel{\text{def}}{=} (a, 0), \\ \iota_B : B &\rightarrow A \oplus B, b \mapsto \iota_B(b) \stackrel{\text{def}}{=} (0, b). \end{aligned}$$

With these maps, we have two split-exact sequences:

$$0 \longrightarrow A \xrightarrow{\iota_A} A \oplus B \xrightarrow{\pi_B} B \longrightarrow 0$$

$$0 \longrightarrow B \xrightarrow{\iota_B} A \oplus B \xrightarrow{\pi_A} A \longrightarrow 0$$

Both of these induce split-exact sequences in $\mathcal{A}$ and this implies the result by the splitting lemma.

4. By statement (3), we see that $h_n(0 \oplus 0) \simeq h_n(0) \oplus h_n(0) \simeq h_n(0)$ so we're done.

5. Consider the split-exact sequence:

$$0 \longrightarrow A \xrightarrow{\iota} A^\sim \xrightarrow{p} \mathbb{C} \longrightarrow 0$$

associated to the unitization. For each $n \in \mathbb{Z}$, we obtain a sequence:

$$0 \longrightarrow h_n(A) \xrightarrow{h_n(\iota)} h_n(A^\sim) \xrightarrow{h_n(p)} h_n(\mathbb{C}) \longrightarrow 0$$

which is split-exact. But note that $h_n(\mathbb{C}) \simeq 0$ so we see that $h_n(\iota)$ is injective & surjective ■

It is also reasonable for us to establish that C^* -homology theories are additive on morphisms. For this to make sense, we need to introduce the notion of orthogonality of $\star$ -homomorphisms. This is known and quite standard.

Terminology 4.17. Let $A, A', B \in C^*\text{Alg}$, let $f \in C^*\text{Alg}(A, B)$ and $g \in C^*\text{Alg}(A', B)$. f and g are orthogonal iff:

$$f(A)g(A') = 0.$$

In this case, we write that $f \perp g$.

We just said that we wish to establish that C^* -homology theories are additive on morphisms but the next result, which does do what we said, does not deal with C^* -homology theories. We will see, in Proposition 4.27, that homotopy-invariant and half-exact functors are the “correct” way to form (one-sided) C^* -homology theories.

Proposition 4.18

Let $A, B \in C^*\text{Alg}$ and let $f, g \in C^*\text{Alg}(A, B)$ so that $f \perp g$.

1. The pointwise sum $f + g$ is a $\star$ -homomorphism.
2. If $F : C^*\text{Alg} \rightarrow \mathcal{A}$ is homotopy-invariant and half-exact, then it is also split-exact.
3. If $F : C^*\text{Alg} \rightarrow \mathcal{A}$ is homotopy-invariant and half-exact and $A, B \in \text{dom}(F)$, then:

$$F(f + g) = F(f) + F(g).$$

That is, split-exact functors are additive on orthogonal morphisms.

Proof. We will prove each statement one-by-one.

1. The sum of bounded linear maps is known to be bounded. Next, we observe that:

$$\begin{aligned} \forall a \in A : (f + g)(a^\star) &= f(a^\star) + g(a^\star) \\ &= f(a)^\star + g(a)^\star \\ &= (f(a) + g(a))^\star \\ &= (f + g)(a)^\star, \end{aligned}$$

so $f + g$ preserves the involution. Finally, we show that it preserves the product:

$$\begin{aligned} \forall a, b \in A : (f + g)(ab) &= f(ab) + g(ab) \\ &= f(a)f(b) + g(a)g(b) \\ &= f(a)f(b) + f(a)g(b) + g(a)f(b) + g(a)g(b) \\ &= f(a)(f(b) + g(b)) + g(a)(f(b) + g(b)) \\ &= (f(a) + g(a))(f(b) + g(b)) \\ &= (f + g)(a)(f + g)(b), \end{aligned}$$

where we are allowed to add the mixed terms since they are just zero as $f \perp g$.

2. We will imitate the argument in Proposition 4.16. Consider a split-exact sequence in $C^*\text{Alg}$.

$$0 \longrightarrow A \xrightarrow{\phi} B \xleftarrow[\alpha]{\psi} C \longrightarrow 0$$

This implies that $\psi \circ \alpha = \text{Id}_C$. Now, let $n \in \mathbb{Z}$ be fixed. By functoriality, we conclude that $F(\psi)$ is surjective. But this means that its range is just $F(C)$. On the other hand, the split-exact sequence above gives us a long-exact sequence with boundary homomorphisms $(\partial_n : F_n(C) \rightarrow F_{n-1}(A))_{n \geq 1}$. Since ∂_n connects to $F_n(\psi)$ and the long-exact sequence is exact at $F_n(C)$ for $n \geq 1$, it follows that $\ker(\partial_n) = F_n(C)$ for $n \geq 1$. That is, $\partial_n = 0$ for $n \geq 1$. The boundary homomorphisms in the resulting long-exact sequence are zero, so we obtain:

$$0 \longrightarrow F(A) \xrightarrow{F(\phi)} F(B) \xleftarrow[F(\alpha)]{F(\psi)} F(C) \longrightarrow 0$$

where we do have surjectivity of $F(\psi)$ due to the argument we gave above.

3. We observe that:

$$\text{ran}(f + g) \subseteq \text{ran}(f) \oplus \text{ran}(g),$$

where the internal sum in B on the right-hand side is a direct sum because $\text{ran}(f) \cap \text{ran}(g)$ is trivial. Indeed, if $y \in \text{ran}(f) \cap \text{ran}(g)$, then $y = f(a)$ and $y = g(b)$ for $a, b \in A$. So, we have the following computation:

$$\begin{aligned} \|y\|^2 &= \|yy^\star\| \\ &= \|f(a)g(b^\star)\| \\ &= 0, \end{aligned}$$

from which we conclude that $y = 0$. Since $\text{ran}(f), \text{ran}(g) \in \text{dom}(F)$, we see that their direct sum $\text{ran}(f) \oplus \text{ran}(g) \in \text{dom}(F)$. So, we may replace B by $\text{ran}(f) \oplus \text{ran}(g)$ and note that since F is split-exact:

$$F(B) \simeq F(\text{ran}(f)) \oplus F(\text{ran}(g)).$$

Now, $F(f)$ is a map into $F(\text{ran}(f))$ and $F(g)$ is a map into $F(\text{ran}(g))$. Now, we use the universal property of the product to induce a unique morphism $\beta : F(A) \rightarrow F(\text{ran}(f)) \oplus F(\text{ran}(g))$ such that the obvious diagram commutes. But now, we see that $F(f + g)$ and $F(f) + F(g)$ would make the same diagram commute. ■

Our next goal is to establish that homology theories are “essentially” formed by just one functor and the suspension. In other words, if we have h_n , then we can calculate it using h_{n-1} . Before we state this precisely, here’s a construction.

Construction 4.19. Let $(h_n)_{n \in \mathbb{Z}}$ be a C^* -homology theory, let $A \in C^*\text{Alg}$ & consider the sequence:

$$0 \longrightarrow \text{Sus}(A) \longrightarrow \text{Cone}(A) \longrightarrow A \longrightarrow 0$$

as introduced in [Construction 2.8](#). This gives us the following data:

1. A sequence $(\sigma_A : h_n(A) \rightarrow h_{n-1}(\text{Sus}(A)))_{n \in \mathbb{Z}}$ of morphisms and;
2. A long-exact sequence:

$$\dots \longrightarrow h_n(\text{Sus}(A)) \longrightarrow h_n(\text{Cone}(A)) \longrightarrow h_n(A) \xrightarrow{\sigma_A} h_{n-1}(\text{Sus}(A)) \longrightarrow \dots$$

such that the boundary homomorphisms are natural with respect to morphisms of exact sequences.

We refer to each σ_A as a suspension map.

With this construction at hand, we have the following key result about the structure of C^* -homology theories.

Proposition 4.20

Let $(h_n : C^*\text{Alg} \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a C^* -homology theory & let $A \in \text{dom}(h_0)$.

1. If $B \in \text{dom}(h_0)$ and $f \in C^*\text{Alg}(A, B)$ so that $f = 0$, then $h_n(f) = 0$ for each $n \in \mathbb{Z}$.
2. $\forall n \in \mathbb{Z} : h_n(\text{Cone}(A)) = 0$.

In particular, the suspension map σ_A is an isomorphism.

Proof. We’ll prove each of these statements one-by-one.

1. Observe that we can factorize f as follows:

$$A \rightarrow 0 \rightarrow B.$$

Applying each h_n to f , we get that:

$$h_n(f) = h_n(0 \rightarrow B) \circ h_n(A \rightarrow 0),$$

and since $h_n(0) = 0$, it follows that $h_n(f) = 0$.

2. $\text{Cone}(A)$ is contractible by [Claim 2.12](#), so $\text{Id}_{\text{Cone}(A)}$ is homotopic to 0. This means that:

$$\begin{aligned} 0 &= h_n(0) \\ &= h_n(\text{Id}_{\text{Cone}(A)}) \\ &= \text{Id}_{h_n(\text{Cone}(A))}, \end{aligned}$$

and this only happens when $h_n(\text{Cone}(A)) = 0$.

Now, let $n \in \mathbb{N}$ and consider the suspension map:

$$\sigma_A : h_n(A) \rightarrow h_{n-1}(\text{Sus}(A)),$$

within the following long-exact sequence:

$$\dots \longrightarrow h_n(\text{Sus}(A)) \longrightarrow h_n(\text{Cone}(A)) \longrightarrow h_n(A) \xrightarrow{\sigma_A} h_{n-1}(\text{Sus}(A)) \longrightarrow \dots$$

Now, we’ll show that σ_A is an isomorphism by showing that it is bijective.

1. We see that $h_n(\text{Cone}(A)) = 0$, so σ_A is injective since the sequence above is exact.
2. We see that $h_{n-1}(\text{Cone}(A)) = 0$, so σ_A is surjective since the sequence above is exact.

It follows that σ_A is bijective so we’re done. ■

So, the result of [Proposition 4.20](#) motivates the following bit of notation.

Notation 4.21. Let $F : C^*\text{Alg} \rightarrow \mathcal{A}$ be a functor and define:

$$\forall n \in \mathbb{N}_0 : F_n \stackrel{\text{def}}{=} F \circ \text{Sus}^n.$$

This is known as the n -th iterant of F . We refer to this as the suspension spectrum of F .

We have a remark about [Notation 4.21](#).

Remark 4.22. For this to make sense, we need to ensure that Sus lands inside the domain of F . This is easy to ensure by restricting Sus to the domain of F , since admissible subcategories are closed under suspension^a.

^aSee statement (2) of [Proposition 4.7](#).

We can already prove some basic facts about the suspension spectrum of some functor. Before we do this, let us also introduce the notion of C^* -stability. We will need this when discussing the Bott Periodicity Theorem later.

Terminology 4.23. Let $F : C^*Alg \rightarrow \mathcal{A}$ be a functor, let $A \in \text{dom}(F)$ and let $e \in \mathcal{K}$ be a rank-one projection. Define:

$$p_A : A \rightarrow A \otimes \mathcal{K}, \quad a \mapsto p_A(a) \stackrel{\text{def}}{=} a \otimes e.$$

F is C^* -stable iff $F(p_A)$ is an isomorphism^a for each $A \in \text{dom}(F)$.

^aThis makes sense because $A \otimes \mathcal{K} \in \text{dom}(F)$ by Proposition 4.7.

The next result, which extends what is in the literature, gives us permanence properties of the suspension spectrum.

Proposition 4.24

Let $F : C^*Alg \rightarrow \mathcal{A}$ be a functor.

1. If F is homotopy-invariant, then its suspension spectrum consists of homotopy-invariant functors.
2. If F is half-exact, then its suspension spectrum consists of half-exact functors.
3. If F is split-exact, then its suspension spectrum consists of split-exact functors.
4. If F is C^* -stable, then its suspension spectrum consists of C^* -stable functors.

Proof. Let $\text{dom}(F)$ be denoted by $\mathcal{C}$ throughout the proofs of these statements. Now, it is sufficient to prove that $F \circ \text{Sus}$ satisfies these properties, since this can be extended to the rest of the spectrum by induction.

1. Let $A, B \in \mathcal{C}$ and let $f, g \in \mathcal{C}(A, B)$ be homotopic, with $h : A \rightarrow \text{Cyl}(B)$ being a homotopy between them. We apply Sus to this morphism and let $\alpha : \text{Sus} \circ \text{Cyl} \Rightarrow \text{Cyl} \circ \text{Sus}$ be the natural isomorphism of **Claim 2.5**. Now, we wish to define a homotopy from $\text{Sus}(f)$ to $\text{Sus}(g)$ and we do this as follows:

$$H \stackrel{\text{def}}{=} \alpha_B \circ \text{Sus}(h).$$

Let us show that this is a homotopy from $\text{Sus}(f)$ to $\text{Sus}(g)$. If this is the case, then we can conclude by the homotopy-invariance of F that $F(\text{Sus}(f)) = F(\text{Sus}(g))$ and this implies that $F \circ \text{Sus}$ is, of course, homotopy-invariant. Indeed, let $i \in \{0, 1\}$ and consider the evaluation map $p_i^{\text{Sus}(B)} : \text{Cyl}(\text{Sus}(B)) \rightarrow \text{Sus}(B)$. We need to determine the exact relationship between $p_i^{\text{Sus}(B)}$ and $\text{Sus}(p_i^B)$. In fact, we have the following claim.

Claim. $p_i^{\text{Sus}(B)} \circ \alpha_B = \text{Sus}(p_i^B)$.

This was proved in **Claim 2.6**. Now, we have the following computation:

$$\begin{aligned} p_i^{\text{Sus}(B)} \circ H &= p_i^{\text{Sus}(B)} \circ (\alpha_B \circ \text{Sus}(h)) \\ &= (p_i^{\text{Sus}(B)} \circ \alpha_B) \circ \text{Sus}(h) \\ &= \text{Sus}(p_i^B) \circ \text{Sus}(h) \\ &= \text{Sus}(p_i^B \circ \text{Sus}(h)), \end{aligned}$$

which evaluates to $\text{Sus}(f)$ or $\text{Sus}(g)$ depending on whether i is chosen to be 0 or 1.

2. Since Sus is an exact functor, it takes an exact sequence to an exact sequence, so F takes that exact sequence to a half-exact sequence. In other words, $F \circ \text{Sus}$ is half-exact.
3. Since Sus is an exact functor, it takes a split-sequence to a split-exact sequence, so F takes that split-exact sequence to a split-exact sequence. In other words, $F \circ \text{Sus}$ is split-exact.
4. Let $G \stackrel{\text{def}}{=} - \otimes \mathcal{K}$ be defined on $\mathcal{C}$. We start with the following claim.

Claim. The maps p_A form the components of a natural transformation from $\text{Id}_{\mathcal{C}}$ to G .

Again, this makes sense because the codomain of G lands in $\mathcal{C}$ by the strength of Proposition 4.7. Now, let $A, B \in \mathcal{C}$ and let $f \in \mathcal{C}(A, B)$. Then, we need to show that the following diagram:

$$\begin{array}{ccc} A & \xrightarrow{p_A} & A \otimes \mathcal{K} \\ f \downarrow & & \downarrow f \otimes \text{Id}_{\mathcal{K}} \\ B & \xrightarrow{p_B} & B \otimes \mathcal{K} \end{array}$$

commutes. This is a calculation as follows:

$$\begin{aligned} \forall a \in A : (f \otimes \text{Id}_{\mathcal{K}} \circ p_A)(a) &= f \otimes \text{Id}_{\mathcal{K}}(p_A(a)) \\ &= f \otimes \text{Id}_{\mathcal{K}}(a \otimes e) \\ &= f(a) \otimes e \\ &= p_B(f(a)) \\ &= (p_B \circ f)(a), \end{aligned}$$

as was desired. This proves the claim.

If the natural transformation of the preceding claim is denoted by $p : \text{Id}_{\mathcal{C}} \Rightarrow G$, then let us consider the

left-whiskered natural transformation $q \stackrel{\text{def}}{=} p \circ \text{Sus} : \text{Sus} \Rightarrow G \circ \text{Sus}$. That is, if $A \in \mathcal{C}$, then this is defined as:

$$q_A \stackrel{\text{def}}{=} p_{\text{Sus}(A)}.$$

What we want to show is that $F(\text{Sus}(p_A))$ is an isomorphism and if it was the case that $\text{Sus}(p_A)$ could be “replaced” by q_A , then we would be done. Let us observe that if $A \in \mathcal{C}$, then we have a computation:

$$\begin{aligned} \text{Sus}(A \otimes \mathcal{K}) &\simeq \mathcal{C}_0(\mathbb{R}) \otimes (A \otimes \mathcal{K}) \\ &\simeq (\mathcal{C}_0(\mathbb{R}) \otimes A) \otimes \mathcal{K} \\ &\simeq \text{Sus}(A) \otimes \mathcal{K}. \end{aligned}$$

This simply uses the associativity of the tensor product and since this is natural, we see that the above establishes a natural isomorphism $\alpha : \text{Sus} \circ G \Rightarrow G \circ \text{Sus}$. Now, we are naturally led to define the right-whiskering:

$$r \stackrel{\text{def}}{=} \text{Sus} \circ p : \text{Sus} \Rightarrow \text{Sus} \circ G,$$

of p . This leads us to the following claim.

| **Claim.** $\alpha \circ r = q$.

If this claim is true, then we note that $r_A = \text{Sus}(p_A)$ for each $A \in \mathcal{C}$. So:

$$\begin{aligned} F(q_A) &= F(\alpha_A \circ r_A) \\ &= F(\alpha_A) \circ F(r_A) \\ &= F(\alpha_A) \circ F(\text{Sus}(p_A)). \end{aligned}$$

Since $F(q_A) = F(p_{\text{Sus}(A)})$ is an isomorphism (due to the C^* -stability of F) and $F(\alpha_A)$ is an isomorphism, it follows that $F(\text{Sus}(p_A))$ is an isomorphism and this shows that $F \circ \text{Sus}$ is C^* -stable. It remains to show the claim. To prove this, we need to carefully compute the left-hand side. So, let $A \in \mathcal{C}$ and let $f \in \text{Sus}(A)$:

$$\begin{aligned} (\alpha \circ r)_A(f) &= \alpha_A(r_A(f)) \\ &= \alpha_A(\text{Sus}(p_A)(f)) \\ &= \alpha_A((\text{Id}_{\text{Sus}(\mathcal{C})} \otimes p_A)(f)) \end{aligned}$$

We cannot progress further without making further assumptions on f . In fact, it suffices to show that q_A and $(\alpha \circ r)_A$ coincide on a dense subset of $\text{Sus}(A)$ and we choose the dense subset in [Lemma 2.15](#). Indeed, let $f = ga$ for some $g \in \mathcal{C}_0(\mathbb{R}) = \text{Sus}(\mathbb{C})$ and $a \in A$. Continuing the computation of $(\alpha \circ r)_A(f)$ above, we have that:

$$\begin{aligned} (\alpha \circ r)_A(f) &= \alpha_A((\text{Id}_{\text{Sus}(\mathcal{C})} \otimes p_A)(f)) \\ &= \alpha_A((\text{Id}_{\text{Sus}(\mathcal{C})} \otimes p_A)(ga)) \\ &= \alpha_A(g \otimes p_A(a)) \\ &= \alpha_A(g \otimes (a \otimes e)) \\ &= (g \otimes a) \otimes e \\ &= ga \otimes e \\ &= e \otimes_f \\ &= q_A(f) \end{aligned}$$

which proves the claim. ■

We should, now, address the following natural question.

| **Question 4.25.** Let $F : C^*\text{Alg} \rightarrow \mathcal{A}$ be a functor. When is its suspension spectrum a C^* -homology theory?

We will give an answer to this through the proof of Proposition 4.27. That result was known for functors defined on $C^*\text{Alg}$ and we extend it to functors defined on admissible subcategories. To prove it, we will need a technical lemma.

Lemma 4.26. [[18](#), Corollary 3.7.4, p.93] Consider the following exact sequence of C^* -algebras:

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0$$

and let $B \in C^*\text{Alg}$. If A/I is nuclear or B is nuclear, then the tensored sequence:

$$0 \longrightarrow I \otimes_{\min} B \longrightarrow A \otimes_{\min} B \longrightarrow (A/I) \otimes_{\min} B \longrightarrow 0$$

is exact.

We are ready to discuss the desired proposition.

Proposition 4.27. The Barratt-Puppe Sequence

Let $F : C^*\text{Alg} \rightarrow \mathcal{A}$ be homotopy-invariant & half-exact. Its suspension spectrum is a one-sided C^* -homology theory.

Proof. Suppose we have our extension of C^* -algebras in $\text{dom}(F)$:

$$0 \longrightarrow I \xrightarrow{\iota_1} A \xrightarrow{\pi_1} A/I \longrightarrow 0$$

The proof is predicated around the following claim.

Claim. There exists an extension:

$$0 \longrightarrow \text{Sus}(A/I) \xrightarrow{\iota_2} \text{cone}(\pi_1) \xrightarrow{\pi_2} A \longrightarrow 0$$

with the following properties:

1. There exists a $\star$ -homomorphism $\alpha_1 : I \rightarrow \text{cone}(\pi_1)$ which induces an isomorphism on F_n for all $n \in \mathbb{N}$.
2. The composite $\pi_2 \circ \alpha_1$ induces the same map on F_n as the inclusion ι_1 for each $n \in \mathbb{N}$.

Let's start by noting that this claim makes sense in $\text{dom}(F)$. Indeed, since $A/I \in \text{dom}(F)$, we see that $\text{Sus}(A/I) \in \text{dom}(F)$. So, if the extension in the claim exists, then $\text{cone}(\pi_1) \in \text{dom}(F)$ since $\text{dom}(F)$ is extension-closed. So, the properties also make sense. At this point, the proof splits into two parts. In the first part, we'll check that we actually obtain the given long-exact sequence provided that the claim holds. In the second part, we'll prove the claim.

1. We obtain a sequence $(G_n)_{n \in \mathbb{N}}$ of exact sequences by the following specification:

$$\begin{array}{ccccccccccc} G_1 : & 0 & \longrightarrow & I & \xrightarrow{\iota_1} & A & \xrightarrow{\pi_1} & A/I & \longrightarrow & 0 \\ & & & & & & & & & \\ G_2 : & 0 & \longrightarrow & \text{Sus}(A/I) & \xrightarrow{\iota_2} & \text{cone}(\pi_1) & \xrightarrow{\pi_2} & A & \longrightarrow & 0 \\ & & & & & & & & & \\ G_3 : & 0 & \longrightarrow & \text{Sus}(A) & \xrightarrow{\iota_3} & \text{cone}(\pi_2) & \xrightarrow{\pi_3} & \text{cone}(\pi_1) & \longrightarrow & 0 \\ & & & & & & & & & \\ & \vdots & & \vdots & & \vdots & & \vdots & & \vdots \end{array}$$

That is, G_1 is the extension we started with and for $n \in \mathbb{N}$, we have that the extension G_{n+1} is obtained from the extension G_n by the mapping cone construction in the claim. Now, we apply F to each of the sequences above and utilize the fact that F is a half-exact functor to obtain the following exact sequences:

$$\begin{array}{ccccccc} F(G_1) : & F(I) & \xrightarrow{F(\iota_1)} & F(A) & \xrightarrow{F(\pi_1)} & F(A/I) \\ & & & & & \\ F(G_2) : & F(\text{Sus}(A/I)) & \xrightarrow{F(\iota_2)} & F(\text{cone}(\pi_1)) & \xrightarrow{F(\pi_2)} & F(A) \\ & & & & & \\ F(G_3) : & F(\text{Sus}(A)) & \xrightarrow{F(\iota_3)} & F(\text{cone}(\pi_2)) & \xrightarrow{F(\pi_3)} & F(\text{cone}(\pi_1)) \\ & & & & & \\ & \vdots & & \vdots & & \vdots \end{array}$$

Now, the idea is to just do the following “replacements”:

$$F(\text{cone}(\pi_1)) \rightsquigarrow F(I)$$

$$F(\text{cone}(\pi_2)) \rightsquigarrow F(\text{Sus}(A/I)) = F_1(A/I)$$

and glue the resulting sequences once we've done the replacements. This will actually yield the first part of the given long-exact sequence and continuing the process as we have done above will yield the rest of it. So, let's illustrate why we can carry out the replacements above. Indeed, the claim gives us a commutative diagram:

$$\begin{array}{ccccccc} F(\text{Sus}(A/I)) & \xrightarrow{F(\iota_2)} & F(\text{cone}(\pi_1)) & \xrightarrow{F(\pi_2)} & F(A) & \xrightarrow{F(\pi_1)} & F(A/I) \\ \parallel & & \downarrow F(\alpha_1)^{-1} & & \parallel & & \parallel \\ F(\text{Sus}(A/I)) & \xrightarrow{F(\alpha_1)^{-1} \circ F(\iota_2)} & F(I) & \xrightarrow{F(\iota_1)} & F(A) & \xrightarrow{F(\pi_1)} & F(A/I) \end{array}$$

where the sequence on top is exact, so the sequence on the bottom is also exact. Now, we need to play a similar game with $F(G_3)$. Indeed, we have the following commutative diagram:

$$\begin{array}{ccccccc} F(\text{Sus}(A)) & \xrightarrow{F(\iota_3)} & F(\text{cone}(\pi_2)) & \xrightarrow{F(\pi_3)} & F(\text{cone}(\pi_1)) \\ \parallel & & \downarrow F(\alpha_2)^{-1} & & \downarrow F(\alpha_1)^{-1} \\ F_1(A) & \xrightarrow{F(\alpha_2)^{-1} \circ F(\iota_3)} & F_1(A/I) & \xrightarrow{F(\alpha_1)^{-1} \circ F(\pi_3) \circ F(\alpha_2)} & F(I) \end{array}$$

The top sequence is just $F(G_3)$ and this is exact, so the sequence on the bottom is exact. On the other hand, we can only sew this to the sequence we obtained from $F(G_1)$ and $F(G_2)$ if both morphisms from $F_1(A/I)$ to

$F(I)$ coincide. These coinciding means precisely that the following equality is satisfied:

$$F(\alpha_1)^{-1} \circ F(\pi_3) \circ F(\alpha_2) = F(\alpha_1)^{-1} \circ F(\iota_2),$$

which is equivalent to the statement that:

$$F(\pi_3) \circ F(\alpha_2) = F(\iota_2).$$

By functoriality, the left-hand side is just $F(\pi_3 \circ \alpha_2)$ and this translates to property (2) of the claim above. So, we can sew this short exact sequence to the one obtained from $F(G_1)$ and $F(G_2)$.

2. For this, we use the explicit form of the mapping cone defined in [Construction 4.6](#). We have that:

$$\text{cone}(\pi_1) = \{(f, a) \in \text{Cone}(A/I) \times A : f(0) = 0 \wedge f(1) = \pi_1(a)\}.$$

So, we define the three maps as follows:

$$\iota_2 : \text{Sus}(A/I) \rightarrow \text{cone}(\pi_1), \quad f \mapsto \iota_2(f) \stackrel{\text{def}}{=} (f, 0),$$

$$\pi_2 : \text{cone}(\pi_1) \rightarrow A, \quad (f, a) \mapsto \pi_2(f, a) \stackrel{\text{def}}{=} a,$$

$$\alpha_1 : I \rightarrow \text{cone}(\pi_1), \quad i \mapsto \alpha_1(i) \stackrel{\text{def}}{=} (0, i).$$

The fact that $\pi_2 \circ \alpha_1$ induces the same map as the inclusion is clear. Now, we also have an extension:

$$0 \longrightarrow I \xrightarrow{\alpha_1} \text{cone}(\pi_1) \longrightarrow \text{Cone}(A/I) \longrightarrow 0$$

with the second map being the projection to $\text{Cone}(A/I)$. By [Claim 2.12](#), we know that $\text{Cone}(A/I)$ is contractible. This means that $F(\alpha_1)$ is surjective. To show injectivity, we will replace I with a C^* -algebra Q so that I and Q are homotopy-equivalent and Q fits into an exact sequence with $\text{Cone}(I)$ on the left. So, we define:

$$Q \stackrel{\text{def}}{=} \{f \in \text{Cone}(A) : f(0) \in I\}.$$

We have the following claim.

| **Claim 4.3.** Q and I are homotopy equivalent. This is proved on Page 113.

Let us consider the following exact sequence:

$$0 \longrightarrow \text{Cone}(I) \longrightarrow Q \longrightarrow \text{cone}(\pi_1) \longrightarrow 0$$

where the first map is just an inclusion of functions and the second map is defined as:

$$k : f \mapsto (\pi_1 \circ f, f(1)).$$

The fact that these are $\star$ -homomorphisms will follow by an explicit computation and it also uses the fact that π_1 is a $\star$ -homomorphism. Since $I \in \text{dom}(F)$, we see that $\text{Cone}(I) \in \text{dom}(F)$ by Proposition 4.7. By our argument earlier in the proof, we know that $\text{cone}(\pi_1) \in \text{dom}(F)$. Since Q fits into the extension above, we see that $Q \in \text{dom}(F)$. Now, we use the contractibility of $\text{Cone}(I)$ to conclude that $F(k)$ is injective. Let us establish a formula which connects the map k to α_1 . If we define, as in the proof of [Claim 4.3](#), the map:

$$\psi : I \rightarrow Q, \quad j \mapsto \psi(j),$$

where $\psi(j)$ is the constant function j on $[0, 1]$. Composing this with k , we see that:

$$\begin{aligned} \forall j \in I : (k \circ \psi)(i) &= k(\psi(i)) \\ &= (\pi_1 \circ \psi(i), \psi(i)(1)) \\ &= (0, i) \\ &= \alpha_1(i). \end{aligned}$$

So, we conclude that $k \circ \psi = \alpha_1$ and this means that:

$$\begin{aligned} F(\alpha_1) &= F(k \circ \psi) \\ &= F(k) \circ F(\psi). \end{aligned}$$

Since $F(k)$ is injective and $F(\psi)$ is an isomorphism, we conclude that $F(\alpha_1)$ is injective.

3. Of course, the claim requires that the actual conclusion be true for each F_n and not just for F_0 . Using the fact that the suspension is an exact functor, we see that iterated suspension of each exact sequence that appears above produces an exact sequence. By the argument in statement (1) of Proposition 4.24, iterated suspensions also preserve homotopies, in the sense that if two morphisms are homotopic, then their suspensions are also homotopic. In particular, iterated suspensions of Q and I are also homotopy-equivalent. Finally, we note that each F_n in the suspension spectrum is also a homotopy-invariant and half-exact functor.
4. Finally, we have to argue that the boundary maps obtained are natural with respect to morphisms of exact sequences. One notes that the boundary maps are formed from the exact sequence of the claim and also the exact sequence for Q . Suppose, then, that we have the following morphism of exact sequences in $C^*\text{Alg}$:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \xrightarrow{\iota_1} & A & \xrightarrow{\pi_1} & A/I & \longrightarrow & 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \\ 0 & \longrightarrow & J & \xrightarrow{\iota_2} & B & \xrightarrow{\pi_2} & B/J & \longrightarrow & 0 \end{array}$$

Now, consider the following diagram:

$$\begin{array}{ccccccc}
0 & \longrightarrow & \text{Sus}(A/I) & \xrightarrow{i_1} & \text{cone}(\pi_1) & \xrightarrow{p_1} & A \longrightarrow 0 \\
& & \downarrow \text{Sus}(\gamma) & & \downarrow & & \downarrow \beta \\
0 & \longrightarrow & \text{Sus}(B/J) & \xrightarrow{i_2} & \text{cone}(\pi_2) & \xrightarrow{p_2} & B \longrightarrow 0
\end{array}$$

So, what's happening here is that we start with two exact sequences which have a morphism between them. Each of these exact sequences form another exact sequence which is the one from the claim made earlier in the proof of this result. Now, we want to form a morphism between those exact sequences. We define:

$$\hat{\pi} : \text{cone}(\pi_1) \rightarrow \text{cone}(\pi_2), \quad (f, a) \mapsto \hat{\pi}(f, a) \stackrel{\text{def}}{=} (\text{Cone}(\gamma)(f), \beta(a)).$$

Let us make sure that this makes sense first:

$$\begin{aligned}
\forall (f, a) \in \text{cone}(\pi_1) : \text{Cone}(\gamma)(f)(1) &= (\gamma \circ f)(1) \\
&= \gamma(f(1)) \\
&= \gamma(\pi_1(a)) \\
&= (\gamma \circ \pi_1)(a) \\
&= (\pi_2 \circ \beta)(a) \\
&= \pi_2(\beta(a)),
\end{aligned}$$

which shows, alongside the fact that $\text{Cone}(\gamma)(f)(0) = 0$, that $\hat{\pi}$ makes sense. Of course, it is also clearly a $\star$ -homomorphism so it remains to be checked that it gives us a commutative diagram above. We see that:

$$\begin{aligned}
\forall f \in \text{Sus}(A/I) : (\hat{\pi} \circ i_1)(f) &= \hat{\pi}(f, 0) \\
&= (\text{Cone}(\gamma)(f), 0) \\
&= i_2(\text{Cone}(\gamma)(f)) \\
&= i_2(\text{Sus}(\gamma)(f)),
\end{aligned}$$

where this works because the suspension “pinches” at both ends of $[0, 1]$ while the cone just “pinches” at one end of $[0, 1]$. The commutativity of the right square is actually evident, based on the definition of $\hat{\pi}$. Similarly, we want to form the following diagram and show that it commutes:

$$\begin{array}{ccccccc}
0 & \longrightarrow & \text{Cone}(I) & \longrightarrow & Q_A & \longrightarrow & \text{cone}(\pi_1) \longrightarrow 0 \\
& & \downarrow \text{Cone}(\alpha) & & \downarrow & & \downarrow \hat{\pi} \\
0 & \longrightarrow & \text{Cone}(J) & \longrightarrow & Q_B & \longrightarrow & \text{cone}(\pi_2) \longrightarrow 0
\end{array}$$

Again, from the two exact sequences we started with in this part of the proof, we form the exact sequences involving the auxiliary C^* -algebra Q (we have labeled this for each exact sequence) and the purpose is to relate these resulting exact sequences. Indeed, the map in the middle is defined as follows:

$$\hat{q} \stackrel{\text{def}}{=} \text{Cone}(\beta)|_{Q_A}.$$

We need to check that this maps into Q_B . Indeed:

$$\begin{aligned}
\forall f \in Q_A : \hat{q}(f)(0) &= \text{Cone}(\beta)|_{Q_A}(f)(0) \\
&= (\beta \circ f)(0) \\
&= \beta(f(0)) \\
&= \beta(\iota_1(f(0))) \\
&= \iota_2(\alpha(f(0))),
\end{aligned}$$

where the last term belongs to J . In the middle, we use the fact that if $f \in Q_A$, then $f(0) \in J$. Clearly, $\hat{q}$ is a $\star$ -homomorphism and it can be used to ensure that our diagram just above commutes. However, this is sufficient to establish naturality of the boundary morphisms. After all, the boundary morphisms are formed precisely using the morphisms in these exact sequences and what we have established above is that the two boundary morphisms formed from the original exact sequence must fit into some commuting diagram. This commuting diagram will be precisely the required naturality square.

This completes the proof. ■

We have a remark about this result.

Remark 4.28. For functors defined on all of $C^*\text{Alg}$, this result can be found in [80, Proposition 8.4, p.34-35] with a reasonably detailed proof. We provide more details and explicitly show that this result can be adapted to our requirements. This requires us to take care of many more details than if we were just working on $C^*\text{Alg}$. Also, we are aware that we are being slightly loose with our argument for naturality. However, the last part just requires one to concatenate appropriate diagrams and identify the relevant boundary morphisms. Once these identifications are made, all of the maps that we defined earlier in the proof can be used to construct the desired naturality square.

Having formulated these basic results, we now come to the main test result; the Bott Periodicity Theorem.

Theorem 4.29. The Bott Periodicity Theorem

Let $F : C^*Alg \rightarrow \mathcal{A}$ be a homotopy-invariant, half-exact and C^* -stable functor^a. Then, $F \simeq F \circ \text{Sus}^2$.

^aFrom now on, Notation 4.21 is in effect.

Here is some context for this theorem.

Remark 4.30. The Bott Periodicity Theorem, of course, has an extensive history that we will not discuss here since it is not relevant for us; one could also discuss the operator-algebraic version and the topological version(s) separately, alongside attempts to obtain periodicity for algebraic K-theory. Nevertheless, we mention that the rather general theorem above was proved by Cuntz in [7, Volume 1046, Theorem 4.4, p.63] for functors satisfying the listed properties and defined on all of C^*Alg . A more extensive proof with details is provided also in (Nigel) Higson's PhD thesis [42, Theorem 2.1.14, p.28]. The main contribution here is to go through Cuntz's proof & keep track of the auxiliary algebras introduced along the way, making sure that they are always contained in the domain of the functor F .

Let us start working towards Cuntz's proof of this theorem. A major step in that proof is to work with the *reduced* Toeplitz extension. In the next construction, we will study the Toeplitz extension and its reduced version. The main content of the construction lies in showing that the auxiliary C^* -algebras are in admissible subcategories of C^*Alg .

Construction 4.31. Let us consider the standard Toeplitz extension as follows:

$$0 \longrightarrow \mathcal{K} \xrightarrow{i_1} \mathcal{T} \xrightarrow{p_1} \mathcal{C}(S^1) \longrightarrow 0$$

For a detailed discussion of this extension and the proof of its existence, we strongly recommend the treatment of [52, Chapter 6.8, p.76-79]. We will briefly recall the essential points for your convenience. The key is to view $\mathcal{T}$ as the universal C^* -algebra generated by an isometry and $\mathcal{C}(S^1)$ as the universal C^* -algebra generated by a unitary. In that case, we can see that the map p_1 is induced by mapping our isometry of $\mathcal{T}$ (the generator) to the unitary of $\mathcal{C}(S^1)$ (the generator). If $v \in \mathcal{T}$ is the generator, then it will become the case that:

$$\mathcal{K} \simeq \langle 1_{\mathcal{T}} - vv^* \rangle,$$

with the right-hand side being an ideal in $\mathcal{T}$. The “reason” for this isomorphism is explained in the aforementioned notes very well, so we will not dwell on it. What we will do is define:

$$e_{00} \stackrel{\text{def}}{=} 1_{\mathcal{T}} - vv^*.$$

This will be used in our proof of the Bott Periodicity Theorem and we will recall it when necessary; it will be a very useful formula. Now, from the Toeplitz extension, we can construct the following diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_1} & \mathcal{T} & \xrightarrow{p_1} & \mathcal{C}(S^1) \longrightarrow 0 \\ & & & & \downarrow \varepsilon = \text{ev}_1 \circ p_1 & & \downarrow \text{ev}_1 \\ & & & & \mathbb{C} & \xlongequal{\quad} & \mathbb{C} \end{array}$$

where ev_1 is the evaluation map at $1 \in S^1$. Now, we put in $\ker(\varepsilon)$ into the diagram:

$$\begin{array}{ccccccc} & & & & 0 & & \\ & & & & \downarrow & & \\ & & & & \mathcal{T}_0 \stackrel{\text{def}}{=} \ker(\varepsilon) & & \\ & & & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_1} & \mathcal{T} & \xrightarrow{p_1} & \mathcal{C}(S^1) \longrightarrow 0 \\ & & & & \downarrow \varepsilon = \text{ev}_1 \circ p_1 & & \downarrow \text{ev}_1 \\ & & & & \mathbb{C} & \xlongequal{\quad} & \mathbb{C} \end{array}$$

Now, we have the following claim.

Claim. A copy of $\mathcal{K}$ is contained in $\mathcal{T}_0$, which we will call the reduced Toeplitz C^* -algebra. This is nuclear.

Let $T \in \mathcal{K}$. Then, we need to show that $i_1(T) \in \mathcal{T}_0 = \ker(\varepsilon)$. For this to work, we need to show that:

$$\varepsilon(i_1(T)) = 0.$$

But now, we see that:

$$\text{ev}_1(p_1(i_1(T))) = 0,$$

which proves the result. Finally, nuclearity follows from the fact that it sits between two C^* -algebras which are known to be nuclear; $\mathcal{K}$ is an AF algebra so it is nuclear^a and $\mathcal{C}_0(S^1 \setminus \{0\})$ is commutative so it is nuclear^b.

In light of this result, we can place the reduced Toeplitz C^* -algebra in an extension as follows:

$$\begin{array}{ccccccc}
& & & 0 & & & \\
& & & \downarrow & & & \\
0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_2} & \mathcal{T}_0 & \xrightarrow{p_2} & \mathcal{C}_0(S^1 \setminus \{1\}) \longrightarrow 0 \\
& & \parallel & & \downarrow & & \downarrow \\
0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_1} & \mathcal{T} & \xrightarrow{p_1} & \mathcal{C}(S^1) \longrightarrow 0 \\
& & & & \downarrow \varepsilon & & \downarrow \text{ev}_1 \\
& & & & \mathbb{C} & \xlongequal{\quad} & \mathbb{C}
\end{array}$$

where the morphisms are chosen so that the given squares commute. The extension obtained involving $\mathcal{T}_0$ is known as the reduced Toeplitz extension. There are three facts worth noting:

1. i_2 is just i_1 .
2. p_2 is just p_1 restricted to $\mathcal{T}_0$.
3. $\mathcal{C}(S^1 \setminus \{1\})$ is just $\mathcal{C}_0((0, 1))$.

Now, the key point lies in the following results.

Claim. Let $\mathcal{C}$ be an admissible subcategory of $C^*\text{Alg}$. Then, every C^* -algebra in the diagram above lies in $\mathcal{C}$.

We just need to check this for $\mathcal{T}_0$ and $\mathcal{C}(S^1 \setminus \{1\})$, since every other C^* -algebra above has been shown to lie in admissible subcategories. We note that $\mathcal{C}(S^1 \setminus \{1\}) \simeq \mathcal{C}_0(\mathbb{R}) = \text{Sus}(\mathbb{C})$ so this certainly lies in $\mathcal{C}$. Since $\mathcal{T}_0$ fits into the middle of an exact sequence where the other algebras belong to $\mathcal{C}$, we see that $\mathcal{T}_0 \in \mathcal{C}$.

We also need the following result.

Claim. Let $\mathcal{C}$ be an admissible subcategory of $C^*\text{Alg}$, let $A \in \mathcal{C}$ and suppose that B is any of the C^* -algebras appearing in the diagram above. Then, $A \otimes B \in \mathcal{C}$; every B above is nuclear so the notation is unambiguous.

We need to check this result for all of the algebras individually.

1. Let $B = \mathbb{C}$. Then, $A \otimes \mathbb{C} \simeq A$ so $A \otimes B \in \mathcal{C}$.
2. Let $B = \mathcal{K}$. We know that $M_{n \times n}(A) \in \mathcal{C}$ for each $n \in \mathbb{N}$. Now, we can write $A \otimes \mathcal{K}$ as an inductive limit of full matrix algebras over A and since $\mathcal{C}$ is closed under inductive limits, it follows that $A \otimes \mathcal{K} \in \mathcal{C}$.
3. Let $B = \mathcal{C}(S^1)$. Then, $A \otimes B \simeq \mathcal{C}(S^1, A)$ and we can fit this into a pullback diagram:

$$\begin{array}{ccc}
\mathcal{C}(S^1, A) & \longrightarrow & \text{Cyl}(A) \\
\downarrow & & \downarrow (\text{ev}_0, \text{ev}_1) \\
A & \xrightarrow{\Delta_A} & A \oplus A
\end{array}$$

Now, $A \oplus A \in \mathcal{C}$ and $\text{Cyl}(A) \in \mathcal{C}$, so we conclude that $\mathcal{C}(S^1, A) \in \mathcal{C}$.

4. Let $B = \mathcal{T}$. Let us tensor the Toeplitz exact sequence on the right by A and note that by [Lemma 4.26](#), the resulting sequence is exact since $\mathcal{C}(S^1)$ is nuclear. Now, we know that $\mathcal{K} \otimes A \in \mathcal{C}$ and $\mathcal{C}(S^1, A) \in \mathcal{C}$. So, we use the extension property of nuclearity^c to conclude that $\mathcal{T} \otimes A \in \mathcal{C}$.
5. Let $B = \mathcal{C}_0(S^1 \setminus \{1\})$. This is just $\text{Sus}(\mathbb{C})$, so tensoring it with A produces $\text{Sus}(A)$ and this lies in $\mathcal{C}$.
6. Let $B = \mathcal{T}_0$. The same argument as we used for $\mathcal{T}$ will work.

This proves the claim.

This completes our desired construction.

^aSee [\[18, Proposition 2.4.1, p. 39, Theorem 3.8.7, p.104\]](#). The first result stated in the citation shows nuclearity of $\mathcal{K}$ through a different definition. The second result shows that this definition is equivalent to the characterization we use for the claim.

^bSee [\[18, Proposition 2.4.2, p. 38-39, Theorem 3.8.7, p.104\]](#).

^cThis was discussed (and referenced) in [Example 4.10](#).

Next, we need two technical lemmas; the first is about tensor products.

Lemma 4.32  Let $A_i, B_i \in C^*\text{Alg}$ and $\phi_i \in C^*\text{Alg}(A_i, B_i)$ for $i \in \{1, 2\}$. TFSH:

1. If each ϕ_i is injective, then so is $\phi_1 \otimes_{\min} \phi_2$.
2. If each ϕ_i is surjective, then so is $\phi_1 \otimes_{\min} \phi_2$.

Proof. We will prove each statement one-by-one.

1. Let $B_1 \subseteq \mathcal{B}(H_1)$ and $B_2 \subseteq \mathcal{B}(H_2)$ be faithfully represented on $\mathbb{C}$ -Hilbert spaces by the Gelfand-Naimark-Segal Theorem. Composing these faithful $\star$ -representations with each ϕ_i , we see that each ϕ_i forms a faithful $\star$ -representation of each A_i . Now, let $x \in A_1 \otimes_{\min} A_2$ be a mixed tensor, represented as follows:

$$x = \sum_{i=1}^m a_i \otimes b_i.$$

Next, we have the following computation:

$$\begin{aligned} \|(\phi_1 \otimes_{\min} \phi_2)(x)\| &= \left\| \sum_{i=1}^m \phi_1(a_i) \otimes \phi_2(b_i) \right\| \\ &= \|x\|, \end{aligned}$$

where we have used the definition of the minimal C^* -norm for the second equality above. This computation implies that $\phi_1 \otimes_{\min} \phi_2$ is isometric on the mixed tensors and since every element of $A_1 \otimes_{\min} A_2$ is a limit of mixed tensors, it follows that $\phi_1 \otimes_{\min} \phi_2$ is isometric on $A_1 \otimes_{\min} A_2$.

2. We use the fact that $\phi_1 \otimes_{\mathbb{C}} \phi_2$ is surjective^a. This implies that $\text{ran}(\phi_1 \otimes_{\min} \phi_2)$ is dense in $B_1 \otimes_{\min} B_2$. But this is also closed in $B_1 \otimes_{\min} B_2$, since $\star$ -homomorphisms have closed ranges. So, $\phi_1 \otimes_{\min} \phi_2$ is surjective. ■

^aTake an element $y \in B_1 \otimes_{\mathbb{C}} B_2$ and write it as a linear combination of simple tensors. Each simple tensor certainly has a preimage in $A_1 \otimes_{\mathbb{C}} A_2$ since ϕ_1 and ϕ_2 are surjective, so we can use that to construct a preimage for y .

Next, we also need a lemma about paths of self-adjoint unitaries.

Lemma 4.33 Let $A \in {}^1C^*\text{Alg}$ and let $u \in A$ be a self-adjoint unitary. There is a path of unitaries from u to 1_A .

Proof. The proof just involves explicitly constructing a path. Indeed, we define:

$$\phi : [0, 1] \rightarrow A, \quad t \mapsto \phi(t) \stackrel{\text{def}}{=} \frac{1}{2}(1_A + u) + \frac{1}{2} \exp(\pi i t)(u - 1_A).$$

Let us check that this works.

| **Claim 4.4.** ϕ is a path of unitaries from u to 1_A .

This is proved on Page 113. This completes the proof. ■

At this stage, we are ready to prove the Bott Periodicity Theorem.

Proof of Theorem 4.29. We tensor the reduced Toeplitz extension by A so we get an extension^a:

$$0 \longrightarrow \mathcal{K} \otimes A \longrightarrow \mathcal{T}_0 \otimes A \xrightarrow{p_2 \otimes \text{Id}_A} \text{Sus}(A) \longrightarrow 0$$

Applying the Barratt-Puppe sequence, we obtain a long-exact sequence:

$$\dots \longrightarrow F_1(\text{Sus}(A)) \longrightarrow F_0(\mathcal{K} \otimes A) \longrightarrow F_0(\mathcal{T}_0 \otimes A) \longrightarrow F_0(\text{Sus}(A))$$

where F_n is defined by Notation 4.21 for each $n \in \mathbb{N}_0$. Now, we have the boundary map:

$$\partial : F_1(\text{Sus}(A)) \rightarrow F_0(\mathcal{K} \otimes A)$$

squeezed in-between $F_0(\mathcal{T}_0 \otimes A)$ and $F_1(\mathcal{T}_0 \otimes A)$. On the other hand, we also notice that:

$$\begin{aligned} F_1(\text{Sus}(A)) &= F_0(\text{Sus}^2(A)) = F_2(A), \\ F_0(\mathcal{K} \otimes A) &\simeq F_0(A), \end{aligned}$$

where the second isomorphism is induced from the C^* -stability. In fact, the proof of statement (4) of Proposition 4.24 showed that this second isomorphism is induced from a natural transformation p from the identity to right-tensoring by $\mathcal{K}$, so let α be the right-whiskering of F_0 with p . Now, we consider the composition $\alpha \circ \partial$. This is a natural transformation and it would be a natural isomorphism provided that for each $A \in \text{dom}(F)$, we have that:

$$F_0(\mathcal{T}_0 \otimes A) = F_1(\mathcal{T}_0 \otimes A) \simeq 0.$$

So, our next goal is to prove the following claim.

| **Claim.** $\forall i \in \{0, 1\} : F_i(\mathcal{T}_0 \otimes A) \simeq 0$.

The rest of the proof is quite complicated and technical. We will essentially show this by a series of subclaims which act as reductions of the claim. Let us start by introducing the following auxiliary C^* -algebra:

$$\widehat{\mathcal{T}} \stackrel{\text{def}}{=} \mathcal{K} \otimes \mathcal{T}_0 + \mathcal{T} \otimes \mathbb{C}1_{\mathcal{T}} \subseteq \mathcal{T} \otimes \mathcal{T}.$$

In fact, we need to prove that this is a C^* -algebra.

| **Subclaim.** $\widehat{\mathcal{T}}$ is a C^* -algebra.

This will follow from three facts:

1. $\mathcal{K} \otimes \mathcal{T}_0$ is a closed ideal in $\mathcal{T} \otimes \mathcal{T}$. This follows from the fact that both $\mathcal{K}$ and $\mathcal{T}_0$ include into $\mathcal{T}$ as ideals, so we just apply Lemma 4.32 to those inclusions to obtain an inclusion of $\mathcal{K} \otimes \mathcal{T}_0$ into $\mathcal{T} \otimes \mathcal{T}$. This is an ideal simply because $\mathcal{K}$ and $\mathcal{T}_0$ are both ideals in $\mathcal{T}$; refer back to the (reduced) Toeplitz extension.

2. $\mathcal{T} \otimes \mathbb{C}\text{Id}_{\mathcal{T}}$ is a C^* -subalgebra of $\mathcal{T} \otimes \mathcal{T}$. This is clear.
3. The sum of a C^* -subalgebra and a closed ideal is a C^* -subalgebra. Let $A \in C^*\text{Alg}$, let $B \subseteq A$ be a C^* -subalgebra and let $I \subseteq A$ be a closed ideal. If $\pi : A \rightarrow A/I$ is the canonical projection, then $\pi(B)$ is closed in A/I since B is a C^* -algebra and π is a $\star$ -homomorphism. On the other hand, we have that:

$$B + I = \pi^{\leftarrow}(\pi(B)),$$

and so, we conclude that $B + I$ is closed.

This proves the subclaim.

Now, we wish to fit $\widehat{\mathcal{T}}$ into an extension. By construction, $\mathcal{K} \otimes \mathcal{T}_0$ is an ideal in $\widehat{\mathcal{T}}$. We note that $\mathcal{T} \otimes \mathbb{C}\text{Id}_{\mathcal{T}} \cap \mathcal{K} \otimes \mathcal{T}_0 = \{0\}$ since the former contains the identity and $\mathcal{T}_0$ does not contain the identity since it is a codimension 1 closed ideal in $\mathcal{T}$. So, we see that $\widehat{\mathcal{T}}/(\mathcal{K} \otimes \mathcal{T}_0) \simeq \mathcal{T}$. Therefore, $\widehat{\mathcal{T}}$ fits into the following extension:

$$0 \longrightarrow \mathcal{K} \otimes \mathcal{T}_0 \longrightarrow \widehat{\mathcal{T}} \longrightarrow \mathcal{T} \longrightarrow 0$$

This leads us to another subclaim.

Subclaim. $\widehat{\mathcal{T}}$ is nuclear, lies in $\text{dom}(F)$ and $\widehat{\mathcal{T}} \otimes B \in \text{dom}(F)$ when $B \in \text{dom}(F)$.

Since both $\mathcal{T}_0$ and $\mathcal{K}$ are nuclear, we see that $\mathcal{K} \otimes \mathcal{T}_0$ is nuclear. Since $\mathcal{T}$ is nuclear, it follows that $\widehat{\mathcal{T}}$ is nuclear. Now, using the same extension, the fact that $\mathcal{K} \otimes \mathcal{T}_0 \in \text{dom}(F)$ by the last claim of [Construction 4.31](#) and the fact that $\mathcal{T} \in \text{dom}(F)$, we conclude that $\widehat{\mathcal{T}} \in \text{dom}(F)$. Finally, let $B \in \text{dom}(F)$. Using the nuclearity of $\mathcal{T}$, we can tensor the extension above by B . Applying [Lemma 4.26](#), we obtain an extension:

$$0 \longrightarrow \mathcal{K} \otimes \mathcal{T}_0 \otimes B \longrightarrow \widehat{\mathcal{T}} \otimes B \longrightarrow \mathcal{T} \otimes B \longrightarrow 0$$

Using the last claim of [Construction 4.31](#), we see that $\mathcal{K} \otimes \mathcal{T}_0 \otimes B \in \text{dom}(F)$ and $\mathcal{T} \otimes B \in \text{dom}(F)$. So, it follows that $\widehat{\mathcal{T}} \otimes B \in \text{dom}(F)$ and we are done.

Since $\mathcal{K} \otimes \mathcal{T}_0 \in \text{dom}(F)$ and $\mathcal{T} \in \text{dom}(F)$, we conclude that $\widehat{\mathcal{T}} \in \text{dom}(F)$. Moreover, $\mathcal{T}$ is nuclear so we can tensor the extension above by A and by applying [Lemma 4.26](#), we obtain an extension:

$$0 \longrightarrow \mathcal{K} \otimes \mathcal{T}_0 \otimes A \longrightarrow \widehat{\mathcal{T}} \otimes A \longrightarrow \mathcal{T} \otimes A \longrightarrow 0$$

We want to note that all three algebras here lie in $\text{dom}(F)$; the first and third one lie in $\text{dom}(F)$ due to the third claim of [Construction 4.31](#) and the middle one lies in $\text{dom}(F)$ since $\text{dom}(F)$ is extension-closed. Now, we note that this sequence actually splits. Indeed, we can define the following $\star$ -homomorphism:

$$k : \mathcal{T} \otimes A \rightarrow \widehat{\mathcal{T}} \otimes A, \quad b \otimes a \mapsto k(b \otimes a) \stackrel{\text{def}}{=} b \otimes 1_{\mathcal{T}} \otimes a.$$

This is clearly a section for the map $\widehat{\mathcal{T}} \otimes A \rightarrow \mathcal{T} \otimes A$. If we put this arrow into the diagram above and also include $\mathcal{T}_0 \otimes A$ into its stabilization as usual, then we obtain the following commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{K} \otimes \mathcal{T}_0 \otimes A & \xrightarrow{i} & \widehat{\mathcal{T}} \otimes A & \xleftarrow{k} & \mathcal{T} \otimes A \longrightarrow 0 \\ & & \uparrow p & & \nearrow j & & \\ & & \mathcal{T}_0 \otimes A & & & & \end{array}$$

where we explicitly note that p is defined by the formula $a \otimes b \mapsto e_{00} \otimes ab$, with e_{00} having been defined at the start of [Construction 4.31](#). This leads us to our second subclaim.

Subclaim. For $t \in \{0, 1\}$, we have that $F_t(j)$ is injective.

Since $F = F_0$ is homotopy-invariant, half-exact and C^* -stable, we can deduce that the same is true for F_1 by the strength of [Proposition 4.24](#). We note that homotopy-invariance & half-exactness implies, by statement (2) of [Proposition 4.18](#), that F_0 is split-exact and this means that F_1 is also split-exact. Now, let $t \in \{0, 1\}$ and note that $F_t(p)$ is an isomorphism since F_t is C^* -stable, while $F_t(i)$ is injective because F_t is split-exact and our sequence above is split-exact. So, it follows that $F_t(j)$ is injective.

So, if we want to show our main claim, it suffices to prove that:

$$\forall i \in \{0, 1\} : F_i(j) = 0.$$

To prove this, we will construct another map from $\mathcal{T}_0 \otimes A \rightarrow \widehat{\mathcal{T}} \otimes A$. Now, we need a general result.

Claim 4.5. Let $B \in {}^1C^*\text{Alg}$ and let $u \in B$ be an isometry. Then, the map:

$$\text{Ad}(u) : B \rightarrow B, \quad x \mapsto \text{Ad}(u)(x) \stackrel{\text{def}}{=} u x u^*,$$

is a $\star$ -homomorphism. This is proved on [Page 113](#).

If v is the isometry generating $\mathcal{T}$, then we define:

$$\psi : \widehat{\mathcal{T}} \otimes A^{\sim} \rightarrow \widehat{\mathcal{T}} \otimes A^{\sim},$$

as $\text{Ad}(v \otimes 1_{\mathcal{T}} \otimes 1_{A^{\sim}})$. The input into $\text{Ad}(-)$ is an isometry since v is an isometry, so ψ is a $\star$ -homomorphism by

Claim 4.5. Now, we observe that $\psi(\widehat{\mathcal{T}} \otimes A) \subseteq \widehat{\mathcal{T}} \otimes A$; this is clear by directly using the definition of ψ and noting that the conjugating isometry has $1_{A^\sim}$ as its last entry. Now, we have our next subclaim.

Subclaim. $\psi \perp j$. That is, ψ and j are orthogonal^a.

^aRecall that this notion was introduced in Terminology 4.17.

We observe that, explicitly, we have that:

$$\forall x \in \mathcal{T}_0 \otimes A : j(x)\psi(x) = (e_{00} \otimes x)(v \otimes 1_{\mathcal{T}} \otimes 1_{A^\sim})x(v^* \otimes 1_{\mathcal{T}} \otimes 1_{A^\sim}),$$

where e_{00} is a rank-one projection in $\mathcal{K}$ and is used to define p explicitly. Now, if we multiply out the first two terms, then we see that we need to evaluate $e_{00}(v \otimes 1_{\mathcal{T}})$. This is the same as taking a unilateral shift on $l^2(\mathbb{N})$ and projecting down to the first coordinate; in that case, we get that this must be 0 and this proves the subclaim.

Now, we construct the map:

$$\psi_1 \stackrel{\text{def}}{=} \psi \circ k|_{\mathcal{T}_0 \otimes A},$$

and this acts from $\mathcal{T}_0 \otimes A$ to $\widehat{\mathcal{T}} \otimes A$. By the subclaim above, we deduce that:

$$\psi_1 \perp j.$$

Let $\psi_0 \stackrel{\text{def}}{=} j + \psi_1$; this is a $\star$ -homomorphism, by statement (1) of Proposition 4.18. Since F_0 and F_1 are both homotopy-invariant and half-exact, we can apply statement (4) of the aforementioned result to conclude that:

$$\forall i \in \{0, 1\} : F_i(\psi_0) = F_i(j) + F_i(\psi_1).$$

This equation leads us to our next reduction.

Subclaim. It suffices to prove that ψ_0 and ψ_1 are homotopic $\star$ -homomorphisms.

If we do this, then the homotopy-invariance of each F_i implies that:

$$F_i(\psi_0) = F_i(\psi_1),$$

which implies that $F_i(j) = 0$ for $i \in \{0, 1\}$, which is exactly what we wanted to show.

Now, the rest of the proof involves constructing a homotopy from ψ_0 to ψ_1 . Firstly, let us show that we can reduce to the case where $A = \mathbb{C}$. To prove this, we need the result of the following claim.

Claim 4.6. Let $A, B \in \text{NucC}^*\text{Alg}$, let $C \in \text{C}^*\text{Alg}$ and let $f, g \in \text{NucC}^*\text{Alg}(A, B)$ be homotopic. Then:

$$f \otimes \text{Id}_C \simeq_h g \otimes \text{Id}_C.$$

That is, tensoring by C preserves homotopies. This is proved on Page 113.

Now, ψ_0 and ψ_1 both depend on A , so we may reflect this dependence by writing them as ψ_0^A and ψ_1^A . If we, then, run through the explicit construction of these maps, we will see that:

$$\forall i \in \{0, 1\} : \psi_i^A = \psi_i^{\mathbb{C}} \otimes \text{Id}_C.$$

So, if we can show that $\psi_0^{\mathbb{C}}$ and $\psi_1^{\mathbb{C}}$ are homotopic, then **Claim 4.6** will give us a homotopy between ψ_0^A and ψ_1^A . Therefore, we assume that $A = \mathbb{C}$. If that is the case, then ψ_0 and ψ_1 are maps $\mathcal{T}_0 \rightarrow \widehat{\mathcal{T}}$. Since $\mathcal{T}_0$ fits into a split-exact sequence with $\mathcal{T}$ as a codimension 1 closed ideal, it is the case that:

$$\mathcal{T}_0 \simeq \mathcal{T}.$$

So, we can extend ψ_0 and ψ_1 uniquely to $\mathcal{T}$; we denote these extensions by π_0 and π_1 . We do this so that we can construct our homotopy through universal property by simply specifying it on the generator v of $\mathcal{T}$. We define:

$$v_0 \stackrel{\text{def}}{=} \pi_0(v),$$

$$v_1 \stackrel{\text{def}}{=} \pi_1(v).$$

Therefore, constructing a homotopy from π_0 to π_1 is the same as specifying a path of isometries from v_0 to v_1 . Since π_0 and π_1 are unital, let us observe that v_0 and v_1 are isometries. Now, we have the final subclaim of this proof.

Subclaim. We can find unitaries u_0, u_1 such that:

$$\forall i \in \{0, 1\} : v_i = u_i(v \otimes 1_{\mathcal{T}}).$$

Moreover, these unitaries are self-adjoint.

Let us momentarily postpone the proof of this subclaim and, instead, construct our path of isometries from v_0 to v_1 . If the subclaim holds, then **Lemma 4.33** gives a path of unitaries $(u_t)_{t \in [0,1]}$ from u_0 to u_1 . Now, we simply define:

$$\forall t \in [0, 1] : v_t \stackrel{\text{def}}{=} u_t(v \otimes 1_{\mathcal{T}}).$$

It is readily checked by an explicit computation that each v_t is an isometry. Moreover, this is a path because $(u_t)_{t \in [0,1]}$ is a path and multiplication is a continuous map. Finally, this is clearly a path from v_0 to v_1 , given the content of the subclaim. So, modulo this subclaim, this completes the proof of the Bott Periodicity Theorem.

Let us prove the subclaim. For this, we need to simply come up with explicit formulas for u_0 and u_1 . Motivating these formulas requires a fairly extended discussion where we embed $\mathcal{T} \otimes \mathcal{T}$ into $\mathcal{B}(l^2(\mathbb{N}_0 \times \mathbb{N}_0))$ and study the action

of each v_i on the canonical orthonormal basis. This would provide motivation for the formulas that we will present:

$$u_0 \stackrel{\text{def}}{=} v^2(v^\star)^2 \otimes 1_{\mathcal{T}} + e_{00}v^\star \otimes v + ve_{00} \otimes v^\star + e_{00} \otimes e_{00}$$

$$u_1 \stackrel{\text{def}}{=} v^2(v^\star)^2 \otimes 1_{\mathcal{T}} + e_{00}v^\star \otimes 1_{\mathcal{T}} + ve_{00} \otimes 1_{\mathcal{T}},$$

where $e_{00} = 1 - vv^\star$. Now, there are several statements that need to be shown and all of them require, unfortunately, very detailed computations. We will carry them out in the next few claims just for u_0 and v_0 . However, we stress that the philosophy is exactly the same with u_1 and v_1 . For the sake of brevity, we will write 1 whenever we mean $1_{\mathcal{T}}$ and this will not cause us issues. With that said, let us proceed towards our desired results.

| **Claim 4.7.** $\pi_0(v) = v^2v^\star \otimes 1 + e_{00} \otimes v$.

| **Claim 4.8.** $\pi_0(v) = u_0(v \otimes 1)$

| **Claim 4.9.** u_0 is a self-adjoint unitary.

Again, similar calculations go through for u_1 and v_1 , so we're done. ■

^aThis is an application of [Lemma 4.26](#), where the lemma applies since $\mathcal{C}_0((0, 1))$ is nuclear.

Through our work above, we hope that we have demonstrated that many results involving axiomatic (co)homology on $C^\star$ -algebras can be deduced formally and they can also be understood on various interesting subcategories using the notion of fibrant subcategory. Our last order of business in the main body of this thesis is to address the Spanier-Whitehead construction. In the next subsection, we will provide a proof of that result using machinery from higher category theory.

4.2 Towards The Spanier-Whitehead Construction

Our goal is to prove the following theorem.

Theorem 4.34. The Spanier-Whitehead Construction

Let $\mathcal{C}$ be a category of fibrant objects. Then, $\mathbf{SHo}(\mathcal{C})$ admits the structure of a triangulated category.

Indeed, this is the first portion of [Theorem 3.79](#). Let us discuss the strategy for proving this theorem.

Remark 4.35. Let $\mathcal{C}$ be a finitely bicomplete category of fibrant objects. Then:

1. We will take the ∞ -categorical localization of $\mathcal{C}$ at its weak equivalences; call this $\mathcal{D}$.
2. We will show that $\mathcal{D}$ is a pointed ∞ -category which is finitely bicomplete.
3. We will define the Spanier-Whitehead category:

$$\mathbf{SW}(\mathcal{D}),$$

which is supposed to be the ∞ -categorical analogue of the same construction as the stable homotopy category.

4. We will show that $\mathbf{SW}(\mathcal{D})$ is a stable ∞ -category.

5. Since the homotopy category of a stable ∞ -category is a triangulated category, the theorem will follow quickly.

Our main encyclopaedic references for higher category theory are [\[57\]](#), [\[56\]](#), [\[58\]](#) and [\[24\]](#) & we will usually refer to these resources for specific results with full proofs (though we will sometimes reference the more educational resources we'll mention in just a moment if they are clearer). Since we will need a few facts about stable ∞ -categories, we will also find it reasonable to reference [\[55\]](#) since it is much shorter than the aforementioned references. For a very pleasant introduction to ∞ -categories from the viewpoint of quasicategories, see [\[78\]](#). For actually learning this material in a synthetic and model-independent way, we recommend the (in-progress) book project [\[25\]](#).

Before we proceed towards the explicit realization of this strategy, we need to discuss triangulated categories. We will only need the definition for our exposition to be complete, but we will include some basic results about pre-triangulated categories just for intuition. By now, there are neat resources on triangulated category theory; the main reference is (Amnon) Neeman's treatise [\[67\]](#). The main ingredients for specifying a triangulated category are listed as follows:

1. an additive category $\mathcal{T}$,
2. an additive automorphism $\Sigma : \mathcal{T} \rightarrow \mathcal{T}$,
3. a set $\mathcal{E}$ of "exact triangles".

This data is specified, in some sense, out of necessity. Historically, it was recognized that there are many interesting categories, such as the stable homotopy category or the derived category of an abelian category, which are not abelian but seem to be equipped with the data listed above. It was, then, noticed that all of them satisfy a similar list of axioms which seem to give some control over long-exact sequences. Such long-exact sequences are encoded precisely by triangles in an additive category equipped with an additive automorphism, as we will demonstrate in the next construction.

Construction 4.36. Let $\mathcal{T}$ be an additive category and let $\Sigma \in \text{Aut}(\mathcal{T})$ be additive. We define a structure:

$$\Delta(\mathcal{T}, \Sigma),$$

as follows:

1. An object in $\Delta(\mathcal{T}, \Sigma)$, which we call a triangle for Σ , is a sequence of morphisms:

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

We refer to x, y, z as the vertices of the triangle while the morphisms are the sides.

2. If we have two triangles for Σ , then a morphism between them is represented by the dotted arrows below:

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \xrightarrow{g} & z & \xrightarrow{h} & x[1] \\ \vdots & & \vdots & & \vdots & & \downarrow \Sigma(a) \\ x' & \xrightarrow{f'} & y' & \xrightarrow{g'} & z' & \xrightarrow{h'} & x'[1] \end{array}$$

such that the diagram commutes. In other words, a morphism is a triple (a, b, c) consisting of morphisms between the vertices such that the diagram above commutes. The composition of morphisms is defined exactly as one expects and clearly, the composite of two morphisms yields a morphism between triangles.

This data turns $\Delta(\mathcal{T}, \Sigma)$ into a category.

To give the desired definition, we need one piece of terminology and some notation.

Terminology 4.37. Let $\mathcal{C}$ be a category and let $\mathcal{D}$ be a full subcategory of $\mathcal{C}$. $\mathcal{D}$ is closed under isomorphisms iff for every $x \in \mathcal{D}$ and for every $y \in \mathcal{C}$ which is isomorphic to x , we have that $y \in \mathcal{D}$.

Notation 4.38. Let $\mathcal{C}$ be a category, let $x \in \mathcal{C}$ and let $T \in \text{Fun}(\mathcal{C})$. We define:

$$x[1] \stackrel{\text{def}}{=} T(x).$$

In this case, T will also be known as a shift/suspension^a. Indeed, we define:

$$\forall n \in \mathbb{Z} : x[n] \stackrel{\text{def}}{=} T^n(x),$$

where this is defined for negative values of n because T has an inverse.

^aThis only makes sense when T comes as part of the data.

With this notation in mind, we are ready to define Neeman pre-triangulated categories.

Definition 4.39

A (Neeman) pre-triangulated category is a triple $(\mathcal{T}, \Sigma, \text{Ex}\Delta(\mathcal{T}, \Sigma))$ consisting of:

1. an additive category $\mathcal{T}$,
2. an additive automorphism $\Sigma : \mathcal{T} \rightarrow \mathcal{T}$ and,
3. a full subcategory $\text{Ex}\Delta(\mathcal{T}, \Sigma)$ of $\Delta(\mathcal{T}, \Sigma)$ which is closed under isomorphisms,

such that the following properties hold:

TR1: Let $x, y \in \mathcal{T}$ and let $f \in \mathcal{T}(x, y)$. Then, the following triangle:

$$x \xrightarrow{\text{Id}_x} x \longrightarrow 0 \longrightarrow x[1]$$

lies in $\text{Ex}\Delta(\mathcal{T}, \Sigma)$. Moreover, for any $z \in \mathcal{T}$, there exists a triangle which fits f :

$$x \xrightarrow{f} y \longrightarrow z \longrightarrow x[1]$$

within $\text{Ex}\Delta(\mathcal{T}, \Sigma)$.

TR2: The following triangle:

$$x \xrightarrow{u} y \xrightarrow{v} z \xrightarrow{w} x[1]$$

lies in $\text{Ex}\Delta(\mathcal{T}, \Sigma)$ iff the following triangle:

$$y \xrightarrow{v} z \xrightarrow{w} x[1] \xrightarrow{-u[1]} y[1]$$

lies in $\text{Ex}\Delta(\mathcal{T}, \Sigma)$.

TR3: If we have a commutative diagram:

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \xrightarrow{g} & z & \xrightarrow{h} & x[1] \\ \downarrow a & & \downarrow b & & \downarrow c & & \downarrow a[1] \\ x' & \xrightarrow{f'} & y' & \xrightarrow{g'} & z' & \xrightarrow{h'} & x'[1] \end{array}$$

of solid arrows with the rows being triangles in $\text{Ex}\Delta(\mathcal{T}, \Sigma)$, then there exists a morphism $c : z \rightarrow z'$ such that the triple (a, b, c) forms a morphism of triangles in $\text{Ex}\Delta(\mathcal{T}, \Sigma)$.

$\text{Ex}\Delta(\mathcal{T}, \Sigma)$ is known as the full subcategory of distinguished/exact triangles.

Let us introduce some terminology that we have found useful for proofs.

Terminology 4.40. Let $\mathcal{T}$ be a pre-triangulated category, let $x, y \in \mathcal{T}$ and let $f \in \mathcal{C}(x, y)$ fit in a exact triangle:

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

Then, the pair (z, g) is the cone of f (or dually, (x, f) is a cocone of g). Suppose we have a exact triangle:

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

By TR2, we have another exact triangle:

$$y \xrightarrow{g} z \xrightarrow{h} x[1] \xrightarrow{-\Sigma(f)} y[1]$$

This will be known as the shifted triangle^a. Moreover, if we have a commutative diagram:

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \xrightarrow{g} & z & \xrightarrow{h} & x[1] \\ \downarrow a & & \downarrow b & & & & \downarrow \Sigma(a) \\ x' & \xrightarrow{f'} & y' & \xrightarrow{g'} & z' & \xrightarrow{h'} & x'[1] \end{array}$$

where the rows are exact triangles, then the pair (a, b) is called a partial morphism of exact triangles.

^aOne might even say suspended triangle but “shift” works better here.

As promised above, we will present some (known) results which demonstrate how this theory is used.

Proposition 4.41

Let $\mathcal{T}$ be a pre-triangulated category and consider the following exact triangle:

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

Then, $g \circ f = 0$ and $h \circ g = 0$.

Proof. By working with the shifted triangle, it suffices to prove that $g \circ f = 0$. Next, consider the diagram:

$$\begin{array}{ccccccc} x & \xrightarrow{\text{Id}_x} & x & \longrightarrow & 0 & \longrightarrow & x[1] \\ \downarrow \text{Id}_x & & \downarrow f & & \downarrow \text{dashed} & & \downarrow \text{Id}_{x[1]} \\ x & \xrightarrow{f} & y & \xrightarrow{g} & z & \xrightarrow{h} & x[1] \end{array}$$

Over here, we applied TR3 to the solid diagram above. This induces a morphism $0 \rightarrow z$ and this has to be the unique morphism from 0 to z . So, the composition $x \rightarrow 0 \rightarrow z$ is just the zero morphism and this means that $g \circ f = 0$. ■

One may have noticed that TR3 is very specific about which morphism is excluded from the diagram and “filled in”. However, one can show that variants of TR3 hold where the other two morphisms are excluded and then “filled in”.

Proposition 4.42

Let $\mathcal{T}$ be a pre-triangulated category and suppose we have two commutative diagrams:

$$\begin{array}{ccccccc} x & \longrightarrow & y & \longrightarrow & z & \longrightarrow & x[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ x' & \longrightarrow & y' & \longrightarrow & z' & \longrightarrow & x[1]' \end{array} \quad \begin{array}{ccccccc} x & \longrightarrow & y & \longrightarrow & z & \longrightarrow & x[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ x' & \longrightarrow & y' & \longrightarrow & z' & \longrightarrow & x[1]' \end{array}$$

where the rows are exact triangles. Then, we can “fill in” the missing morphisms so we get the following:

$$\begin{array}{ccccccc} x & \longrightarrow & y & \longrightarrow & z & \longrightarrow & x[1] \\ \downarrow & & \downarrow \text{dashed} & & \downarrow & & \downarrow \\ x' & \longrightarrow & y' & \longrightarrow & z' & \longrightarrow & x[1]' \end{array} \quad \begin{array}{ccccccc} x & \longrightarrow & y & \longrightarrow & z & \longrightarrow & x[1] \\ \downarrow \text{dashed} & & \downarrow & & \downarrow & & \downarrow \text{dashed} \\ x' & \longrightarrow & y' & \longrightarrow & z' & \longrightarrow & x[1]' \end{array}$$

commutative diagrams.

Proof. If we shift each row of the right diagram to the left via TR2, then we get a commutative diagram:

$$\begin{array}{ccccccc} y & \longrightarrow & z & \longrightarrow & x[1] & \longrightarrow & y[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ y' & \longrightarrow & z' & \longrightarrow & x[1]' & \longrightarrow & y[1]' \end{array}$$

with each row being an exact triangle. By TR3, we obtain a morphism $x[1] \rightarrow x[1]'$ such that the resulting diagram above commutes. Since Σ is an automorphism, we can find a morphism $x \rightarrow x'$ such that the aforementioned right diagram commutes. So, the result holds for the diagram on the right (in the statement) and we apply it to the diagram on the left by first shifting each row to the left & obtaining the diagram on the right. ■

With some of these results in place, we can talk about cohomological functors on pre-triangulated categories.

Terminology 4.43. Let $\mathcal{A}$ be an abelian category, let $\mathcal{T}$ be a pre-triangulated category and let $F \in \text{Fun}(\mathcal{T}, \mathcal{A})$. F is cohomological iff for every exact triangle $x \rightarrow y \rightarrow z \rightarrow x[1]$, we have that the sequence:

$$F(x) \longrightarrow F(y) \longrightarrow F(z)$$

is exact in $\mathcal{A}$.

The intuition is that cohomological functors are a recipe for turning short exact sequences into long exact sequences. Since pre-triangulated categories are additive (and therefore, enriched in abelian groups), we may view the (co)representable functors on such categories as functors into AbGrp . Since AbGrp is an abelian category, the next result makes sense.

Proposition 4.44. Triangulated Yoneda Lemma

Let $\mathcal{T}$ be a pre-triangulated category and let $x \in \mathcal{T}$. Then, the (co)representable functors:

$$\begin{aligned} \mathcal{T}(x, -) : \mathcal{T} &\rightarrow \text{AbGrp}, \\ \mathcal{T}(-, x) : \mathcal{T}^{\text{op}} &\rightarrow \text{AbGrp}, \end{aligned}$$

are cohomological.

Proof. We will only prove this for $\mathcal{T}(x, -)$ as the result for $\mathcal{T}(-, x)$ follows analogously. Consider an exact triangle:

$$x' \xrightarrow{f} y' \xrightarrow{g} z' \xrightarrow{h} x[1]'$$

and consider the sequence:

$$\mathcal{T}(x, x') \xrightarrow{\mathcal{T}(x, f)} \mathcal{T}(x, y') \xrightarrow{\mathcal{T}(x, g)} \mathcal{T}(x, z')$$

This is a sequence of abelian groups and to check exactness, we need to check that:

$$\text{ran}(\mathcal{T}(x, f)) = \ker(\mathcal{T}(x, g)).$$

We'll prove each inclusion separately.

$\subseteq$: Since $g \circ f = 0$, it follows that:

$$0 = \mathcal{T}(x, g \circ f) = \mathcal{T}(x, g) \circ \mathcal{T}(x, f),$$

which means that $\text{ran}(\mathcal{T}(x, f)) \subseteq \ker(\mathcal{T}(x, g))$.

$\supseteq$: For the other inclusion, let $u \in \ker(\mathcal{T}(x, g))$. Then, $u \in \mathcal{T}(x, y')$ and $g \circ u = 0$. Now, we need a $v \in \mathcal{T}(x, x')$ such that $f \circ v = u$. The trick is to fit all of this data in a commutative diagram with a partial morphism between two exact triangles. In this case, the following diagram fulfills the desired requirement:

$$\begin{array}{ccccccc} x & \xrightarrow{\text{Id}_x} & x & \longrightarrow & 0 & \longrightarrow & x[1] \\ & & \downarrow u & & \downarrow & & \\ x' & \xrightarrow{f} & y' & \xrightarrow{g} & z' & \xrightarrow{h} & x[1]' \end{array}$$

Indeed, this diagram commutes since $g \circ u = 0$. So, now, we use Proposition 4.42 to fill in the diagram above with a morphism $v : x \rightarrow x'$ so that the diagram commutes. But this means precisely that:

$$u = f \circ v = \mathcal{T}(x, f)(v),$$

as was desired.

This completes the proof. ■

We also have a triangulated version of the Five Lemma.

Proposition 4.45. Triangulated Five Lemma

Let $\mathcal{T}$ be a pre-triangulated category and consider a morphism in $\text{Ex}\Delta(\mathcal{T}, \Sigma)$ as follows:

$$\begin{array}{ccccccc} x & \longrightarrow & y & \longrightarrow & z & \longrightarrow & x[1] \\ \downarrow a & & \downarrow b & & \downarrow c & & \downarrow a[1] \\ x' & \longrightarrow & y' & \longrightarrow & z' & \longrightarrow & x[1]' \end{array}$$

If a and b are isomorphisms, then so is c .

Proof. The key is to use the (co)representable functors, which are cohomological, to reduce this problem to the Five Lemma for abelian groups. For $w \in \mathcal{T}$, we apply the functor $\mathcal{T}(w, -)$ so that we get the following large diagram:

$$\begin{array}{ccccccccc} \mathcal{T}(w, x) & \longrightarrow & \mathcal{T}(w, y) & \longrightarrow & \mathcal{T}(w, z) & \longrightarrow & \mathcal{T}(w, x[1]) & \longrightarrow & \mathcal{T}(w, y[1]) \\ \downarrow \mathcal{T}(w, a) & & \downarrow \mathcal{T}(w, b) & & \downarrow \mathcal{T}(w, c) & & \downarrow \mathcal{T}(w, a[1]) & & \downarrow \mathcal{T}(w, b[1]) \\ \mathcal{T}(w, x') & \longrightarrow & \mathcal{T}(w, y') & \longrightarrow & \mathcal{T}(w, z') & \longrightarrow & \mathcal{T}(w, x'[1]) & \longrightarrow & \mathcal{T}(w, y'[1]) \end{array}$$

This is a commutative diagram of abelian groups where the rows are exact since $\mathcal{T}(w, -)$ is a cohomological functor. Since the solid vertical morphisms are isomorphisms, we can apply the Five Lemma to conclude that $\mathcal{T}(w, c)$ is an isomorphism. Now, we can use the Yoneda Lemma to conclude that c is an isomorphism. ■

As a corollary, we obtain a useful characterization of when a morphism is an isomorphism in a pre-triangulated category.

Claim 4.10. Let $\mathcal{T}$ be a pre-triangulated category and let $f : x \rightarrow y$ be a morphism in $\mathcal{T}$. TFAE:

1. f is an isomorphism.
2. The triangle $x \xrightarrow{f} y \rightarrow 0 \rightarrow x[1]$ is exact.

This is proved on Page 115.

We have a remark about our work above.

Remark 4.46. We note that there are treatments of triangulated category theory where one assumes that the underlying shift is not an automorphism but only an *auto-equivalence*. It is not completely clear if this makes a huge difference for the purposes of the theory. Nevertheless, this is related to some of the applications of the Spanier-Whitehead construction which we discussed in Remark 3.81. Indeed, one can “fully invert” the shift (if it is only an auto-equivalence) precisely via the Spanier-Whitehead construction.

We have worked with pre-triangulated categories so far. The next step is to define triangulated categories.

Definition 4.47

Let $\mathcal{T}$ be a pre-triangulated category. It is (Neeman) triangulated iff for any three exact triangles:

$$x \xrightarrow{u} y \longrightarrow z \longrightarrow x[1]$$

$$x \xrightarrow{v \circ u} y' \longrightarrow z' \longrightarrow x[1]$$

$$y \xrightarrow{v} y' \longrightarrow y'' \longrightarrow y[1]$$

in $\mathcal{T}$, there exist morphisms $f \in \mathcal{T}(z, z')$, $g \in \mathcal{T}(z', y'')$ and $h \in \mathcal{T}(y'', z[1])$ such that the diagram:

$$\begin{array}{ccccccc} x & \xrightarrow{u} & y & \longrightarrow & z & \longrightarrow & x[1] \\ \parallel & & \downarrow v & & \downarrow f & & \parallel \\ x & \xrightarrow{v \circ u} & y' & \longrightarrow & z' & \longrightarrow & x[1] \\ \downarrow u & & \parallel & & \downarrow g & & \downarrow u[1] \\ y & \xrightarrow{v} & y' & \longrightarrow & y'' & \longrightarrow & y[1] \\ \downarrow & & \downarrow & & \parallel & & \downarrow \\ z & \xrightarrow{\quad f \quad} & z' & \xrightarrow{\quad g \quad} & y'' & \xrightarrow{\quad h \quad} & z[1] \end{array}$$

commutes and the bottom row is an exact triangle. This is known as TR4 or the octahedral axiom.

At this stage, we will not continue further development of triangulated category theory since we will not need its finer points here. Instead, we will move straight to the original problem of interest; proving [Theorem 4.34](#). Let us start implementing the strategy in [Remark 4.35](#) and for that, we will first discuss ∞ -categorical localizations.

Definition 4.48

Let $\mathcal{C}$ be an ∞ -category and let W be a set of morphisms in $\mathcal{C}$. The Dwyer-Kan localization (DK-localization) of $\mathcal{C}$ at W is an ∞ -category $\mathcal{C}_W$ and a functor $L : \mathcal{C} \rightarrow \mathcal{C}_W$ which induces an equivalence of ∞ -categories^a:

$$\mathrm{Fun}(\mathcal{C}_W, \mathcal{D}) \rightarrow \mathrm{Fun}^W(\mathcal{C}, \mathcal{D}), \quad f \mapsto f \circ L,$$

for any ∞ -category $\mathcal{D}$. The Dwyer-Kan localization is also defined for 1-categories with respect to some set of morphisms; we simply pass to the ∞ -category associated to the given 1-category by the nerve & take the DK-localization with respect to the set of morphisms obtained after passing through the nerve.

^a $\mathrm{Fun}^W(\mathcal{C}, \mathcal{D})$ is the full subcategory of $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ spanned by those functors $f : \mathcal{C} \rightarrow \mathcal{D}$ which invert W .

We have a further remark about this definition.

Remark 4.49. The existence of Dwyer-Kan localizations in the model of quasicategories is established in [24, Proposition 7.1.3, p.321-322]; in the synthetic framework for ∞ -category theory, the existence is put forth as an axiom. One point which we would like to mention, which is emphasized in [24, Remark 7.1.4, p.322], is that the localization can be chosen so that the objects of $\mathcal{C}$ are the same as the objects of the localization, so the functor L is just the identity at the level of objects. This seems to rely on the intricacies of the construction of the localization.

We have deliberately chosen different notation for the Dwyer-Kan localization to distinguish it from the 1-categorical localization. These two are related to each other in a very simple way which can be described via the homotopy category construction. We will need this construction later, so it is best to discuss it now. Roughly speaking, the homotopy category kills all data about higher homotopies in an ∞ -category and produces a 1-category from the “killed” data.

Construction 4.50. Let $N : \mathrm{Cat}_1 \rightarrow \mathrm{Cat}_\infty$ be the nerve. It has a left adjoint:

$$\mathbf{Ho}(-) : \mathrm{Cat}_\infty \rightarrow \mathrm{Cat}_1,$$

which we will refer to as the homotopy category functor.

The “construction” above is purely synthetic; we know that such a left adjoint will exist within a model for ∞ -categories, so we can synthetically impose the existence of such a left adjoint. For the sake of being somewhat concrete in this discussion, let us describe how this functor is constructed via quasicategories, a model for ∞ -categories.

Remark 4.51. This discussion is primarily inspired by material found in [78, p.30-34]. The key point is that if the term “ ∞ -category” means “quasicategory”, then there is a concrete construction of the nerve and there is a concrete construction of the homotopy category of a quasicategory; we will assume some familiarity with the former. For the latter, the key is to note that it’s possible to define a notion of homotopy between 1-simplices (read: morphisms) in

a quasicategory. Let us denote this relation by $\simeq_h$. If X is a quasicategory, then we simply define:

$$\mathbf{Ho}(X),$$

as follows:

1. The objects are the elements of X_0 .
2. If $x, y \in X_0$, then:

$$\mathbf{Ho}(X)(x, y) \stackrel{\text{def}}{=} \text{hom}_X(x, y) / \simeq_h.$$

The composition, which is defined as one expects, is well-defined.

Note that this is a 1-category because we have defined it to be so. More importantly, it gives us a direct construction of the homotopy category from a quasicategory^a. As is shown in the resource above, this construction can be assembled into a functor and it is this functor which forms a left adjoint to the nerve.

^aAs (Charles) Rezk shows in the aforementioned notes, one can construct a 1-category out of any simplicial set.

The next result tells us how we can obtain the 1-categorical localization from the Dwyer-Kan localization.

Proposition 4.52

Let $\mathcal{C}$ be a 1-category and let W be a set of morphisms in $\mathcal{C}$. Then, $\mathbf{Ho}(\mathcal{C}_W) \simeq \mathcal{C}[W^{-1}]$.

Proof. We first note that, in this case, we have that:

$$\mathcal{C}_W \stackrel{\text{def}}{=} N(\mathcal{C})_{N(W)},$$

where the right-hand side is the Dwyer-Kan localization of the ∞ -category $N(\mathcal{C})$ with respect to the set of morphisms $N(W)$; we mentioned this in [Definition 4.48](#). We would like to make sense of the following calculation:

$$\begin{aligned} \text{Fun}(\mathbf{Ho}(\mathcal{C}_W), \mathcal{D}) &\simeq \text{Fun}(\mathcal{C}_W, N(\mathcal{D})) \\ &\simeq \text{Fun}(N(\mathcal{C})_{N(W)}, N(\mathcal{D})) \\ &\simeq \text{Fun}^{N(W)}(N(\mathcal{C}), N(\mathcal{D})) \\ &\simeq N(\text{Fun}^W(\mathcal{C}, \mathcal{D})) \\ &\simeq \text{Fun}^W(\mathcal{C}, \mathcal{D}) \\ &\simeq \text{Fun}(\mathcal{C}[W^{-1}], \mathcal{D}). \end{aligned}$$

This calculation implies the result if it makes sense. For this to work, we will impose that $\mathcal{D}$ is a 1-category. Now, the first step is just the nerve-homotopy adjunction of [Construction 4.50](#). In the second step, we use the definition of $\mathcal{C}_W$. In the third step, we use the universal property of $\mathcal{C}_W$. In the fourth step, we note that $\mathcal{C}$ and $\mathcal{D}$ are 1-categories so functors between them are identified with the ∞ -functors between their associated ∞ -categories; we use this also in the fifth step. Finally, we use the universal property of the 1-categorical localization. ■

Now, we have the following result.

Proposition 4.53

Let $\mathcal{C}$ be a finitely complete 1-category of fibrant objects. Then, $\mathcal{C}_W$ is a pointed and finitely complete ∞ -category.

Proof. This is proved in [\[24, Proposition 7.5.6, p.370-371\]](#). The assumption that $\mathcal{C}$ is a 1-category of fibrant objects is necessary, because this will imply that its nerve has the structure of an ∞ -category with weak equivalences and fibrations; this is what is necessary to use the given proposition. ■

So, that completes the first two steps of the strategy outlined in [Remark 4.35](#). Now, we will move on to the third step. For this step, we will present the construction of the Spanier-Whitehead category for pointed, finitely cocomplete ∞ -categories. Since this starting situation is not what we want, we will adapt it to our situation by duality.

Construction 4.54 ([\[58, Construction C.1.1.1, p.1943\]](#)). Let $\mathcal{C}$ be a pointed, finitely cocomplete ∞ -category. We define:

$$\mathbf{SW}(\mathcal{C}),$$

by the following colimit in^a Cat_∞ :

$$\mathcal{C} \xrightarrow{\Sigma} \mathcal{C} \xrightarrow{\Sigma} \dots$$

In this case, we refer to $\mathbf{SW}(\mathcal{C})$ as the Spanier-Whitehead category of $\mathcal{C}$. This construction gives a canonical map:

$$\mathcal{C} \rightarrow \mathbf{SW}(\mathcal{C}),$$

which is the “inclusion” of $\mathcal{C}$ in its Spanier-Whitehead stabilization^b.

^aNote that $\Sigma : \mathcal{C} \rightarrow \mathcal{C}$ is the suspension functor.

^bIt is not the same construction as the usual stabilization procedure that produces Sp .

In fact, the Spanier-Whitehead category is the universal way to form a stable ∞ -category from a pointed, finitely cocomplete ∞ -category. To state this formula, we need to introduce (one-sided) exact functors.

Terminology 4.55. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor.

1. F is left-exact iff it preserves final objects^a and pullbacks.

2. F is right-exact iff it preserves initial objects and pushouts.
3. F is exact iff it preserves the zero object and sends exact sequences^b in $\mathcal{C}$ to exact sequences in $\mathcal{D}$.

We denote by the three symbols:

$$\mathrm{Fun}^{\mathrm{lex}}(\mathcal{C}, \mathcal{D}), \mathrm{Fun}^{\mathrm{rex}}(\mathcal{C}, \mathcal{D}), \mathrm{Fun}^{\mathrm{ex}}(\mathcal{C}, \mathcal{D}),$$

the full subcategories of $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ spanned by left-exact, right-exact and exact functors.

^aIf they exist, of course.

^bRecall that these are null-sequences which are fiber-cofiber sequences.

With this terminology, we can state the result.

Proposition 4.56. [58, Proposition C.1.1.7, p.1945]

Let $\mathcal{C}$ be a pointed, finitely cocomplete ∞ -category. TFSH:

1. $\mathbf{SW}(\mathcal{C})$ is a stable ∞ -category.
2. If $\mathcal{D}$ is a stable ∞ -category, then the canonical map $\mathcal{C} \rightarrow \mathbf{SW}(\mathcal{C})$ induces an equivalence:

$$\mathrm{Fun}^{\mathrm{ex}}(\mathbf{SW}(\mathcal{C}), \mathcal{D}) \rightarrow \mathrm{Fun}^{\mathrm{rex}}(\mathcal{C}, \mathcal{D}),$$

of ∞ -categories.

Now, we can proceed towards our Spanier-Whitehead category for pointed, finitely complete ∞ -categories.

Construction 4.57. Let $\mathcal{C}$ be a pointed, finitely complete ∞ -category. We define:

$$\mathbf{SW}(\mathcal{C})_0 \stackrel{\mathrm{def}}{=} \mathbf{SW}(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}},$$

to be the Spanier-Whitehead category of $\mathcal{C}$.

The following is an immediate corollary.

Corollary 4.58

Let $\mathcal{C}$ be a pointed, finitely complete ∞ -category. Then, $\mathbf{SW}(\mathcal{C})_0$ is stable.

Proof. We see that:

$$\mathbf{SW}(\mathcal{C})_0 = \mathbf{SW}(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}}.$$

Observe that $\mathbf{SW}(\mathcal{C}^{\mathrm{op}})$ is a stable ∞ -category and since the opposite of a stable ∞ -category is a stable ∞ -category, we see that $\mathbf{SW}(\mathcal{C})_0$ is also a stable ∞ -category. This completes the proof. ■

With these results, we have completed steps (3) and (4) of the strategy outlined in Remark 4.35. We are ready to work towards the completion of the last step. This requires us to simply state a result.

Theorem 4.59. J. Lurie, 2009, [55, Theorem 3.11, p.10-12]

Let $\mathcal{C}$ be a stable ∞ -category. Then, $\mathbf{Ho}(\mathcal{C})$ has the structure of a triangulated category^a.

^aIn the cited resource, the additive structure (along with other preliminary results) is established in the same section.

The theorem requires further clarification. For instance, it is necessary to state the triangulated structure explicitly.

Construction 4.60 (J. Lurie, [55, Definition 3.8, p.9]). Let $\mathcal{C}$ be a stable ∞ -category and suppose we have a diagram:

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

in $\mathbf{Ho}(\mathcal{C})$. This diagram is a distinguished triangle iff there exists a commutative diagram:

$$\begin{array}{ccccc} x & \xrightarrow{\tilde{f}} & y & \xrightarrow{\quad} & 0 \\ \downarrow & & \downarrow \tilde{g} & & \downarrow \\ 0' & \xrightarrow{\quad} & z & \xrightarrow{\tilde{h}} & w \end{array}$$

such that the following conditions are satisfied:

1. $0, 0'$ are zero objects in $\mathcal{C}$,
2. Both squares are pushouts in $\mathcal{C}$,
3. the morphisms $\tilde{f}$ and $\tilde{g}$ represent f & g respectively,
4. the map $h : z \rightarrow x[1]$ is the composition of the homotopy class of $\tilde{h}$ with the isomorphism:

$$w \simeq x[1],$$

determined by the outer rectangle.

A comment about (4) above: both squares are pushouts so the entire rectangle is a pushout but this is just the pushout diagrama for $x[1]$ by definition, so there does exist an isomorphism between w and $x[1]$. Effectively, we see that the triangulated structure is given by fiber-cofiber sequences in $\mathcal{C}$.

We are finally able to establish the Spanier-Whitehead construction for categories of fibrant objects.

Proof of Theorem 4.34. The strategy for the proof has been discussed in Remark 4.35 but let us repeat it here:

1. We will take the ∞ -categorical localization of $\mathcal{C}$ at its weak equivalences; call this $\mathcal{D}$.
2. We will show that $\mathcal{D}$ is a pointed ∞ -category which is finitely bicomplete.
3. We will define the Spanier-Whitehead category:

$$\mathbf{SW}(\mathcal{D}),$$

which is supposed to be the ∞ -categorical analogue of the same construction as the stable homotopy category.

4. We will show that $\mathbf{SW}(\mathcal{D})$ is a stable ∞ -category.
5. Since the homotopy category of a stable ∞ -category is a triangulated category, the theorem will follow quickly.

Each step of this construction has an associated result/definition which handles it, so we are done in that regard. On the other hand, we also have to ensure that $\mathbf{Ho}(\mathbf{SW}(\mathcal{C})_0)$ is the same as the category $\mathbf{SHo}(\mathcal{C})$ defined in Construction 3.78. In fact, this follows from the usual, explicit description of the homotopy category of the Spanier-Whitehead construction as given in [58, Remark C.1.1.2, p.1943-1944]. The only difference is that, in that case, one works with the finitely cocomplete ∞ -categories so the suspension is the endofunctor which is inverted, whereas our construction above inverts the loop functor so this is what appears as part of the mapping spaces. ■

Here is a remark about our work.

Remark 4.61. To some extent, Theorem 4.34 is somewhat unsatisfactory in the sense that there are few indications as to whether or not gives the “correct” triangulated category. One indication would be that an object $(x, n) \in \mathbf{SHo}(\mathcal{C})$ actually gets mapped to $(x, n+1)$; this seems to be intuitively true based on the construction but seems to be difficult to verify formally. Moreover, we have barely utilized the fact that we start with a category of fibrant objects; the main place we used it was in Proposition 4.53. Indeed, while we do obtain a triangulated structure, it is not completely clear how the given suspension relates to the loop on $\mathbf{Ho}(\mathcal{C})$.

This brings us to the end of the main content of this thesis. In the next section, we will discuss some open questions and future projects which are prompted by the work done in this thesis.

5 Open Questions & Further Possibilities

In this thesis, we have had the opportunity to come into contact with questions/conjectures of interest to people in the community of operator algebras. Indeed, we have even been able to address some of these questions/conjectures to varying degrees of success. In the process of doing this thesis, we were able to concoct many interesting questions which extend results we have proven here and could be the basis for future work. Our goal is to state these problems and some strategies that we came up with to resolve them. Note that this part of the thesis is exploratory and very informal, so our ideas may not work out. Nevertheless, we still do try to present precise statements and concrete strategies, while explaining possible obstructions (which might be done rather informally).

5.1 An Outlook on Lifting Problems

We will start by giving an outlook on lifting problems; this will primarily focus on the work done in Section 2. Let us start by noting that Corollary 2.33 gives us a way for Schochet fibrations to be semisplit surjections. This raises a question.

| **Question 5.1.** Is there a way to produce Schochet fibrations from semisplit surjections in G - $C^*\text{Alg}$?

A naive, positive answer to this is provided by Proposition 3.47; we can just factorize out semisplit surjection into a composition of a Schochet fibration and a weak equivalence. However, this is not interesting as it applies to all morphisms in G - $C^*\text{Alg}$ and it does not tell us if the entire semisplit surjection is, itself, a Schochet fibration. In the next remark, we will discuss the extent to which it is possible to conclude that a semisplit surjection is a Schochet fibration and if there are indications for the conditions that might be necessary for that to happen.

Remark 5.2. With semisplit surjections, we do obtain a “weakened” version of the Schochet fibration property. So, let $A, B \in C^*\text{Alg}$ and let $f \in C^*\text{Alg}(A, B)$ be a semisplit surjection with g being a ccp split. Now, we consider:

$$\begin{array}{ccc} & & A \\ & \nearrow p & \uparrow f \\ D & \xrightarrow{(h_t)_{t \in [0,1]}} & B \end{array}$$

where we have that $f \circ p = h_0$. Now, we define:

$$\forall t \in [0, 1] : H_t \stackrel{\text{def}}{=} g \circ h_t.$$

For each $t \in [0, 1]$, we see that H_t is a ccp map. We also have that:

$$\begin{aligned} \forall t \in [0, 1] : f \circ H_t &= f \circ (g \circ h_t) \\ &= (f \circ g) \circ h_t \\ &= h_t. \end{aligned}$$

Finally, we see that:

$$H_0 = g \circ h_0 = g \circ (f \circ p) = (g \circ f) \circ p.$$

So, with a semisplit surjection, we naturally obtain a kind of “partial ccp homotopy lift”. The semisplit condition does not “solve” the homotopy lifting problem above but there is a sense in which we do get some nice lifting behavior. It has been known for some time, due to the work of (Man-Duen) Choi, (Edward) Effros and others, that lifting problems for C^* -algebras are best studied when working with ccp maps. So, it might be reasonable to pursue homotopy lifting problems in that context as well. It should be noted that the part of operator algebra which deals with operator spaces and morphisms between them is known informally as “noncommutative functional analysis” or, sometimes more accurately, as “noncommutative Banach space theory”. As far as we are aware, homotopical methods are not extremely pervasive in the study of Banach spaces (though homological methods certainly are) and there is some doubt in this perspective being useful in the study of operator spaces. Nevertheless, the recent development of K-theory as an invariant for operator spaces in [87] and [88] implies that this is worth investigating more seriously.

In Section 2.2, our discussion of lifting problems centered around the desire to better understand Schochet C^* -algebras and obtain more examples/non-examples of such C^* -algebras. One reason why we think it is easier to probe these is because the Calkin projection is defined between two C^* -algebras which have a lot more structure than the generic surjective $\star$ -homomorphism. In particular, it would be useful to address the following question.

| **Question 5.3.** Let $A \in C^*\text{Alg}$ be Schochet. Is it the case that any of the following C^* -algebras:

1. $\text{Cyl}(A)$,
2. $\text{Cone}(A)$,
3. $\text{Sus}(A)$,
4. $M_{n \times n}(A)$ for $n \in \mathbb{N}$,

are Schochet? Moreover, if $A, B \in C^*\text{Alg}$, then under what conditions is it the case that:

$$A \otimes_{\max} B,$$

is Schochet? What about the minimal tensor product?

Of course, the C^* -algebras in the list above are secretly just tensor products of specific nuclear C^* -algebras with A , so knowing the answer to the first question might very well be useful for dealing with the second question. We also defined,

through the reduced group C^* -algebra, the Schochet property for LCH groups. On the other hand, we do not have too many examples of groups which are (not) Schochet. This leads us into the following question.

| **Question 5.4.** Let $n \geq 2$. Is $\text{Free}(n)$ a Schochet group?

Let us discuss this in a remark.

Remark 5.5. We already know the answer to this question when $n = 1$; see [Example 2.103](#). It would also be desirable to relate the Schochet property to other analytic properties of infinite groups, such as amenability or Kazhdan's property (T). At the moment, it is unclear how we can do this, as these other properties do not have lifting characterizations. One thing which we want to make clear is that it is highly desirable to **not** proceed via semiprojectivity. Indeed, semiprojectivity is much stronger than being Schochet, so we actually expect there to be many more Schochet C^* -algebras which are not semiprojective, even if it is hard to construct explicit examples. On the other hand, semiprojectivity could be a useful tool for simply constructing “quick” examples. For instance, it might be relatively easy to show that $C_{\text{red}}^*(G)$ is semiprojective for some concrete group G and so, G would be Schochet. It might, then, be useful to conduct a systematic study of semiprojectivity for reduced group C^* -algebras^a.

^aWe could ask the same selection of questions for the maximal group C^* -algebra. Similarly, we could ask the same selection of questions for groupoid C^* -algebras or graph C^* -algebras. At the moment, it is not clear that this helps us at all.

This leads us naturally into the analogue of [Question 5.3](#) for approximately Schochet C^* -algebras.

| **Question 5.6.** What are some other permanence properties of approximately Schochet, separable C^* -algebras?

Let us discuss this in a remark.

Remark 5.7. In [\[89, Theorem 4.9, p.13\]](#), Thiel establishes several permanence properties of the category of approximately semiprojective C^* -algebras. In particular, knowing which constructions under which the approximate Schochet property is closed might be a way to distinguish between these two categories, thereby giving a resolution to Blackadar's Semiprojectivity Conjecture ([Conjecture 2.114](#)). Actually, this is somewhat similar to another existing obstruction in the literature that is worth mentioning here. In [\[23\]](#), it is proved that every approximately semiprojective C^* -algebra satisfies the so-called *asymptotic homotopy lifting property*; this is essentially the same property that we have been working with except that we replace the morphisms by asymptotic morphisms which are already known to play a significant role in E-Theory. Indeed, if one can find an example of a C^* -algebra which does not satisfy the asymptotic homotopy lifting theorem, then Blackadar's conjecture is false.

This discussion of permanence properties is particularly relevant for our partial resolution of [Question 2.51](#) in [Theorem 2.110](#). We note that even with regards to direct sums, we only know that the direct sum of two unital Schochet C^* -algebras is Schochet ([Corollary 2.94](#)). However, our partial resolution of Conway's question required the use of a difficult result ([Proposition 2.111](#)) due to Enders. In general, it would be quite desirable to know if the Schochet property is closed under general pullbacks since that would lead to a proof that is independent of the result by Enders. Also, we did not discuss the space in statement (3) of Conway's question, so let us partially address that in the next remark.

Remark 5.8. Let $X = [-1, 0] \cup \{2^{-n} : n \in \mathbb{N}_0\}$. Then, we can write:

$$\mathcal{C}(X) \simeq \mathcal{C}([-1, 0]) \oplus_{\mathbb{C}} \mathbf{c} = \{(f, (a_n)_{n \in \mathbb{N}_0}) \in \mathcal{C}([-1, 0]) \oplus \mathbf{c} : f(0) = \lim_{n \rightarrow \infty} a_n\},$$

where $\mathbf{c}$ is the C^* -algebra of all convergent sequences indexed by $\mathbb{N}_0$. In fact, we cannot use [Proposition 2.111](#) directly because $\mathbf{c}$ is not semiprojective; see [\[13, Example 6.6, p.301-302\]](#). Let us put $\mathcal{C}(X)$ into a pullback:

$$\begin{array}{ccc} \mathcal{C}(X) & \xrightarrow{\pi_1} & \mathbf{c} \\ \pi_1 \downarrow & & \downarrow \lim_{n \rightarrow \infty} \\ \mathcal{C}([-1, 0]) & \xrightarrow{\text{ev}_0} & \mathbb{C} \end{array}$$

We note that $\mathbf{c}$ is Schochet, because its Gelfand spectrum is just $\mathbb{N}_0^+$ and this is certainly a Schochet space. So, it follows that the direct sum $\mathcal{C}([-1, 0]) \oplus \mathbf{c}$ is a Schochet C^* -algebra. For the moment, it is not known if the Schochet property passes to C^* -subalgebras under any conditions. So, the answer to this question remains unclear.

Regarding [Question 2.65](#), we were able to give a simple characterization of when this question has a positive answer; see [Corollary 2.71](#). This characterization certainly does have its roots in questions whose answers are not known. Nevertheless, we do not think that the characterization itself makes the question any easier to answer. We have two possible strategies that could be useful in finally addressing this open problem and we discuss them in the next two remarks.

Remark 5.9. In our first strategy, the goal is to find a marked C^* -algebra (A, I) such that:

1. A/I has a normal element which is not normal-liftable.
2. Normal elements in A/I with finite spectrum are liftable to A .
3. Every normal element in A/I can be approximated by a sequence of normal elements with finite spectrum.

In fact, the second statement is known to be true generally; see [\[81, Proposition 4.4, p. 12\]](#). The third statement is not generally true but [\[39, Corollary 3.12, p.363\]](#) provides a nice set of conditions which allow us to conclude that a given normal element is approximated by normal elements with finite spectrum. To be precise, we have the claim.

Claim. Let A be a unital C^* -algebra of real rank zero^a and stable rank one^b and let $a \in A$ be a normal element such that for every proper closed ideal $I \subseteq A$, we have that $\pi(a) - \lambda(1_A + I)$ is invertible & homotopic to $1_A + I$ in $\text{GL}(A/I)$. Then, a can be written as the limit of a sequence of finite-spectrum normal elements.

^aIf $B \in C^*\text{Alg}$, then B is of real rank zero iff the finite-spectrum self-adjoint elements of B are dense in $B_{\text{s.a.}}$.

^bIf $B \in C^*\text{Alg}$, then B is of stable rank one iff $\text{GL}(B^\sim)$ is dense in B . If B is unital, then we may forego the unitization.

Even though real rank zero and stable rank one has been verified for many C^* -algebras, the second condition seems to be difficult to check. So, with this strategy, both statements (1) and (3) seem to be significant obstructions.

Remark 5.10. In our second strategy, the goal is to use the following result.

Claim. Let $A \in C^*\text{Alg}$ have property (IR), let I be a closed ideal in A and let $x \in A/I$ be normal. For every $\epsilon > 0$, there exists a normal element $a \in A$ such that $\|\pi(a) - x\| < \epsilon$.

This was proved in [82, Corollary 4.5, p. 129] and for this discussion, the definition of the property (IR) will not matter (though the reader can find it in Definition 3.1, p.125 of the aforementioned reference). What matters is that every normal element in a quotient of A can be approximated by a sequence of elements which are normal-liftable. If there is an example of A for which there is a normal element in a quotient which does not lift, then this is a counterexample for Question 2.65. In [82, p.125], some examples of C^* -algebras with property (IR) are provided:

1. C^* -algebras of stable rank one,
2. simple von Neumann algebras,
3. simple, purely infinite unital C^* -algebras.

In light of Example 2.74, we are not going to find counterexamples among the simple C^* -algebras with property (IR)^a. So, it seems that the next option is to check for counterexamples among C^* -algebras of stable rank one.

^aIt is mentioned in the aforementioned paper that it is not known if every simple C^* -algebra has property (IR).

We aim to pursue both strategies stated in Remarks 5.9 and 5.10 within future work. It is clear that there is an obstruction which is common to both strategies; this has to do with the normal-liftability of elements in a quotient C^* -algebra. Indeed, even though the strategy in Remark 5.10 looks more promising, there does not seem to be a systematic way of finding reasonable candidates. Let us address this in the next remark and a possible strategy for overcoming it.

Remark 5.11. Let (A, I) be a marked C^* -algebra and let $b \in A/I$ be normal. Our problem is to determine when there exists a normal $a \in A$ such that $\pi(a) = b$. In the proof of Theorem 2.56, we have already demonstrated that it is not possible to guarantee the existence of a normal lift even to a unitary in A/I (assuming it is unital, of course). The key points in that argument are summarized abstractly as follows:

1. Find a Fredholm element a of A of non-zero Fredholm index.
2. The normal Fredholm elements of A have Fredholm index 0.
3. If $\pi(a)$ had a normal lift b , then $b - a \in I$ and their Fredholm indices must be the same.
4. Since their Fredholm indices are not the same, we are done.

With both strategies in Remarks 5.9 and 5.10, we would want to find A and I with A having stable rank one so that the argument above goes through for some reasonable notion of “Fredholm Index”. The real content of our proposed argument above is in statements (1) and (2). Statement (2) is meant to be a hard restriction on the normal elements so we have complete knowledge of their Fredholm indices. Statement (1) requires an actual construction. In fact, we can modify this strategy so that it is a bit more relaxed and we obtain the following list.

1. Consider the set of all “Fredholm indices” of normal elements in A .
2. Find a Fredholm element a of A whose “Fredholm index” does not lie in the set in statement (1).
3. By the argument above, we see that $\pi(a)$ cannot have a normal lift.

With the first list, the most difficult step seems to be step (1). On the other hand, if a construction like the one we used for the proof of Proposition 2.59 exists, then we might be able to use that to construct our desired Fredholm element with non-zero index. In that case, statement (2) becomes the main obstruction.

In the remark above, we did not specify what we have meant by “Fredholm index”. Moreover, we did not really give further information about how we could resolve the obstruction posed by statement (2) of the first list, nor did we address the second list. This is because these problems are, in fact, intimately related to the following questions.

Question 5.12. Let $A \in C^*\text{Alg}$ and let $\mathcal{K}$ be the compact operators on $l^2(\mathbb{N})$. We define^a:

$$\begin{aligned}\mathcal{K}(H_A) &\stackrel{\text{def}}{=} A \otimes \mathcal{K}, \\ \mathcal{B}(H_A) &\stackrel{\text{def}}{=} \text{Max}^1(A \otimes \mathcal{K}) \\ \mathcal{Q}(H_A) &\stackrel{\text{def}}{=} \mathcal{B}(H_A)/\mathcal{K}(H_A).\end{aligned}$$

In this case, we can define the generalized Calkin projection $\pi_A : \mathcal{B}(H_A) \rightarrow \mathcal{Q}(H_A)$ and this is certainly a $\star$ -homomorphism. For which A is it the case that this morphism is a Schochet fibration?

^aIf $B \in C^*\text{Alg}$, then $\text{Max}^1(B)$ is the multiplier algebra of B ; it is the smallest unital C^* -algebra containing B as an essential ideal. Of course, if $I \subseteq B$ is a closed ideal, it is essential iff for any $a \in A$, if $aI = 0$ then $a = 0$.

Question 5.13. Let $X \in \text{Ban}_{\mathbb{C}}$ and consider the Banach-Calkin projection:

$$\pi_X : \mathcal{B}(X) \rightarrow \mathcal{Q}(X),$$

where $\mathcal{Q}(X) \stackrel{\text{def}}{=} \mathcal{B}(X)/\mathcal{K}(X)$ is a $\mathbb{C}$ -Banach algebra. When is this a Schochet fibration in BanAlg ?

These questions are both of the same flavor, though it is clear that [Question 5.12](#) is probably more accessible. Nevertheless, both of these are actually natural extensions of [Conjecture 2.39](#) and one may note that our work (and also the work of Loring & Exel in [\[37, Example 6.3, p. 153\]](#)) actually resolves the [Question 5.12](#) in the special case where $A = \mathbb{C}$. So, this implies that there is a lot to be done for that question. It is possibly true that any C^* -algebra A which is tensorially absorbed by $\mathcal{K}$ will yield a generalized Calkin projection which is not a Schochet fibration, though we do not have a solid proof of this yet. Nevertheless, it would be interesting to find more examples. Now, the notation used in [Question 5.12](#) is meant to be suggestive, because it implies that there is a notion of a “Fredholm element $a \in \mathcal{B}(H_A)$ ”.

Indeed, there is a generalized Fredholm theory developed for this algebra in [\[94, Chapter 17, p.277-292\]](#) using the theory of Hilbert C^* -modules and K-theory, introducing the so-called *generalized Fredholm index*. This generalized index has many nice properties, which is a good thing in relation to our discussion in [Remark 5.11](#). In fact, we even have the following nice result which characterizes when Fredholm elements of the multiplier have non-zero index.

Theorem 5.14. J.A. Mingo, J.S. Spielberg, [\[65, Corollary 1, p.192\]](#)

Let $A \in C^*\text{Alg}$. TFAE:

1. There exists a $\star$ -homomorphism $\phi : \mathcal{C}_0(D) \rightarrow \mathcal{K}(H_A)$ with:

$$K_0(\phi) \neq 0,$$

where D is the open unit disk in $\mathbb{C}$.

2. There is a normal Fredholm element in $\mathcal{B}(H_A)$ with non-zero index.

Going back to list (1) in [Remark 5.11](#), if we wish to use the universal C^* -algebra:

$$A = C^*(x | xx^* = x^*x \wedge \|x\| \leq 1),$$

to show that π_A is not a Schochet fibration and if we wish to use the strategy in [Theorem 2.56](#), then we need it to be the case that all normal Fredholm elements in $\mathcal{B}(H_A)$ have trivial index. So, this means that $K_0(\phi)$ is trivial for each $\star$ -homomorphism $\phi : \mathcal{C}_0(D) \rightarrow \mathcal{K}(H_A)$. Now, we will use the C^* -stability of K_0 to conclude that:

$$K_0(\mathcal{K}(H_A)) \simeq K_0(A).$$

So, one way in which we can ensure that the above holds is to take a C^* -algebra A with trivial K_0 -group. Now, there is a sense in which this does partially help us with the problem in [Remark 5.10](#) (and also the one in [Remark 5.9](#)) but we would, now, have to take the additional step of determining when it is the case that $\mathcal{B}(H_A)$ has property (IR) and this seems difficult to do. So, while the strategy above seems to yield something that can be worked at, we do also get more hurdles which need to be overcome. On the other hand, we also do have a partial strategy to handle list (2) in [Remark 5.11](#) and it is based on the following remark made by Mingo & Spielberg in their paper.

Remark 5.15 (J. A. Mingo, J. S. Spielberg, [\[65, p.192\]](#)). In [\[37\]](#), a filtration of K-theory is defined for which the second filtration F_2 is the set of all images of b (the Bott element) under the maps $K_0(\phi)$ as in (1) of [Theorem 5.14](#). Thus, the theorem implies that the set of indices of normal Fredholm elements of $\mathcal{B}(H_A)$ is precisely $F_2(A)$.

So, this remark gives us a strategy for handling list (2) in [Remark 5.11](#) if we do not require that normal elements have index 0; there seems to be an explicit way of using methods from algebraic topology to compute such filtrations. Again, the main difficulty is in constructing an element whose index does not lie in the second filtration; such an element cannot have a normal lift. Moreover, we would still be faced with the difficulty of finding conditions on A such that $\mathcal{B}(H_A)$ has property (IR). Now, we also note that Exel and Loring’s work is also relevant in another way.

Theorem 5.16. R. Exel, T. Loring, [\[37, Theorem 5.6, p.151-152\]](#)

Consider the following exact sequence:

$$0 \longrightarrow I \longrightarrow A \xrightarrow{\pi} A/I \longrightarrow 0$$

in $C^*\text{Alg}$, where π is a Schochet fibration. Then:

$$\forall n \geq 0 : \partial(F_n K_n(A/I)) \subseteq F_{n+1} K_{n+1}(I),$$

where $\partial : K_n(A/I) \rightarrow K_{n+1}(I)$ is the boundary morphism.

Indeed, it is precisely this theorem that was used to resolve [Conjecture 2.39](#), though we stress that this does not give an explicit counterexample for homotopy lifting. Nevertheless, this definitely provides one way in which we can check if a given morphism is a Schochet fibration or not, thus giving us another strategy for approaching [Question 5.12](#). Moreover, we note that [Theorem 5.16](#) is phrased in terms of C^* -algebras but the statement makes sense for a broader class of algebras (once a filtration for the K-theory of $\mathbb{C}$ -Banach algebras is defined and satisfies the right properties). So, one might hope for an analogue of the theorem to hold also for $\mathbb{C}$ -Banach algebras. If that is the case, then this would provide a possible strategy for approaching [Question 5.13](#). Actually, for that problem, our approach via generators and relations will not work. Indeed, there does not seem to be a very well-behaved theory of “universal Banach algebras”.

5.2 An Outlook on Abstract Homotopy Theory

In this subsection, we will focus on problems involving pure homotopy theory, though these questions are also motivated by natural questions in the realm of operator algebras. The focus is mainly on the contents of Section 3 and Section 4. One big difference between our approach here and the one we took in Section 5.1 is that, over here, we are driven much more by the development of theory as opposed to solving specific problems. Indeed, there are fewer “open problems” that we can discuss here and any explicit questions which are asked will often be vague enough that a clear-cut answer will not always be possible. Indeed, we illustrate this by discussing Section 4.1 to some extent.

| **Question 5.17.** Is the notion of admissible category, as given in Definition 4.5, correct?

Throughout Section 4.1, we have tried to convince the reader that the answer to this question is a resounding *Yes* and our biggest piece of evidence in this direction is the proof of the Bott Periodicity Theorem. Indeed, Cuntz’s proof of this theorem goes through for functors defined on admissible subcategories of $C^*\text{Alg}$ and which satisfy all of the relevant properties. However, there are actually more results in axiomatic homology beyond the Bott Periodicity Theorem which one would need to test this notion against. Indeed, in [85], there is a slightly more extensive study of (co)homology theories on C^* -algebras which is undertaken, with a view also towards studying cofibre (co)homology theories.

The latter are relevant in the proof of the Mayer-Vietoris Theorem; this is suppose to compute the homology of a pullback. Moreover, we have also not undertaken a serious study of (co)homology theories commuting with (co)limits; it is the case that K -theory commutes with inductive limits and that happens to be an important property which is also related to various simplifications of C^* -stability available in the literature. So, even though we have shown that our results do work in some non-trivial range, a much more systematic study of axiomatic homology on C^* -algebras is certainly necessary. Of course, most of our work in that subsection has been in the non-equivariant case and while that is very important, one would eventually like to extend those results to the equivariant situation. This is where a few things get tricky.

| **Question 5.18.** Let $G \in \text{Grp}(\text{LCH})$. What does it mean for a full subcategory $\mathcal{C}$ of $G\text{-}C^*\text{Alg}$ to be admissible?

The situation here becomes more tricky, as one would like to have a definition which accounts also for the group. For instance, in [62, p.10], it is implied that one would like to be able to consider functors defined on, say, the category of separable nuclear $G\text{-}C^*$ -algebras with an amenable (or proper) action. It is not clear how this can be done.

| **Question 5.19.** Let $G \in \text{Grp}(\text{LCH})$. What does it mean for a functor $F : G\text{-}C^*\text{Alg} \rightarrow \mathcal{A}$ to be C^* -stable?

A definition for this is proposed in [62, Definition 36, p.13] but it is difficult to see how this could be used to prove, for instance, an equivariant version of the Bott Periodicity Theorem via the approach of Cuntz. In general, the equivariant setting brings with it additional phenomena, additional functors and additional complications that need to be accounted for & the questions above only reflect the difficulties on the surface. Of course, we can also consider the following question.

| **Question 5.20.** Can we extend our results on axiomatic (co)homology to $G\text{-BanAlg}$?

We note that there are certain references and papers which do treat results in axiomatic (co)homology in relation to (equivariant) Banach algebras; examples of this are certainly in [70] and [71]. Nevertheless, these resources (and others) typically prove what is necessary, in this case focusing on developing Banach KK -theory. So, it would be nice for there to be systematic treatments of axiomatic (co)homology on $G\text{-BanAlg}$. A good focal point for all of this would actually be Schochet’s paper, [85]. Indeed, even in our development of basic homotopy theory, we have not extended **all** of his results to the equivariant setting or to the setting of Banach algebras, so it seems that there is potential for a lot of work in that direction. Moreover, even though it is not the case that there are any interesting (co)homology theories on proper, full subcategories of $G\text{-BanAlg}$, it might still be interesting to determine which subcategories of Banach algebras are of “homotopical type”. Indeed, the novelty in our definition of an admissible subcategory is that all such subcategories are actually fibrant subcategories of $C^*\text{Alg}$ and, therefore, are themselves categories of fibrant objects. One would like to achieve the same result for certain subcategories of, say, BanAlg & possibly $G\text{-BanAlg}$.

One reason why it would be good to systematically further develop homotopy theory on categories of operator algebras is that it gives us explicit ideas for working with categories of fibrant objects. Some of our work has definitely led to us believing that many results that are true about model categories should also be true about categories of fibrant objects, possibly under appropriate examples. Our introduction of, say, h -categories of fibrant objects in Definition 3.62 was certainly inspired by existing homotopical structures on categories of operator algebras. On the other hand, we were not able to take this extremely far due to space constraints in this thesis. Let us illustrate this by the following remark.

Remark 5.21. Let $A \in C^*\text{Alg}$. We define a map:

$$\tau : [0, 1] \times [0, 1] \rightarrow [0, 1] \times [0, 1] , (s, t) \mapsto \tau(s, t) \stackrel{\text{def}}{=} (t, s).$$

This can be used to construct the following isomorphism:

$$\Psi_A : \text{Cyl}(\text{Cyl}(A)) \rightarrow \text{Cyl}(\text{Cyl}(A)) , f \mapsto \Psi_A(f),$$

where we define:

$$\forall t, s \in [0, 1] : \Psi_A(f)(t)(s) \stackrel{\text{def}}{=} f(s)(t).$$

This is a $\star$ -isomorphism essentially because τ is a homeomorphism. Bearing this in mind, if $t \in [0, 1]$, then we would

like to know the relationship between $\text{Cyl}(\alpha_t^A)$ and $\alpha_t^{\text{Cyl}(A)}$. Let us simply compute them separately:

$$\forall \phi \in \text{Cyl}(\text{Cyl}(A)) : \forall s \in [0, 1] : \text{Cyl}(\alpha_t^A)(\phi)(s) = (\alpha_t^A \circ \phi)(s) = \phi(s)(t).$$

On the other hand:

$$\forall \phi \in \text{Cyl}(\text{Cyl}(A)) : \alpha_t^{\text{Cyl}(A)}(\phi)(s) = \phi(t)(s).$$

In other words, it seems to be the case that these two are the same, up to time-exchange. That is:

$$\alpha_t^{\text{Cyl}(A)} = \text{Cyl}(\alpha_t^A) \circ \Psi_A.$$

This formula, of course, motivates what we need very well.

In fact, the last formula in [Remark 5.21](#) can be made sense of in a h -category of fibrant objects.

Definition 5.22

Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects. This is a good h -category of fibrant objects iff T preserves pullbacks and there exists a natural isomorphism $\beta : T^2 \Rightarrow T^2$ such that for every $x \in \mathcal{C}$ and each $i \in \{0, 1\}$, we have:

$$\alpha_i^{T(x)} = T(\alpha_i^x) \circ \beta_x.$$

We refer to β as the time-exchange isomorphism.

In fact, there are other structures that are obviously available on C^*Alg that we have not even begun to use. For instance, the cone and the suspension functors are also available & one could attempt to model them on a h -category of fibrant objects. This may seem like a blind development of the theory but it is partly motivated by some real questions.

Question 5.23. Let $\mathcal{C}$ be a category of fibrant objects. Does $W_{\mathcal{C}}$ coincide with $W_{\mathcal{C},h}$?

Question 5.24. Let $\mathcal{C}$ be a h -category of fibrant objects. Does $\text{Fib}_{\mathcal{C}}$ coincide with the internal fibrations?

Question 5.25. Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects and let $x \in \mathcal{C}$. When are the evaluation morphisms:

$$\alpha_0^x, \alpha_1^x : T(x) \rightarrow x,$$

internal fibrations in $\mathcal{C}$?

We should note that these questions are not trivial. Indeed, even in a model category, the answer to [Question 5.23](#) is a *Yes* if we restrict to the bifibrant objects but it is the abstract analogue of the Whitehead Theorem (see [\[11, Theorem 2.3.1, p.121-122\]](#) for an accessible proof). Moreover, constructions of the type similar to what we have done have also been used in other contexts as well. For example, in [\[24, Chapter 2.4, p.50-68\]](#), Cisinski uses variants of such constructions to construct cofibrantly-generated model structures on categories of presheaves of sets over a fixed small category. These constructions are, then, used to throughout the rest of the book to construct model structures that are needed to develop a coherent theory of $(\infty, 1)$ -categories. One might imagine that the right analogue of such constructions might be useful for constructing model structures on, say, $G\text{-C}^*\text{Alg}$ or $G\text{-BanAlg}$. Indeed, in the former case, this is known to be an impossibility if we insist on the standard choice of weak equivalences and fibrations. But it would be interesting to have access to other category-of-fibrant-objects structures on $G\text{-C}^*\text{Alg}$ or $G\text{-BanAlg}$.

Speaking of model structures, we mentioned in [Remark 3.23](#) that Joachim and Datta have successfully constructed a model structure on the category of locally multiplicative convex C^* -algebras which extends the category of fibrant objects structure on C^*Alg , except that the weak equivalences here are equivariant KK-equivalences. Of course, the use of KK-equivalences is not particularly surprising, since one would like to have a characterization of Kasparov's KK-category as a localization of C^*Alg . Nevertheless, we can apply the same set of questions as above to this setting as well.

Question 5.26. Is it possible to equip^a C^*Alg with interesting model structures?

^aOr relevant, possibly equivariant, extensions of it.

Of course, one could extend this question to (equivariant) $\mathbb{C}$ -Banach algebras.

Question 5.27. Is it possible to adapt Datta and Joachim's construction of a model structure to^a $G\text{-BanAlg}$?

^aOr, again, extensions of it.

Such questions can, in general, be asked about categories of fibrant objects.

Question 5.28. Let $\mathcal{C}$ be a category of fibrant objects. Under what conditions on $\mathcal{C}$ is it the case that there exists a pointed model category $\mathcal{M}$ such that its subcategory of fibrant objects, defined in [Construction 3.28](#), is just $\mathcal{C}$?

It does not seem that these questions have been tackled in the literature. Our hope is that further research in homotopy theory and operator algebras can be stimulated by working on this intersection. As the reader can see, there are plenty of questions that one can ask and many possibilities for answering those questions or variations of them. With the hope that we can potentially address many of these questions in future work, we bid you adieu.

A Miscellaneous Proofs

Section 1

Proof. 1.1. THIS IS A TEST BOX.

Section 2

Proof. 2.1. All we have to do is check continuity, so let $t \in [0, 1]$ and let $(t_n)_{n \in \mathbb{N}}$ be a sequence in $[0, 1]$ which converges to t . We have to show that $(gf)(t_n) \rightarrow (gf)(t)$. So, we have to show that $gf(t_n) \rightarrow gf(t)$. Since f is continuous, we see that $f(t_n) \rightarrow f(t)$ so the sequence $(f(t_n))_{n \in \mathbb{N}}$ lies eventually in every nbhd of $f(t)$. Now, we select a nbhd U of $gf(t)$ and note that there, then, exist two nbhds V_1, V_2 of g and $f(t)$ respectively such that:

$$\forall h \in V_1 : \forall a \in V_2 : ha \in U.$$

This just follows from the continuity of the action and also from the subbasis description of the product topology. Now, $(f(t_n))_{n \in \mathbb{N}}$ eventually lies in V_2 , so we see that $(gf(t_n))_{n \in \mathbb{N}}$ eventually lies in U and we're done.

Proof. 2.2. The proof that these are C^* -algebras is completely routine. It remains to be shown that we have a valid action by automorphisms. Let's deal with the cylinder because that's where everything is happening; we need to show that action above is continuous and acts via automorphisms of $\text{Cyl}_G(A)$.

1. Let $g \in G$ and consider the map:

$$\phi_g : \text{Cyl}_G(A) \rightarrow \text{Cyl}_G(A), \quad f \mapsto \phi_g(f) \stackrel{\text{def}}{=} gf.$$

Well, it's clear that ϕ_g maps into $\text{Cyl}_G(A)$. Now, we need to show that this is a $\star$ -homomorphism.

(a) Let $f_1, f_2 \in \text{Cyl}_G(A)$ and let $\lambda \in \mathbb{C}$. Then, we have the following computation:

$$\begin{aligned} \forall t \in [0, 1] : \phi_g(\lambda f_1 + f_2)(t) &= g(\lambda f_1(t) + f_2(t)) \\ &= g(\lambda f_1(t)) + gf_2(t) \\ &= \lambda(gf_1(t)) + gf_2(t) \\ &= (\lambda(gf_1) + gf_2)(t), \end{aligned}$$

which proves that ϕ_g is linear.

(b) Let $f \in \text{Cyl}_G(A)$. We have the following computation:

$$\begin{aligned} \forall t \in [0, 1] : \phi_g(f^\star)(t) &= (gf^\star)(t) \\ &= gf^\star(t) \\ &= gf(t)^\star \\ &= gf(t)^\star \\ &= (gf)(t)^\star \\ &= (gf)^\star(t), \end{aligned}$$

which proves that ϕ_g is involution-preserving.

(c) Let $f_1, f_2 \in \text{Cyl}_G(A)$. We have the following computation:

$$\begin{aligned} \forall t \in [0, 1] : \phi_g(f_1 f_2)(t) &= g(f_1 f_2)(t) \\ &= g(f_1(t) f_2(t)) \\ &= (gf_1(t))(gf_2(t)) \\ &= \phi_g(f_1)(t) \phi_g(f_2)(t) \\ &= (\phi_g(f_1) \phi_g(f_2))(t), \end{aligned}$$

which proves that ϕ_g preserves the product.

Throughout the computation above, we've just utilized the properties of the action by automorphisms on A by G . This proves that we have a map $G \rightarrow \text{Aut}(\text{Cyl}_G(A))$ and that this is a group homomorphism follows literally from a routine computation. Now, we have to show continuity.

2. Let $(g, f) \in G \times \text{Cyl}_G(A)$ and let $((g_\delta, f_\delta))_{\delta \in D}$ be a net in $G \times \text{Cyl}_G(A)$ which converges to gf . Then, we see that $g_\delta \rightarrow g$ and $f_\delta \rightarrow f$. Next, we have the following computation:

$$\begin{aligned} \forall \delta \in D : \|g_\delta f_\delta - gf\| &= \sup_{t \in [0, 1]} \|g_\delta f_\delta(t) - gf(t)\| \\ &= \sup_{t \in [0, 1]} \|g_\delta f_\delta - gf_\delta(t) + gf_\delta(t) - gf(t)\| \\ &\leq \sup_{t \in [0, 1]} \|g_\delta f_\delta(t) - gf_\delta(t)\| + \sup_{t \in [0, 1]} \|gf_\delta(t) - gf(t)\| \\ &= \sup_{t \in [0, 1]} \|g_\delta f_\delta(t) - gf_\delta(t)\| + \sup_{t \in [0, 1]} \|g(f_\delta(t) - f(t))\| \\ &\leq \sup_{t \in [0, 1]} \|g_\delta f_\delta(t) - gf_\delta(t)\| + \sup_{t \in [0, 1]} \|g\| \|f_\delta(t) - f(t)\|, \end{aligned}$$

where $\|g\|$ is the operator norm of the automorphism $a \mapsto ga$. So, the second term above goes to 0 and now,

we have to argue that the first one goes to 0 as well. Let us simplify that term more:

$$\begin{aligned}
\sup_{t \in [0,1]} \|g_\delta f_\delta(t) - g f_\delta(t)\| &= \sup_{t \in [0,1]} \|g_\delta f_\delta(t) - g_\delta f(t) + g_\delta f(t) - g f(t)\| \\
&\leq \sup_{t \in [0,1]} \|g_\delta f_\delta(t) - g_\delta f(t)\| + \sup_{t \in [0,1]} \|g_\delta f(t) - g f(t)\| \\
&\leq \sup_{t \in [0,1]} \|g_\delta(f_\delta - f)(t)\| + \sup_{t \in [0,1]} \|g_\delta f(t) - g f(t)\| \\
&= \sup_{t \in [0,1]} \|g_\delta\| \cdot \|f_\delta(t) - f(t)\| + \sup_{t \in [0,1]} \|g_\delta f(t) - g f(t)\|.
\end{aligned}$$

Since G is locally compact, g has a compact nbhd so the net of automorphisms is eventually uniformly bounded, so the first term above vanishes in the limit. Now, $f([0, 1])$ is compact since f is continuous and g has a compact nbhd. So, if we restrict the action $G \times A \rightarrow A$ to the product of the compact nbhd of g and $f([0, 1])$, we obtain a uniformly continuous map & it is this uniform continuity which ensures that the second term converges.

This completes the proof.

Proof. 2.3. This is a $\star$ -homomorphism by the explicit formula. For equivariance, we have:

$$\begin{aligned}
\forall g \in G : \forall \phi \in \text{Cyl}_G(A) : p_t^A(g\phi) &= (g\phi)(t) \\
&= g\phi(t) \\
&= gp_t^A(\phi),
\end{aligned}$$

as was desired. For the second statement, the explicit formula for each p_t^A tells us that this is contractive.

Proof. 2.4. Consider the following diagram:

$$\begin{array}{ccc}
\text{Cyl}_G(A) & \xrightarrow{\alpha_t^A} & A \\
\text{Cyl}_G(f) \downarrow & & \downarrow f \\
\text{Cyl}_G(B) & \xrightarrow{\alpha_t^B} & B
\end{array}$$

Let's show that this commutes:

$$\begin{aligned}
\forall \phi \in \text{Cyl}_G(A) : (f \circ \alpha_t^A)(\phi) &= f(\alpha_t^A(\phi)) \\
&= f(\phi(t)) \\
&= (f \circ \phi)(t) \\
&= (\text{Cyl}_G(f)(\phi))(t) \\
&= \alpha_t^B(\text{Cyl}_G(f)(\phi)) \\
&= (\alpha_t^B \circ \text{Cyl}_G(f))(\phi).
\end{aligned}$$

This computation reflects the naturality of each α_t so we're done. In the case that $A \in G\text{-BanAlg}$, the argument does not change as we only use the algebraic properties of the maps above and those do hold in this case.

Proof. 2.5. Let $A \in C^*\text{Alg}$. We need to define a $\star$ -isomorphism:

$$\alpha_A : \mathcal{C}_0(\mathbb{R}, \mathcal{C}([0, 1], A)) \rightarrow \mathcal{C}([0, 1], \mathcal{C}_0(\mathbb{R}, A)), \quad f \mapsto \alpha_A(f),$$

where we define:

$$\alpha_A(f) : [0, 1] \rightarrow \mathcal{C}_0(\mathbb{R}, A), \quad s \mapsto (t \mapsto \alpha_A(f)(s)(t) \stackrel{\text{def}}{=} f(t)(s)).$$

It is clear that this is well-defined, because we are simply interchanging the roles of the variables in the function f and this function satisfies the requisite properties. Moreover, it is also clear that α_A is a $\star$ -homomorphism, since its action on functions in $\mathcal{C}_0(\mathbb{R}, \mathcal{C}([0, 1], A))$ is defined point-wise. The formula used to define α_A also tells us that this is bijective, since we can construct an explicit inverse by rearranging variables on functions in $\mathcal{C}([0, 1], \mathcal{C}_0(\mathbb{R}, A))$. Finally, we need to prove naturality, so let $A, B \in C^*\text{Alg}$, let $f \in C^*\text{Alg}(A, B)$ and consider the following diagram:

$$\begin{array}{ccc}
(\text{Sus} \circ \text{Cyl})(A) & \xrightarrow{\alpha_A} & (\text{Cyl} \circ \text{Sus})(A) \\
(\text{Sus} \circ \text{Cyl})(f) \downarrow & & \downarrow (\text{Cyl} \circ \text{Sus})(f) \\
(\text{Sus} \circ \text{Cyl})(B) & \xrightarrow{\alpha_B} & (\text{Cyl} \circ \text{Sus})(B)
\end{array}$$

commutes. So, let $\phi \in (\text{Sus} \circ \text{Cyl})(A)$, let $t \in [0, 1]$ and let $s \in \mathbb{R}$. We consider the top part of the diagram:

$$\begin{aligned}
[(\text{Cyl} \circ \text{Sus})(f)(\alpha_A(\phi))](t)(s) &= [\text{Cyl}(\text{Sus}(f))(\alpha_A(\phi))](t)(s) \\
&= [\text{Sus}(f) \circ \alpha_A(\phi)](t)(s) \\
&= \text{Sus}(f)(\alpha_A(\phi)(t))(s) \\
&= (f \circ \alpha_A(\phi)(t))(s) \\
&= f(\alpha_A(\phi)(t)(s)) \\
&= f(\phi(s)(t)).
\end{aligned}$$

For the bottom part of the diagram, this eventually computes out as follows:

$$\begin{aligned}
[\alpha_B \circ (\text{Sus} \circ \text{Cyl})(f)](\phi)(t)(s) &= [\alpha_B((\text{Sus} \circ \text{Cyl})(f)(\phi))](t)(s) \\
&= [\alpha_B(\text{Sus}(\text{Cyl}(f))(\phi))](t)(s) \\
&= [\alpha_B(\text{Cyl}(f) \circ \phi)](t)(s) \\
&= (\text{Cyl}(f) \circ \phi)(s)(t) \\
&= \text{Cyl}(f)(\phi(s))(t) \\
&= (f \circ \phi(s))(t) \\
&= f(\phi(s)(t)),
\end{aligned}$$

which gives us the desired result.

Proof. 2.6. Let $f \in \text{Sus}_G(\text{Cyl}_G(A))$, let $i \in [0, 1]$ and let $t \in \mathbb{R}$. Then, we have the following calculation:

$$\begin{aligned}
[(p_i^{\text{Sus}_G(A)} \circ \alpha_A)(f)](t) &= [p_i^{\text{Sus}_G(A)}(\alpha_A(f))](t) \\
&= \alpha_A(f)(i)(t) \\
&= f(t)(i) \\
&= (p_i^A \circ f)(t) \\
&= \text{Sus}(p_i^A)(f)(t),
\end{aligned}$$

which proves the result.

Proof. 2.7. We will deal with each sequence separately.

1. Clearly, ι is injective. For the surjectivity of p_1^A , let $a \in A$ and let $f \in \mathcal{C}[0, 1]$ such that $f(1) = 1$. Define:

$$fa : [0, 1] \rightarrow A, \quad t \mapsto (fa)(t) \stackrel{\text{def}}{=} f(t)a.$$

Observe that $(fa)(1) = f(1)a = a$. But this just means that:

$$p_1^A(fa) = a,$$

so p_1^A is surjective. Finally, we need to show that $\text{ran}(\iota) = \ker(p_1^A)$. Indeed, the fact that:

$$\text{ran}(\iota) \subseteq \ker(p_1^A),$$

follows from the exactness of the sequence. On the other hand, let $\phi \in \text{Cyl}_G(A)$ such that $p_1^A(\phi) = 0$. This means that $\phi(1) = 0$. That is, $\phi \in \text{Cone}_G(A)$. So, $\phi \in \text{ran}(\iota)$ because $\iota(\phi) = 0$.

2. The proof proceeds by almost the same argument as above; one literally just has to replace specific symbols by the appropriate one to get a proof applicable to the second sequence.

This completes the proof. The same argument works also for $A \in G\text{-BanAlg}$.

Proof. 2.8. We will check linearity first:

$$\begin{aligned}
\forall a, b \in A : \forall \lambda \in \mathbb{C} : \forall t \in [0, 1] : h(\lambda a + b)(t) &= H_{\lambda a + b}(t) \\
&= \phi_t(\lambda a + b) \\
&= \lambda \phi_t(a) + \phi_t(b) \\
&= \lambda H_a(t) + H_b(t) \\
&= \lambda h(a)(t) + h(b)(t) \\
&= (\lambda h(a) + h(b))(t).
\end{aligned}$$

Doing similar calculations with the preservation involving the involution and products, we obtain that h is a $\star$ -homomorphism. Showing equivariance is a computation:

$$\begin{aligned}
\forall g \in G : \forall a \in A : \forall t \in [0, 1] : h(ga)(t) &= H_{ga}(t) \\
&= \pi_t(ga) \\
&= g\phi_t(a) \\
&= gH_a(t) \\
&= gh(a)(t),
\end{aligned}$$

which implies that h is also equivariant.

Proof. 2.9. Let $A, B \in G\text{-C}^*\text{Alg}$ and let $f_1, f_2, f_3 \in G\text{-C}^*\text{Alg}(A, B)$. We define:

$$s_A : A \rightarrow \text{Cyl}_G(A),$$

which takes $a \in A$ and maps it to the constant function $s_A(a)$ on $[0, 1]$ which takes the value a .

1. We define:

$$h : A \rightarrow \text{Cyl}_G(B), \quad a \mapsto h(a) \stackrel{\text{def}}{=} s_B(f_1(a)).$$

By definition, one observes that h is a homotopy from f_1 to itself so that proves reflexivity.

2. Let $f_1 \simeq_h f_2$ and let $h : A \rightarrow \text{Cyl}_G(B)$ be the homotopy from f_1 to f_2 . We define:

$$\forall a \in A : \forall t \in [0, 1] : H(a)(t) \stackrel{\text{def}}{=} h(a)(1 - t).$$

This defines a map $H : A \rightarrow \text{Cyl}_G(B)$. For each $a \in A$, we note that $H(a)$ is simply the precomposition of $t \mapsto 1 - t$ on $[0, 1]$ and the map $t \mapsto h(a)(t)$. Since both of these maps are continuous, we see that H maps into

$\text{Cyl}_G(B)$. The fact that H is an equivariant $\star$ -homomorphism follows from the fact that h is an equivariant $\star$ -homomorphism. Finally, the fact that this works follows by direct computation^a.

3. Let $f_1 \simeq_h f_2$ and $f_2 \simeq_h f_3$, with equivariant $\star$ -homomorphisms $h_1 : A \rightarrow \text{Cyl}_G(B)$ and $h_2 : A \rightarrow \text{Cyl}_G(B)$ implementing the respective homotopies. Now, let $a \in A$ and let us define the following map:

$$H(a) : [0, 1] \rightarrow A, \quad t \mapsto H(a)(t) \stackrel{\text{def}}{=} \begin{cases} h_1(a)(2t) & \text{if } 0 \leq t < \frac{1}{2}, \\ h_2(a)(2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1. \end{cases}$$

This is a function which is continuous everywhere except possibly at $t = \frac{1}{2}$; it is continuous even at that point because the left and right limits of $H(a)$ exist at $t = \frac{1}{2}$ & they coincide. So, we can define a map:

$$H : A \rightarrow \text{Cyl}_G(A), \quad a \mapsto H(a).$$

Due to the explicit definition of H , we see that it must be a $\star$ -homomorphism. Moreover, it gives precisely a homotopy from f_1 to f_3 which passes through f_2 ; this follows from its explicit definition.

This completes the proof.

^aOver here, we use the fact that h is a homotopy from f_1 to f_2 .

Proof. 2.10. This is clear from the definition of isomorphism and the reflexivity of homotopy.

Proof. 2.11. Let's note that:

$$\begin{aligned} \forall \phi \in \text{Cyl}_G(A) : (s \circ p_0^A)(\phi) &= s(p_0^A(\phi)) = s(\phi(0)), \\ \forall a \in A : (p_0^A \circ s)(a) &= p_0^A(s(a)) = s(a)(0) = a. \end{aligned}$$

We see that $p_0^A \circ s = \text{Id}_A$ and now, we need to construct a homotopy:

$$h : \text{Cyl}_G(A) \rightarrow \text{Cyl}_G(\text{Cyl}_G(A)).$$

So, let $\phi \in \text{Cyl}_G(A)$ and consider the following map:

$$h(\phi) : [0, 1] \rightarrow \text{Cyl}_G(A), \quad t \mapsto h(\phi)(t),$$

where we further define:

$$h(\phi)(t) : [0, 1] \rightarrow A, \quad r \mapsto h(\phi)(t)(r) \stackrel{\text{def}}{=} \phi(rt).$$

If h is a well-defined $\star$ -homomorphism, then it is easily seen that this formula gives the homotopy from $d_0 \circ s$ to Id_A . So, we just need to show that everything works. We see that $h(\phi)(t)$ is continuous for each $t \in [0, 1]$; this follows from the continuity of ϕ . So, $h(\phi)$ does, indeed, map into A^I . For the continuity of $h(\phi)$, we observe that it can be curried into a map $[0, 1] \times [0, 1] \rightarrow A$ with $(t, r) \mapsto \phi(rt)$ and this is continuous since ϕ is continuous. Now, uncurrying it produces a continuous map since every space involved above is a metric space. So, we conclude that the function h is well-defined and now, showing that it is a $\star$ -homomorphism comes down to verifying the axioms of a $\star$ -homomorphism and showing that the functions obtained on both sides agree. So, p_0^A is a homotopy inverse of s .

Proof. 2.12. Let's show that $\text{Id}_{\text{Cone}_G(A)}$ is homotopic to 0. For each $t \in [0, 1]$, we have a morphism:

$$\phi_t : \text{Cone}_G(A) \rightarrow \text{Cone}_G(A).$$

Let $f \in \text{Cone}_G(A)$. Then, we will define a function $\phi_t(f) \in \text{Cone}_G(A)$ as follows:

$$\forall s \in [0, 1] : \phi_t(f)(s) \stackrel{\text{def}}{=} f(ts).$$

It is clear that $\phi_t(f)$ is continuous since it is just a composition of two continuous functions and we see that:

$$\phi_t(f)(0) = f(t \cdot 0) = f(0) = 0,$$

so ϕ_t is a well-defined function into $\text{Cone}_G(A)$. Now, we need to check that it works.

1. Note that $\phi_0(f)(s) = f(s)$ for every $s \in [0, 1]$. That is, $\phi_0(f) = f$ so $\phi_0 = \text{Id}_{\text{Cone}_G(A)}$.
2. Note that $\phi_1(f)(s) = (1 - 1)f(s) = 0$. That is, $\phi_1(f) = 0$, so $\phi_1 = 0$.
3. Let us check that each ϕ_t is a $\star$ -homomorphism.

(a) Let $\lambda \in \mathbb{C}$ and $f, g \in \text{Cone}_G(A)$. Then:

$$\begin{aligned} \forall s \in [0, 1] : \phi_t(\lambda f + g)(s) &= (\lambda f + g)(ts) \\ &= (\lambda f)(ts) + g(ts) \\ &= \lambda f(ts) + g(ts) \\ &= \lambda \phi_t(f)(s) + \phi_t(g)(s) \\ &= (\lambda \phi_t(f) + \phi_t(g))(s), \end{aligned}$$

from which we conclude that ϕ_t is linear.

(b) Let $f, g \in \text{Cone}_G(A)$. Then:

$$\begin{aligned} \forall s \in [0, 1] : \phi_t(fg)(s) &= (fg)(ts) \\ &= f(ts)g(ts) \\ &= \phi_t(f)(s)\phi_t(g)(s) \\ &= (\phi_t(f)\phi_t(g))(s), \end{aligned}$$

which proves that ϕ_t preserves products.

(c) Let $f \in \text{Cone}_G(A)$. Then:

$$\begin{aligned}\forall s \in [0, 1] : \phi_t(f^*)(s) &= f^*(ts) \\ &= \overline{f(ts)} \\ &= \overline{\phi_t(f)(s)} \\ &= \phi_t(f)^*(s),\end{aligned}$$

which proves that ϕ_t preserves involutions.

If A was just a G -Banach algebra, then we have to show continuity of each ϕ_t . We have that:

$$\begin{aligned}\forall f \in \text{Cone}_G(A) : \|\phi_t(f)\| &= \sup_{s \in [0, 1]} |\phi_t(f)(s)| \\ &= \sup_{s \in [0, 1]} (1-t)|f(s)| \\ &= (1-t)\|f\|,\end{aligned}$$

which proves the result.

4. Let $f \in \text{Cone}_G(A)$ and define:

$$H_f : [0, 1] \rightarrow \text{Cone}_G(A), \quad t \mapsto H_f(t) \stackrel{\text{def}}{=} \phi_t(f).$$

We need to check that this is continuous. Let $t, s \in [0, 1]$ and consider the following estimate:

$$\begin{aligned}\|H_f(t) - H_f(s)\| &= \|\phi_t(f) - \phi_s(f)\| \\ &= \sup_{u \in [0, 1]} \|(1-t)f(u) - (1-s)f(u)\| \\ &\leq \sup_{u \in [0, 1]} |t-s| \cdot \|f(u)\| \\ &\leq |t-s| \cdot \|f\|.\end{aligned}$$

This calculation proves continuity.

This completes the proof.

Proof. 2.13. We will prove each statement one-by-one.

1. We have the following formal calculation:

$$\begin{aligned}g_1 &\simeq_h g_1 \circ \text{Id}_B \\ &\simeq_h g_1 \circ (f \circ g_2) \\ &\simeq_h (g_1 \circ f) \circ g_2 \\ &\simeq_h \text{Id}_A \circ g_2 \\ &\simeq_h g_2,\end{aligned}$$

which proves the desired result.

2. We have the following computation:

$$\begin{aligned}g_1 &\simeq_h \text{Id}_A \circ g_1 \\ &\simeq_h (g_2 \circ f) \circ g_1 \\ &\simeq_h g_2 \circ (f \circ g_1) \\ &\simeq_h g_2 \circ \text{Id}_B \\ &\simeq_h g_2,\end{aligned}$$

as was desired. To prove the last statement, we note that:

$$\begin{aligned}g_1 \circ f &\simeq_h g_2 \circ f \\ &\simeq_h \text{Id}_A,\end{aligned}$$

so we're done.

3. There are three cases to deal with:

- (a) If f and g are homotopy equivalences, then $g \circ f$ has a left and right homotopy inverse. So, $g \circ f$ has a homotopy inverse.
- (b) Let f and $g \circ f$ be homotopy equivalences. Let t_1 be the homotopy inverse of f and let t_2 be the homotopy inverse of $g \circ f$. This means that $f \circ t_2$ is the right homotopy inverse of g . On the other hand, $t_2 \circ g$ is the left homotopy inverse of f . So, $t_2 \circ g$ is homotopic to t_1 . Now, we have a calculation:

$$\begin{aligned}(f \circ t_2) \circ g &= f \circ (t_2 \circ g) \\ &\simeq_h f \circ t_1 \\ &\simeq_h \text{Id},\end{aligned}$$

which means that g has a left homotopy inverse and we're done.

- (c) Let g and $g \circ f$ be homotopy equivalences. Let t_1 be the homotopy inverse of g and let t_2 be the homotopy inverse of $g \circ f$. This means that f has a left homotopy inverse given by $t_1 \circ g$. Moreover, $f \circ t_2$ is the

right homotopy inverse of g , so this means that $f \circ t_2 \simeq_h t_1$. Now, we have a calculation:

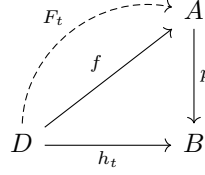
$$\begin{aligned} f \circ (t_2 \circ g) &= (f \circ t_2) \circ g \\ &\simeq_h t_1 \circ g \\ &\simeq_h \text{Id}, \end{aligned}$$

so f has a right homotopy inverse and we're done.

This completes the proof of property (3).

Proof. 2.14.

(1) $\Rightarrow$ (2): Indeed, we have the following diagram:



where $(F_t : D \rightarrow A)_{t \in [0,1]}$ is a homotopy with $F_0 = f$. Now, we define:

$$\forall x \in D : \forall t \in [0, 1] : H(x)(t) \stackrel{\text{def}}{=} F_t(x).$$

Clearly, $H : D \rightarrow \text{Cyl}_G(A)$ is a homotopy such that:

$$\begin{aligned} \forall x \in D : (p_0 \circ H)(x) &= p_0(H(x)) \\ &= H(x)(0) \\ &= F_0(x) \\ &= f(x), \end{aligned}$$

which means that $p_0 \circ H = f$. On the other hand:

$$\begin{aligned} \forall x \in D : \forall t \in [0, 1] : [(\text{Cyl}_G(p) \circ H)(x)](t) &= \text{Cyl}_G(p)(H(x))(t) \\ &= (p \circ H(x))(t) \\ &= p(H(x)(t)) \\ &= p(F_t(x)) \\ &= (p \circ F_t)(x) \\ &= h_t(x), \end{aligned}$$

as was desired.

(2) $\Rightarrow$ (1): This follows simply by reversing the argument above.

Proof. 2.15. This is a homotopy of p by the following computation:

$$\begin{aligned} p_0^A \circ H &= p_0^A \circ \text{Cyl}_G(p) \circ s \\ &= p \circ p_0^D \circ s \\ &= p \circ \text{Id}_D \\ &= p. \end{aligned}$$

Next, we observe that this is the required homotopy lift:

$$\begin{aligned} \text{Cyl}_G(f) \circ H &= \text{Cyl}_G(f) \circ \text{Cyl}_G(p) \circ s \\ &= \text{Cyl}_G(f \circ p) \circ s \\ &= \text{Cyl}_G(p_0^0 \circ h) \circ s \\ &= \text{Cyl}_G(p_0^0) \circ \text{Cyl}_G(h) \circ s \\ &= 0 \\ &= h, \end{aligned}$$

as was desired.

Proof. 2.16. Let us check that this is a homotopy of g :

$$\begin{aligned} p_0^A \circ H &= p_0^A \circ \text{Cyl}_G(f^{-1}) \circ h \\ &= f^{-1} \circ p_0^B \circ h \\ &= f^{-1} \circ (f \circ g) \\ &= g. \end{aligned}$$

Next, we show that this is our desired homotopy lift:

$$\begin{aligned} \text{Cyl}_G(f) \circ H &= \text{Cyl}_G(f) \circ (\text{Cyl}_G(f^{-1}) \circ h) \\ &= \text{Cyl}_G(\text{Id}_B) \circ h \\ &= \text{Id}_{\text{Cyl}_G(B)} \circ h \\ &= h, \end{aligned}$$

as was desired.

Proof. 2.17. Indeed, this is a computation as follows:

$$\begin{aligned}
 \forall t \in [0, 1] : (f \circ \pi_1)(j(\phi, \psi)(t)) &= f(\pi_1(j(\phi, \psi)(t))) \\
 &= f(\phi(t)) \\
 &= g(\psi(t)) \\
 &= g(\pi_2(j(\phi, \psi)(t))) \\
 &= (g \circ \pi_2)(j(\phi, \psi)(t)),
 \end{aligned}$$

so $j(\phi, \psi)$ is valued in P .

Proof. 2.18. We start with the following computation.

$$\begin{aligned}
 \forall (\phi, \psi) \in Q : (i \circ j)(\phi, \psi) &= i(j(\phi, \psi)) \\
 &= (\pi_1 \circ j(\phi, \psi), \pi_2 \circ j(\phi, \psi)) \\
 &= (\phi, \psi),
 \end{aligned}$$

which shows that $i \circ j = \text{Id}_Q$. Similarly:

$$\begin{aligned}
 \forall \phi \in \text{Cyl}_G(P) : \forall t \in [0, 1] (j \circ i)(\phi)(t) &= j(i(\phi))(t) \\
 &= j(\pi_1 \circ \phi, \pi_2 \circ \phi)(t) \\
 &= ((\pi_1 \circ \phi)(t), (\pi_2 \circ \phi)(t)) \\
 &= \phi(t),
 \end{aligned}$$

which implies that $j \circ i = \text{Id}_{\text{Cyl}_G(P)}$.

Proof. 2.19. Consider the following pullback diagram:

$$\begin{array}{ccc}
 P & \xrightarrow{\pi_A} & A \\
 \pi_C \downarrow & & \downarrow f \\
 C & \xrightarrow{g} & B
 \end{array}$$

where f is a GDR and let $F : A \rightarrow \text{Cyl}_G(A)$ be a $\star$ -homomorphism as in [Definition 2.34](#). Now, we want to define a $\star$ -homomorphism $G : P \rightarrow \text{Cyl}_G(P)$ which satisfies the defining conditions of a GDR. Recall that:

$$P = A \times_B C,$$

and define:

$$\forall (a, c) \in A \times_B C : \forall t \in [0, 1] : G(a, c)(t) \stackrel{\text{def}}{=} (F(a)(t), c).$$

Now, we need to check that this works.

1. Clearly, G defines a $\star$ -homomorphism; this follows from F being a $\star$ -homomorphism.
2. We need to ensure that G maps into $\text{Cyl}_G(P)$ so let $(a, c) \in P$ and let $t \in [0, 1]$. Then, we know that:

$$p_t^B \circ \text{Cyl}_G(f) \circ F = f.$$

So, now, we have the following computation:

$$\begin{aligned}
 f(F(a)(t)) &= (f \circ F(a))(t) \\
 &= p_t^B(f \circ F(a)) \\
 &= p_t^B(\text{Cyl}_G(f)(F(a))) \\
 &= (p_t^B \circ \text{Cyl}_G(f) \circ F)(a) \\
 &= f(a) \\
 &= g(c),
 \end{aligned}$$

which proves that G maps into $\text{Cyl}_G(P)$.

3. Now, we need to start verifying the actual conditions of a GDR.

(a) Let $(a, c) \in P$. Then, we have the following computation:

$$\begin{aligned}
 (p_0^P \circ G)(a, c) &= p_0^P(G(a, c)) \\
 &= G(a, c)(0) \\
 &= (F(a)(0), c) \\
 &= ((p_0^A \circ F)(a), c) \\
 &= (\text{Id}_A(a), c) \\
 &= (a, c),
 \end{aligned}$$

which proves that $p_0^P \circ G = \text{Id}_P$.

(b) Let $(a, c) \in P$ and let $t \in [0, 1]$. Then, we have the following computation:

$$\begin{aligned}
 (p_t^C \circ \text{Cyl}_G(\pi_C) \circ G)(a, c) &= p_t^C(\text{Cyl}_G(\pi_C)(G(a, c))) \\
 &= p_t^C(\pi_C \circ G(a, c)) \\
 &= (\pi_C \circ G(a, c))(t) \\
 &= \pi_C(G(a, c)(t)) \\
 &= \pi_C(F(a)(t), c) \\
 &= c \\
 &= \pi_C(a, c),
 \end{aligned}$$

as was desired.

(c) Observe that:

$$\ker(\pi_C) = \{(a, c) \in P : c = 0\}.$$

Now, we have the following computation:

$$\begin{aligned}
 \forall (a, c) \in \ker(\pi_C) : (p_1^P \circ G)(a, c) &= p_1^P(G(a, c)) \\
 &= G(a, c)(1) \\
 &= (F(a)(1), c).
 \end{aligned}$$

Since $(a, c) \in \ker(\pi_C)$, we have that $c = 0$. But if $c = 0$, then $g(c) = 0$ so $f(a) = 0$ and this means that $a \in \ker(f)$. Since $p_1^A \circ F$ maps $\ker(f)$ to 0, it follows that $(p_1^P \circ F)(\ker(\pi_C)) = 0$.

So, we see that π_C must be a GDR.

This completes the proof.

Proof. 2.20.

(1) $\Rightarrow$ (2): Let $(\phi_t : A \rightarrow B)_{t \in [0, 1]}$ be a homotopy from f to g . We define:

$$\psi : A \times [0, 1] \rightarrow B, (a, t) \mapsto \psi(a, t) \stackrel{\text{def}}{=} \phi_t(a).$$

For each $t \in [0, 1]$, we clearly have a $\star$ -homomorphism, with the evaluation at 0 and 1 giving f & g respectively. Next, we prove that ψ is continuous; let $(b, s) \in A \times [0, 1]$. Then, we compute as follows:

$$\begin{aligned}
 \forall (a, t) \in A \times [0, 1] : \|\psi(a, t) - \psi(b, s)\| &= \|\phi_t(a) - \phi_s(b)\| \\
 &= \|\phi_t(a) - \phi_t(b) + \phi_t(b) - \phi_s(b)\| \\
 &\leq \|\phi_t(a - b)\| + \|\phi_t(b) - \phi_s(b)\| \\
 &\leq \|a - b\| + \|\phi_t(b) - \phi_s(b)\|.
 \end{aligned}$$

So, what we get is the following estimate:

$$\forall (a, t) \in A \times [0, 1] : \|\psi(a, t) - \psi(b, s)\| \leq \|a - b\| + \|\phi_t(b) - \phi_s(b)\|.$$

If we allow for (a, t) to be close to (b, s) , then we see that the first term on the right-hand side above goes to 0 and the second term goes to zero due to the additional continuity condition on our homotopy.

(2) $\Rightarrow$ (1): Let $\psi : A \times [0, 1] \rightarrow B$ be a $\star$ -homotopy from f to g . We define, for each $t \in [0, 1]$, the map:

$$\phi_t : A \rightarrow B, a \mapsto \phi_t(a) \stackrel{\text{def}}{=} \psi(a, t).$$

This is a $\star$ -homomorphism, by the definition of a $\star$ -homotopy. Now, let $a \in A$ be fixed and define:

$$H_a : [0, 1] \rightarrow B, t \mapsto H_a(t) \stackrel{\text{def}}{=} \phi_t(a) = \psi(a, t).$$

The continuity of ψ implies that H_a is continuous, so we're done.

Proof. 2.21. This is clear, since we can write homotopies as families of morphisms or as a single map into a cylinder. The naturality of the square in statement (2) gives us a correspondence between the two diagrams of interest.

Proof. 2.22.

(1) $\Rightarrow$ (2): Let $\phi : A \times [0, 1] \rightarrow \mathcal{Q}(H)$ be a $\star$ -homotopy and let $f \in C^*\text{Alg}(A, \mathcal{B}(H))$ such that:

$$\pi \circ f = \phi(\cdot, 0).$$

We can construct a homotopy $(h_t : A \rightarrow \mathcal{Q}(H))_{t \in [0, 1]}$ from ϕ so that we have a lifting problem:

$$\begin{array}{ccc}
 & & \mathcal{B}(H) \\
 & \nearrow f & \downarrow \pi \\
 A & \xrightarrow{(h_t)_{t \in [0, 1]}} & \mathcal{Q}(H)
 \end{array}$$

and since A is Schochet, we may simply lift our homotopy to some homotopy $(H_t : A \rightarrow \mathcal{B}(H))_{t \in [0, 1]}$. Now, this can be turned into a $\star$ -homotopy which satisfies the requirements of the C^* -CHP property.

(2) $\Rightarrow$ (1): We simply run the argument for (1) $\Rightarrow$ (2) in reverse.

Proof. 2.23. Let $t = 0$. On the generator u , we have that:

$$\begin{aligned} H(u)(0) &= s(h_0(u)) \\ &= f(u), \end{aligned}$$

while on the generator 1, we have that:

$$\begin{aligned} H(1)(0) &= S_0 \\ &= f(1), \end{aligned}$$

which proves that $H_0 = f$. Next, we have the following computation:

$$\begin{aligned} \forall t \in [0, 1] : \pi(H(u)(t)) &= \pi(s(h_t(u))) \\ &= (\pi \circ s)(h_t(u)) \\ &= h_t(u), \end{aligned}$$

and similarly, we see that:

$$\begin{aligned} \forall t \in [0, 1] : \pi(H(1)(t)) &= \pi(S_t) \\ &= h_t(1), \end{aligned}$$

so we conclude that $\pi \circ H_t = h_t$.

Proof. 2.24. By construction, the unitized family of maps still is a family of $\star$ -homomorphisms so it is sufficient to show the point-norm continuity condition. Let $x = a + \lambda 1_{A^\sim} \in A^\sim$ and define the following map:

$$H_x : [0, 1] \rightarrow B^\sim, \quad t \mapsto H_x(t) \stackrel{\text{def}}{=} \phi_t^\sim(x) = \phi_t(a) + \lambda 1_{B^\sim}.$$

Now, a direct calculation which compares $H_x(t)$ with $H_x(s)$ for $t, s \in [0, 1]$ will show that the continuity of H_x reduces to the continuity of the map $t \mapsto H_t(a)$ and we know that this is continuous, so we're done.

Proof. 2.25. We will verify this for the $C^\star$ -algebra A . There are three statements to prove:

1. Let us prove that ψ^A is a $\star$ -homomorphism. Indeed, linearity is clear. Next:

$$\begin{aligned} \forall a, a' \in A : \psi^A(aa') &= P^1 f(aa', 0) P^1 \\ &= P^1 f(a, 0) f(a', 0) P^1 \\ &= P^1 f(a, 0) f(1_A, 0) f(1_A, 0) f(a', 0) P^1 \\ &= P^1 f(a, 0) P^1 \cdot P^1 f(a', 0) P^1 \\ &= \psi^A(a) \psi^A(a'). \end{aligned}$$

Next, we have to show that this preserves the involution:

$$\begin{aligned} \forall a \in A : \psi^A(a^\star) &= P^1 f(a^\star, 0) P^1 \\ &= P^1 f(a, 0)^\star P^1 \\ &= (P^1 f(a, 0) P^1)^\star \\ &= \psi^A(a)^\star, \end{aligned}$$

which proves that ψ^A is a $\star$ -homomorphism.

2. Let us prove that each Φ_t^A is a $\star$ -homomorphism. Linearity is clear and let's show products are preserved:

$$\begin{aligned} \forall a, a' \in A : \Phi_t^A(aa') &= \pi(U_1(t))^\star h_t(aa', 0) \pi(U_1(t)) \\ &= \pi(U_1(t))^\star h_t(a, 0) h_t(a', 0) \pi(U_1(t)) \\ &= \pi(U_1(t))^\star h_t(a, 0) h_t(1_A, 0) h_t(a', 0) \pi(U_1(t)) \\ &= \pi(U_1(t))^\star h_t(a, 0) Q_1(t) h_t(a', 0) \pi(U_1(t)) \\ &= \pi(U_1(t))^\star h_t(a, 0) \pi(P_t^1) h_t(a', 0) \pi(U_1(t)) \\ &= \pi(U_1(t))^\star h_t(a, 0) \pi(U_1(t)) \pi(U_1(t))^\star h_t(a', 0) \pi(U_1(t)) \\ &= \Phi_t^A(a) \Phi_t^A(a'), \end{aligned}$$

where we have used some of the facts derived earlier. Next, let us show that this preserves the involution:

$$\begin{aligned} \forall a \in A : \Phi_t^A(a^\star) &= \pi(U_1(t))^\star h_t(a^\star, 0) \pi(U_1(t)) \\ &= \pi(U_1(t))^\star h_t(a, 0)^\star \pi(U_1(t)) \\ &= (\pi(U_1(t))^\star h_t(a, 0) \pi(U_1(t)))^\star \\ &= \Phi_t^A(a)^\star, \end{aligned}$$

which proves that Φ_t^A is a $\star$ -homomorphism.

3. Let us show that $(\Phi_t^A)_{t \in [0, 1]}$ is a homotopy. This follows from the fact that $t \mapsto h_t(a, 0)$ is continuous for each $a \in A$, the fact that $(U_1(t))_{t \in [0, 1]}$ is a path, the fact that the involution is continuous and the fact that multiplication in a $C^\star$ -algebra which is continuous.

This proves the claim. Since things are defined in the “same” way for B , we are done.

Proof. 2.26. Again, let us show this for the C^* -algebra A . So, let $t = 0$. Then:

$$\begin{aligned}\forall a \in A : \Phi_0^A(a) &= \pi(U_1(0))^* h_0(a, 0) \pi(U_1(0)) \\ &= \pi(U_1(0))^* \pi(f(a, 0)) \pi(U_1(0)) \\ &= \pi(U_1(0))^* f(a, 0) U_1(0) \\ &= \pi(P_0^1 f(a, 0) P_0^1) \\ &= \pi(\psi^A(a)) \\ &= (\pi \circ \psi^A)(a),\end{aligned}$$

as was desired. Since these maps are defined “in the same way” for B , we are done.

Proof. 2.27. Let $t = 0$. We see that:

$$\begin{aligned}\forall (a, b) \in A \oplus B : \Psi_0(a, b) &= U_1(0) \Psi_0^A(a) U_1(0)^* + U_2(0) \Psi_0^B(b) U_2(0)^* \\ &= P_0^1 \psi^A(a) P_0^1 + P_0^2 \psi^B(b) P_0^2 \\ &= P_0^1 f(a, 0) P_0^1 + P_0^2 f(0, b) P_0^2 \\ &= f(1_A, 0) f(a, 0) f(1_A, 0) + f(0, 1_B) f(0, b) f(0, 1_B) \\ &= f(a, 0) + f(0, b) \\ &= f(a, b),\end{aligned}$$

which implies that $\Psi_0 = f$. Next, let $t \in [0, 1]$ and $(a, b) \in A \oplus B$. We have a computation:

$$\begin{aligned}(\pi \circ \Psi_t)(a, b) &= \pi(\Psi_t(a, b)) \\ &= \pi(U_1(t) \Psi_t^A(a) U_1(t)^* + U_2(t) \Psi_t^B(b) U_2(t)^*) \\ &= \pi(U_1(t)) \pi(\Psi_t^A(a)) \pi(U_1(t))^* + \pi(U_2(t)) \pi(\Psi_t^B(b)) \pi(U_2(t))^* \\ &= \pi(U_1(t)) \Phi_t^A(a) \pi(U_1(t))^* + \pi(U_2(t)) \Phi_t^B(b) \pi(U_2(t))^* \\ &= \pi(U_1(t) U_1(t)^*) h_t(a, 0) \pi(U_1(t) U_1(t)^*) + \pi(U_2(t) U_2(t)^*) h_t(0, b) \pi(U_2(t) U_2(t)^*) \\ &= \pi(P^1) h_t(a, 0) \pi(P^1) + \pi(P^2) h_t(0, b) \pi(P^1) \\ &= h_t(1_A, 0) h_t(a, 0) h_t(1_A, 0) + h_t(0, 1_B) h_t(0, b) h_t(0, 1_B) \\ &= h_t(a, 0) + h_t(0, b) \\ &= h_t(a, b),\end{aligned}$$

which shows that $(\Psi_t)_{t \in [0, 1]}$ is a solution to the original lifting problem. In the computation above, we just put in the formulas for $\Phi_t^A(a)$ and $\Phi_t^B(b)$, and simplified the resulting expression using the facts derived earlier.

Section 3

Proof. 3.1. This involves just reversing the arrows in the diagram associated with the retraction.

Proof. 3.2. We’ll verify each statement one-by-one.

1. Consider the following diagram of three composable arrows:

$$w \xrightarrow{f} x \xrightarrow{g} y \xrightarrow{h} z$$

where $g \circ f$ and $h \circ g$ are isomorphisms. This implies that g has a right & left inverse so it is an isomorphism. This means that both h and f are isomorphisms. This implies that $h \circ g \circ f$ is an isomorphism and we’re done.

2. Consider the following diagram of two composable arrows:

$$x \xrightarrow{f} y \xrightarrow{g} z$$

where any two of f, g and $g \circ f$ are in W . Now, we have to deal with a few cases:

- (a) Suppose that $f, g \in W$. Then, we consider the following diagram of composable arrows:

$$x \rightarrow y \rightarrow y \rightarrow z,$$

where the first map is f , the second map is the identity and the third is g . In that case, we see that $\text{Id}_y \circ f$ and $g \circ \text{Id}_y$ belong to W , so we can use the 2-out-of-6 property to conclude that $g \circ \text{Id}_y \circ f = g \circ f \in W$.

- (b) Let $f, g \circ f \in W$. Then, we consider the following diagram of composable arrows:

$$x \rightarrow x \rightarrow y \rightarrow z,$$

where the first map is the identity, the second is f and the third is g . Now, we note that $f \circ \text{Id}_x \in W$ and $g \circ f \in W$ so the 2-out-of-6 property allows us to conclude that $g \in W$ (among other facts, of course).

- (c) Let $g, g \circ f \in W$. Then, we consider the following diagram of composable arrows:

$$x \rightarrow y \rightarrow z \rightarrow z,$$

where the first map is f , the second is g and the third is the identity. So, we note that $g \circ f \in W$ and $\text{Id}_z \circ g \in W$ so the 2-out-of-6 property lets us conclude that $f \in W$ and we’re done.

This completes the proof of the given statement.

3. Let f be an isomorphism. Then, we consider the following diagram of composable arrows:

$$\bullet \rightarrow \bullet \rightarrow \bullet \rightarrow \bullet,$$

where the first arrow is f , the second one is f^{-1} and the third is f . Observe that $f^{-1} \circ f$ and $f \circ f^{-1}$ belong to W since they are identity arrows. By the 2-out-of-6 property, we conclude $f \in W$ and so does f^{-1} .

This completes the proof of the overall result.

Proof. 3.3. Let's suppose that we have a diagram:

$$w \xrightarrow{f} x \xrightarrow{g} y \xrightarrow{h} z$$

where $g \circ f$ and $h \circ g$ lies in W . Then, we note that we have 6 morphisms:

1. $F(f)$,
2. $F(g)$,
3. $F(h)$,
4. $F(g \circ f) = F(g) \circ F(f)$,
5. $F(h \circ g) = F(h) \circ F(g)$,
6. $F(h \circ g \circ f) = F(h) \circ F(g) \circ F(f)$,

where $F(g \circ f)$ and $F(h \circ g)$ are isomorphisms. But now, we can apply the 2-out-of-6 property for the isomorphisms in $\mathcal{D}$ to conclude that all 6 of the morphisms listed above are isomorphisms. But this implies that $f, g, h, h \circ g \circ f \in W$ so W satisfies the 2-out-of-6 property.

Proof. 3.4. The localization functor with respect to W satisfies the hypothesis of preceding result, so we're done.

Proof. 3.5. This is clear by the definition, since taking opposites just reverses the directions of the arrows.

Proof. 3.6. Since $\mathcal{C}$ has a final object 0 , we see that it admits empty products. Next, it's enough to show that $\mathcal{C}$ admits binary products so let $x, y \in \mathcal{C}$. Then, we consider the diagram:

$$x \rightarrow 0 \leftarrow y$$

If we take the pullback of this diagram, then we get $x \times y$ by the universal property of the pullback. Indeed, $\mathcal{C}$ is closed under such pullbacks because both maps in this pullback diagram are fibrations.

Proof. 3.7. Let us verify axioms (2) – (4) of Definition 3.31, since axiom (1) is given.

Axiom 2: This is clear.

Axiom 3: This is clear.

Axiom 4: Let us verify the Gluing Lemma. We consider the following commutative diagram in $\mathcal{C}$:

$$\begin{array}{ccccc} b & \xrightarrow{\text{red}} & a & \xleftarrow{\text{blue}} & c \\ \downarrow & & \downarrow & & \downarrow \\ b' & \xrightarrow{\text{red}} & a' & \xleftarrow{\text{blue}} & c' \end{array}$$

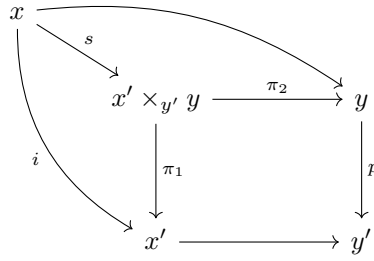
where the red arrows are fibrations and the vertical arrows are isomorphisms. Let's extend the diagram:

$$\begin{array}{ccccc} & & b \times_a c & & \\ & \swarrow & & \searrow & \\ b & \xrightarrow{\text{red}} & a & \xleftarrow{\text{blue}} & c \\ \downarrow & & \downarrow & & \downarrow \\ b' & \xrightarrow{\text{red}} & a' & \xleftarrow{\text{blue}} & c' \\ & \swarrow & & \searrow & \\ & & b' \times_{a'} c' & & \end{array}$$

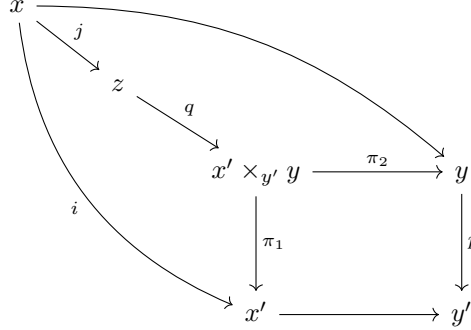
So, the two triangles are the pullback diagrams and so, the entire diagram above commutes. Now, the three vertical arrows are isomorphisms, so we can compose the inverses $b' \rightarrow b$ and $c' \rightarrow c$ with $b' \times_{a'} c' \rightarrow b'$ & $b' \times_{a'} c' \rightarrow c'$. So, this projects $b' \times_{a'} c'$ onto b and c respectively while keeping the diagram commutative, so we obtain a natural map from $b' \times_{a'} c' \rightarrow b \times_a c$. This map can be checked to be the explicit inverse of the natural map $b \times_a c \rightarrow b' \times_{a'} c'$, essentially using the universal property of the pullback. We do not carry out this computation explicitly as it is besides the point.

This completes the proof.

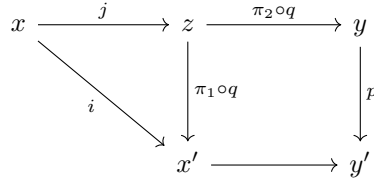
Proof. 3.8. Consider the following pullback diagram:



Now, we want to factorize s as follows:

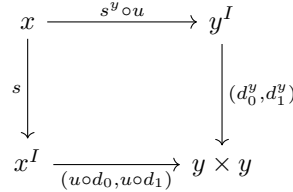


where $s = q \circ j$, with q being a fibration and j being a weak equivalence. Now, we form the following diagram:

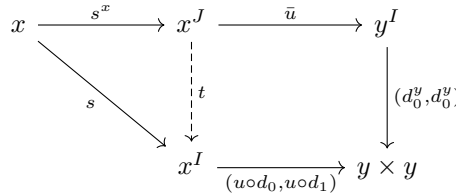


Define $t = \pi_1 \circ q$. Since π_1 and q are fibrations, where π_1 is a fibration because p is a fibration & fibrations are preserved by base extensions. So, t is a fibration. On the other hand, $i = t \circ j$ is a weak equivalence and j is a weak equivalence, so it follows that t is a weak equivalence and we're done.

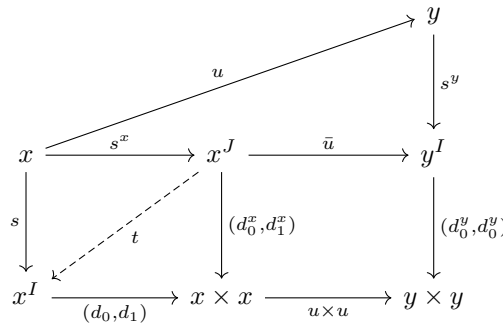
Proof. 3.9. The key is to use the boundary of the diagram given in the statement of this result and apply the preceding result repeatedly until we get the desired diagram. We start with the following situation:



Since s is a weak equivalence and (d_0^y, d_1^y) is a fibration, we can apply the preceding result. This gives us the diagram:



where t is a trivial fibration. We re-draw this diagram as follows:



where we note that (d_0^x, d_1^x) is defined as $(d_0, d_1) \circ t$. This implies that the diagram above commutes as a whole; if we were to remove (d_0^x, d_1^x) , then we simply obtain the diagrams we had previously and those certainly commute.

Proof. 3.10. If f, g are homotopic, then there is a weak equivalence t such that:

$$f \circ t \simeq_{h,r} g \circ t.$$

So, applying the first statement, we see that:

$$\begin{aligned}\gamma(f) \circ \gamma(t) &= \gamma(f \circ t) \\ &= \gamma(g \circ t) \\ &= \gamma(g) \circ \gamma(t),\end{aligned}$$

where we can cancel $\gamma(t)$ since it is an isomorphism. So, it is sufficient for us to prove the first statement. In that case, let y^I be a path space and let $h : x \rightarrow y^I$ be a right-homotopy from f to g . Then, we see that:

$$\begin{aligned}f &= d_0 \circ h, \\ g &= d_1 \circ h.\end{aligned}$$

Now, we have a calculation^a:

$$\begin{aligned}\gamma(f) &= \gamma(d_0 \circ h) \\ &= \gamma(d_0) \circ \gamma(h) \\ &= \gamma(d_0) \circ \gamma(d_1)^{-1} \circ \gamma(d_1) \circ \gamma(h) \\ &= \gamma(d_0) \circ \gamma(d_1)^{-1} \circ \gamma(d_1 \circ h) \\ &= \gamma(d_0) \circ \gamma(d_1)^{-1} \circ \gamma(g).\end{aligned}$$

Now, we need to show that:

$$\gamma(d_0) = \gamma(d_1).$$

By definition of the path space, there is a morphism $s : x \rightarrow x^I$ as part of the data of a path space such that:

$$\begin{aligned}d_0 \circ s &= \text{Id}_x, \\ d_1 \circ s &= \text{Id}_x.\end{aligned}$$

Applying γ to both sides in both equations, we have that:

$$\begin{aligned}\gamma(d_0) \circ \gamma(s) &= \gamma(d_0 \circ s) \\ &= \gamma(\text{Id}_x) \\ &= \gamma(d_1) \circ \gamma(s).\end{aligned}$$

Since γ inverts weak equivalences and s is a weak equivalence, we can cancel $\gamma(s)$ and we're done.

^aThis is justified since d_1 is a trivial fibration so its image in $\mathbf{Ho}(\mathcal{C})$ is an isomorphism.

Proof. 3.11. We simply apply **Claim 3.3** to the quotient functors.

Proof. 3.12. We start with the following computation:

$$\begin{aligned}\alpha_0^x \circ k &= \alpha_0^x \circ (w \circ \hat{h}) \\ &= (\alpha_0^x \circ w) \circ \hat{h} \\ &= \bar{d}_f \circ \hat{h} \\ &= g.\end{aligned}$$

This shows that k is a homotopy of g . Next, we have the following computation:

$$\begin{aligned}T(f) \circ k &= T(f) \circ (w \circ \hat{h}) \\ &= (T(f) \circ w) \circ \hat{h} \\ &= \bar{f} \circ \hat{h} \\ &= h,\end{aligned}$$

as was desired.

Section 4

Proof. 4.1. Let's start with the following computation:

$$\begin{aligned}\forall \phi \in \text{Cyl}(A) : (j \circ i)(\phi) &= j(i(\phi)) \\ &= j(\phi(0), \phi) \\ &= \phi,\end{aligned}$$

so $j \circ i = \text{Id}_{\text{Cyl}(A)}$. Similarly, we have that:

$$\begin{aligned}\forall (a, \phi) \in \text{cyl}(\text{Id}_A) : (i \circ j)(a, \phi) &= i(j(a, \phi)) \\ &= i(\phi) \\ &= (\phi(0), \phi) \\ &= (a, \phi)\end{aligned}$$

where we use actual concrete form of the mapping cylinder.

Proof. 4.2. This is clear if one goes through the axioms, essentially because both $\mathcal{C}$ and $\mathcal{D}$ are admissible.

Proof. 4.3. We define two obvious maps:

$$\phi : Q \rightarrow I, \quad f \mapsto \phi(f) \stackrel{\text{def}}{=} f(0),$$

$$\psi : I \rightarrow Q, \quad j \mapsto \psi(j),$$

where $\psi(j)$ is the constant function j on $[0, 1]$. These maps are well-defined and we note that:

$$\forall f \in Q : (\psi \circ \phi)(f) = f(0),$$

and:

$$\forall j \in I : (\phi \circ \psi)(j) = j,$$

which means that $\phi \circ \psi = \text{Id}_I$. Now, let $f \in Q$ and define:

$$h(f) : [0, 1] \rightarrow Q, \quad t \mapsto (s \mapsto h(f)(t)(s) = f(ts)).$$

By currying, one can see that $h(f)$ is a continuous map (this can also be proved by hand). So, we have constructed a function $h : Q \rightarrow \text{Cyl}(Q)$ and this will be a $\star$ -homomorphism. This is a homotopy from $\psi \circ \phi$ to Id_Q .

Proof. 4.4. We will go through all of the properties one-by-one.

1. ϕ is clearly continuous.

$$2. \quad \phi(0) = \frac{1}{2}(1_A + u) + \frac{1}{2}(u - 1_A) = u.$$

$$3. \quad \phi(1) = \frac{1}{2}(1_A + u) + \frac{1}{2}(-1)(u - 1_A) = 1_A.$$

4. Let us show that for $t \in [0, 1]$, we have that $\phi(t)$ is a unitary. Indeed, we have a calculation:

$$\begin{aligned} \phi(t)\phi(t)^* &= \left(\frac{1}{2}(1_A + u) + \frac{1}{2}\exp(\pi it)(u - 1_A) \right) \left(\frac{1}{2}(1_A + u) + \frac{1}{2}\exp(-\pi it)(u - 1_A) \right) \\ &= \frac{1}{4}(1_A + u)^2 + \frac{1}{4}\exp(\pi it)(u^2 - 1_A) + \frac{1}{2}\exp(-\pi it)(u^2 - 1_A) + \frac{1}{4}(u - 1_A)^2 \\ &= \frac{1}{4}(1_A + u)^2 + \frac{1}{4}(u - 1_A)^2 \\ &= \frac{1}{4}(1_A^2 + 2u + u^2 + u^2 - 2u + 1_A^2) \\ &= 1_A, \end{aligned}$$

as was desired. A similar calculation reveals that:

$$\forall t \in [0, 1] : \phi(t)^* \phi(t) = 1_A,$$

as was desired.

This proves the result.

Proof. 4.5. Linearity is clear. To show that this is product-preserving, we have a computation:

$$\begin{aligned} \forall x, y \in B : \text{Ad}(u)(xy) &= u(xy)u^* \\ &= u(xu^*uy)u^* \\ &= (uxu^*)(uyu^*) \\ &= \text{Ad}(u)(x)\text{Ad}(u)(y), \end{aligned}$$

where we have used the fact that u is an isometry in the middle of the computation. Next, we have:

$$\begin{aligned} \forall x \in B : \text{Ad}(u)(x^*) &= ux^*u^* \\ &= u(ux)^* \\ &= (uxu^*)^* \\ &= \text{Ad}(u)(x)^*, \end{aligned}$$

which shows that $\text{Ad}(u)$ preserves the involution.

Proof. 4.6. Let $(h_t : A \rightarrow B)_{t \in [0, 1]}$ be a homotopy from f to g . We define:

$$\forall t \in [0, 1] : H_t \stackrel{\text{def}}{=} h_t \otimes \text{Id}_C.$$

This gives us a family $(H_t : A \otimes C \rightarrow B \otimes C)_{t \in [0, 1]}$ of $\star$ -homomorphisms which start at $f \otimes \text{Id}_C$ and end at $g \otimes \text{Id}_C$. It remains to show that this is a point-norm continuous family. It is sufficient to do this on the simple tensors of $A \otimes_{\mathbb{C}} B$, since we can extend it to elements of $A \otimes C$ by a approximation. Consider the map:

$$\phi : [0, 1] \rightarrow B \otimes C, \quad t \mapsto \phi(t) \stackrel{\text{def}}{=} H_t(a \otimes c) = h_t(a) \otimes c.$$

Now, let us prove continuity of this map by an estimate as follows:

$$\begin{aligned} \forall t, s \in [0, 1] : \|\phi(t) - \phi(s)\| &= \|h_t(a) \otimes c - h_s(a) \otimes c\| \\ &= \|(h_t(a) - h_s(a)) \otimes c\| \\ &= \|h_t(a) - h_s(a)\| \cdot \|c\|, \end{aligned}$$

where we have used the fact that the tensor norm is a cross norm. Now, we use the continuity of the map $r \mapsto h_r(a)$ (which is continuous since $(h_t : A \rightarrow B)_{t \in [0, 1]}$ is a homotopy) to conclude that when t and s are close, then $\phi(t)$ and $\phi(s)$ are close because $h_t(a)$ and $h_s(a)$ are close. So, this proves continuity.

Proof. 4.7. We recall that π_0 is the unitization of ψ_0 . So:

$$\begin{aligned}
\pi_0(v) &= 1_{\widehat{\mathcal{T}}} + \psi_0(v - 1) \\
&= 1 \otimes 1 + j(v - 1) + \psi_1(v - 1) \\
&= 1 \otimes 1 + e_{00} \otimes (v - 1) + (v \otimes 1)((v - 1) \otimes 1)(v^* \otimes 1) \\
&= 1 \otimes 1 + e_{00} \otimes (v - 1) + v(v - 1)v^* \otimes 1 \\
&= 1 \otimes 1 + (1 - vv^*) \otimes (v - 1) + (v^2v^* - vv^*) \otimes 1 \\
&= 1 \otimes 1 - 1 \otimes 1 + 1 \otimes v - vv^* \otimes v + vv^* \otimes 1 + v^2v^* \otimes 1 - vv^* \otimes 1 \\
&= 1 \otimes v - vv^* \otimes v + v^2v^* \otimes 1 \\
&= (1 - vv^*) \otimes v + v^2v^* \otimes 1 \\
&= e_{00} \otimes v + v^2v^* \otimes 1,
\end{aligned}$$

as was desired.

Proof. 4.8. We just need to compute the right-hand side using our formula:

$$\begin{aligned}
u_0(v \otimes 1) &= (v^2(v^*)^2 \otimes 1 + e_{00}v^* \otimes v + ve_{00} \otimes v^* + e_{00} \otimes e_{00})(v \otimes 1) \\
&= v^2v^* \otimes 1 + e_{00} \otimes v + ve_{00}v \otimes v^* + e_{00}v \otimes e_{00} \\
&= v^2v^* \otimes 1 + e_{00} \otimes v \\
&= \pi_0(v).
\end{aligned}$$

where we use the fact that $e_{00}v = 0$ in the middle of the computation to simplify it significantly.

Proof. 4.9. The self-adjointness of u_0 is actually obvious and is deduced just by staring at the formula. So, the non-trivial bit is showing that $u_0^2 = 1$. In fact, this is a direct calculation but let us simplify it through the following long list of 1-line computations as follows:

$$\begin{aligned}
(v^2(v^*)^2 \otimes 1)(v^2(v^*)^2 \otimes 1) &= v^2(v^*)^2 \otimes 1 \\
(v^2(v^*)^2 \otimes 1)(e_{00}v^* \otimes v) &= 0 \\
(v^2(v^*)^2 \otimes 1)(ve_{00} \otimes v^*) &= 0 \\
(v^2(v^*)^2 \otimes 1)(e_{00} \otimes e_{00}) &= 0 \\
(e_{00}v^* \otimes v)(v^2(v^*)^2 \otimes 1) &= 0 \\
(e_{00}v^* \otimes v)(e_{00} \otimes v^*) &= 0 \\
(e_{00}v^* \otimes v)(ve_{00} \otimes v^*) &= e_{00}^2 \otimes vv^* = e_{00} \otimes vv^* \\
(e_{00}v^* \otimes v)(e_{00} \otimes e_{00}) &= 0 \\
(ve_{00} \otimes v^*)(v^2(v^*)^2 \otimes 1) &= 0 \\
(ve_{00} \otimes v^*)(e_{00}v^* \otimes v) &= ve_{00}v^* \otimes 1 \\
(ve_{00} \otimes v^*)(ve_{00} \otimes v^*) &= 0 \\
(ve_{00} \otimes v^*)(e_{00} \otimes e_{00}) &= 0 \\
(e_{00} \otimes e_{00})(v^2(v^*)^2 \otimes 1) &= 0 \\
(e_{00} \otimes e_{00})(e_{00}v^* \otimes v) &= 0 \\
(e_{00} \otimes e_{00})(ve_{00} \otimes v^*) &= 0 \\
(e_{00} \otimes e_{00})(e_{00} \otimes e_{00}) &= e_{00} \otimes e_{00},
\end{aligned}$$

where we have repeatedly used the fact that e_{00} is a projection and that:

$$e_{00}v = v^*e_{00} = 0,$$

in these 16 computations. Now, this means that:

$$u_0^2 = v^2(v^*)^2 \otimes 1 + e_{00} \otimes vv^* + ve_{00}v^* \otimes 1 + e_{00} \otimes e_{00}.$$

Let us simplify each term on the right-hand side. Here's the simplification of the second term:

$$\begin{aligned}
e_{00} \otimes vv^* &= 1 - vv^* \otimes vv^* \\
&= 1 \otimes vv^* - vv^* \otimes vv^*.
\end{aligned}$$

Here's the simplification of the third term:

$$\begin{aligned}
ve_{00}v^* \otimes 1 &= vv^* - v^2(v^*)^2 \otimes 1 \\
&= vv^* \otimes 1 - v^2(v^*)^2 \otimes 1.
\end{aligned}$$

Finally, we have a simplification of the third term:

$$\begin{aligned}
e_{00} \otimes e_{00} &= 1 - vv^* \otimes 1 - vv^* \\
&= 1 \otimes 1 - vv^* \otimes 1 - 1 \otimes vv^* + vv^* \otimes vv^*.
\end{aligned}$$

Combining everything and noting that everything cancels out except for $1 \otimes 1$ and so, u_0^2 is the identity.

Proof. 4.10.

(1) $\Rightarrow$ (2): Consider the following commutative diagram:

$$\begin{array}{ccccccc}
 x & \xrightarrow{f} & y & \longrightarrow & 0 & \longrightarrow & x[1] \\
 \downarrow \text{Id}_x & & \downarrow f^{-1} & & \downarrow & & \downarrow \\
 x & \xrightarrow{\text{Id}_x} & x & \longrightarrow & 0 & \longrightarrow & x[1]
 \end{array}$$

Note that the rows are exact triangles in $\mathcal{T}$. Since the bottom row belongs to the set of exact triangles of $\mathcal{T}$ by assumption and since the set of exact triangles is closed under isomorphisms, we see that the top row must also be an exact triangle.

(2) $\Rightarrow$ (1): We have the following diagram:

$$\begin{array}{ccccccc}
 x & \xrightarrow{f} & y & \longrightarrow & 0 & \longrightarrow & x[1] \\
 \downarrow \text{Id}_x & & & & \downarrow & & \downarrow \\
 x & \xrightarrow{\text{Id}_x} & x & \longrightarrow & 0 & \longrightarrow & x[1]
 \end{array}$$

Now, we use Proposition 4.42 to find a morphism $g : y \rightarrow x$ such that the above is a commutative diagram with g filled into it. This implies that $g \circ f = \text{Id}_x$. Now, consider the following diagram:

$$\begin{array}{ccccccc}
 y & \longrightarrow & y & \longrightarrow & 0 & \longrightarrow & \Sigma y \\
 & & \downarrow & & \downarrow & & \\
 x & \xrightarrow{f} & y & \longrightarrow & 0 & \longrightarrow & \Sigma x
 \end{array}$$

By the aforementioned result, we can find a morphism $h : y \rightarrow x$ such that $f \circ h = \text{Id}_y$. But this means that f is a split epimorphism and a split epimorphism, from which it follows that it is an isomorphism.

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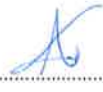
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Selbstständigkeitserklärung

Ich erkläre hiermit, dass ich die vorliegende Arbeit selbstständig verfasst und noch nicht für andere Prüfungen eingereicht habe. Sämtliche Quellen einschließlich Internetquellen, die unverändert oder abgewandelt wiedergegeben werden, insbesondere Quellen für Texte, Grafiken, Tabellen, Bilder sowie die Nutzung von generativer Künstlicher Intelligenz für die Erstellung von Texten und Abbildungen, sind als solche kenntlich gemacht. Mir ist bekannt, dass bei Verstößen gegen diese Grundsätze ein Verfahren wegen Täuschungsversuchs bzw. Täuschung eingeleitet wird.

Berlin, 30/07/2026, 

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